STAT 444: Advanced Regression

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Part I

Foundations

Chapter 1

Linear Algebra for Regression

Assuming that you are a quant or someone that aspires to be one, you have probably encountered the bulk of the concepts covered in these introductory sections. If you are confident in Linear Algebra I & II, feel free to skip these chapters and move directly to Ordinary Least Squares.

1.1 Vector spaces, subspaces, and projections

You have most likely encountered vector spaces back in Linear Algebra I & II. I attended the **University of Waterloo** so I am referring to MATH136 and MATH235/245. In MATH235, you most likely recall the 10 axioms of which vector spaces retain in addition to an operation of addition and scalar multiplication.

Definition 1.1 (Vector Space over $\mathbb{R}$)**.** A *Vector Space* over $\mathbb{R}$ is a set $\mathbb{V}$ with an operation of addition and scalar multiplication such that for all $\vec{x}, \vec{y}, \vec{z} \in \mathbb{V}$, and scalars $c, d \in \mathbb{R}$

(i) If $\vec{x}, \vec{y} \in \mathbb{V}$, then $\vec{x} + \vec{y} \in \mathbb{V}$.
(ii) For all $\vec{x}, \vec{y} \in \mathbb{V}$, $\vec{x} + \vec{y} = \vec{y} + \vec{x}$.
(iii) For all $\vec{x}, \vec{y}, \vec{z} \in \mathbb{V}$, $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$.
(iv) There exists a vector $\vec{0} \in \mathbb{V}$, called the *zero vector*, such that $\vec{x} + \vec{0} = \vec{x}$ for every $\vec{x} \in \mathbb{V}$.
(v) For every $\vec{x} \in \mathbb{V}$, there exists a vector $-\vec{x} \in \mathbb{V}$ such that $\vec{x} + (-\vec{x}) = \vec{0}$.
(vi) For every scalar c and every $\vec{x} \in \mathbb{V}$, $c\vec{x} \in \mathbb{V}$.
(vii) For all scalars c, d and every $\vec{x} \in \mathbb{V}$, $c(d\vec{x}) = (cd)\vec{x}$.
(viii) For all scalars c, d and every $\vec{x} \in \mathbb{V}$, $(c + d)\vec{x} = c\vec{x} + d\vec{x}$.
(ix) For every scalar c and all $\vec{x}, \vec{y} \in \mathbb{V}$, $c(\vec{x} + \vec{y}) = c\vec{x} + c\vec{y}$.
(x) For every $\vec{x} \in \mathbb{V}$, $1\vec{x} = \vec{x}$.

It is important to note that elements in a vector space are referred to as vectors even though $\mathbb{V}$ may not be $\mathbb{R}^n$. By vector space over $\mathbb{R}$, we simply mean that our scalars are real numbers. $\mathbb{R}^n$ is a vector space with the standard vector addition and scalar multiplication. $M_{m\times n}(\mathbb{R})$ is a vector space with the standard vector addition and scalar multiplication. $P_n(\mathbb{R})$ is a vector space with the standard vector addition and scalar multiplication just to list a few. As a result, we have a couple pretty useful theorems that make things easier for us to define.

Theorem 1.2 (Zero Vector and Additive Inverse)**.** *Let $\mathbb{V}$ be a vector space. Then,*

(i) $0\vec{x} = \vec{0}$ *for all* $\vec{x} \in \mathbb{V}$
(ii) $(-1)\vec{x} = -\vec{x}$ *for all* $\vec{x} \in \mathbb{V}$
(iii) $c(\vec{0}) = \vec{0}$ *for all* $c \in \mathbb{R}$.

Proof. For (i), distributivity gives

$$0\vec{x} = (0+0)\vec{x} = 0\vec{x} + 0\vec{x}.$$

Adding the inverse of $0\vec{x}$ to both sides yields $0\vec{x} = \vec{0}$. For (ii),

$$\vec{x} + (-1)\vec{x} = (1 + (-1))\vec{x} = 0\vec{x} = \vec{0},$$

so $(-1)\vec{x}$ is the additive inverse of $\vec{x}$.

For (iii), suppose that we have a scalar $c \in \mathbb{R}$. Then,

$$\begin{aligned}
\vec{0} + \vec{0} &= \vec{0} && \text{by Axiom 4,} \\
c(\vec{0} + \vec{0}) &= c\vec{0} && \text{multiply both sides by } c \\
c\vec{0} + c\vec{0} &= c\vec{0} && \text{by Axiom 9} \\
(c\vec{0} + c\vec{0}) + (-c\vec{0}) &= c\vec{0} + (-c\vec{0}) && \text{add } (-c\vec{0}) \text{ to both sides} \\
c\vec{0} + (c\vec{0} + (-c\vec{0})) &= c\vec{0} + (-c\vec{0}) && \text{by Axiom 3} \\
c\vec{0} + \vec{0} &= \vec{0} && \text{by Axiom 5} \\
c\vec{0} &= \vec{0} && \text{by Axiom 4.}
\end{aligned}$$

Using the fact that $\mathbb{V}$ is a vector space, then we can determine the zero vector by picking any vector $\vec{x} \in \mathbb{V}$ and computing $0\vec{x}$. For any $\vec{x} \in \mathbb{V}$, we can determine $-\vec{x}$ by computing $(-1)\vec{x}$. □

From the 10 Axioms, it becomes apparent that it is hard to characterize vector spaces succinctly and in a way that is interpretable. Moreover, most vector spaces are too large and complicated to understand all at once. Therefore, we introduce the concept of subspaces. Consider the vector space $\mathbb{R}^{1000}$. A general vector $(x_1, x_2, \ldots, x_{1000})$ is nearly impossible to visualize since even $\mathbb{R}^4$ is beyond geometric intuition. So rather than trying to understand the entire space, we can look for meaningful subspaces that behave nicely. For now we define conditions for subspaces to exist. We have the following theorem.

Theorem 1.3 (Subspace Test)**.** *Let $\mathbb{V}$ be a vector space and let $U \subseteq \mathbb{V}$.*

U is a subspace of $\mathbb{V}$ if,

(i) $\vec{0} \in U$
(ii) if $\vec{x}, \vec{y} \in U$*, then* $\vec{x} + \vec{y} \in U$
(iii) if $\vec{x} \in U$ *and* $c \in \mathbb{R}$ *then* $c\vec{x} \in U$.

It is important to note that every subspace U of a vector space is itself a vector space under the same operations of vector addition and scalar multiplication. Observe that Axioms 2, 3, 7, 8, 9, 10 are

inherited from V since (i) is simply Axiom 4, (ii) is Axiom 1 and (iii) is Axiom 6. Axiom 5 clearly follows from Axiom 6.

Theorem 1.4. *Let $\mathbb{V}$ be a vector space and $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$. Then,*

$$U = \{c_1\vec{v}_1 + \cdots + c_k\vec{v}_k \mid c_1, \ldots, c_k \in \mathbb{R}\}$$

is also a subspace of $\mathbb{V}$.

Proof. Using the subspace test we see that,

(i) By definition, $\vec{0}$ is in U

$$\vec{0} = 0\vec{v}_1 + \cdots + 0\vec{v}_k \in U.$$

(ii) Let $\vec{x}, \vec{y} \in U$. Then there are scalars $c_1, \ldots, c_k, d_1, \ldots, d_k \in \mathbb{R}$ such that,

$$\vec{x} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k, \qquad \vec{y} = d_1\vec{v}_1 + \cdots + d_k\vec{v}_k.$$

Then, their sum is,

$$\begin{aligned}\vec{x} + \vec{y} &= (c_1\vec{v}_1 + \cdots + c_k\vec{v}_k) + (d_1\vec{v}_1 + \cdots + d_k\vec{v}_k) \\ &= (c_1 + d_1)\vec{v}_1 + \cdots + (c_k + d_k)\vec{v}_k \in U.\end{aligned}$$

Therefore, U is closed under vector addition.

(iii) For scalar $c \in \mathbb{R}$,

$$c\vec{x} = c(c_1\vec{v}_1 + \cdots + c_k\vec{v}_k) = c(c_1)\vec{v}_1 + \cdots + c(c_k)\vec{v}_k \in U.$$

Therefore, U is closed under scalar multiplication.

Therefore, we conclude that U is a subspace of $\mathbb{V}$. □

We refer to the subspace constructed in the preceding theorem as the **Span**. That is, each vector in the space is called a linear combination of $\vec{v}_1, \ldots, \vec{v}_k$. Therefore, the span of any finite collection of vectors is a subspace of $\mathbb{V}$.

Definition 1.5 (Span.)**.** Let $\mathbb{V}$ be a vector space and let $\vec{v}_1, \ldots, \vec{v}_k \in \mathbb{V}$.

We define the span as the set,

$$\text{Span}\{\vec{v}_1, \ldots, \vec{v}_k\} = \{c_1\vec{v}_1 + \cdots + c_k\vec{v}_k \mid c_1, \ldots, c_k \in \mathbb{R}\}.$$

Theorem 1.6 (Span is the Smallest Containing Subspace.)**.** *Let $\vec{v}_1, \ldots, \vec{v}_k \in \mathbb{V}$ and define $U = \text{Span}\{\vec{v}_1, \ldots, \vec{v}_k\}$. Then U is a subspace of $\mathbb{V}$ and if W is any subspace of $\mathbb{V}$ containing $\vec{v}_1, \ldots, \vec{v}_k$ then $U \subseteq W$.*

Proof. We have already shown that U is a subspace. It remains to prove that U is contained in every subspace containing $\vec{v}_1, \ldots, \vec{v}_k$. Suppose that W is a subspace and that $\vec{v}_1, \ldots \vec{v}_k \in W$. Since W is closed under scalar multiplication for any scalars $c_1, \ldots, c_k \in \mathbb{R}$ then $c_1\vec{v}_1, \ldots, c_k\vec{v}_k \in W$. Because W is also closed under vector addition, then $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k \in W$. Therefore, every vector in U is also in W and hence $U \subseteq W$. This proves that U is the smallest subspace containing $\vec{v}_1, \ldots, \vec{v}_k$. □

Observe that from our vector space axioms, we are able to add vectors and multiply them by scalars, but they do not by themselves provide notions of length, angle or orthogonality. These properties and ideas arise from an inner product.

Definition 1.7 (Inner Products in $\mathbb{R}^n$)**.** For vectors,

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \qquad \vec{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n,$$

their standard inner product is,

$$\langle \vec{x}, \vec{y} \rangle = \vec{x}^\top \vec{y} = \sum_{i=1}^{n} x_i \cdot y_i.$$

The inner product takes two vectors and returns a scalar.

Definition 1.8 (Euclidean Norm)**.** The *Euclidean Norm* of a vector $\vec{x} \in \mathbb{R}^n$ is defined as,

$$\|\vec{x}\| = \sqrt{\langle \vec{x}, \vec{x} \rangle} = \sqrt{x_1^2 + \cdots + x_n^2}.$$

Because $\|\vec{x}\|^2 = \langle \vec{x}, \vec{x} \rangle$, the squared norm is often more conveniently written than the norm itself. The standard inner product has several useful properties.

Proposition 1.9 (Properties of Inner Product)**.** *For all $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and $c \in \mathbb{R}$,*

(i) (Symmetry) $\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$.
(ii) (Linearity) $\langle c\vec{x} + \vec{y}, \vec{z} \rangle = c\langle \vec{x}, \vec{z} \rangle + \langle \vec{y}, \vec{z} \rangle$.
(iii) (Positive Definiteness) $\langle \vec{x}, \vec{x} \rangle \geq 0$, with equality if and only if $\vec{x} = \vec{0}$.

Proof. (i) To prove symmetry, let $\vec{x}, \vec{y} \in \mathbb{R}^n$. By the definition of the standard inner product,

$$\langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^{n} x_i y_i = \sum_{i=1}^{n} y_i x_i = \langle \vec{y}, \vec{x} \rangle,$$

where the second equality follows from the commutativity of multiplication in $\mathbb{R}$.

(ii) To prove linearity in the first argument, let $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and let $c \in \mathbb{R}$. Then

$$\begin{aligned}\langle c\vec{x} + \vec{y}, \vec{z}\rangle &= \sum_{i=1}^{n}(cx_i + y_i)z_i \\ &= \sum_{i=1}^{n}(cx_iz_i + y_iz_i) = \sum_{i=1}^{n} cx_iz_i + \sum_{i=1}^{n} y_iz_i \\ &= c\sum_{i=1}^{n} x_iz_i + \sum_{i=1}^{n} y_iz_i = c\langle \vec{x}, \vec{z}\rangle + \langle \vec{y}, \vec{z}\rangle.\end{aligned}$$

(iii) To prove positive definiteness, let $\vec{x} \in \mathbb{R}^n$. By the definition of the standard inner product,

$$\langle \vec{x}, \vec{x}\rangle = \sum_{i=1}^{n} x_ix_i = \sum_{i=1}^{n} x_i^2.$$

Since $x_i^2 \geq 0$ for every $i \in \{1, \ldots, n\}$, it follows that

$$\langle \vec{x}, \vec{x}\rangle = \sum_{i=1}^{n} x_i^2 \geq 0.$$

Moreover,

$$\begin{aligned}\langle \vec{x}, \vec{x}\rangle = 0 &\iff \sum_{i=1}^{n} x_i^2 = 0 \\ &\iff x_i^2 = 0 \text{ for every } i \in \{1, \ldots, n\} \\ &\iff x_i = 0 \text{ for every } i \in \{1, \ldots, n\} \\ &\iff \vec{x} = \vec{0}.\end{aligned}$$

Therefore, $\langle \vec{x}, \vec{x}\rangle \geq 0$, with equality if and only if $\vec{x} = \vec{0}$.

□

The concept of orthogonality will come up in these notes repeatedly so it is convenient to introduce orthogonality proceeding inner products.

Definition 1.10 (Orthogonal Vectors.)**.** Two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ are said to be orthogonal if $\langle \vec{x}, \vec{y}\rangle = 0$ where we write $\vec{x} \perp \vec{y}$. Orthogonality generalizes the familiar notion of perpendicular vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$.

A useful result follows from orthogonality.

Theorem 1.11 (Pythagorean Theorem.)**.** *Suppose that $\vec{x}, \vec{y} \in \mathbb{R}^n$ are orthogonal. Then,*

$$\|\vec{x} + \vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2.$$

Proof. Using the definition of the norm,

$$\begin{aligned}\|\vec{x}+\vec{y}\|^2 &= \langle \vec{x}+\vec{y}, \vec{x}+\vec{y}\rangle \\ &= \langle \vec{x},\vec{x}\rangle + \langle \vec{x},\vec{y}\rangle + \langle \vec{y},\vec{x}\rangle + \langle \vec{y},\vec{y}\rangle \\ &= \|\vec{x}\|^2 + \langle \vec{x},\vec{y}\rangle + \langle \vec{y},\vec{x}\rangle + \|\vec{y}\|^2.\end{aligned}$$

Since $\vec{x} \perp \vec{y}$, $\langle \vec{x},\vec{y}\rangle = \langle \vec{y},\vec{x}\rangle = 0$. Therefore,

$$\|\vec{x}+\vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2.$$

□

We now consider the simplest projection problem.

Let $\vec{y} \in \mathbb{R}^n$, let $\vec{v} \neq \vec{0}$, and define the one-dimensional subspace

$$U = \text{span}\{\vec{v}\}.$$

Every vector in U has the form $c\vec{v}$ for some $c \in \mathbb{R}$. We seek the value of c for which $c\vec{v}$ is the closest point in U to $\vec{y}$. Geometrically, the error vector must be orthogonal to the subspace, so

$$\begin{aligned}\langle \vec{y} - c\vec{v}, \vec{v}\rangle = 0 &\iff \langle \vec{y},\vec{v}\rangle - c\langle \vec{v},\vec{v}\rangle = 0 \\ c\langle \vec{v},\vec{v}\rangle &= \langle \vec{y},\vec{v}\rangle \\ \implies c &= \frac{\langle \vec{y},\vec{v}\rangle}{\langle \vec{v},\vec{v}\rangle} = \frac{\langle \vec{y},\vec{v}\rangle}{\|\vec{v}\|^2},\end{aligned}$$

where $\langle \vec{v},\vec{v}\rangle = \|\vec{v}\|^2 > 0$ since $\vec{v} \neq \vec{0}$. Thus, the projection of $\vec{y}$ onto the line spanned by $\vec{v}$ is

$$\text{proj}_{\vec{v}}(\vec{y}) = \frac{\langle \vec{y},\vec{v}\rangle}{\|\vec{v}\|^2}\,\vec{v}.$$

Definition 1.12 (Orthogonal Projection onto a Vector.)**.** Let $\vec{y},\vec{v} \in \mathbb{R}^n$ where $\vec{v} \neq \vec{0}$.

The orthogonal projection of $\vec{y}$ onto $\vec{v}$ is,

$$\text{proj}_{\vec{v}}(\vec{y}) = \frac{\langle \vec{y},\vec{v}\rangle}{\langle \vec{v},\vec{v}\rangle} \cdot \vec{v} = \frac{\vec{v}^\top \vec{y}}{\vec{v}^\top \vec{v}}\vec{v}.$$

The projection is not merely a scalar but a vector lying in the line spanned by $\vec{v}$. The scalar $\frac{\langle \vec{y},\vec{v}\rangle}{\|\vec{v}\|}$ is referred to as the scalar component of $\vec{y}$ in the direction of $\vec{v}$, whereas $\frac{\langle \vec{y},\vec{v}\rangle}{\langle \vec{v},\vec{v}\rangle}\vec{v}$ is the vector projection. When $\vec{v}$ is a unit vector so that its norm is 1, the expression simplifies to,

$$\text{proj}_{\vec{v}}(\vec{y}) = \langle \vec{y},\vec{v}\rangle \vec{v}.$$

Theorem 1.13. *Let $\vec{v} \neq \vec{0}$. Every vector $\vec{y} \in \mathbb{R}^n$ can therefore be written $\vec{y} = \hat{\vec{y}} + \vec{e}$ where $\hat{\vec{y}} \in Span\{\vec{v}\}$ and therefore $\vec{e} \perp \vec{v}$. This decomposition is unique and $\hat{\vec{y}} = \text{proj}_{\vec{v}}(\vec{y})$.*

Proof. Define,

$$\hat{\vec{y}} = \frac{\langle \vec{y}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}, \qquad \vec{e} = \vec{y} - \hat{\vec{y}}.$$

Since $\hat{\vec{y}}$ is clearly a scalar multiple of $\vec{v}$, we have $\hat{\vec{y}} \in \text{Span}\{\vec{v}\}$. Furthermore, we see that,

$$\langle \vec{e}, \vec{v} \rangle = \left\langle \vec{y} - \frac{\langle \vec{y}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}, \vec{v} \right\rangle.$$

Expanding by linearity of inner product,

$$\langle \vec{e}, \vec{v} \rangle = \langle \vec{y}, \vec{v} \rangle - \frac{\langle \vec{y}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \langle \vec{v}, \vec{v} \rangle = 0.$$

Therefore we see that $\vec{e} \perp \vec{v}$. To see that this is unique, suppose that $\vec{y} = \hat{\vec{y_1}} + \vec{e}_1$ and $\vec{y} = \hat{\vec{y_2}} + \vec{e}_2$ where $\hat{\vec{y_1}}, \hat{\vec{y_2}} \in \text{Span}\{\vec{v}\}$ and both $\vec{e}_1, \vec{e}_2$ are orthogonal to $\vec{v}$.

Subtracting the two representations,

$$\hat{\vec{y_1}} - \hat{\vec{y_2}} = \vec{e_2} - \vec{e_1}.$$

The left-hand side belongs to $\text{Span}\{\vec{v}\}$ while the right-hand side is orthogonal to $\vec{v}$. Hence, the common vector belongs to the line and is also orthogonal to the line. The only vector with both properties is $\vec{0}$. Therefore $\hat{\vec{y_1}} - \hat{\vec{y_2}} = \vec{0}$ so $\hat{\vec{y_1}} = \hat{\vec{y_2}}$. It follows that $\vec{e}_1 = \vec{e}_2$, proving uniqueness. It is evident that the projection is characterized not only by orthogonality but also by minimum-distance property. □

Theorem 1.14. *Let $U = Span\{\vec{v}\}$ and $\vec{v} \neq \vec{0}$ and let $\hat{\vec{y}} = \text{proj}_{\vec{v}}(\vec{y})$. Then, for every $\vec{u} \in U$,*

$$\|\vec{y} - \hat{\vec{y}}\| \leq \|\vec{y} - \vec{u}\|.$$

Equality holds if and only if $\vec{u} = \hat{\vec{y}}$.

Proof. Let $\hat{\vec{y}} = \text{proj}_U(\vec{y})$, and let $\vec{u} \in U$ be arbitrary. Since both $\hat{\vec{y}}$ and $\vec{u}$ belong to U, the vector $\hat{\vec{y}} - \vec{u}$ also belongs to U. Moreover, by the defining property of the orthogonal projection, the error vector $\vec{y} - \hat{\vec{y}}$ is orthogonal to every vector in U, and hence

$$(\vec{y} - \hat{\vec{y}}) \perp (\hat{\vec{y}} - \vec{u}).$$

Now observe that,

$$\vec{y} - \vec{u} = (\vec{y} - \hat{\vec{y}}) + (\hat{\vec{y}} - \vec{u}).$$

Since the two vectors on the right-hand side are orthogonal, the Pythagorean Theorem gives

$$\begin{aligned} \|\vec{y} - \vec{u}\|^2 &= \|\vec{y} - \hat{\vec{y}}\|^2 + \|\hat{\vec{y}} - \vec{u}\|^2 \\ &\geq \|\vec{y} - \hat{\vec{y}}\|^2. \end{aligned}$$

Taking square roots of both sides yields

$$\|\vec{y} - \vec{u}\| \geq \|\vec{y} - \hat{\vec{y}}\|.$$

Equality holds if and only if

$$\|\hat{\vec{y}} - \vec{u}\|^2 = 0,$$

which is equivalent to $\hat{\vec{y}} - \vec{u} = \vec{0}$, or simply $\vec{u} = \hat{\vec{y}}$.

Therefore, $\hat{\vec{y}}$ is the unique vector in U closest to $\vec{y}$. □

It is evident that the orthogonal projection of a vector onto a subspace is the unique vector in that subspace that minimizes the Euclidean distance.

1.2 Rank, null spaces, and identifiability

Recall that a set $S = \{\vec{v}_1, \ldots, \vec{v}_k\}$ in a vector space $\mathbb{V}$ is said to be linearly dependent if there exist scalars $c_1, \ldots, c_k \in \mathbb{R}$, not all zero, such that $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$. S is linearly independent if the only solution to $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ is $c_1 = \cdots = c_k = 0$ i.e., the trivial solution. These terms and concepts should be familiar to you.

A spanning set tells us that every vector in a vector space can be written as a linear combination of the vectors in the set. However, this alone does not tell us whether that representation is unique. The following theorem makes this clear.

Theorem 1.15 (Unique Representation Theorem)**.** *Suppose that $\mathbb{V}$ is a vector space with spanning set $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$. Then, every vector in $\mathbb{V}$ can be uniquely expressed as a linear combination of the vectors in S if and only if S is linearly independent.*

Proof. Assume that the spanning set S is linearly independent and let $\vec{x} \in \mathbb{V}$. Then $\vec{x} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$ for some $c_1, \ldots, c_k \in \mathbb{R}$. Suppose that $\vec{x} = d_1\vec{v}_1 + \cdots + d_k\vec{v}_k$ for some $d_1, \ldots, d_k \in \mathbb{R}$. Then it follows that,

$$c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = d_1\vec{v}_1 + \cdots + d_k\vec{v}_k.$$

So therefore,

$$(c_1 - d_1)\vec{v}_1 + \cdots + (c_k - d_k)\vec{v}_k = \vec{0}.$$

Since S is linearly independent, $c_i - d_i = 0$ for all $i = 1, \ldots, k$ meaning that $c_i = d_i$ for all $i = 1, \ldots, k$. Thus, $\vec{x} \in \mathbb{V}$ has a unique representation as a linear combination of $\vec{v}_1, \ldots, \vec{v}_k$. Now, if S is linearly dependent, there exist scalars $c_1, \ldots, c_k \in \mathbb{R}$ not all zero such that,

$$\vec{0} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k.$$

However, $\vec{0} = 0\vec{v}_1 + \cdots + 0\vec{v}_k$.

So, $\vec{0}$ does not have a unique representation as a linear combination of $\vec{v}_1, \ldots, \vec{v}_k$. □

More often than not, a spanning set for a vector space contains redundant vectors and is therefore larger than necessary. Instead, we seek a minimal spanning set that is still capable of representing every vector in the space. Such a set is called the basis of the vector space. Since every basis is a

subset of the vector space, it also spans a subspace. However, because it spans the entire vector space, that subspace is in fact the vector space itself. Consequently, a basis provides the most efficient representation of a vector space by eliminating all linear redundancy while preserving its ability to represent every vector uniquely.

Theorem 1.16 (Basis.)**.** *Let $\mathbb{V}$ be a vector space and let $B = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$. If the set B spans $\mathbb{V}$ and B is linearly independent, then B is a Basis for $\mathbb{V}$.*

If $\mathbb{V} = \{\vec{0}\}$ then B is the empty set, denoted $\emptyset$.

In fact, since every basis of a vector space has the same number of vectors, the dimension of the vector space, we'll call it $\mathbb{V}$ is denoted as $\dim(\mathbb{V}) = n$ where n is the number of vectors in basis. Suppose that we now have another set $S = \{\vec{v}_1, \ldots, \vec{v}_m\} \subseteq \mathbb{V}$ where $m > n$. Because every vector in S lies in an n-dimensional space, it is impossible for all m vectors to be linearly independent. We can conclude that S is therefore linearly dependent.

Theorem 1.17. *Suppose that $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ is a basis for vector space $\mathbb{V}$. If $C = \{\vec{w}_1, \ldots, \vec{w}_l\} \subseteq \mathbb{V}$ where $l > k$, then C is linearly dependent.*

Proof. Because B is a basis, for each $j = 1, \ldots, l$ there are scalars $a_{1j}, \ldots, a_{kj}$ such that

$$\vec{w}_j = \sum_{i=1}^{k} a_{ij}\vec{v}_i.$$

Let $A = [a_{ij}] \in \mathbb{R}^{k \times l}$. The homogeneous system $A\vec{c} = \vec{0}$ has more unknowns than equations, so row reduction leaves at least one free variable and hence admits a non-zero solution $\vec{c} = (c_1, \ldots, c_l)^\top$. Therefore,

$$\sum_{j=1}^{l} c_j\vec{w}_j = \sum_{i=1}^{k} \left(\sum_{j=1}^{l} a_{ij}c_j \right) \vec{v}_i = \vec{0}.$$

Since the coefficients $c_1, \ldots, c_l$ are not all zero, C is linearly dependent. □

Obviously, if we have two bases for a given vector space, then the two bases must have the same number of vectors otherwise they would not be minimal spanning sets for the vector space.

Theorem 1.18. *If $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ and $C = \{\vec{w}_1, \ldots, \vec{w}_l\}$ are both bases for the vector space $\mathbb{V}$, then $k = l$ i.e., they contain the same number of vectors.*

Proof. Since B is a basis for $\mathbb{V}$ and $C \subseteq \mathbb{V}$ is linearly independent, then $l \leq k$. Similarly, since C is a basis for $\mathbb{V}$ and $B \subseteq \mathbb{V}$ is linearly independent, then $k \leq l$. Hence we have that $k = l$. □

Given that we have a formal definition for the basis of a vector space, we can now define the dimension of a vector space. This is quite remarkable as all valid bases of a vector space have the same number of vectors.

Definition 1.19 (Dimension.)**.** Suppose that $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ is a basis for the vector space $\mathbb{V}$. Then, the dimension of $\mathbb{V}$ is k which we denote $\dim(\mathbb{V}) = k$. If the vector space is the zero vector, then the dimension of the vector space is simply 0.

For example, the dimension of $\mathbb{R}^n$ is simply n i.e., the standard basis of $\mathbb{R}^n$. The dimension of the vector space of $m \times n$ matrices denoted $\mathbb{M}_{m\times n}(\mathbb{R})$ is simply $m \times n$. Given a polynomial vector space of polynomials of degree up to n, denoted $\mathbb{P}_n(\mathbb{R})$, such a vector space has dimension $n + 1$ since we include the constant.

Interestingly enough, if we know the dimension of a vector space $\mathbb{V}$ prior to doing anything with spanning sets with respect to the vector space, we can make quite a few statements regarding the linear dependence/independence of such a spanning set simply by knowing how many vectors are in the span.

Theorem 1.20. *Let $\mathbb{V}$ be an n-dimensional vector space. Then,*

(i) A set of more than n vectors must be linearly dependent,
(ii) A set of fewer than n vectors cannot span $\mathbb{V}$ and,
(iii) A set with n vectors of $\mathbb{V}$ is a spanning set for $\mathbb{V}$ if and only if it is linearly independent.

Suppose that we have a linearly independent set of vectors $C = \{\vec{v}_1, \ldots, \vec{v}_k\}$ that contain no redundancy, then we can keep adding suitable vectors for the independent set of vectors until we get a complete coordinate system for the entire space. More importantly, the original vectors $\vec{v}_1, \ldots, \vec{v}_k$ are preserved and we need not replace them nor alter them.

Geometrically in $\mathbb{R}^3$, one non-zero vector determines one direction. If we can find a second vector not lying on that line and then a third not lying in the plane generated by the first two, the resulting three vectors form a basis of $\mathbb{R}^3$. Similarly, if we begin with two non-parallel vectors, they span a plane, and we only need to add one vector outside of that plane. Remarkably, this is the local independence that can also be promoted to a global representation of a vector space. A linearly independent set may initially describe only part of the space, but it can always be embedded into a basis that describes every vector in that space uniquely.

Theorem 1.21 (Extending a Linearly Independent Set.)**.** *Suppose we have an n-dimensional vector space and $C = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$ is a linearly independent set with $k < n$. Then, there are $\vec{w}_{k+1}, \ldots, \vec{w}_n \in \mathbb{V}$ such that $B = \{\vec{v}_1, \ldots, \vec{v}_k, \vec{w}_{k+1}, \ldots, \vec{w}_n\}$ is a basis for $\mathbb{V}$.*

Proof. Let $C = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$ with $k < n$ be linearly independent where $\dim(\mathbb{V}) = n$. Then clearly $\text{Span}C \neq \mathbb{V}$ so there exists $\vec{w}_{k+1} \in \mathbb{V}$ such that $\vec{w}_{k+1} \notin \text{Span}C$.

Let $c_1, \ldots, c_k \in \mathbb{R}$ be scalars and consider,

$$c_1\vec{v}_1 + \cdots + c_k\vec{v}_k + c_{k+1}\vec{w}_{k+1} = \vec{0}.$$

Suppose that $c_{k+1} \neq 0$ meaning that,

$$\vec{w}_{k+1} = -\frac{c_1}{c_{k+1}}\vec{v}_1 - \cdots - \frac{c_k}{c_{k+1}}\vec{v}_k \in \text{Span}C,$$

which directly contradicts $\vec{w}_{k+1} \notin \text{Span}C$ meaning that $c_{k+1} = 0$. Therefore,

$$c_1\vec{v}_1 + \cdots + c_k\vec{v}_k + c_{k+1}\vec{w}_{k+1} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0},$$

and we have that $c_1 = \cdots = c_k = 0$ since C is linearly independent. Hence $\{\vec{v}_1, \ldots, \vec{v}_k, \vec{w}_{k+1}\}$ is linearly independent. If $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_k, \vec{w}_{k+1}\} = \mathbb{V}$, then we are done.

If not, we continue until we get a linearly independent spanning set for $\mathbb{V}$ which must happen when our set contains n vectors. □

Having developed the notions of span, linear independence, basis and dimension, we are now prepared to study how one vector space may be mapped into another while preserving its linear structure. A linear transformation $L : \mathbb{V} \to \mathbb{W}$ is completely determined by its action on a basis of $\mathbb{V}$, since the image of every vector follows from linearity. Two subspaces naturally arise from such a transformation, that is, the kernel consisting of all vectors in $\mathbb{V}$ that are mapped to the zero vector, and the image, consisting of all vectors in $\mathbb{W}$ that can be produced by L. We refer to the dimension of the image as the rank of the linear transformation L, while the dimension of the kernel is called its nullity. These quantities describe respectively how many independent directions are preserved by the transformation and how many are effectively lost.

As we will see in later sections, this distinction is fundamental: the columns of a design matrix define a linear transformation from coefficient space to observation space, and a non-trivial kernel means that different coefficient vectors produce the same fitted values. Consequently, the rank and null space determine whether the model parameters are uniquely identifiable.

Definition 1.22 (Linear Transformation.)**.** Let $\mathbb{V}$ and $\mathbb{W}$ be vector spaces. A function $L : \mathbb{V} \to \mathbb{W}$ is called a linear mapping if for all vectors $\vec{x}, \vec{y} \in \mathbb{V}$ and all $s, t \in \mathbb{R}$,

$$L(s\vec{x} + t\vec{y}) = sL(\vec{x}) + tL(\vec{y}).$$

We call $\mathbb{V}$ the domain of L and $\mathbb{W}$ the codomain of L. If $\mathbb{V} = \mathbb{W}$ then we call L a linear operator.

Proposition 1.23 (Uniqueness of a Linear Operator)**.** *If $L, M : \mathbb{V} \to \mathbb{W}$ are linear mappings, then $L = M$ when $L(\vec{x}) = M(\vec{x})$ for all $\vec{x} \in \mathbb{V}$.*

Since we are now discussing two vector spaces $\mathbb{V}$ and $\mathbb{W}$, we write $\vec{0}_{\mathbb{V}}$ to indicate the zero vector of $\mathbb{V}$ and $\vec{0}_{\mathbb{W}}$ to indicate the zero vector of $\mathbb{W}$. We now define two important linear mappings.

Definition 1.24 (Zero Map and Identity)**.** Let $\mathbb{V}, \mathbb{W}$ be vector spaces. The **zero mapping** is the linear mapping $Z : \mathbb{V} \to \mathbb{W}$ defined by $Z(\vec{x}) = \vec{0}_{\mathbb{W}}$ for every $\vec{x} \in \mathbb{V}$. The **identity mapping** $\text{Id}_{\mathbb{V}} : \mathbb{V} \to \mathbb{V}$ is the linear operator defined by $\text{Id}_{\mathbb{V}}(\vec{x}) = \vec{x}$ for every $\vec{x} \in \mathbb{V}$.

Definition 1.25. Let $L : \mathbb{V} \to \mathbb{W}$ be a linear mapping. Then, the *nullspace* of L is,

$$\text{Null}(L) = \{\vec{x} \in \mathbb{V} \mid L(\vec{x}) = \vec{0}_{\mathbb{W}}\}.$$

The range of L is,

$$\text{Range}(L) = \{L(\vec{x}) \mid \vec{x} \in \mathbb{V}\}.$$

By definition, we see that $\text{Null}(L) \subseteq \mathbb{V}$ and $\text{Range}(L) \subseteq \mathbb{W}$.

Theorem 1.26. *If $L : \mathbb{V} \to \mathbb{W}$ is a linear mapping then,*

(i) $L(\vec{0}_{\mathbb{V}}) = \vec{0}_{\mathbb{W}}$
(ii) $\text{null}(L)$ *is a subspace of* $\mathbb{V}$*,*
(iii) $\text{range}(L)$ *is a subspace of* $\mathbb{W}$*.*

Proof. (i) Since $L : \mathbb{V} \to \mathbb{W}$ is linear,

$$L(\vec{0}_{\mathbb{V}}) = L(0\vec{x}) = 0L(\vec{x}) = \vec{0}_{\mathbb{W}}$$

for any $\vec{x} \in \mathbb{V}$.

(ii) Using the subspace test, since $L(\vec{0}_{\mathbb{V}}) = \vec{0}_{\mathbb{W}}$ we have that $\vec{0}_{\mathbb{V}} \in \text{null}(L)$. Again, let $\vec{x}, \vec{y} \in \text{null}(L)$. Then we know that $L(\vec{x}) = \vec{0}_{\mathbb{W}}$ and $L(\vec{y}) = \vec{0}_{\mathbb{W}}$. Since,

$$L(\vec{x} + \vec{y}) = L(\vec{x}) + L(\vec{y}) = \vec{0}_{\mathbb{W}} + \vec{0}_{\mathbb{W}} = \vec{0}_{\mathbb{W}},$$

then we have that $\vec{x} + \vec{y} \in \text{null}(L)$ so $\text{null}(L)$ is closed under vector addition.
Suppose that we have a scalar $s \in \mathbb{R}$ then,

$$L(s\vec{x}) = sL(\vec{x}) = s\vec{0}_{\mathbb{W}} = \vec{0}_{\mathbb{W}},$$

so $s\vec{x} \in \text{null}(L)$ and $\text{null}(L)$ is closed under scalar multiplication.
Therefore we have shown that $\text{null}(L)$ is a subspace of $\mathbb{V}$.

(iii) We again apply subspace test. Since $L(\vec{0}_{\mathbb{V}}) = \vec{0}_{\mathbb{W}}$, then we conclude that $\vec{0}_{\mathbb{W}} \in \text{range}(L)$. Suppose that $\vec{y_1}, \vec{y_2} \in \text{range}(L)$. Then there are $\vec{x_1}, \vec{x_2} \in \mathbb{V}$ such that $\vec{y_1} = L(\vec{x_1})$ and $\vec{y_2} = L(\vec{x_2})$. Since we know that,

$$\vec{y_1} + \vec{y_2} = L(\vec{x_1}) + L(\vec{x_2}) = L(\vec{x_1} + \vec{x_2}),$$

we know that $\vec{y_1} + \vec{y_2} \in \text{range}(L)$ so $\text{range}(L)$ is closed under vector addition. Take the scalar $s \in \mathbb{R}$. Then we know that,

$$s\vec{y_1} = sL(\vec{x_1}) = L(s\vec{x_1}),$$

so $s\vec{y_1} \in \text{range}(L)$ and $\text{range}(L)$ is closed under scalar multiplication.
Therefore we have shown that $\text{range}(L)$ is a subspace of $\mathbb{W}$.

□

Having shown that $\text{null}(L)$ is a subspace of the domain $\mathbb{V}$ and that $\text{range}(L)$ is a subspace of the codomain $\mathbb{W}$, we may now study their dimensions. These dimensions measure two complementary aspects of a linear mapping. The dimension of $\text{range}(L)$ called the rank of L counts the number of linearly independent directions in the codomain that can actually be produced by the transformation. The dimension of $\text{null}(L)$ called the nullity of L counts the number of independent directions in the domain that are collapsed to the zero vector. Indeed if $L(\vec{x}) = L(\vec{y})$ then $L(\vec{x} - \vec{y}) = \vec{0}_{\mathbb{W}}$ so $\vec{x} - \vec{y} \in \text{null}(L)$.

Thus, a non-trivial nullspace means that distinct input vectors may produce the same output whereas L is one-to-one precisely when $\text{null}(L) = \{\vec{0}_{\mathbb{V}}\}$.

The basis-extension theorem now becomes especially useful. Starting with a basis for null(L), we may extend it to a basis for all of $\mathbb{V}$. The images of the additional basis vectors then form a basis for range(L). This formulation gives us the Rank-Nullity theorem.

Definition 1.27. Let $L : \mathbb{V} \to \mathbb{W}$ be a linear mapping.

The rank of L is $\text{rank}(L) = \dim(\text{range}(L))$. The nullity of L is $\text{nullity}(L) = \dim(\text{null}(L))$.

Theorem 1.28. *Let $\mathbb{V}, \mathbb{W}$ be vector spaces with dim$(\mathbb{V}) = n$.*

If $L : \mathbb{V} \to \mathbb{W}$ is a linear mapping then,

$$\text{rank}(L) + \text{nullity}(L) = \dim(\mathbb{V}).$$

Given the Rank-Nullity theorem, we are now able to characterize when a linear mapping preserves all the information contained in its domain. If $\text{null}(L) = \{\vec{0}_{\mathbb{V}}\}$ then $\text{nullity}(L) = 0$ and hence $\text{rank}(L) = \dim(\mathbb{V})$. Therefore, no non-zero direction in $\mathbb{V}$ is collapsed to a zero vector and L is one-to-one. If, in addition $\dim(\mathbb{V}) = \dim(\mathbb{W}) < \infty$ then $\text{rank}(L) = \dim(\mathbb{W})$ so $\text{range}(L) = \mathbb{W}$ and L is also onto. Therefore, between finite-dimension vector spaces of equal dimension, a linear mapping is one-to-one if and only if it is onto.

A linear mapping preserving both properties establishes a perfect correspondence between the two vector space, that is, every vector in $\mathbb{V}$ is mapped to a distinct vector in $\mathbb{W}$ and every vector in $\mathbb{W}$ is obtained from exactly one vector in $\mathbb{V}$.

As we will define now, such a mapping is called an isomorphism. An isomorphic vector space may contain entirely different objects but possess the same linear structure and therefore be regarded as equivalent from the perspective of linear algebra.

Definition 1.29. A linear mapping $L : \mathbb{V} \to \mathbb{W}$ is one-to-one (or injective) if for any $\vec{v}_1, \vec{v}_2 \in \mathbb{V}$,

$$L(\vec{v}_1) = L(\vec{v}_2) \implies \vec{v}_1 = \vec{v}_2.$$

Notice this is equivalent to saying $\vec{v}_1 \neq \vec{v}_2 \implies L(\vec{v}_1) \neq L(\vec{v}_2)$.

Example 1.30. For example, the linear map $L : \mathbb{R}^2 \to \mathbb{R}^3$ defined as,

$$L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix},$$

is one-to-one. We can see this by setting,

$$L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = L\left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) \implies \begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix} = \begin{bmatrix} y_2 \\ y_1 \\ y_1 + y_2 \end{bmatrix}.$$

So, $x_1 = y_1$ and $x_2 = y_2$ hence L is one-to-one. However, $L : \mathbb{M}_{2\times 2}(\mathbb{R}) \to \mathbb{P}_2(\mathbb{R})$ defined as,

$$L\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a + bx + (c + d)x^2,$$

is not one-to-one. If two matrices have the same image, then

$$a_1 + b_1x + (c_1 + d_1)x^2 = a_2 + b_2x + (c_2 + d_2)x^2.$$

Thus, equality of the images implies $a_1 = a_2$, $b_1 = b_2$, and $c_1 + d_1 = c_2 + d_2$, but it does not imply $c_1 = c_2$ and $d_1 = d_2$.

For example,

$$L\left(\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\right) = 1 + x + x^2 = L\left(\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\right),$$

although the two input matrices are distinct. Hence, L is not one-to-one.

Theorem 1.31. *A linear mapping $L : \mathbb{V} \to \mathbb{W}$ is one-to-one if and only if* $\text{null}(L) = \{\vec{0}_{\mathbb{V}}\}$.

Proof. Assume that L is one-to-one. If $\vec{x} \in \text{null}(L)$, then $L(\vec{x}) = \vec{0}_{\mathbb{W}} = L(\vec{0}_{\mathbb{V}})$.

Since L is one-to-one, then $\vec{x} = \vec{0}_{\mathbb{V}}$ so $\text{null}(L) = \{\vec{0}_{\mathbb{V}}\}$.

Now assume that $\text{null}(L) = \{\vec{0}_{\mathbb{V}}\}$. If $L(\vec{v}_1) = L(\vec{v}_2)$ then,

$$\vec{0}_{\mathbb{W}} = L(\vec{v}_1) - L(\vec{v}_2) = L(\vec{v}_1 - \vec{v}_2),$$

so $\vec{v}_1 - \vec{v}_2 \in \text{null}(L) = \{\vec{0}_{\mathbb{V}}\}$. Hence we see that $\vec{v}_1 - \vec{v}_2 = \vec{0}_{\mathbb{V}}$.

Thus $\vec{v}_1 = \vec{v}_2$ so L is one-to-one. □

Definition 1.32. A linear mapping $L : \mathbb{V} \to \mathbb{W}$ is onto (or surjective) if for every $w \in \mathbb{W}$ there exists a $v \in \mathbb{V}$ such that $L(\vec{v}) = \vec{w}$ that is if $\text{range}(L) = \mathbb{W}$.

Example 1.33. Consider the linear map $L : \mathbb{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}$ defined by $L\left(\begin{bmatrix}a & b\\ c & d\end{bmatrix}\right) = a + d$.

L is clearly onto because,

$$L\left(\begin{bmatrix}x & 0\\ 0 & 0\end{bmatrix}\right) = x + 0 = x.$$

Example 1.34. Consider the linear map defined by $L : \mathbb{R}^3 \to \mathbb{P}_5(\mathbb{R})$,

$$L\left(\begin{bmatrix}a\\ b\\ c\end{bmatrix}\right) = a(1 + x^3) + b(1 + x^2 + x^5) + cx^4.$$

Since $\text{range}(L) = \text{span}\{1 + x^3, 1 + x^2 + x^5, x^4\}$ we have that,

$$\text{rank}(L) = \dim \text{range}(L) \leq 3 < 6 = \dim \mathbb{P}_5(\mathbb{R}).$$

So, L is not onto.

Proposition 1.35. *Suppose that* $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$ *and let* $L : \mathbb{V} \to \mathbb{W}$ *be a linear mapping.*

(i) If L *is one-to-one and* $\{\vec{v}_1, \ldots, \vec{v}_k\}$ *is linearly independent, then* $\{L(\vec{v}_1), \ldots, L(\vec{v}_k)\}$ *is linearly independent.*

(ii) If L *is onto and* $\text{span}\{\vec{v}_1, \ldots, \vec{v}_k\} = \mathbb{V}$ *then* $\text{span}\{L(\vec{v}_1), \ldots, L(\vec{v}_k)\} = \mathbb{W}$.

Proof. (i) Suppose that we have scalars $c_1, \ldots, c_k \in \mathbb{R}$ and consider,

$$\begin{aligned} c_1 L(\vec{v}_1) + \cdots + c_k L(\vec{v}_k) &= \vec{0}_{\mathbb{W}} \\ L(c_1\vec{v}_1 + \cdots + c_k\vec{v}_k) &= \vec{0}_{\mathbb{W}} \\ c_1\vec{v}_1 + \cdots + c_k\vec{v}_k &\in \text{null}(L) \\ c_1\vec{v}_1 + \cdots + c_k\vec{v}_k &= \vec{0}_{\mathbb{V}}. \end{aligned}$$

Since we know that $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is linearly independent, then $c_1 = \cdots = c_k = 0$ so $\{L(\vec{v}_1), \ldots, L(\vec{v}_k)\}$ is linearly independent as well.

(ii) Let $\vec{w} \in \mathbb{W}$. Since L is onto, there is a $\vec{v} \in \mathbb{V}$ so that $L(\vec{v}) = \vec{w}$. Since we know that $\{\vec{v}_1, \ldots, \vec{v}_k\}$ spans our vector space $\mathbb{V}$ there are $c_1, \ldots, c_k \in \mathbb{R}$ such that $\vec{v} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$. Therefore $L(\vec{v}) = L(c_1\vec{v}_1 + \cdots + c_k\vec{v}_k)$ which means that $\vec{w} = c_1 L(\vec{v}_1) + \cdots + c_k L(\vec{v}_k)$ since L is a linear mapping. Therefore, $\text{span}\{L(\vec{v}_1), \ldots, L(\vec{v}_k)\} = \mathbb{W}$.

□

Theorem 1.36. *Consider a linear map* $L : \mathbb{V} \to \mathbb{W}$. *$L$ has an inverse mapping* $L^{-1} : \mathbb{W} \to \mathbb{V}$ *if and only if L is both one-to-one and onto.*

Definition 1.37. Let $\mathbb{V}, \mathbb{W}$ be vector spaces and let $L : \mathbb{V} \to \mathbb{W}$ be the linear mapping between the two vector spaces. If L is one-to-one and onto, then L is called an isomorphism and the vector spaces $\mathbb{V}$ and $\mathbb{W}$ are then said to be isomorphic.

Example 1.38. Prove that $\mathbb{R}^3$ and $\mathbb{P}_2(\mathbb{R})$ are isomorphic by constructing an explicit isomorphism L. Define

$$L : \mathbb{R}^3 \to \mathbb{P}_2(\mathbb{R}), \qquad L\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = a + bx + cx^2.$$

(i) We show that L is a linear mapping. For $\begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$ and $\begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix}$ in $\mathbb{R}^3$ and scalars $s, t \in \mathbb{R}$,

$$\begin{aligned} L\left(s\begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} + t\begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix} \right) &= L\left(\begin{bmatrix} sa_1 + ta_2 \\ sb_1 + tb_2 \\ sc_1 + tc_2 \end{bmatrix} \right) \\ &= (sa_1 + ta_2) + (sb_1 + tb_2)x + (sc_1 + tc_2)x^2 \\ &= s(a_1 + b_1x + c_1x^2) + t(a_2 + b_2x + c_2x^2) \\ &= sL\left(\begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} \right) + tL\left(\begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix} \right). \end{aligned}$$

(ii) We show L is one-to-one. Let $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \text{null}(L)$. Then,

$$0 + 0x + 0x^2 = L\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = a + bx + cx^2.$$

Clearly, $a = b = c = 0$ therefore $\text{null}(L) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$ showing L is one-to-one.

(iii) We show that L is also onto. Suppose we let $a + bx + cx^2 \in \mathbb{P}_2(\mathbb{R})$. Since,

$$L\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = a + bx + cx^2.$$

Therefore we see that L is onto.

Hence we conclude that L is an isomorphism, so $\mathbb{R}^3$ and $\mathbb{P}_2(\mathbb{R})$ are isomorphic.

Proposition 1.39. *If $L : \mathbb{V} \to \mathbb{W}$ is an isomorphism and $\{\vec{v}_1, \ldots, \vec{v}_n\}$ is a basis for $\mathbb{V}$, then $\{L(\vec{v}_1), \ldots, L(\vec{v}_n)\}$ is a basis for $\mathbb{W}$.*

Proof. If L is an isomorphism, then we know that L is one-to-one which implies that $\{L(\vec{v}_1), \ldots, L(\vec{v}_n)\}$ is a linearly independent set. Since L is onto then we know that $\{L(\vec{v}_1), \ldots, L(\vec{v}_n)\}$ spans $\mathbb{W}$. Hence, $\{L(\vec{v}_1), \ldots, L(\vec{v}_n)\}$ is a basis for $\mathbb{W}$ by definition. □

Theorem 1.40. *If $\mathbb{V}$ and $\mathbb{W}$ are finite-dimensional vector spaces then $\mathbb{V}$ and $\mathbb{W}$ are isomorphic if and only if $\dim(\mathbb{V}) = \dim(\mathbb{W})$.*

Theorem 1.41. *If $\mathbb{V}$ and $\mathbb{W}$ are n-dimensional vector spaces over $\mathbb{R}$, then a linear mapping $L : \mathbb{V} \to \mathbb{W}$ is one-to-one if and only if L is onto.*

1.3 Orthogonality and least-squares geometry

In the previous section, we saw that isomorphisms tell us when two vector spaces possess the same structure i.e., preservation of vector addition, scalar multiplication, linear independence, bases, and dimension. However, these concepts alone do not provide any notion of length, angle, distance, or closeness. Such geometric structure becomes essential when we wish to compare vectors, decompose one vector into components, or determine the point in a subspace that best approximates a given vector.

Definition 1.42. A set of vectors $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{R}^n$ is an orthogonal set if $\vec{v}_i \cdot \vec{v}_j = 0$ for $i \neq j$.

Geometrically, vectors in orthogonal sets point in mutually perpendicular directions meaning none contain any component in the direction of another. This means that algebraically, the cross terms vanish in inner-product calculations and for any scalars $c_1, \ldots, c_k$,

$$\|c_1\vec{v}_1 + \cdots + c_k\vec{v}_k\|^2 = \sum_{i=1}^{k} c_i^2 \|\vec{v}_i\|^2.$$

Consequently, every orthogonal set of non-zero vectors is automatically linearly independent.

Theorem 1.43. *If $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{R}^n$ is an orthogonal set of non-zero vectors, then it is linearly independent.*

Proof. For scalars $c_1, \ldots, c_k \in \mathbb{R}$ consider,

$$c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}.$$

Then for $1 \leq i \leq k$,

$$\vec{v}_i \cdot (c_1\vec{v}_1 + \cdots + c_k\vec{v}_k) = \vec{v}_i \cdot \vec{0},$$

so expanding the above on the left gives,

$$\underbrace{c_1(\vec{v}_i \cdot \vec{v}_1)}_{=0} + \cdots + \underbrace{c_{i-1}(\vec{v}_i \cdot \vec{v}_{i-1})}_{=0} + \underbrace{c_i(\vec{v}_i \cdot \vec{v}_i)}_{=\|\vec{v}_i\|^2} + \underbrace{c_{i+1}(\vec{v}_i \cdot \vec{v}_{i+1})}_{=0} + \cdots + \underbrace{c_k(\vec{v}_i \cdot \vec{v}_k)}_{=0} = 0,$$

which simplifies to,

$$c_i\|\vec{v}_i\|^2 = 0.$$

Since we know that $\vec{v}_i \cdot \vec{v}_j = 0$ for $j \neq i$ and since $\vec{v}_i \neq \vec{0}$, we see that $\|\vec{v}_i\| \neq 0$. It follows that $c_i = 0$ for $1 \leq i \leq k$. Therefore we see that $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is a linearly independent set. □

When an orthogonal set of nonzero vectors also spans the entire vector space, it forms an *orthogonal basis*. Such a basis is convenient as it allows us to compute the coordinates of any vector independently.

Definition 1.44. An orthogonal set B that is a basis for a subspace $U \subseteq \mathbb{R}^n$ is referred to as an orthogonal basis of U.

Suppose that $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ is an orthogonal basis for subspace $U \subseteq \mathbb{R}^n$. For $\vec{u} \in U$, there are unique real scalars $c_1, \ldots, c_k \in \mathbb{R}$ such that,

$$\vec{u} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k.$$

For $1 \leq i \leq k$ we can express this as,

$$\vec{v}_i \cdot \vec{u} = \vec{v}_i \cdot (c_1\vec{v}_1 + \cdots + c_k\vec{v}_k).$$

Similarly, we arrive at $\vec{v}_i \cdot \vec{u} = c_i\|\vec{v}_i\|^2$ which gives, $c_i = \dfrac{\vec{u} \cdot \vec{v}_i}{\|\vec{v}_i\|^2}$.

Example 1.45. Given the following orthogonal basis,

$$B = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\} \subseteq \mathbb{R}^2,$$

we would like to express the vector $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ as a linear combination of the vectors in B. We use the formulation we have derived above for c_i which gives the linear combination,

$$\vec{x} = \frac{\begin{bmatrix} 3 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\|^2} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{\begin{bmatrix} 3 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix}}{\left\| \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\|^2} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \frac{7}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

In fact, the same orthogonality that allows each coefficient c_i to be recovered independently also greatly simplifies the norm of a sum. In general, expanding the squared norm of a sum produces cross-terms of the form $\vec{v}_i \cdot \vec{v}_j$. For an orthogonal set, every such cross term vanishes. Consequently, the usual Pythagorean theorem extends from two perpendicular vectors to any finite orthogonal set.

Theorem 1.46. *Let $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{R}^n$ be an orthogonal set. Then,*

$$\|\vec{v}_1 + \cdots + \vec{v}_k\|^2 = \|\vec{v}_1\|^2 + \cdots + \|\vec{v}_k\|^2.$$

Proof. The result follows algebraically from the properties of the inner product.

Suppose we have an orthogonal set of vectors $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{R}^n$.

$$\begin{aligned}
\|\vec{v}_1 + \cdots + \vec{v}_k\|^2 &= \langle \vec{v}_1 + \cdots + \vec{v}_k, \vec{v}_1 + \cdots + \vec{v}_k \rangle \\
&= \sum_{i=1}^{k} \sum_{j=1}^{k} \langle \vec{v}_i, \vec{v}_j \rangle \\
&= \sum_{i=1}^{k} \langle \vec{v}_i, \vec{v}_i \rangle + \sum_{1 \leq i,j \leq k, i \neq j} \langle \vec{v}_i, \vec{v}_j \rangle \\
&= \sum_{i=1}^{k} \langle \vec{v}_i, \vec{v}_i \rangle \\
&= \sum_{i=1}^{k} \|\vec{v}_i\|^2.
\end{aligned}$$

This follows since $\langle \vec{v}_i, \vec{v}_j \rangle = 0$ whenever $i \neq j$. Therefore,

$$\|\vec{v}_1 + \cdots + \vec{v}_k\|^2 = \|\vec{v}_1\|^2 + \cdots + \|\vec{v}_k\|^2.$$

□

Notice that, while an orthogonal basis already allows each coordinate to be computed independently, the coefficient formula still contains the normalization factor $\|v_i\|^2$. We may eliminate this factor by scaling each basis vector to have unit length. Since multiplying a non-zero vector by a non-zero scalar does not change the direction it spans, normalizing an orthogonal basis preserves both orthogonality and the subspace it spans.

Definition 1.47 (Orthonormal Set.)**.** A set of vectors $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{R}^n$ is an orthonormal set if it is an orthogonal set and each $\vec{v}_i$ is a unit vector i.e., $\|\vec{v}_i\| = 1$.

Proposition 1.48. *We make note of several special conditions of orthonormal sets.*

(i) An orthonormal set cannot contain the zero vector since $\|\vec{0}\| = 0 \neq 1$. Hence, every orthonormal set is linearly independent.

(ii) Given an orthogonal set $\{\vec{v}_1, \ldots, \vec{v}_k\}$ of non-zero vectors, we may produce an orthonormal set by normalizing $\vec{v}_1, \ldots, \vec{v}_k$.

Definition 1.49. An orthonormal set B that is a basis for subspace $U \subseteq \mathbb{R}^n$ is called an orthonormal basis for U.

Suppose that we have $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ which is an orthonormal basis for a subspace $U \subseteq \mathbb{R}^n$. For $\vec{u} \in U$, there are unique scalars $c_1, \ldots, c_k \in \mathbb{R}$ such that,

$$\vec{u} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k.$$

Since we know that every orthonormal set is an orthogonal set, we obtain, $c_i = \dfrac{\vec{u} \cdot \vec{v}_i}{\|\vec{v}_i\|^2}$.

Since we know that $\|\vec{v}_i\| = 1$ then $c_i = \vec{u} \cdot \vec{v}_i$.

Example 1.50. Given the following orthonormal basis,

$$B = \left\{ \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}, \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix} \right\} \subseteq \mathbb{R}^2,$$

we would like to express $\vec{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ as a linear combination of the vectors. Using the new formulation of $c_i = \vec{u} \cdot \vec{v}_i$ that we have formulated above, we get,

$$\vec{x} = \left(\begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} \right) \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} + \left(\begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix} \right) \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix} = \frac{7}{\sqrt{5}} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} - \frac{1}{\sqrt{5}} \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}.$$

Consider the same basis $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ for $U \subseteq \mathbb{R}^n$ and let $\vec{u} \in U$.

Then, we see that,

$$[\vec{u}]_B = \begin{bmatrix} \frac{\vec{u} \cdot \vec{v}_1}{\|\vec{v}_1\|^2} \\ \vdots \\ \frac{\vec{u} \cdot \vec{v}_k}{\|\vec{v}_k\|^2} \end{bmatrix}, \qquad \text{if } B \text{ is an orthogonal basis for } U,$$

$$[\vec{u}]_B = \begin{bmatrix} \vec{u} \cdot \vec{v}_1 \\ \vdots \\ \vec{u} \cdot \vec{v}_k \end{bmatrix}, \qquad \text{if } B \text{ is an orthonormal basis for } U.$$

Suppose that $B = \{\vec{v}_1, \vec{v}_2\}$ is an orthonormal basis for $\mathbb{R}^2$ and let $P = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix} \in \mathbb{M}_{2\times 2}(\mathbb{R})$. Then $P^\top = \begin{bmatrix} \vec{v}_1^\top \\ \vec{v}_2^\top \end{bmatrix}$. Expanding $P^\top P$ we get the identity matrix,

$$P^\top P = \begin{bmatrix} \vec{v}_1^\top \\ \vec{v}_2^\top \end{bmatrix} \begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix} = \begin{bmatrix} \vec{v}_1^\top \vec{v}_1 & \vec{v}_1^\top \vec{v}_2 \\ \vec{v}_2^\top \vec{v}_1 & \vec{v}_2^\top \vec{v}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Clearly we see that $P^\top = P^{-1}$. We define such matrix as the *orthogonal matrix*.

In fact, orthonormal bases admit a particularly useful matrix representation. If the vectors of an orthonormal basis are placed as the columns of a square matrix P, then the entries of $P^\top P$ are precisely the pairwise inner products of those columns. Thus, orthonormality is equivalent to the matrix identity $P^\top P = I_n$ which motivates the definition of an orthogonal matrix.

Definition 1.51. A matrix $P \in \mathbb{M}_{n\times n}(\mathbb{R})$ such that $P^\top = P^{-1}$ is called an orthogonal matrix.

Example 1.52. For any $\theta \in \mathbb{R}$, the set,

$$\left\{ \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}, \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \right\}$$

is an orthonormal set and a basis in $\mathbb{R}^2$. Similarly, we see that,

$$P = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

performs a counterclockwise rotation through angle θ about the origin and is an orthogonal matrix. Therefore, we conclude that,

$$P^{-1} = P^\top = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

is also an orthogonal matrix.

Theorem 1.53. *Let $P \in \mathbb{M}_{n\times n}(\mathbb{R})$. The following statements are equivalent,*

(i) *P is an orthogonal matrix.*
(ii) *The columns of P form an orthonormal basis for $\mathbb{R}^n$.*
(iii) *The rows of P form an orthonormal basis for $\mathbb{R}^n$.*

Theorem 1.54. *Let $P, Q \in \mathbb{M}_{n\times n}(\mathbb{R})$ be orthogonal matrices and let $\vec{x}, \vec{y} \in \mathbb{R}^n$.*

(i) $(P\vec{x}) \cdot (P\vec{y}) = \vec{x} \cdot \vec{y}$.
(ii) $\|P\vec{x}\| = \|\vec{x}\|$.
(iii) $\det P = \pm 1$.
(iv) *All real eigenvalues of P are* ± 1.
(v) *PQ is an orthogonal matrix.*

Proof. (i) The proof for (i) follows directly by properties of orthogonal matrices,

$$(P\vec{x}) \cdot (P\vec{y}) = (P\vec{x})^\top P\vec{y} = \vec{x}^\top P^\top P\vec{y} = \vec{x}^\top I\vec{y} = \vec{x}^\top \vec{y} = \vec{x} \cdot \vec{y}.$$

(ii) Similar to (i) the proof follows directly from properties of orthogonal matrices,

$$\|P\vec{x}\|^2 = (P\vec{x}) \cdot (P\vec{x}) = \vec{x} \cdot \vec{x} = \|\vec{x}\|^2 \implies \|P\vec{x}\| = \|\vec{x}\|.$$

(iii) Using properties of the transpose of orthogonal matrices,

$$\begin{aligned} P^\top = P^{-1} &\implies \det(P^\top) = \det(P^{-1}) \\ &\implies \det(P) = \frac{1}{\det(P)} \\ &\implies (\det(P))^2 = 1 \\ &\implies \det(P) = \pm 1. \end{aligned}$$

(iv) If $\lambda \in \mathbb{R}$ is an eigenvalue of P with corresponding eigenvector $\vec{x} \neq \vec{0}$, then $P\vec{x} = \lambda\vec{x}$ so $\|P\vec{x}\| = \|\lambda\vec{x}\| = |\lambda|\|\vec{x}\|$. However by (ii), we know that $\|P\vec{x}\| = \|\vec{x}\|$ so $\|\vec{x}\| = |\lambda|\|\vec{x}\|$. Since we know that $\vec{x} \neq \vec{0}$, we have that $|\lambda| = 1$ so $\lambda = \pm 1$.

(v) Similarly, this follows by properties of orthogonal matrices.

$$(PQ)^\top (PQ) = (Q^\top P^\top)(PQ) = Q^\top I Q = Q^\top Q = I,$$

so we know that PQ is orthogonal.

□

We introduce the notions of projections and perpendicular lines.

Suppose that $\vec{x}, \vec{y} \in \mathbb{R}^n$ with $\vec{y} \neq \vec{0}$. Then we define,

$$\text{proj}_{\vec{y}}\, \vec{x} = \frac{\vec{x} \cdot \vec{y}}{\|\vec{y}\|^2} \cdot \vec{y}, \qquad \text{perp}_{\vec{y}}\, \vec{x} = \vec{x} - \text{proj}_{\vec{y}}\, \vec{x}.$$

A common decomposition of a vector is therefore given by,

$$\vec{x} = \text{proj}_{\vec{y}}\, \vec{x} + \text{perp}_{\vec{y}}\, \vec{x}.$$

If we let $U = \text{span}\{\vec{y}\}$ then we can view projecting the vector $\vec{x}$ onto $\vec{y}$ as projecting $\vec{x}$ onto the span U that is,

$$\vec{x} = \underbrace{\text{proj}_{\vec{y}}\, \vec{x}}_{\in U} + \underbrace{\text{perp}_{\vec{y}}\, \vec{x}}_{\text{orthogonal to } U}.$$

Now, for any subspace $U \subseteq \mathbb{R}^n$ and any $\vec{x} \in \mathbb{R}^n$, we'd like $\vec{x} = \text{proj}_U\, \vec{x} + \text{perp}_U\, \vec{x}$ where $\text{proj}_U\, \vec{x}$ is a vector in U and $\text{perp}_U\, \vec{x}$ is a vector orthogonal to U. It remains for us to characterize the definitions of $\text{proj}_U\, \vec{x}$ and $\text{perp}_U\, \vec{x}$. We first define what it means to be orthogonal to a subspace.

Definition 1.55. Let U be a subspace of $\mathbb{R}^n$. Then, a vector $\vec{x} \in \mathbb{R}^n$ is orthogonal to the subspace U if $\vec{x} \cdot \vec{u} = 0$ for every vector $\vec{u} \in U$. The set of all vectors in $\mathbb{R}^n$ orthogonal to U is defined as the orthogonal complement,

$$U^\perp = \{\vec{x} \in \mathbb{R}^n \mid \vec{x} \cdot \vec{u} = 0 \text{ for every } \vec{u} \in U\}.$$

This is quite a verbose definition since it requires us to show that $\vec{x} \in U^\perp$ by showing $\vec{x} \cdot \vec{u} = 0$ for every $\vec{u} \in U$. The next theorem shows we only need to show $\vec{x}$ is orthogonal to the vectors in any spanning set of the subspace U.

Theorem 1.56. *Let $\{\vec{v}_1, \ldots, \vec{v}_k\}$ be a spanning set for a subspace $U \subseteq \mathbb{R}^n$. Then $\vec{x} \in U^\perp$ if and only if $\vec{x} \cdot \vec{v}_i = 0$ for all $i = 1, \ldots, k$.*

Proof. Since $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is a spanning set for U then we have that $\vec{v}_1, \ldots \vec{v}_k \in U$. Assume first that $\vec{x} \in U^\perp$. Then it follows that $\vec{x} \cdot \vec{u} = 0$ for every $\vec{u} \in U$. In particular, we know that $\vec{x} \cdot \vec{v}_i = 0$ for

every $i = 1, \ldots, k$. Now assume that $\vec{x} \cdot \vec{v}_i = 0$ for every $i = 1, \ldots, k$ and let $\vec{u} \in U$. Then there are scalars $c_1, \ldots, c_k \in \mathbb{R}$ such that $\vec{u} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$. Then,

$$\begin{aligned}\vec{x} \cdot \vec{u} &= \vec{x} \cdot (c_1\vec{v}_1 + \cdots + c_k\vec{v}_k) \\ &= c_1(\vec{x} \cdot \vec{v}_1) + \cdots + c_k(\vec{x} \cdot \vec{v}_k) \\ &= c_1(0) + \cdots + c_k(0) \\ &= 0.\end{aligned}$$

So, we conclude that $\vec{x} \in U^\perp$. □

Quite remarkably, we can write every vector in $\mathbb{R}^n$ uniquely as the sum of a component in U and a component orthogonal to U and orthogonal bases of these two subspaces combine to form an orthogonal basis for the entire space.

Example 1.57. We wish to find $U^\perp$ for the subspace

$$U = \text{span}\left\{\begin{bmatrix}1\\-1\\1\\-1\end{bmatrix}, \begin{bmatrix}1\\0\\1\\0\end{bmatrix}\right\}.$$

Clearly, we are given a spanning set and thus must find $\vec{x} \in \mathbb{R}^4$ such that,

$$\begin{bmatrix}1\\-1\\1\\-1\end{bmatrix} \cdot \begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} = x_1 - x_2 + x_3 - x_4 = 0, \qquad \begin{bmatrix}1\\0\\1\\0\end{bmatrix} \cdot \begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} = x_1 + x_3 = 0.$$

This is a homogeneous system of equations, which we solve by row reduction:

$$\begin{bmatrix}1 & -1 & 1 & -1\\1 & 0 & 1 & 0\end{bmatrix} \longrightarrow \begin{bmatrix}1 & 0 & 1 & 0\\0 & 1 & 0 & 1\end{bmatrix} \implies x_1 = -s, \qquad x_2 = -t, \qquad x_3 = s, \qquad x_4 = t,$$

Therefore, we see that,

$$\vec{x} = s\begin{bmatrix}-1\\0\\1\\0\end{bmatrix} + t\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}, \quad s, t \in \mathbb{R} \implies U^\perp = \text{span}\left\{\begin{bmatrix}-1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\-1\\0\\1\end{bmatrix}\right\}.$$

This also confirms directly that $U^\perp$ is a subspace of $\mathbb{R}^4$.

Theorem 1.58. *Let U be a k-dimensional subspace of $\mathbb{R}^n$. Then,*

(i) *$U^\perp$ is a subspace of $\mathbb{R}^n$.*
(ii) $\dim(U^\perp) = n - k$.
(iii) $U \cap U^\perp = \{\vec{0}\}$.
(iv) *If $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is an orthogonal basis for U and $\{\vec{v}_{k+1}, \ldots, \vec{v}_n\}$ is an orthogonal basis for $U^\perp$, then $\{\vec{v}_1, \ldots, \vec{v}_n\}$ is an orthogonal basis for $\mathbb{R}^n$.*

Proof. (i) The zero vector belongs to $U^\perp$. If $\vec{x}, \vec{y} \in U^\perp$ and $a, b \in \mathbb{R}$, then for every $\vec{u} \in U$,

$$(a\vec{x} + b\vec{y}) \cdot \vec{u} = a(\vec{x} \cdot \vec{u}) + b(\vec{y} \cdot \vec{u}) = 0.$$

Thus, $a\vec{x} + b\vec{y} \in U^\perp$, so $U^\perp$ is a subspace.

(ii) Let $\{\vec{u}_1, \ldots, \vec{u}_k\}$ be a basis for U and define $T : \mathbb{R}^n \to \mathbb{R}^k$ by

$$T(\vec{x}) = \begin{bmatrix} \vec{u}_1 \cdot \vec{x} \\ \vdots \\ \vec{u}_k \cdot \vec{x} \end{bmatrix}.$$

Its kernel is exactly $U^\perp$. The Gram matrix $G = [\vec{u}_i \cdot \vec{u}_j]_{i,j=1}^k$ is positive definite because, for every non-zero $\vec{c} \in \mathbb{R}^k$,

$$\vec{c}^\top G \vec{c} = \left\| \sum_{i=1}^k c_i \vec{u}_i \right\|^2 > 0.$$

Hence G is invertible, which shows that T is onto: for any $\vec{y} \in \mathbb{R}^k$, set $\vec{c} = G^{-1}\vec{y}$ and $\vec{x} = \sum_{i=1}^k c_i \vec{u}_i$; then $T(\vec{x}) = \vec{y}$. Therefore $\operatorname{rank}(T) = k$, and the Rank–Nullity Theorem gives

$$\dim(U^\perp) = \dim(\operatorname{null}(T)) = n - k.$$

(iii) If $\vec{x} \in U \cap U^\perp$, then $\vec{x} \cdot \vec{x} = 0$ because $\vec{x} \in U^\perp$ and $\vec{x} \in U$. Hence $\vec{x} = \vec{0}$, so $U \cap U^\perp = \{\vec{0}\}$.

(iv) Each vector in U is orthogonal to each vector in $U^\perp$. Thus, the union of the two stated orthogonal bases is an orthogonal set of n non-zero vectors in $\mathbb{R}^n$. It is therefore linearly independent, and any linearly independent set of n vectors in $\mathbb{R}^n$ is a basis.

□

Consider a subspace U of $\mathbb{R}^n$ with an orthogonal basis $\{\vec{v}_1, \ldots, \vec{v}_k\}$ and let $\{\vec{v}_{k+1}, \ldots, \vec{v}_n\}$ be an orthogonal basis for $U^\perp$. Then $\{\vec{v}_1, \ldots, \vec{v}_n\}$ is an orthogonal basis for $\mathbb{R}^n$. For any $\vec{x} \in \mathbb{R}^n$,

$$\vec{x} = \underbrace{\frac{\vec{x} \cdot \vec{v}_1}{\|\vec{v}_1\|^2}\vec{v}_1 + \cdots + \frac{\vec{x} \cdot \vec{v}_k}{\|\vec{v}_k\|^2}\vec{v}_k}_{\in U} + \underbrace{\frac{\vec{x} \cdot \vec{v}_{k+1}}{\|\vec{v}_{k+1}\|^2}\vec{v}_{k+1} + \cdots + \frac{\vec{x} \cdot \vec{v}_n}{\|\vec{v}_n\|^2}\vec{v}_n}_{\in U^\perp}.$$

Definition 1.59. Let $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ be an orthogonal basis for a k-dimensional subspace U of $\mathbb{R}^n$, and let $\vec{x} \in \mathbb{R}^n$. The projection of $\vec{x}$ onto U is

$$\text{proj}_U \vec{x} = \frac{\vec{x} \cdot \vec{v}_1}{\|\vec{v}_1\|^2}\vec{v}_1 + \cdots + \frac{\vec{x} \cdot \vec{v}_k}{\|\vec{v}_k\|^2}\vec{v}_k,$$

and the projection of $\vec{x}$ perpendicular to U is

$$\text{perp}_U \vec{x} = \vec{x} - \text{proj}_U \vec{x}.$$

Notice that we have defined $\text{perp}_U \vec{x}$ in a way that does not require an orthogonal basis for $U^\perp$. We thus ensure that $\text{perp}_U \vec{x} \in U^\perp$. For $1 \leq i \leq k$,

$$\begin{aligned}
\vec{v}_i \cdot \text{perp}_U \vec{x} &= \vec{v}_i \cdot \left(\vec{x} - \frac{\vec{x} \cdot \vec{v}_1}{\|\vec{v}_1\|^2}\vec{v}_1 - \cdots - \frac{\vec{x} \cdot \vec{v}_k}{\|\vec{v}_k\|^2}\vec{v}_k \right) \\
&= \vec{v}_i \cdot \vec{x} - \frac{\vec{x} \cdot \vec{v}_i}{\|\vec{v}_i\|^2}(\vec{v}_i \cdot \vec{v}_i) \\
&= \vec{v}_i \cdot \vec{x} - \vec{v}_i \cdot \vec{x} \\
&= 0.
\end{aligned}$$

Therefore, $\text{perp}_U \vec{x} \in U^\perp$ since it is orthogonal to every $\vec{v}_i \in B$.

Example 1.60. Consider the subspace spanned by the following set,

$$U = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} \right\} \quad \text{and} \quad \vec{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

We wish to compute $\text{proj}_U \vec{x}$ and $\text{perp}_U \vec{x}$. We define the following vectors,

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}.$$

Clearly, since we know that $\vec{v}_1 \cdot \vec{v}_2 = 0$ and $B = \{\vec{v}_1, \vec{v}_2\}$ is an orthogonal basis for our subspace U, we now can compute,

$$\text{proj}_U \vec{x} = \frac{\vec{x} \cdot \vec{v}_1}{\|\vec{v}_1\|^2}\,\vec{v}_1 + \frac{\vec{x} \cdot \vec{v}_2}{\|\vec{v}_2\|^2}\,\vec{v}_2 = -\frac{3}{10}\begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix} - \frac{4}{4}\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -\frac{13}{10} \\ -\frac{7}{10} \\ \frac{2}{5} \\ \frac{8}{5} \end{bmatrix},$$

$$\text{perp}_U \vec{x} = \vec{x} - \text{proj}_U \vec{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} -\frac{13}{10} \\ -\frac{7}{10} \\ \frac{2}{5} \\ \frac{8}{5} \end{bmatrix} = \begin{bmatrix} \frac{23}{10} \\ \frac{27}{10} \\ \frac{13}{5} \\ \frac{12}{5} \end{bmatrix}.$$

Geometrically, $\text{proj}_U \vec{x}$ should be the closest vector in U to $\vec{x}$.

Theorem 1.61. *Let U be a subspace of $\mathbb{R}^n$. For any $\vec{x} \in \mathbb{R}^n$ the vector in U that is closest to $\vec{x}$ is $\text{proj}_U \vec{x}$. That is, for any $\vec{v} \in U$ with $\vec{v} \neq \text{proj}_U \vec{x}$ then,*

$$\|\vec{x} - \text{proj}_U \vec{x}\| < \|\vec{x} - \vec{v}\|.$$

Proof. Let $\vec{v} \in U$ with $\vec{v} \neq \text{proj}_U \vec{x}$. We decompose the vector $\vec{x} - \vec{v}$ as

$$\vec{x} - \vec{v} = \Big(\vec{x} - \text{proj}_U \vec{x}\Big) + \Big(\text{proj}_U \vec{x} - \vec{v}\Big).$$

By the definition of the perpendicular projection,

$$\vec{x} - \text{proj}_U \vec{x} = \text{perp}_U \vec{x}.$$

Since $\text{perp}_U \vec{x} \in U^\perp$, it follows that $\vec{x} - \text{proj}_U \vec{x} \in U^\perp$. On the other hand, both $\text{proj}_U \vec{x}$ and $\vec{v}$ belong to U. Since U is a subspace and is therefore closed under subtraction,

$$\text{proj}_U \vec{x} - \vec{v} \in U.$$

Every vector in $U^\perp$ is orthogonal to every vector in U. Therefore,

$$\Big(\vec{x} - \text{proj}_U \vec{x}\Big) \cdot \Big(\text{proj}_U \vec{x} - \vec{v}\Big) = 0.$$

Hence, the vectors $\vec{x} - \text{proj}_U \vec{x}$ and $\text{proj}_U \vec{x} - \vec{v}$ are orthogonal.

We may therefore apply the Pythagorean Theorem to obtain

$$\begin{aligned}\|\vec{x} - \vec{v}\|^2 &= \left\|\Big(\vec{x} - \text{proj}_U \vec{x}\Big) + \Big(\text{proj}_U \vec{x} - \vec{v}\Big)\right\|^2 \\ &= \|\vec{x} - \text{proj}_U \vec{x}\|^2 + \|\text{proj}_U \vec{x} - \vec{v}\|^2.\end{aligned}$$

Since $\vec{v} \neq \text{proj}_U \vec{x}$, we have $\text{proj}_U \vec{x} - \vec{v} \neq \vec{0}$, and therefore $\|\text{proj}_U \vec{x} - \vec{v}\|^2 > 0$. It follows that

$$\|\vec{x} - \vec{v}\|^2 > \|\vec{x} - \text{proj}_U \vec{x}\|^2.$$

Because both norms are nonnegative, taking square roots preserves the strict inequality. Thus,

$$\|\vec{x} - \vec{v}\| > \|\vec{x} - \text{proj}_U \vec{x}\|.$$

Therefore, $\text{proj}_U \vec{x}$ is the unique vector in U that is closest to $\vec{x}$. □

We have seen the importance of orthogonal (orthonormal) bases and how simple it is to work with them. But how do we actually find an orthogonal (orthonormal) basis for a subspace $U \subseteq \mathbb{R}^n$?

Before we define the procedure, we first introduce the notion of modifying each vector without changing the subspace spanned by the original set.

Theorem 1.62. *If* $\vec{v}_1, \ldots, \vec{v}_k \in \mathbb{R}^n$ *then,*

$$\text{span}\{\vec{v}_1, \ldots, \vec{v}_k\} = \text{span}\{\vec{v}_1, \ldots, \vec{v}_{k-1}, \vec{v}_k + t_1\vec{v}_1 + \cdots + t_{k-1}\vec{v}_{k-1}\},$$

for any $t_1, \ldots, t_{k-1} \in \mathbb{R}$.

Proof. Let $t_1, \ldots, t_{k-1} \in \mathbb{R}$, and define

$$A = \text{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$$

and

$$B = \text{span}\left\{\vec{v}_1, \ldots, \vec{v}_{k-1}, \vec{v}_k + t_1\vec{v}_1 + \cdots + t_{k-1}\vec{v}_{k-1}\right\}.$$

Let $\vec{x} \in A$. Then, for some $c_1, \ldots, c_k \in \mathbb{R}$,

$$\vec{x} = c_1\vec{v}_1 + \cdots + c_{k-1}\vec{v}_{k-1} + c_k\vec{v}_k.$$

We may rewrite this expression as

$$\begin{aligned}\vec{x} &= (c_1 - c_k t_1)\vec{v}_1 + \cdots + (c_{k-1} - c_k t_{k-1})\vec{v}_{k-1} + c_k\vec{v}_k + c_k t_1\vec{v}_1 + \cdots + c_k t_{k-1}\vec{v}_{k-1} \\ &= (c_1 - c_k t_1)\vec{v}_1 + \cdots + (c_{k-1} - c_k t_{k-1})\vec{v}_{k-1} + c_k\left(\vec{v}_k + t_1\vec{v}_1 + \cdots + t_{k-1}\vec{v}_{k-1}\right).\end{aligned}$$

Thus, $\vec{x}$ is a linear combination of the vectors spanning B, so $\vec{x} \in B$. Therefore,

$$A \subseteq B.$$

Now let $\vec{y} \in B$. Then, for some $d_1, \ldots, d_k \in \mathbb{R}$,

$$\vec{y} = d_1\vec{v}_1 + \cdots + d_{k-1}\vec{v}_{k-1} + d_k\left(\vec{v}_k + t_1\vec{v}_1 + \cdots + t_{k-1}\vec{v}_{k-1}\right).$$

Expanding and collecting like terms gives

$$\begin{aligned}\vec{y} &= d_1\vec{v}_1 + \cdots + d_{k-1}\vec{v}_{k-1} + d_k\vec{v}_k + d_k t_1\vec{v}_1 + \cdots + d_k t_{k-1}\vec{v}_{k-1} \\ &= (d_1 + d_k t_1)\vec{v}_1 + \cdots + (d_{k-1} + d_k t_{k-1})\vec{v}_{k-1} + d_k\vec{v}_k.\end{aligned}$$

Hence, $\vec{y}$ is a linear combination of $\vec{v}_1, \ldots, \vec{v}_k$, so $\vec{y} \in A$. Therefore,

$$B \subseteq A.$$

Since $A \subseteq B$ and $B \subseteq A$, it follows that

$$A = B.$$

□

Theorem 1.63 (Gram-Schmidt Procedure)**.** *Let $\{\vec{w}_1, \ldots, \vec{w}_k\}$ be a basis for a subspace U of $\mathbb{R}^n$.*

(i) Let $\vec{v}_1 = \vec{w}_1$ and let $U_1 = \text{span}\{\vec{v}_1\} = \text{span}\{\vec{w}_1\}$.

(ii) Let $\vec{v}_2$ be defined as,

$$\vec{v}_2 = \text{perp}_{U_1} \vec{w}_2 = \vec{w}_2 - \frac{\vec{w}_2 \cdot \vec{v}_1}{\|\vec{v}_1\|^2} \cdot \vec{v}_1,$$

and let $U_2 = \text{span}\{\vec{v}_1, \vec{v}_2\} = \text{span}\{\vec{w}_1, \vec{w}_2\}$.

$$\vdots$$

(iii) At step i, after $i-1$ steps, $\{\vec{v}_1, \ldots, \vec{v}_{i-1}\}$ is an orthogonal set and $U_{i-1} = \text{span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}\} = \text{span}\{\vec{w}_1, \ldots, \vec{w}_{i-1}\}$. Then let,

$$\vec{v}_i = \text{perp}_{U_{i-1}} \vec{w}_i = \vec{w}_i - \frac{\vec{w}_i \cdot \vec{v}_1}{\|\vec{v}_1\|^2}\vec{v}_1 - \cdots - \frac{\vec{w}_i \cdot \vec{v}_{i-1}}{\|\vec{v}_{i-1}\|^2}\vec{v}_{i-1}.$$

After step k, $\{\vec{v}_1, \ldots, \vec{v}_k\}$ will be an orthogonal basis for U.

Note that at the k-th step, we replace $\vec{w}_k$ by

$$\vec{v}_k = \vec{w}_k - \sum_{i=1}^{k-1} \frac{\vec{w}_k \cdot \vec{v}_i}{\|\vec{v}_i\|^2}\vec{v}_i,$$

which differs from $\vec{w}_k$ only by a linear combination of the preceding vectors. Therefore $\text{span}\{\vec{w}_1, \ldots, \vec{w}_k\} = \text{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$ while the projection terms are chosen so that $\vec{v}_k \cdot \vec{v}_i = 0$ for every $i < k$. Repeating this replacement produces an orthogonal set spanning exactly the same subspace as the original vectors. If the original set is linearly independent, then none of the resulting vectors is zero.

To introduce the notion of least squares, we will introduce the method of least squares using the concepts we have used so far. Let $A = \begin{bmatrix} \vec{v}_1 & \cdots & \vec{v}_n \end{bmatrix} \in \mathbb{M}_{m\times n}(\mathbb{R})$ such that $\vec{v}_1, \ldots, \vec{v}_n \in \mathbb{R}^m$. Then we denote the column space of A as $\text{Col}(A) = \{A\vec{x} \mid \vec{x} \in \mathbb{R}^n\} = \text{span}\{\vec{v}_1, \ldots, \vec{v}_n\}$. Similarly, the nullspace of A is $\text{null}(A) = \{\vec{x} \in \mathbb{R}^n \mid A\vec{x} = \vec{0}\}$. The row space of A is $\text{row}(A) = \{A^\top \vec{x} \mid \vec{x} \in \mathbb{R}^m\} = \text{Col}(A^\top)$ and the left nullspace of A is $\text{null}(A^\top) = \{\vec{x} \in \mathbb{R}^m \mid A^\top \vec{x} = \vec{0}\}$.

Recall that $\text{Col}(A)$ and $\text{null}(A^\top)$ are subspaces of $\mathbb{R}^m$ while $\text{row}(A)$ and $\text{null}(A)$ are subspaces of $\mathbb{R}^n$ where m denotes the number of rows and n denotes the number of columns.

For $A \in \mathbb{M}_{m\times n}(\mathbb{R})$ we also have that,

$$A = \begin{bmatrix} \vec{r}_1^\top \\ \vdots \\ \vec{r}_m^\top \end{bmatrix} \implies A\vec{x} = \begin{bmatrix} \vec{r}_1^\top \\ \vdots \\ \vec{r}_m^\top \end{bmatrix} \vec{x} = \begin{bmatrix} \vec{r}_1 \cdot \vec{x} \\ \vdots \\ \vec{r}_m \cdot \vec{x} \end{bmatrix},$$

for some vectors $\vec{r}_1, \ldots, \vec{r}_m \in \mathbb{R}^n$ and any vector $\vec{x} \in \mathbb{R}^n$. It follows that $\text{null}(A) = \{\vec{x} \in \mathbb{R}^n \mid \vec{x} \cdot \vec{r} = 0 \quad \forall \vec{r} \in \text{row}(A)\} = (\text{row}(A))^\perp$. Similarly, we have that $\text{null}(A^\top) = \{\vec{x} \in \mathbb{R}^m \mid \vec{x} \cdot \vec{c} = 0 \quad \forall \vec{c} \in \text{Col}(A)\} = (\text{Col}(A))^\perp$.

Example 1.64. Suppose we have the vectors,

$$\vec{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \vec{w}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \qquad \text{and} \qquad \vec{w}_3 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}.$$

Using the Gram-Schmidt Orthogonalization procedure, we wish to find an orthogonal basis for $U = \text{span}\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$.

(i) Let $\vec{v}_1 = \vec{w}_1$ and $U_1 = \text{span}\{\vec{v}_1\} = \text{span}\{\vec{w}_1\}$.

(ii) Computing,

$$\begin{aligned} \text{perp}_{U_1} \vec{w}_2 = \vec{w}_2 - \text{proj}_{U_1} \vec{w}_2 &= \vec{w}_2 - \frac{\vec{w}_2 \cdot \vec{v}_1}{\|\vec{v}_1\|^2} \vec{v}_1 \\ &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix} - \frac{3}{3} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}. \end{aligned}$$

We let $\vec{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$. Then we see that $\vec{v}_2 \cdot \vec{v}_1 = 0$. We now let $U_2 = \text{span}\{\vec{v}_1, \vec{v}_2\} = \text{span}\{\vec{w}_1, \vec{w}_2\}$,

$$\begin{aligned} \text{span}\{\vec{w}_1, \vec{w}_2\} &= \text{span}\{\vec{v}_1, \vec{w}_2\} \\ &= \text{span}\left\{ \vec{v}_1,\ \vec{w}_2 - \frac{\vec{w}_2 \cdot \vec{v}_1}{\|\vec{v}_1\|^2} \vec{v}_1 \right\} \\ &= \text{span}\{\vec{v}_1, \vec{v}_2\}. \end{aligned}$$

(iii) Computing,

$$\begin{aligned}\operatorname{perp}_{U_2}\vec{w}_3 &= \vec{w}_3 - \operatorname{proj}_{U_2}\vec{w}_3\\ &= \vec{w}_3 - \frac{\vec{w}_3\cdot\vec{v}_1}{\|\vec{v}_1\|^2}\,\vec{v}_1 - \frac{\vec{w}_3\cdot\vec{v}_2}{\|\vec{v}_2\|^2}\,\vec{v}_2\\ &= \begin{bmatrix}1\\2\\-1\\3\end{bmatrix} - \frac{6}{3}\begin{bmatrix}1\\1\\0\\1\end{bmatrix} - \frac{1}{2}\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}\\ &= \begin{bmatrix}-1\\\frac{1}{2}\\-1\\\frac{1}{2}\end{bmatrix}.\end{aligned}$$

We let $\vec{v}_3 = \begin{bmatrix}-2\\1\\-2\\1\end{bmatrix}$. So then we see that $\vec{v}_3\cdot\vec{v}_2 = 0 = \vec{v}_3\cdot\vec{v}_1$ and $U = \operatorname{span}\{\vec{w}_1,\vec{w}_2,\vec{w}_3\} = \operatorname{span}\{\vec{v}_1,\vec{v}_2,\vec{v}_3\}$, hence we have that $\{\vec{v}_1,\vec{v}_2,\vec{v}_3\}$ is an orthogonal basis for U.
Note that our spanning equality follows because we have that,

$$\begin{aligned}\operatorname{span}\{\vec{w}_1,\vec{w}_2,\vec{w}_3\} &= \operatorname{span}\{\vec{v}_1,\vec{v}_2,\vec{w}_3\}\\ &= \operatorname{span}\left\{\vec{v}_1,\ \vec{v}_2,\ \vec{w}_3 - \frac{\vec{w}_3\cdot\vec{v}_1}{\|\vec{v}_1\|^2}\,\vec{v}_1 - \frac{\vec{w}_3\cdot\vec{v}_2}{\|\vec{v}_2\|^2}\,\vec{v}_2\right\}\\ &= \operatorname{span}\{\vec{v}_1,\vec{v}_2,\vec{v}_3\}.\end{aligned}$$

Suppose that we have conducted an experiment that measures a quantity y at times t. We obtain the data points $(t_1,y_1),\ldots,(t_m,y_m)$. Plotting the points, we observe that the data have approximately the shape of a quadratic polynomial of the form $y = a + bt + ct^2$.

An exact fit would require the system

$$\begin{aligned}y_1 &= a + bt_1 + ct_1^2\\ &\vdots\\ y_m &= a + bt_m + ct_m^2,\end{aligned}$$

which is typically inconsistent unless every data point lies exactly on one quadratic curve.

If we re-write the system into vector form,

$$\underbrace{\begin{bmatrix}y_1\\\vdots\\y_m\end{bmatrix}}_{=\vec{y}} \overset{?}{=} \begin{bmatrix}a+bt_1+ct_1^2\\\vdots\\a+bt_m+ct_m^2\end{bmatrix} = \underbrace{\begin{bmatrix}1 & t_1 & t_1^2\\\vdots & \vdots & \vdots\\1 & t_m & t_m^2\end{bmatrix}}_{:=X}\underbrace{\begin{bmatrix}a\\b\\c\end{bmatrix}}_{=\vec{a}}$$

in order to obtain the candidate system $\vec{y} = X\vec{a}$. In the typical inconsistent case, no such $\vec{a}$ satisfies the system exactly. Instead, we seek to find $\vec{a}$ such that $X\vec{a}$ is as close as possible to $\vec{y}$; that is, we minimize the norm $\|\vec{y} - X\vec{a}\|$. Since $X\vec{a} \in \text{Col}(X)$, the closest vector in $\text{Col}(X)$ to $\vec{y}$ is $\text{proj}_{\text{Col}(X)}\,\vec{y}$ i.e., we solve the consistent system,

$$X\vec{a} = \text{proj}_{\text{Col}(X)}\,\vec{y}.$$

It follows that we can re-write $\vec{y} - X\vec{a}$ as,

$$\vec{y} - X\vec{a} = \vec{y} - \text{proj}_{\text{Col}(X)}\,\vec{y} = \text{perp}_{\text{Col}(X)}\,\vec{y} \in (\text{Col}(X))^{\perp}.$$

Since the orthogonal complement of the column space of X is the nullspace of $X^\top$, we write $\vec{y} - X\vec{a} \in \text{null}(X^\top)$. By definition of nullspace we can write,

$$X^\top(\vec{y} - X\vec{a}) = \vec{0} \implies X^\top X\vec{a} = X^\top\vec{y}.$$

Note that all solutions to $X^\top X\vec{a} = X^\top\vec{y}$ minimize the distance $\|\vec{y} - X\vec{a}\|$ which is why we call this the Method of Least Squares since,

$$\begin{aligned}
\|\vec{y} - X\vec{a}\|^2 &= \left\| \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} - \left(a \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + b \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix} + c \begin{bmatrix} t_1^2 \\ \vdots \\ t_m^2 \end{bmatrix} \right) \right\|^2 \\
&= \sum_{i=1}^{m} \left(y_i - \left(a + bt_i + ct_i^2 \right) \right)^2 \\
&= \sum_{i=1}^{m} (\text{err}_i)^2,
\end{aligned}$$

where $\text{err}_i = y_i - (a + bt_i + ct_i^2)$ for $i = 1, \ldots, m$.

Example 1.65. We wish to find the best fitting equation of the form $y = a + bt + ct^2$ for the data,

t	-1	0	1	3
y	4	3	5	9

We have the following system,

$$\begin{aligned}
(-1, 4): &\quad a + b(-1) + c(-1)^2 = 4 &\implies &\quad a - b + c = 4, \\
(0, 3): &\quad a + b(0) + c(0)^2 = 3 &\implies &\quad a = 3, \\
(1, 5): &\quad a + b(1) + c(1)^2 = 5 &\implies &\quad a + b + c = 5, \\
(3, 9): &\quad a + b(3) + c(3)^2 = 9 &\implies &\quad a + 3b + 9c = 9.
\end{aligned}$$

This gives the matrix equation

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \\ 9 \end{bmatrix}.$$

If we define the following matrices/vectors for our least squares set up,

$$\vec{y} = \begin{bmatrix} 4 \\ 3 \\ 5 \\ 9 \end{bmatrix}, \qquad \vec{a} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \qquad X = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{bmatrix},$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 3 \\ 1 & 0 & 1 & 9 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 3 & 11 \\ 3 & 11 & 27 \\ 11 & 27 & 83 \end{bmatrix},$$

$$X^T \vec{y} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 3 \\ 1 & 0 & 1 & 9 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 21 \\ 28 \\ 90 \end{bmatrix}.$$

We can now solve the system $X^\top X\vec{a} = X^\top \vec{y}$,

$$\begin{bmatrix} 4 & 3 & 11 & 21 \\ 3 & 11 & 27 & 28 \\ 11 & 27 & 83 & 90 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & \frac{201}{55} \\ 0 & 1 & 0 & \frac{83}{220} \\ 0 & 0 & 1 & \frac{21}{44} \end{bmatrix}.$$

So the resulting fitted least squares equation is,

$$p(t) = \frac{201}{55} + \frac{83}{220}t + \frac{21}{44}t^2.$$

In the quadratic polynomial model above, $X^\top X$ is invertible exactly when the data contain at least three distinct values of t. More generally, for a degree-d polynomial design with columns $1, t, \ldots, t^d$, invertibility holds exactly when the data contain at least $d+1$ distinct values of t. For an arbitrary design matrix, $X^\top X$ is invertible if and only if the columns of X are linearly independent.

That is, for $X^\top X$ invertible, we may solve the system $X^\top X\vec{a} = X^\top \vec{y}$ which if we solve for the coefficient vector $\vec{a}$ we get,

$$X^\top X\vec{a} = X^\top \vec{y} \iff \vec{a} = (X^\top X)^{-1} X^\top \vec{y}.$$

Example 1.66. We wish to determine a, b to obtain the best-fitting equation of the form

$$y = a + bt$$

for the data points

$$(-2, 1), \qquad (1, 2), \qquad (2, 1).$$

Obtaining our full system of equations we get,

$$\begin{aligned}(-2,1): &\quad a+b(-2)=1 &\implies &\quad a-2b=1,\\ (1,2): &\quad a+b(1)=2 &\implies &\quad a+b=2,\\ (2,1): &\quad a+b(2)=1 &\implies &\quad a+2b=1.\end{aligned}$$

We put the systems into matrices and solve using the $\vec{a}$ formulation we derived above,

$$\underbrace{\begin{bmatrix}1 & -2\\ 1 & 1\\ 1 & 2\end{bmatrix}}_{X}\underbrace{\begin{bmatrix}a\\ b\end{bmatrix}}_{\vec{a}}=\underbrace{\begin{bmatrix}1\\ 2\\ 1\end{bmatrix}}_{\vec{y}}.$$

Substituting our values and simplifying we get,

$$\begin{gathered}X^TX\vec{a}=X^T\vec{y}\\ \implies \quad \begin{bmatrix}1 & 1 & 1\\ -2 & 1 & 2\end{bmatrix}\begin{bmatrix}1 & -2\\ 1 & 1\\ 1 & 2\end{bmatrix}\begin{bmatrix}a\\ b\end{bmatrix}=\begin{bmatrix}1 & 1 & 1\\ -2 & 1 & 2\end{bmatrix}\begin{bmatrix}1\\ 2\\ 1\end{bmatrix}\\ \implies \quad \begin{bmatrix}3 & 1\\ 1 & 9\end{bmatrix}\begin{bmatrix}a\\ b\end{bmatrix}=\begin{bmatrix}4\\ 2\end{bmatrix}.\end{gathered}$$

Therefore,

$$\begin{aligned}\begin{bmatrix}a\\ b\end{bmatrix}&=\frac{1}{26}\begin{bmatrix}9 & -1\\ -1 & 3\end{bmatrix}\begin{bmatrix}4\\ 2\end{bmatrix}\\ &=\frac{1}{26}\begin{bmatrix}34\\ 2\end{bmatrix}\\ &=\begin{bmatrix}\frac{17}{13}\\ \frac{1}{13}\end{bmatrix}.\end{aligned}$$

Hence, the best-fitting equation is

$$p(t)=\frac{17}{13}+\frac{1}{13}t.$$

Definition 1.67. A system of linear equations is **overdetermined** if it has more equations than unknowns.

Example 1.68. For example, if a system has 2 unknowns and 3 equations, then we consider it overdetermined. A system with 2 equations and 2 unknowns is considered determined.

Note that overdetermined systems of the form $A\vec{x}=\vec{b}$ are often inconsistent. In this case, we instead find $\vec{x}$ that minimizes $\|A\vec{x}-\vec{b}\|$. We do this by solving the normal system $A^\top A\vec{x}=A^\top\vec{b}$. If A has full column rank, the minimizer is unique and is given by $\vec{x}=(A^\top A)^{-1}A^\top\vec{b}$.

1.4 Eigenvalues, singular values, and conditioning

Up to this point, orthogonality has been developed using the standard dot product on $\mathbb{R}^n$. However, the essential arguments depend not on the particular form $\vec{x}^\top \vec{y}$ but only on the algebraic and geometric properties satisfied by the formula.

This motivates the abstraction of an *inner product space* in which an inner product equips general vector spaces with notions of length, angle, orthogonality, projection, and the orthonormal basis. Once an inner product is available, we can distinguish operators that interact naturally with the geometry of the space, especially symmetric matrices, and more generally, self-adjoint operators. Their eigenvectors corresponding to distinct eigenvalues are orthogonal, and the Spectral Theorem will show that every real symmetric matrix admits an orthonormal basis of eigenvectors and is therefore orthogonally diagonalizable.

Definition 1.69. Let $\mathbb{V}$ be a vector space. An inner product on $\mathbb{V}$ is a function $\langle \cdot, \cdot \rangle : \mathbb{V} \times \mathbb{V} \to \mathbb{R}$ such that for all $\vec{v}, \vec{w}, \vec{z} \in \mathbb{V}$, and scalars $s, t \in \mathbb{R}$,

(i) $\langle \vec{v}, \vec{v} \rangle \geq 0$, with equality if and only if $\vec{v} = \vec{0}$.
(ii) $\langle \vec{v}, \vec{w} \rangle = \langle \vec{w}, \vec{v} \rangle$.
(iii) $\langle \vec{v}, s\vec{w} + t\vec{z} \rangle = s\langle \vec{v}, \vec{w} \rangle + t\langle \vec{v}, \vec{z} \rangle$.

Property (i) is also referred to as positive definiteness. (ii) shows the symmetric nature of inner products and (iii) states right-linearity of the inner product. $\mathbb{V}$ together with an inner product is called an inner product space. Note that $\langle \cdot, \cdot \rangle : \mathbb{V} \times \mathbb{V} \to \mathbb{R}$ means that the inner product takes two vectors from $\mathbb{V}$ and returns a real number. Over $\mathbb{R}$, (ii) and (iii) together imply left-linearity as well. Hence, real inner products are bilinear, that is, they are linear in both arguments.

Example 1.70. Suppose we define $\langle \vec{x}, \vec{y} \rangle$ by $4x_1y_1 + 3x_2y_2$ on $\mathbb{R}^2$. For any $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^2$ and scalars $s, t \in \mathbb{R}$, we have

(i) $\langle \vec{x}, \vec{x} \rangle = 4x_1^2 + 3x_2^2 \geq 0$, with equality if and only if $\vec{x} = \vec{0}$;
(ii) $\langle \vec{x}, \vec{y} \rangle = 4x_1y_1 + 3x_2y_2 = 4y_1x_1 + 3y_2x_2 = \langle \vec{y}, \vec{x} \rangle$; and
(iii)

$$\begin{aligned} \langle \vec{x},\, s\vec{y} + t\vec{z} \rangle &= 4x_1(sy_1 + tz_1) + 3x_2(sy_2 + tz_2) \\ &= s(4x_1y_1 + 3x_2y_2) + t(4x_1z_1 + 3x_2z_2) \\ &= s\,\langle \vec{x}, \vec{y} \rangle + t\,\langle \vec{x}, \vec{z} \rangle\,. \end{aligned}$$

Therefore, $\langle \vec{x}, \vec{y} \rangle = 4x_1y_1 + 3x_2y_2$ is an inner product on $\mathbb{R}^2$.

We define a few key operators useful for future chapters.

Definition 1.71 (Trace.)**.** Let $A = [a_{ij}] \in \mathbb{M}_{n\times n}(\mathbb{R})$. The trace of matrix A is,

$$\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn},$$

the sum of the main diagonal entries.

Theorem 1.72 (Properties of Trace.)**.** *Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be matrices in $\mathbb{M}_{n\times n}(\mathbb{R})$. Then,*

(i) $\operatorname{tr}(A^\top) = \operatorname{tr}(A)$.
(ii) $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$.
(iii) $\operatorname{tr}(sA) = s\operatorname{tr}(A)$ *for any* $s \in \mathbb{R}$.

Proof. Each identity follows immediately by writing the trace as the sum of the diagonal entries and applying the corresponding scalar identity term by term. □

Example 1.73. We wish to show that $\langle A, B\rangle = \operatorname{tr}(A^\top B)$ is an inner product on $\mathbb{M}_{2\times 2}(\mathbb{R})$. Evaluating $\langle A, B\rangle$ for,

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}.$$

For all $A, B, C \in \mathbb{M}_{2\times 2}(\mathbb{R})$ and scalars $s, t \in \mathbb{R}$,

(i) We show that positive definiteness follows,

$$\begin{aligned}
\langle A, A\rangle &= \operatorname{tr}\Big(A^T A\Big) \\
&= \operatorname{tr}\left(\begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}\left(\begin{bmatrix} a_{11}^2 + a_{21}^2 & a_{11}a_{12} + a_{21}a_{22} \\ a_{12}a_{11} + a_{22}a_{21} & a_{12}^2 + a_{22}^2 \end{bmatrix}\right) \\
&= a_{11}^2 + a_{12}^2 + a_{21}^2 + a_{22}^2 \geq 0.
\end{aligned}$$

(ii) $\langle A, B\rangle = \operatorname{tr}(A^\top B) = \operatorname{tr}((A^\top B)^\top) = \operatorname{tr}(B^\top A) = \langle B, A\rangle$.

(iii) By properties of matrix multiplication and right-linearity we get,

$$\begin{aligned}
\langle A, sB + tC\rangle &= \operatorname{tr}\Big(A^T(sB + tC)\Big) \\
&= \operatorname{tr}\Big(sA^T B + tA^T C\Big) \\
&= s\operatorname{tr}\Big(A^T B\Big) + t\operatorname{tr}\Big(A^T C\Big) \\
&= s\langle A, B\rangle + t\langle A, C\rangle.
\end{aligned}$$

Hence $\langle A, B\rangle = \operatorname{tr}(A^\top B)$ is an inner product on $\mathbb{M}_{2\times 2}(\mathbb{R})$. For the matrices above,

$$\langle A, B\rangle = \operatorname{tr}(A^T B) = \operatorname{tr}\left(\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}\right) = \operatorname{tr}\left(\begin{bmatrix} 3 & 5 \\ 2 & 2 \end{bmatrix}\right) = 3 + 2 = 5.$$

Proposition 1.74. *For $A = [a_{ij}], B = [b_{ij}] \in \mathbb{M}_{m\times n}(\mathbb{R})$, the Frobenius inner product is*

$$\langle A, B\rangle_{\mathrm{F}} \coloneqq \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ij} = \operatorname{tr}(A^\top B).$$

This defines an inner product on $\mathbb{M}_{m\times n}(\mathbb{R})$.

Proof. Since $A^\top B$ is an $n \times n$ matrix,

$$\begin{aligned}\operatorname{tr}(A^\top B) &= \sum_{j=1}^{n}(A^\top B)_{jj}\\ &= \sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}b_{ij}\\ &= \langle A, B\rangle_{\mathrm{F}}.\end{aligned}$$

Symmetry and bilinearity follow term by term from the corresponding properties in $\mathbb{R}$. Finally,

$$\langle A, A\rangle_{\mathrm{F}} = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2 \geq 0,$$

with equality if and only if every $a_{ij} = 0$, equivalently $A = 0$. □

Theorem 1.75. *If $\mathbb{V}$ is an inner product space with induced norm $\|\vec{x}\| = \sqrt{\langle \vec{x}, \vec{x}\rangle}$, and if $\vec{x}, \vec{y} \in \mathbb{V}$ and $t \in \mathbb{R}$, then*

(i) $\|\vec{x}\| \geq 0$ *with equality if and only if* $\vec{x} = \vec{0}$.
(ii) $\|t\vec{x}\| = |t|\|\vec{x}\|$.
(iii) $|\langle \vec{x}, \vec{y}\rangle| \leq \|\vec{x}\|\|\vec{y}\|$ *with equality if and only if* $\{\vec{x}, \vec{y}\}$ *is linearly dependent.*
(iv) $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$.

Let $\mathbb{V}$ be an inner product space and let $\vec{v} \in \mathbb{V}$ be non-zero. Then $\|\vec{v}\| > 0$.

Consider $\vec{w} = \frac{1}{\|\vec{v}\|}\vec{v}$. Then,

$$\|\vec{w}\| = \left\|\frac{1}{\|\vec{v}\|}\vec{v}\right\| = \left|\frac{1}{\|\vec{v}\|}\right| \|\vec{v}\| = \frac{1}{\|\vec{v}\|}\|\vec{v}\| = 1.$$

Therefore $\vec{w}$ is a unit vector called the normalization of $\vec{v}$.

Inner products allow us to define orthogonality without the notion of the standard dot product.

Example 1.76. Suppose that $p(x) = 1 + x \in \mathbb{P}_1(\mathbb{R})$. Using the inner product,

$$\langle p, q\rangle = p(-1)q(-1) + p(1)q(1),$$

we wish to find the normalization of the polynomial p. Since we know that the norm,

$$\|p\|^2 = \big(p(-1)\big)^2 + p(1)^2 = 0^2 + 2^2 = 4,$$

we have that $\|p\| = 2$ so the normalization of p is given as,

$$\frac{1}{2} + \frac{1}{2}x.$$

Theorem 1.77. *Let $\mathbb{V}$ be an inner product space. Two vectors $\vec{v}, \vec{w}$ are orthogonal if $\langle \vec{v}, \vec{w} \rangle = 0$. A set $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$ is an orthogonal set if $\langle \vec{v}_i, \vec{v}_j \rangle = 0$ whenever $i \neq j$.*

Theorem 1.78. *Let $\mathbb{V}$ be an inner product space and let $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$ be an orthogonal set of non-zero vectors. Then S is linearly independent.*

Proof. For scalars $c_1, \ldots, c_k \in \mathbb{R}$ consider $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$.

For any $i \in \{1, \ldots, k\}$, take the inner product with $\vec{v}_i$:

$$0 = \left\langle \vec{v}_i, \sum_{j=1}^{k} c_j \vec{v}_j \right\rangle = \sum_{j=1}^{k} c_j \langle \vec{v}_i, \vec{v}_j \rangle = c_i \|\vec{v}_i\|^2.$$

Since $\vec{v}_i \neq \vec{0}$, we have $\|\vec{v}_i\|^2 > 0$, so $c_i = 0$.

This holds for every i, and therefore S is linearly independent. □

Example 1.79. Using the set $S = \{1 + x, 1 + x^2, x + x^2\}$, we wish to show that such a set is not an orthogonal set in $\mathbb{P}_2(\mathbb{R})$ with the inner product,

$$\langle p, q \rangle = p(0)q(0) + p(1)q(1) + p(2)q(2).$$

Notice that since,

$$\langle 1 + x, 1 + x^2 \rangle = 1(1) + 2(2) + 3(5) = 20 \neq 0,$$

we conclude that S is not an orthogonal set.

Theorem 1.80. *Let $\mathbb{V}$ be an inner product space and let $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$ be an orthogonal set of non-zero vectors. If $\vec{x} \in \text{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$ then,*

$$\vec{x} = \frac{\langle \vec{x}, \vec{v}_1 \rangle}{\|\vec{v}_1\|^2}\vec{v}_1 + \cdots + \frac{\langle \vec{x}, \vec{v}_k \rangle}{\|\vec{v}_k\|^2}\vec{v}_k.$$

Proof. Since $\vec{x} \in \text{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$, there are scalars $c_1, \ldots, c_k \in \mathbb{R}$ such that $\vec{x} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$.

For $i = 1, \ldots, k$,

$$\begin{aligned}
\langle \vec{x}, \vec{v}_i \rangle &= \langle c_1\vec{v}_1 + \cdots + c_k\vec{v}_k,\ \vec{v}_i \rangle \\
&= c_1\langle \vec{v}_1, \vec{v}_i \rangle + \cdots + c_k\langle \vec{v}_k, \vec{v}_i \rangle \\
&= c_i\langle \vec{v}_i, \vec{v}_i \rangle \qquad (\text{since } \{\vec{v}_1, \ldots, \vec{v}_k\} \text{ is an orthogonal set}) \\
&= c_i\|\vec{v}_i\|^2.
\end{aligned}$$

Since $\vec{v}_i \neq \vec{0}$ which means that $\|\vec{v}_i\|^2 \neq 0$ we have that $c_i = \dfrac{\langle \vec{x}, \vec{v}_i \rangle}{\|\vec{v}_i\|^2}$. □

Theorem 1.81. *Let $\mathbb{V}$ be an inner product space and let $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$. Then, S is an orthonormal set if it is an orthogonal set and $\|\vec{v}_i\| = 1$ for all $i = 1, \ldots, k$.*

The preceding theorem defines an orthonormal set as an orthogonal set whose vectors all have unit norm. We exploit these properties in the next theorem. That is, if $\vec{x}$ can be expressed as a linear combination of some vectors $\{\vec{v}_1, \ldots, \vec{v}_k\}$ with some scalars $c_1, \ldots, c_k \in \mathbb{R}$, then taking the inner product with some $\vec{v}_j$ gives,

$$\langle \vec{x}, \vec{v}_j \rangle = \sum_{i=1}^{k} c_i \langle \vec{v}_i, \vec{v}_j \rangle = c_j,$$

because we know that $\langle \vec{v}_i, \vec{v}_j \rangle = 0$ for $i \neq j$ and $\langle \vec{v}_j, \vec{v}_j \rangle = \|\vec{v}_j\|^2 = 1$. Hence, each coordinate is recovered simply by an inner product yielding,

$$\vec{x} = \sum_{j=1}^{k} \langle \vec{x}, \vec{v}_j \rangle \vec{v}_j.$$

Example 1.82. Consider the set containing the following matrices,

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\}$$

which is an orthonormal set in $\mathbb{M}_{2\times 2}(\mathbb{R})$ with $\langle A, B \rangle = \operatorname{tr}(A^\top B)$ since we compute that,

$$\begin{aligned} \left\langle \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle &= \operatorname{tr}\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right) \\ &= \operatorname{tr}\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right) = 0 + 0 = 0. \end{aligned}$$

We can verify that they are unit matrices,

$$\left\| \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\| = \sqrt{1^2 + 0^2 + 0^2 + 0^2} = 1, \quad \left\| \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\| = \sqrt{0^2 + 1^2 + 0^2 + 0^2} = 1.$$

Theorem 1.83. *Let $\mathbb{V}$ be an inner product space and let $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{V}$ be an orthonormal set. If $\vec{x} \in \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$, then $\vec{x} = \langle \vec{x}, \vec{v}_1 \rangle \vec{v}_1 + \cdots + \langle \vec{x}, \vec{v}_k \rangle \vec{v}_k$.*

Definition 1.84. Let $\mathbb{V}$ be an inner product space and let U be a subspace of $\mathbb{V}$ and let $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ be a basis for U.

(i) If B is orthogonal, then B is an orthogonal basis for U

(ii) If B is an orthonormal set, then B is an orthonormal basis for U.

How about projecting a vector $\vec{v}$ in an inner product space $\mathbb{V}$ onto a subspace U of $\mathbb{V}$? We must find an orthogonal basis for U. That is, if U is a subspace of an inner product space $\mathbb{V}$ and $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ is an orthogonal basis for U, then for any $\vec{x} \in \mathbb{V}$,

$$\operatorname{proj}_U \vec{x} = \frac{\langle \vec{x}, \vec{v}_1 \rangle}{\|\vec{v}_1\|^2} \vec{v}_1 + \cdots + \frac{\langle \vec{x}, \vec{v}_k \rangle}{\|\vec{v}_k\|^2} \vec{v}_k.$$

In the Gram-Schmidt procedure, we saw that if $B = \{\vec{w}_1, \ldots, \vec{w}_k\}$ is a basis for subspace U of an inner product space $\mathbb{V}$ and we define,

$$\begin{aligned}
\vec{v}_1 &= \vec{w}_1, \\
\vec{v}_2 &= \vec{w}_2 - \frac{\langle \vec{w}_2, \vec{v}_1 \rangle}{\|\vec{v}_1\|^2} \vec{v}_1, \\
&\vdots \\
\vec{v}_k &= \vec{w}_k - \frac{\langle \vec{w}_k, \vec{v}_1 \rangle}{\|\vec{v}_1\|^2} \vec{v}_1 - \cdots - \frac{\langle \vec{w}_k, \vec{v}_{k-1} \rangle}{\|\vec{v}_{k-1}\|^2} \vec{v}_{k-1},
\end{aligned}$$

then the resulting set of vectors $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is an orthogonal basis for U with $\text{span}\{\vec{v}_1, \ldots, \vec{v}_i\} = \text{span}\{\vec{w}_1, \ldots, \vec{w}_i\}$ for $i = 1, \ldots, k$.

Example 1.85. Consider the polynomial vector space $\mathbb{P}_2(\mathbb{R})$ equipped with the inner product space,

$$\langle p, q \rangle = p(0)q(0) + p(1)q(1) + p(2)q(2).$$

Then we apply the Gram-Schmidt procedure to the basis $B = \{1, x, x^2\}$ of $\mathbb{P}_2(\mathbb{R})$ to obtain an orthogonal basis for $\mathbb{P}_2(\mathbb{R})$. Suppose that we let $p_1(x) = 1, p_2(x) = x$ and $p_3(x) = x^2$. We let $q_1 = p_1$ and then apply the Gram-Schmidt procedure,

$$p_2 - \frac{\langle p_2, q_1 \rangle}{\|q_1\|^2} q_1 = x - \frac{(0)(1) + (1)(1) + 2(1)}{1^2 + 1^2 + 1^2}(1) = x - \frac{3}{3}(1) = x - 1.$$

Taking $q_2(x) = x - 1$ we can compute,

$$\begin{aligned}
p_3 - \frac{\langle p_3, q_1 \rangle}{\|q_1\|^2} q_1 - \frac{\langle p_3, q_2 \rangle}{\|q_2\|^2} q_2 &= x^2 - \frac{0(1) + 1(1) + 4(1)}{1^2 + 1^2 + 1^2}(1) - \frac{0(-1) + 1(0) + 4(1)}{(-1)^2 + 0^2 + 1^2}(x - 1) \\
&= x^2 - \frac{5}{3}(1) - \frac{4}{2}(x - 1) \\
&= x^2 - \frac{5}{3} - 2x + 2 = x^2 - 2x + \frac{1}{3}.
\end{aligned}$$

We can take $q_3(x) = 3x^2 - 6x + 1$. Then $C = \{1, x - 1, 3x^2 - 6x + 1\}$ is an orthogonal basis for $\mathbb{P}_2(\mathbb{R})$.

Example 1.86. Let $U = \text{span}\{1, x + x^2\}$ be a subspace of $\mathbb{P}_2(\mathbb{R})$, equipped with the inner product

$$\langle p, q \rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1).$$

We wish to find $\text{proj}_U(1 + x^2)$. Since $\langle 1, x + x^2 \rangle = 1(0) + 1(0) + 1(2) = 2 \neq 0$, we first apply the Gram-Schmidt procedure.

We let $r_1 = p_1$ and compute,

$$p_2 - \frac{\langle p_2, r_1 \rangle}{\|r_1\|^2} r_1 = x + x^2 - \frac{0(1) + 0(1) + 2(1)}{1^2 + 1^2 + 1^2}(1) = x^2 + x - \frac{2}{3}.$$

So we can take $r_2 = 3x^2 + 3x - 2$ and thus $\{1, 3x^2 + 3x - 2\}$ is an orthogonal basis for U. Now we can compute,

$$\begin{aligned}
\text{proj}_U(1 + x^2) &= \frac{\langle 1 + x^2, 1\rangle}{\|1\|^2}(1) + \frac{\langle 1 + x^2, 3x^2 + 3x - 2\rangle}{\|3x^2 + 3x - 2\|^2}(3x^2 + 3x - 2) \\
&= \frac{2(1) + 1(1) + 2(1)}{1 + 1 + 1}(1) + \frac{2(-2) + 1(-2) + 2(4)}{4 + 4 + 16}(3x^2 + 3x - 2) \\
&= \frac{5}{3}(1) + \frac{2}{24}(3x^2 + 3x - 2) \\
&= \frac{1}{4}x^2 + \frac{1}{4}x + \frac{3}{2}.
\end{aligned}$$

The Gram-Schmidt procedure shows that every finite-dimensional inner product space admits an orthonormal basis and this allows us to represent linear operators in coordinate systems that respect the geometry of the space. We now turn to the problem of choosing such a basis so that the matrix of a linear operator becomes as simple as possible.

In particular, we seek a basis consisting of eigenvectors, since relative to an eigenvector basis the matrix representation of the operator is diagonal. Not every linear operator admits such a basis, however, real symmetric matrices possess a remarkable compatibility with the inner product. That is, eigenvectors corresponding to distinct eigenvalues are orthogonal, and an orthonormal basis of eigenvectors can always be constructed. This leads to the spectral theorem and the orthogonal diagonalization of symmetric matrices.

Let us begin by defining what it means for a matrix to be symmetric.

Definition 1.87. Let A be a matrix in $\mathbb{M}_{n\times n}(\mathbb{R})$.

We call such a matrix symmetric if $A^\top = A$, that is $a_{ji} = a_{ij}$ for $1 \le i, j \le n$.

Example 1.88. Consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 4 \\ -1 & 2 & 7 \\ 4 & 7 & 3 \end{bmatrix}.$$

Observe that by computing the transposes,

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \neq A \qquad B^T = \begin{bmatrix} 1 & -1 & 4 \\ -1 & 2 & 7 \\ 4 & 7 & 3 \end{bmatrix} = B.$$

So we see that B is symmetric and A is not symmetric.

Example 1.89. We aim to diagonalize the matrix,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

We have that the characteristic polynomial is defined by,

$$c_A(\lambda) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The roots of the characteristic polynomial give the eigenvalues $\lambda_1 = 1, \lambda_2 = 3$.

For $\lambda_1 = 1$ we can solve the system $(A - 1 \cdot I)\vec{x} = \vec{0}$,

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \Longrightarrow \vec{x} = \begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Similarly, we can compute the system $(A - 3I)\vec{x} = \vec{0}$,

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \Longrightarrow \vec{x} = \begin{bmatrix} t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Placing the eigenvectors into matrix Q we get,

$$Q = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

where the matrix D contains the corresponding eigenvalues on the diagonal.

Notice that the columns of Q are orthogonal as their inner product is 0, so we can normalize them,

$$\frac{1}{\left\| \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\|} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \qquad \frac{1}{\left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\|} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Thus, with

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix},$$

we obtain the orthogonal diagonalization $P^\top AP = D$.

You may recall from first and second year linear algebra that to diagonalize an $n \times n$ (symmetric) matrix, we first compute the characteristic polynomial using λ and solve for its roots, which are the eigenvalues. Then, for λ_i, we solve the system $(A - \lambda_i I)\vec{x} = \vec{0}$ from which we can obtain the corresponding eigenvector $\vec{v}_i$. Placing the eigenvectors as the columns to some matrix Q and letting the sorted eigenvalues corresponding to those eigenvectors be placed as the diagonals in some matrix D *diagonalizes* matrix A such that $Q^{-1}AQ = D$ or equivalently, $A = QDQ^{-1}$.

If the columns of Q are orthogonal it is possible to normalize them by computing,

$$\frac{1}{\|\vec{v}_i\|}\vec{v}_i, \qquad \forall i = 1, \ldots, k,$$

if we have k eigenvectors and let the normalized eigenvectors be the columns of some new matrix P with the same diagonal matrix D. From this formulation, we can write,

$$P^{-1}AP = D \iff A = PDP^{-1}.$$

Furthermore, since the columns of P form an orthonormal set, P is an orthogonal matrix so $P^{-1} = P^\top$. Hence we can conclude that $P^\top AP = D$ or equivalently $A = PDP^\top$. We have derived a method that avoids computing the inverse and instead computes the transpose. Thus, if every symmetric matrix is diagonalizable, we can find a diagonalizing matrix P such that $P^\top AP = D$.

Theorem 1.90. *If $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ is a symmetric matrix, then every eigenvalue of A is real.*

Proof. Suppose $A\vec{z} = \lambda\vec{z}$ for some nonzero $\vec{z} \in \mathbb{C}^n$ and $\lambda \in \mathbb{C}$. Writing $\vec{z}^*$ for the conjugate transpose,

$$\lambda = \frac{\vec{z}^* A\vec{z}}{\vec{z}^*\vec{z}}.$$

Because A is real and symmetric, $A^* = A$, and therefore

$$\left(\vec{z}^* A\vec{z}\right)^* = \vec{z}^* A^*\vec{z} = \vec{z}^* A\vec{z}.$$

Thus, $\vec{z}^* A\vec{z}$ is real, while $\vec{z}^*\vec{z} > 0$. Hence $\lambda \in \mathbb{R}$. □

Note that we require symmetry otherwise some of our eigenvalues could be non-real.

Lemma 1.91. *A matrix $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ is symmetric if and only if,*

$$\vec{x} \cdot (A\vec{y}) = (A\vec{x}) \cdot \vec{y},$$

for every vector $\vec{x}, \vec{y} \in \mathbb{R}^n$.

Proof. Suppose A is a symmetric $n \times n$ matrix. Then for any vector $\vec{x}, \vec{y} \in \mathbb{R}^n$,

$$\begin{aligned}\vec{x} \cdot (A\vec{y}) &= \vec{x}^\top (A\vec{y}) \\ &= \vec{x}^\top (A^\top \vec{y}) \\ &= (\vec{x}^\top A^\top)\vec{y} \\ &= (A\vec{x})^\top \vec{y} = (A\vec{x}) \cdot \vec{y}.\end{aligned}$$

Now suppose that $\vec{x} \cdot (A\vec{y}) = (A\vec{x}) \cdot \vec{y}$ for every $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then,

$$\vec{x}^\top (A\vec{y}) = (A\vec{x})^\top \vec{y} \implies (\vec{x}^\top A)\vec{y} = (A\vec{x})^\top \vec{y},$$

for every $\vec{y} \in \mathbb{R}^n$. Thus, $\vec{x}^\top A = (A\vec{x})^\top = \vec{x}^\top A^\top$, so

$$A^\top \vec{x} = A\vec{x}$$

for every $\vec{x} \in \mathbb{R}^n$. Thus, $A^\top = A$, that is A is symmetric. □

This leads to a pretty remarkable theorem regarding the eigenvalues of a symmetric matrix.

Theorem 1.92. *If $\vec{v}_1, \vec{v}_2$ are eigenvectors of a symmetric matrix $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ corresponding to distinct eigenvalues λ_1, λ_2, then $\vec{v}_1$ and $\vec{v}_2$ are orthogonal.*

Proof. We have $A\vec{v}_1 = \lambda_1\vec{v}_1$ and $A\vec{v}_2 = \lambda_2\vec{v}_2$ with $\lambda_1 \neq \lambda_2$. Then,

$$\begin{aligned}
\lambda_1(\vec{v}_1 \cdot \vec{v}_2) &= (\lambda_1\vec{v}_1) \cdot \vec{v}_2 && \text{by properties of inner products} \\
&= A\vec{v}_1 \cdot \vec{v}_2 && \text{since } A\vec{v}_1 = \lambda_1\vec{v}_1 \\
&= \vec{v}_1 \cdot A\vec{v}_2 && \text{by the preceding lemma since } A \text{ is symmetric} \\
&= \vec{v}_1 \cdot (\lambda_2\vec{v}_2) && \text{since } A\vec{v}_2 = \lambda_2\vec{v}_2 \\
&= \lambda_2(\vec{v}_1 \cdot \vec{v}_2) && \text{by properties of inner products.}
\end{aligned}$$

Thus,

$$\lambda_1(\vec{v}_1 \cdot \vec{v}_2) = \lambda_2(\vec{v}_1 \cdot \vec{v}_2),$$

that is,

$$(\lambda_1 - \lambda_2)(\vec{v}_1 \cdot \vec{v}_2) = 0.$$

Since $\lambda_1 \neq \lambda_2$, we must have

$$\vec{v}_1 \cdot \vec{v}_2 = 0,$$

so $\vec{v}_1$ and $\vec{v}_2$ are orthogonal. □

Example 1.93. Suppose we have the matrix defined by,

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

From which we can then compute the characteristic polynomial as,

$$\begin{aligned}
c_A(\lambda) &= \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 0 & 1+\lambda & -(1+\lambda) \end{vmatrix}_{R_3 - R_2 \to R_3} = \begin{vmatrix} -\lambda & 1 & 2 \\ 1 & -\lambda & 1-\lambda \\ 0 & 1+\lambda & 0 \end{vmatrix}_{C_3 + C_2 \to C_3} \\
&= -(1+\lambda)\begin{vmatrix} -\lambda & 2 \\ 1 & 1-\lambda \end{vmatrix} = -(1+\lambda)\Big((-\lambda)(1-\lambda) - 2\Big) \\
&= -(\lambda+1)(\lambda^2 - \lambda - 2) \\
&= -(\lambda+1)(\lambda+1)(\lambda-2) \\
&= -(\lambda+1)^2(\lambda-2).
\end{aligned}$$

From this we can see that $\lambda_1 = -1$ with $a_{\lambda_1} = 2$ and $\lambda_2 = 2$ where $a_{\lambda_2} = 1$.

For $\lambda_1 = -1$ we can solve the system $(A + I)\vec{x} = \vec{0}$,

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{x} = \begin{bmatrix} -s - t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

So we have that $\{\vec{w}_1, \vec{w}_2\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $E_{\lambda_1}(A)$ and $g_{\lambda_1} = 2 = a_{\lambda_1}$.

For $\lambda_2 = 2$ we solve the system $(A - 2I)\vec{x} = \vec{0}$,

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow[R_3 - R_2 \to R_3]{R_1 + 2R_2 \to R_1} \begin{bmatrix} 0 & -3 & 3 \\ 1 & -2 & 1 \\ 0 & 3 & -3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix}$$

$$\xrightarrow{-\frac{1}{3}R_2 \to R_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow[R_3 - R_2 \to R_3]{R_1 + 2R_2 \to R_1} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\implies \quad \vec{x} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

So we have that $\{\vec{w}_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ is a basis for $E_{\lambda_2}(A)$ and $g_{\lambda_2} = 1 = a_{\lambda_2}$.

Since we know that $a_{\lambda_1} = g_{\lambda_1}$ and $a_{\lambda_2} = g_{\lambda_2}$, we know that A is diagonalizable.

We can put $\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ into a matrix to get,

$$Q = \begin{bmatrix} \vec{w}_1 & \vec{w}_2 & \vec{w}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \qquad D = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Naturally we see that $Q^{-1}AQ = D$. Now our theorem guarantees that $\vec{w}_1 \cdot \vec{w}_3 = 0$ and $\vec{w}_2 \cdot \vec{w}_3 = 0$ since we know that $\vec{w}_1, \vec{w}_2$ correspond to λ_1 while $\vec{w}_3$ corresponds to λ_2. However notice that,

$$\vec{w}_1 \cdot \vec{w}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = 1 + 0 + 0 = 1 \neq 0.$$

So therefore $\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ is not an orthogonal set. We need to apply Gram-Schmidt orthogonalization.

We let $\vec{v}_1 = \vec{w}_1$ and compute,

$$\vec{w}_2 - \frac{\vec{w}_2 \cdot \vec{v}_1}{\|\vec{v}_1\|^2}\vec{v}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}.$$

So we take $\vec{v}_2 = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$ so $\{\vec{v}_1, \vec{v}_2\}$ is an orthogonal set with $\text{span}\{\vec{v}_1, \vec{v}_2\} = \text{span}\{\vec{w}_1, \vec{w}_2\} = E_{\lambda_1}(A)$. Therefore, we now have that $\vec{w}_3$ is orthogonal to both $\vec{v}_1$ and $\vec{v}_2$. Letting $\vec{v}_3 = \vec{w}_3$, the set $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is an orthogonal basis of eigenvectors for $\mathbb{R}^3$ we can now derive an orthonormal basis of eigenvectors for $\mathbb{R}^3$ by normalizing, that is, we compute the following,

$$\vec{u}_1 = \frac{1}{\|\vec{v}_1\|}\vec{v}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \qquad \vec{u}_2 = \frac{1}{\|\vec{v}_2\|}\vec{v}_2 = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}, \qquad \vec{u}_3 = \frac{1}{\|\vec{w}_3\|}\vec{w}_3 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}.$$

Now, we can define the P matrix,

$$P = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Thus, $P^\top AP = D$ and $P^{-1} = P^\top$.

Definition 1.94. A matrix $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ is orthogonally diagonalizable if there exists an orthogonal matrix P and a diagonal matrix D such that $P^\top AP = D$.

Definition 1.95. Let $A = [a_{ij}] \in \mathbb{M}_{n\times n}(\mathbb{R})$. Then we say A is,

(i) Upper triangular if $a_{ij} = 0$ when $i > j$,
(ii) Lower triangular if $a_{ij} = 0$ when $i < j$.

Observe that the transpose of an upper triangular matrix is a lower triangular matrix. Also note that a diagonal matrix is both upper and lower triangular.

Theorem 1.96 (Triangularization Theorem)**.** *If $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ has all real eigenvalues, then there is an orthogonal matrix P and an upper triangular matrix T such that $P^\top AP = T$.*

Theorem 1.97 (Principal Axis Theorem)**.** *A matrix $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ is symmetric if and only if it is orthogonally diagonalizable.*

Proof. Suppose that A is orthogonally diagonalizable. Then we can find an orthogonal matrix P and a diagonal matrix D such that $P^\top AP = D$. Then $A = PDP^\top$, so

$$A^\top = (PDP^\top)^\top = (P^\top)^\top D^\top P^\top = PDP^\top = A,$$

showing that A is symmetric. Conversely, suppose that A is symmetric. Its eigenvalues are real by the preceding result, so the Triangularization Theorem provides an orthogonal matrix Q and an upper triangular matrix T such that $Q^\top AQ = T$.

Therefore we can compute,

$$T^\top = (Q^\top AQ)^\top = Q^\top A^\top Q = Q^\top AQ = T.$$

Therefore we have shown that T is symmetric. Since T is upper triangular, we know that $T^\top$ is lower triangular. Therefore, we have that T is upper and lower triangular and hence diagonal. Since we know that $Q^\top AQ = T$ then A is by procedure orthogonally diagonalizable. □

Example 1.98. Orthogonally diagonalize

$$A = \begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix},$$

given that $\lambda_1 = 9$ and $\lambda_2 = 0$ are the eigenvalues of A. For $\lambda_1 = 9$, solve

$$(A - 9I)\vec{x} = \vec{0}.$$

Row-reducing $A - 9I$, we obtain

$$\begin{bmatrix} -4 & -4 & -2 \\ -4 & -4 & -2 \\ -2 & -2 & -1 \end{bmatrix} \xrightarrow[R_3 - \frac{1}{2}R_1 \to R_3]{R_2 - R_1 \to R_2} \begin{bmatrix} -4 & -4 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-\frac{1}{4}R_1 \to R_1} \begin{bmatrix} 1 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus, $x_1 + x_2 + \frac{1}{2}x_3 = 0$. Setting $x_2 = s$ and $x_3 = t$ gives

$$\vec{x} = \begin{bmatrix} -s - \frac{1}{2}t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

Scaling the second vector by 2 to eliminate fractions, we may take

$$B_1 = \{\vec{w}_1, \vec{w}_2\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \right\}$$

as a basis for $E_{\lambda_1}(A)$. Let $\vec{v}_1 = \vec{w}_1$. Applying the Gram–Schmidt procedure to $\vec{w}_2$, we compute

$$\vec{w}_2 - \frac{\vec{w}_2 \cdot \vec{v}_1}{\|\vec{v}_1\|^2}\vec{v}_1 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 2 \end{bmatrix}.$$

Scaling this vector by 2, let $\vec{v}_2 = \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix}$. Therefore, $\{\vec{v}_1, \vec{v}_2\}$ is an orthogonal basis for $E_{\lambda_1}(A)$.

For $\lambda_2 = 0$, we will solve $A\vec{x} = \vec{0}$. Row-reducing A, we obtain

$$\begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix} \xrightarrow{R_1+R_2\to R_1} \begin{bmatrix} 1 & 1 & -4 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix} \xrightarrow[R_3+2R_1\to R_3]{R_2+4R_1\to R_2} \begin{bmatrix} 1 & 1 & -4 \\ 0 & 9 & -18 \\ 0 & 0 & 0 \end{bmatrix}$$
$$\xrightarrow{\frac{1}{9}R_2\to R_2} \begin{bmatrix} 1 & 1 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-R_2\to R_1} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus, $x_1 = 2x_3$ and $x_2 = 2x_3$. Setting $x_3 = t$ gives

$$\vec{x} = \begin{bmatrix} 2t \\ 2t \\ t \end{bmatrix} = t\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Hence, $B_2 = \{\vec{v}_3\} = \left\{\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}\right\}$ is a basis for $E_{\lambda_2}(A)$. Moreover,

$$\vec{v}_3 \cdot \vec{v}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = -2 + 2 = 0$$

$$\vec{v}_3 \cdot \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix} = -2 - 2 + 4 = 0.$$

Thus, $\vec{v}_3$ is orthogonal to both $\vec{v}_1$ and $\vec{v}_2$. Normalizing $\vec{v}_1, \vec{v}_2, \vec{v}_3$, we have

$$\vec{u}_1 = \frac{1}{\|\vec{v}_1\|}\vec{v}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \qquad \vec{u}_2 = \frac{1}{\|\vec{v}_2\|}\vec{v}_2 = \begin{bmatrix} -\frac{1}{\sqrt{18}} \\ -\frac{1}{\sqrt{18}} \\ \frac{4}{\sqrt{18}} \end{bmatrix}, \qquad \vec{u}_3 = \frac{1}{\|\vec{v}_3\|}\vec{v}_3 = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}.$$

Thus, we take

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{18}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{18}} & \frac{2}{3} \\ 0 & \frac{4}{\sqrt{18}} & \frac{1}{3} \end{bmatrix}, \qquad D = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so that $P^T AP = D$.

The Principal Axis Theorem shows that every real symmetric matrix can be diagonalized using an orthogonal matrix. Suppose that $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ is symmetric and that

$$P = \begin{bmatrix} \vec{u}_1 & \cdots & \vec{u}_n \end{bmatrix}, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

where $\vec{u}_1, \ldots, \vec{u}_n$ form an orthonormal basis of eigenvectors and $A\vec{u}_i = \lambda_i \vec{u}_i$.

Then the Principal Axis Theorem gives

$$A = PDP^\top.$$

We can expand this matrix product in terms of the columns of P:

$$\begin{aligned} A &= \begin{bmatrix} \vec{u}_1 & \cdots & \vec{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} \vec{u}_1^\top \\ \vdots \\ \vec{u}_n^\top \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1\vec{u}_1 & \cdots & \lambda_n\vec{u}_n \end{bmatrix} \begin{bmatrix} \vec{u}_1^\top \\ \vdots \\ \vec{u}_n^\top \end{bmatrix} \\ &= \lambda_1 \vec{u}_1\vec{u}_1^\top + \cdots + \lambda_n \vec{u}_n\vec{u}_n^\top. \end{aligned}$$

This expansion is called the spectral decomposition of A.

Unlike the matrix form $A = PDP^\top$, the spectral decomposition writes A as a sum of rank-one matrices and makes the action of A along each eigendirection explicit.

Theorem 1.99 (Spectral Decomposition Theorem)**.** *Let $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ be symmetric. Then there exist real eigenvalues $\lambda_1, \ldots, \lambda_n$ and an orthonormal basis of eigenvectors $\{\vec{u}_1, \ldots, \vec{u}_n\}$ for $\mathbb{R}^n$ such that*

$$A = \sum_{i=1}^{n} \lambda_i \vec{u}_i \vec{u}_i^\top.$$

Conversely, if $\{\vec{u}_1, \ldots, \vec{u}_n\}$ is an orthonormal basis for $\mathbb{R}^n$, $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, and

$$A = \sum_{i=1}^{n} \lambda_i \vec{u}_i \vec{u}_i^\top,$$

then A is symmetric and $A\vec{u}_i = \lambda_i \vec{u}_i$ for every $i = 1, \ldots, n$.

Proof. Suppose first that A is symmetric. By the Principal Axis Theorem, there is an orthogonal matrix $P = [\vec{u}_1 \ \cdots \ \vec{u}_n]$ and a diagonal matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ such that

$$A = PDP^\top.$$

Expanding the product column by column gives

$$A = \lambda_1 \vec{u}_1 \vec{u}_1^\top + \cdots + \lambda_n \vec{u}_n \vec{u}_n^\top.$$

Conversely, suppose that $A = \sum_{i=1}^n \lambda_i \vec{u}_i \vec{u}_i^\top$. Since each outer product is symmetric,

$$A^\top = \sum_{i=1}^n \lambda_i \left(\vec{u}_i \vec{u}_i^\top\right)^\top = \sum_{i=1}^n \lambda_i \vec{u}_i \vec{u}_i^\top = A.$$

Furthermore, for every $j = 1, \ldots, n$,

$$\begin{aligned} A\vec{u}_j &= \left(\sum_{i=1}^n \lambda_i \vec{u}_i \vec{u}_i^\top\right) \vec{u}_j \\ &= \sum_{i=1}^n \lambda_i \vec{u}_i \left(\vec{u}_i^\top \vec{u}_j\right) \\ &= \lambda_j \vec{u}_j, \end{aligned}$$

because $\vec{u}_i^\top \vec{u}_j = 0$ when $i \neq j$ and $\vec{u}_j^\top \vec{u}_j = 1$. □

The matrix $\vec{u}_i \vec{u}_i^\top$ appearing in the spectral decomposition has a useful geometric interpretation. For every $\vec{x} \in \mathbb{R}^n$,

$$\vec{u}_i \vec{u}_i^\top \vec{x} = \left(\vec{u}_i^\top \vec{x}\right) \vec{u}_i = \langle \vec{x}, \vec{u}_i \rangle \, \vec{u}_i.$$

Therefore, $\vec{u}_i \vec{u}_i^\top$ is the orthogonal projection matrix onto the one-dimensional subspace $\operatorname{span}\{\vec{u}_i\}$.

Definition 1.100. For an orthonormal eigenvector $\vec{u}_i$, define

$$\Pi_i \coloneqq \vec{u}_i \vec{u}_i^\top.$$

We call Π_i the rank-one spectral projector associated with $\vec{u}_i$. The spectral decomposition can then be written as

$$A = \sum_{i=1}^n \lambda_i \Pi_i.$$

Proposition 1.101. *Let $\{\vec{u}_1, \ldots, \vec{u}_n\}$ be an orthonormal basis for $\mathbb{R}^n$ and let $\Pi_i = \vec{u}_i \vec{u}_i^\top$. Then*

(i) $\Pi_i^\top = \Pi_i$,
(ii) $\Pi_i^2 = \Pi_i$,
(iii) $\Pi_i \Pi_j = 0_{n \times n}$ *whenever* $i \neq j$,
(iv) $\Pi_1 + \cdots + \Pi_n = I_n$.

If, in addition, $A\vec{u}_i = \lambda_i \vec{u}_i$ and A is symmetric, then

$$A\Pi_i = \Pi_i A = \lambda_i \Pi_i.$$

Proof. For each i,

$$\Pi_i^\top = \left(\vec{u}_i\vec{u}_i^\top\right)^\top = \vec{u}_i\vec{u}_i^\top = \Pi_i,$$

which proves (i). Since $\vec{u}_i$ is a unit vector,

$$\Pi_i^2 = \vec{u}_i\vec{u}_i^\top\,\vec{u}_i\vec{u}_i^\top = \vec{u}_i\left(\vec{u}_i^\top\,\vec{u}_i\right)\vec{u}_i^\top = \Pi_i,$$

proving (ii). If $i \neq j$, orthogonality gives

$$\Pi_i\Pi_j = \vec{u}_i\left(\vec{u}_i^\top\,\vec{u}_j\right)\vec{u}_j^\top = 0_{n\times n},$$

which proves (iii). Finally, if $P = [\vec{u}_1\ \cdots\ \vec{u}_n]$, then P is orthogonal and

$$I_n = PP^\top = \begin{bmatrix}\vec{u}_1 & \cdots & \vec{u}_n\end{bmatrix}\begin{bmatrix}\vec{u}_1^\top \\ \vdots \\ \vec{u}_n^\top\end{bmatrix} = \sum_{i=1}^n \vec{u}_i\vec{u}_i^\top = \sum_{i=1}^n \Pi_i.$$

This proves (iv).

If $A\vec{u}_i = \lambda_i\vec{u}_i$, then

$$A\Pi_i = A\vec{u}_i\vec{u}_i^\top = \lambda_i\vec{u}_i\vec{u}_i^\top = \lambda_i\Pi_i.$$

Since A is symmetric,

$$\Pi_i A = \vec{u}_i\vec{u}_i^\top A = \vec{u}_i(A\vec{u}_i)^\top = \lambda_i\Pi_i.$$

□

The spectral projectors separate a vector into its orthogonal eigenvector coordinates. Indeed, since $\sum_{i=1}^n \Pi_i = I_n$, every $\vec{x} \in \mathbb{R}^n$ satisfies

$$\vec{x} = \sum_{i=1}^n \Pi_i\vec{x} = \sum_{i=1}^n \langle\vec{x}, \vec{u}_i\rangle\,\vec{u}_i.$$

Applying A then gives

$$A\vec{x} = \sum_{i=1}^n \lambda_i\,\langle\vec{x}, \vec{u}_i\rangle\,\vec{u}_i.$$

Thus, A first extracts the component of $\vec{x}$ in each eigendirection, scales that component by the corresponding eigenvalue, and adds the scaled components back together.

Corollary 1.102. *Let $A = \sum_{i=1}^n \lambda_i\Pi_i$ be the spectral decomposition of a real symmetric matrix. If $p(t) = a_0 + a_1t + \cdots + a_kt^k$ is a polynomial, then*

$$p(A) = \sum_{i=1}^n p(\lambda_i)\Pi_i.$$

In particular, for every integer $k \geq 1$,

$$A^k = \sum_{i=1}^n \lambda_i^k\Pi_i.$$

If every $\lambda_i \neq 0$, then A is invertible and

$$A^{-1} = \sum_{i=1}^{n} \frac{1}{\lambda_i} \Pi_i.$$

Proof. For every integer $r \geq 1$, the identities $\Pi_i^2 = \Pi_i$ and $\Pi_i \Pi_j = 0_{n\times n}$ for $i \neq j$ imply

$$A^r = \left(\sum_{i=1}^{n} \lambda_i \Pi_i \right)^r = \sum_{i=1}^{n} \lambda_i^r \Pi_i.$$

Since $I_n = \sum_{i=1}^{n} \Pi_i$, substituting these expressions into the polynomial gives

$$p(A) = \sum_{i=1}^{n} \left(a_0 + a_1 \lambda_i + \cdots + a_k \lambda_i^k \right) \Pi_i = \sum_{i=1}^{n} p(\lambda_i) \Pi_i.$$

If every $\lambda_i \neq 0$, then

$$A \left(\sum_{i=1}^{n} \frac{1}{\lambda_i} \Pi_i \right) = \sum_{i=1}^{n} \Pi_i = I_n,$$

and the same calculation on the other side proves the formula for A^{-1}. □

The spectral decomposition also characterizes positive definiteness directly through the eigenvalues.

Corollary 1.103. *Let $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n$. Then*

(i) A is positive semidefinite if and only if $\lambda_i \geq 0$ for every $i = 1, \ldots, n$.
(ii) A is positive definite if and only if $\lambda_i > 0$ for every $i = 1, \ldots, n$.

Proof. Write $\vec{x} = \sum_{i=1}^{n} c_i \vec{u}_i$, where $c_i = \langle \vec{x}, \vec{u}_i \rangle$. Then

$$\vec{x}^\top A \vec{x} = \sum_{i=1}^{n} \lambda_i c_i^2.$$

If every $\lambda_i \geq 0$, the right-hand side is nonnegative for every $\vec{x}$, so A is positive semidefinite.

Conversely, choosing $\vec{x} = \vec{u}_j$ gives

$$\vec{u}_j^\top A \vec{u}_j = \lambda_j.$$

Thus, positive semidefiniteness implies $\lambda_j \geq 0$ for every j. This proves (i). The same argument, using $\vec{x} \neq \vec{0}$ and strict inequalities, proves (ii). □

When an eigenvalue is repeated, the orthonormal basis chosen for its eigenspace need not be unique. The individual rank-one projectors may therefore change with the basis, but their sum over the entire eigenspace is unique.

Theorem 1.104 (Spectral Decomposition by Eigenspaces)**.** *Let $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ be symmetric with distinct eigenvalues $\mu_1, \dots, \mu_s$. For each j, let*

$$B_j = \{\vec{u}_{j1}, \dots, \vec{u}_{jd_j}\}$$

be an orthonormal basis for the eigenspace $E_{\mu_j}(A)$, and define

$$\mathcal{P}_j \coloneqq \sum_{\ell=1}^{d_j} \vec{u}_{j\ell}\vec{u}_{j\ell}^{\top}.$$

Then $\mathcal{P}_j$ is the orthogonal projection onto $E_{\mu_j}(A)$ and

$$A = \sum_{j=1}^{s} \mu_j \mathcal{P}_j, \qquad I_n = \sum_{j=1}^{s} \mathcal{P}_j.$$

Moreover, $\mathcal{P}_j\mathcal{P}_k = 0_{n\times n}$ whenever $j \neq k$. The matrix $\mathcal{P}_j$ depends only on the eigenspace $E_{\mu_j}(A)$ and not on the particular orthonormal basis B_j.

Proof. Eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal. Therefore, the union $B_1 \cup \cdots \cup B_s$ is an orthonormal basis for $\mathbb{R}^n$. Applying the spectral decomposition and grouping together all terms having the same eigenvalue gives

$$A = \sum_{j=1}^{s} \mu_j \left(\sum_{\ell=1}^{d_j} \vec{u}_{j\ell}\vec{u}_{j\ell}^{\top} \right) = \sum_{j=1}^{s} \mu_j \mathcal{P}_j.$$

Similarly, summing all rank-one projectors gives $I_n = \sum_{j=1}^{s} \mathcal{P}_j$. Orthogonality between distinct eigenspaces implies $\mathcal{P}_j\mathcal{P}_k = 0_{n\times n}$ when $j \neq k$.

For every $\vec{x} \in \mathbb{R}^n$,

$$\mathcal{P}_j\vec{x} = \sum_{\ell=1}^{d_j} \left\langle \vec{x}, \vec{u}_{j\ell} \right\rangle \vec{u}_{j\ell},$$

which is the orthogonal projection of $\vec{x}$ onto $E_{\mu_j}(A)$. Since the orthogonal projection onto a fixed subspace is unique, $\mathcal{P}_j$ does not depend on the choice of orthonormal basis for that eigenspace. □

Example 1.105. Consider again the symmetric matrix from the preceding example,

$$A = \begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix}.$$

We found that $\lambda_1 = 9$ has multiplicity two and $\lambda_2 = 0$ has multiplicity one, with orthonormal eigenvectors

$$\vec{u}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \qquad \vec{u}_2 = \begin{bmatrix} -\frac{1}{\sqrt{18}} \\ -\frac{1}{\sqrt{18}} \\ \frac{4}{\sqrt{18}} \end{bmatrix}, \qquad \vec{u}_3 = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}.$$

The spectral projector onto the eigenspace corresponding to $\lambda = 9$ is

$$\begin{aligned} \mathcal{P}_9 &= \vec{u}_1\vec{u}_1^\top + \vec{u}_2\vec{u}_2^\top \\ &= \frac{1}{9}\begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix}. \end{aligned}$$

The spectral projector onto the eigenspace corresponding to $\lambda = 0$ is

$$\mathcal{P}_0 = \vec{u}_3\vec{u}_3^\top = \frac{1}{9}\begin{bmatrix} 4 & 4 & 2 \\ 4 & 4 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Notice that

$$\mathcal{P}_9 + \mathcal{P}_0 = I_3 \qquad \text{and} \qquad \mathcal{P}_9\mathcal{P}_0 = 0_{3\times 3}.$$

Therefore, the spectral decomposition of A by eigenspaces is

$$A = 9\mathcal{P}_9 + 0\mathcal{P}_0 = 9\mathcal{P}_9.$$

For every $\vec{x} \in \mathbb{R}^3$, we may write

$$\vec{x} = \mathcal{P}_9\vec{x} + \mathcal{P}_0\vec{x},$$

and hence

$$A\vec{x} = 9\mathcal{P}_9\vec{x}.$$

Thus, A scales the component of $\vec{x}$ in $E_9(A)$ by 9 and annihilates the component in $E_0(A) = \text{null}(A)$.

The spectral decomposition provides a complete description of a real symmetric matrix in terms of its orthogonal eigenspaces. A general rectangular or nonsymmetric matrix does not necessarily possess an orthonormal basis of eigenvectors. However, the matrices $A^\top A$ and $AA^\top$ are always symmetric and positive semidefinite. In the next section, we apply their spectral decompositions to construct the singular value decomposition for an arbitrary real matrix.

1.5 The singular value decomposition

Suppose we have a matrix $A \in \mathbb{M}_{m\times n}(\mathbb{R})$, interpreted as the matrix representation of a linear mapping $A : \mathbb{R}^n \to \mathbb{R}^m$. Clearly, the input vector belongs to $\mathbb{R}^n$ while the output $A\vec{x}$ belongs to $\mathbb{R}^m$. Because the domain and codomain may have different dimensions, A need not be square and therefore may

not possess eigenvalues or eigenvectors in the usual sense. In contrast to eigendecomposition, the singular value decomposition applies to every real matrix, whether rectangular or square.

When $m \neq n$, the eigenvalue equation $A\vec{v} = \lambda\vec{v}$ is not meaningful: $A\vec{v} \in \mathbb{R}^m$ while $\lambda\vec{v} \in \mathbb{R}^n$, so the two sides belong to different vector spaces. Even when $m = n$, a general square matrix need not possess an orthonormal basis of eigenvectors. For example, a nonsymmetric matrix may have complex eigenvalues, repeated eigenvalues with insufficiently many eigenvectors, or highly nonorthogonal eigenvectors.

The failure of eigendecomposition does not imply that the transformation lacks an interpretable geometric structure. Instead of demanding that a direction be mapped back into itself, we ask whether there are unit vectors $\vec{v} \in \mathbb{R}^n$ and $\vec{u} \in \mathbb{R}^m$ such that $A\vec{v} = \sigma\vec{u}$ for some nonnegative scalar σ. In this equation, A transforms the unit input direction $\vec{v}$ into the unit output direction $\vec{u}$, scaled by the factor σ.

The vectors $\vec{v}, \vec{u}$ need not belong to the same vector space for these to hold.

In the singular value decomposition, we show that we can find orthonormal bases $\{\vec{v}_1, \ldots, \vec{v}_n\} \subseteq \mathbb{R}^n$ and $\{\vec{u}_1, \ldots, \vec{u}_m\} \subseteq \mathbb{R}^m$ such that $A\vec{v}_i = \sigma_i\vec{u}_i$ for every direction not eliminated by A.

Definition 1.106. Let $\Sigma \in \mathbb{M}_{m\times n}(\mathbb{R})$ where we call Σ a rectangular diagonal matrix if $\Sigma_{ij} = 0$ whenever $i \neq j$. Let $q = \min\{m, n\}$, then Σ has the form,

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots \\ 0 & \sigma_2 & \cdots & 0 & \cdots \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & \cdots & \sigma_q & \cdots \\ \vdots & \vdots & & \vdots & \ddots \end{bmatrix}$$

where the only entries that may be nonzero are

$$\Sigma_{11}, \Sigma_{22}, \ldots, \Sigma_{qq}.$$

Notice that regardless of whether A is square, the matrices $A^\top A \in \mathbb{R}^{n\times n}$ and $AA^\top \in \mathbb{R}^{m\times m}$ are always square, symmetric, and positive semidefinite. We can therefore apply the Principal Axis Theorem to both.

Theorem 1.107. *Let $A \in \mathbb{M}_{m\times n}(\mathbb{R})$ then,*

(i) $A^\top A$ is a symmetric positive semidefinite $n \times n$ matrix, and
(ii) $AA^\top$ is a symmetric positive semidefinite $m \times m$ matrix.

Proof. We prove that $A^\top A$ is symmetric.

This follows directly using the transpose of the product,

$$(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A.$$

For any vector $\vec{x} \in \mathbb{R}^n$,

$$\vec{x}^\top A^\top A\vec{x} = (A\vec{x})^\top (A\vec{x}) = \|A\vec{x}\|^2 \geq 0,$$

since a squared norm is always non-negative.

Therefore, $A^\top A$ is positive semidefinite. Similarly,

$$(AA^\top)^\top = AA^\top, \qquad \vec{y}^\top AA^\top \vec{y} = (A^\top \vec{y})^\top (A^\top \vec{y}) = \|A^\top \vec{y}\|^2 \geq 0$$

for every $\vec{y} \in \mathbb{R}^m$. Thus, $AA^\top$ is also symmetric and positive semidefinite. □

Theorem 1.108. *Every eigenvalue of the symmetric matrix $A^\top A$ is non-negative.*

Proof. Suppose that $\vec{v} \neq \vec{0}$ is an eigenvector of $A^\top A$ corresponding to the eigenvalue λ. Then $A^\top A\vec{v} = \lambda\vec{v}$. Multiplying on the left by $\vec{v}^\top$ gives

$$\vec{v}^\top A^\top A\vec{v} = \lambda \vec{v}^\top \vec{v} \iff \vec{v}^\top A^\top A\vec{v} = \|A\vec{v}\|^2 = \lambda\|\vec{v}\|^2.$$

Since $\|\vec{v}\|^2 > 0$, we have

$$\lambda = \frac{\|A\vec{v}\|^2}{\|\vec{v}\|^2} \geq 0.$$

Therefore, every eigenvalue of $A^\top A$ is nonnegative.

The same proof shows that every eigenvalue of $AA^\top$ is nonnegative. □

The zero eigenvalues of $A^\top A$ correspond exactly to the directions annihilated by A.

Theorem 1.109. *Let $A \in \mathbb{M}_{m\times n}(\mathbb{R})$. Then* $\text{null}(A^\top A) = \text{null}(A)$.

Proof. Let $\vec{x} \in \mathbb{R}^n$ such that $\vec{x} \in \text{null}(A)$. Then $A\vec{x} = \vec{0}$. Then,

$$A^\top A\vec{x} = A^\top \vec{0} = \vec{0} \implies \vec{x} \in \text{null}(A^\top A).$$

Therefore, we conclude that $\text{null}(A) \subseteq \text{null}(A^\top A)$.

Conversely suppose that $\vec{x} \in \text{null}(A^\top A)$. Then $\vec{x}$ satisfies $A^\top A\vec{x} = \vec{0}$.

$$\vec{x}^\top A^\top A\vec{x} = \|A\vec{x}\|^2 = 0 \iff A\vec{x} = \vec{0},$$

since we know that a vector has zero norm if and only if it is the zero vector.

Therefore we conclude that $\vec{x} \in \text{null}(A)$ meaning that $\text{null}(A^\top A) \subseteq \text{null}(A)$.

Combining the two inclusions yields $\text{null}(A^\top A) = \text{null}(A)$. □

Corollary 1.110. *Let $A \in \mathbb{M}_{m\times n}(\mathbb{R})$. Then* $\text{rank}(A^\top A) = \text{rank}(A)$.

Proof. Notice that both A and $A^\top A$ have domain $\mathbb{R}^n$. By the preceding theorem,

$$\text{null}(A^\top A) = \text{null}(A) \implies \text{nullity}(A^\top A) = \text{nullity}(A).$$

By Rank-Nullity Theorem,

$$\text{rank}(A^\top A) + \text{nullity}(A^\top A) = n \wedge \text{rank}(A) + \text{nullity}(A) = n.$$

Since the nullities are equal, the ranks must also be equal, hence,

$$\text{rank}(A^\top A) = \text{rank}(A).$$

Similarly, we have that,

$$\text{rank}(AA^\top) = \text{rank}(A^\top) = \text{rank}(A).$$

□

Since $A^\top A$ is real and symmetric, the Principal Axis Theorem guarantees an orthonormal basis of eigenvectors. Suppose that $\vec{v}_1, \ldots, \vec{v}_n$ is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of $A^\top A$. We thus write $A^\top A\vec{v}_i = \lambda_i \vec{v}_i$. Because we know that $A^\top A$ is positive semi-definite, $\lambda_i \geq 0$ which allows us to take the non-negative square root of eigenvalue.

Definition 1.111. Let $A \in \mathbb{M}_{m\times n}(\mathbb{R})$. A non-negative number σ is called a singular value of A if σ^2 is an eigenvalue of $A^\top A$. Thus, if we let $A^\top A\vec{v}_i = \lambda_i \vec{v}_i$ then we define,

$$\sigma_i = \sqrt{\lambda_i}.$$

There are a couple notes we make on singular values,

1. Equivalently, we have that $A^\top A\vec{v}_i = \sigma_i^2 \vec{v}_i$.
2. A unit eigenvector $\vec{v}_i$ of $A^\top A$ is called a right singular vector of A.
3. We arrange the singular values in non-increasing order $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

Theorem 1.112. *If $\vec{v}_i$ is a unit right singular vector corresponding to the singular value σ_i then,*

$$\|A\vec{v}_i\| = \sigma_i.$$

Proof. Since $\vec{v}_i$ is a right singular vector, $A^\top A\vec{v}_i = \sigma_i^2 \vec{v}_i$. Therefore,

$$\|A\vec{v}_i\|^2 = (A\vec{v}_i)^\top (A\vec{v}_i) = \vec{v}_i^\top A^\top A\vec{v}_i.$$

Substituting the eigenvalue equation we have that,

$$\|A\vec{v}_i\|^2 = \vec{v}_i^\top (\sigma_i^2 \vec{v}_i) = \sigma_i^2 \vec{v}_i^\top \vec{v}_i.$$

Since $\vec{v}_i$ is a unit vector, $\vec{v}_i^\top \vec{v}_i = 1$ then $\|A\vec{v}_i\|^2 = \sigma_i^2$. Therefore, both sides are non-negative so taking square roots gives $\|A\vec{v}_i\| = \sigma_i$. Thus, the singular value σ_i is precisely the length of the image of the unit right singular vector $\vec{v}_i$. □

For any i such that $\sigma_i > 0$, we have $\|A\vec{v}_i\| = \sigma_i > 0$, so $A\vec{v}_i \neq \vec{0}$.

We can then normalize $A\vec{v}_i$ to obtain a unit vector in $\mathbb{R}^m$.

Definition 1.113. Let $\vec{v}_i$ be a unit right singular vector corresponding to a positive singular value σ_i. The corresponding left singular vector is defined by,

$$\vec{u}_i = \frac{A\vec{v}_i}{\sigma_i} \iff A\vec{v}_i = \sigma_i \vec{u}_i.$$

The two formulations above are equivalent. Thus, the vector $\vec{v}_i$ is a direction in the domain $\mathbb{R}^n$ while $\vec{u}_i$ is a direction in the codomain $\mathbb{R}^m$. Evidently, the singular value σ_i measures how much A stretches the input direction $\vec{v}_i$ and $\vec{u}_i$ gives the output direction.

Theorem 1.114. *If $\vec{u}_i = \frac{A\vec{v}_i}{\sigma_i}$ and $\sigma_i > 0$ then $\|\vec{u}_i\| = 1$.*

Proof. Using the definition of $\vec{u}_i$,

$$\|\vec{u}_i\|^2 = \left\| \frac{A\vec{v}_i}{\sigma_i} \right\|^2.$$

Since we know that $\sigma_i > 0$, then,

$$\|\vec{u}_i\|^2 = \frac{1}{\sigma_i^2}\|A\vec{v}_i\|^2 = \frac{\sigma_i^2}{\sigma_i^2} = 1,$$

by the preceding result. Therefore $\|\vec{u}_i\| = 1$. □

Theorem 1.115. *Suppose that $\vec{v}_i$ and $\vec{v}_j$ are orthonormal right singular vectors corresponding to positive singular values σ_i and σ_j.*

Then their corresponding left singular vectors satisfy,

$$\langle \vec{u}_i, \vec{u}_j \rangle = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Therefore, the left singular vectors corresponding to positive singular values form an orthonormal set.

Proof. By definition we know that,

$$\vec{u}_i = \frac{A\vec{v}_i}{\sigma_i}, \quad \vec{u}_j = \frac{A\vec{v}_j}{\sigma_j}.$$

Therefore we compute their inner product as,

$$\begin{aligned}
\langle \vec{u}_i, \vec{u}_j \rangle &= \left\langle \frac{A\vec{v}_i}{\sigma_i}, \frac{A\vec{v}_j}{\sigma_j} \right\rangle \\
&= \frac{1}{\sigma_i \sigma_j} \langle A\vec{v}_i, A\vec{v}_j \rangle \\
&= \frac{1}{\sigma_i \sigma_j} (A\vec{v}_i)^\top (A\vec{v}_j) \\
&= \frac{1}{\sigma_i \sigma_j} \vec{v}_i^\top A^\top A \vec{v}_j \\
&= \frac{1}{\sigma_i \sigma_j} \vec{v}_i^\top (\sigma_j^2 \vec{v}_j), \qquad \vec{v}_j \text{ is a right singular vector,} \\
&= \frac{\sigma_j}{\sigma_i} \vec{v}_i^\top \vec{v}_j.
\end{aligned}$$

Because the right singular vectors are orthonormal,

$$\vec{v}_i^\top \vec{v}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j \end{cases}.$$

If $i \neq j$ then $\langle \vec{u}_i, \vec{u}_j \rangle = 0$. Otherwise, if $i = j$, then $\langle \vec{u}_j, \vec{u}_i \rangle = \dfrac{\sigma_i}{\sigma_i} = 1$.

Therefore, the corresponding left singular vectors are orthonormal. □

Interestingly, the right and left singular vectors satisfy two symmetric equations.

Theorem 1.116. *Let $\sigma_i > 0$ and let $\vec{v}_i$ and $\vec{u}_i$ be corresponding right and left singular vectors. Then, $A\vec{v}_i = \sigma_i \vec{u}_i$ and $A^\top \vec{u}_i = \sigma_i \vec{v}_i$.*

Proof. The first equation follows directly from the definition of a right singular vector,

$$\vec{u}_i = \frac{A\vec{v}_i}{\sigma_i}.$$

Multiplying by σ_i gives $A\vec{v}_i = \sigma_i \vec{u}_i$.

For the second equation, substitute the definition of $\vec{u}_i$:

$$\begin{aligned}
A^\top \vec{u}_i = A^\top \left(\frac{A\vec{v}_i}{\sigma_i} \right) &= \frac{1}{\sigma_i} A^\top A \vec{v}_i \\
&= \frac{\sigma_i^2}{\sigma_i} \vec{v}_i \\
&= \sigma_i \vec{v}_i.
\end{aligned}$$

These equations show that $\vec{v}_i \mapsto \sigma_i \vec{u}_i$ and $\vec{u}_i \mapsto \sigma_i \vec{v}_i$. □

Corollary 1.117. *If $\sigma_i > 0$ then $AA^\top \vec{u}_i = \sigma_i^2 \vec{u}_i$.*

Therefore, every left singular vector corresponding to a positive singular value is an eigenvector of $AA^\top$.

Proof. Using $A^\top \vec{u}_i = \sigma_i \vec{v}_i$ meaning we have that,

$$AA^\top \vec{u}_i = A(\sigma_i \vec{v}_i) = \sigma_i A\vec{v}_i = \sigma_i^2 \vec{u}_i.$$

Therefore, $A^\top A\vec{v}_i = \sigma_i^2 \vec{v}_i$ and $AA^\top \vec{u}_i = \sigma_i^2 \vec{u}_i$.

The squared nonzero singular values are simultaneously eigenvalues of $A^\top A$ and $AA^\top$. □

Theorem 1.118. *The matrices $A^\top A$ and $AA^\top$ have the same positive eigenvalues including multiplicities.*

Proof. Suppose that $A^\top A\vec{v} = \lambda \vec{v}$ for some $\lambda > 0$ and $\vec{v} \neq \vec{0}$. Since $\lambda > 0$ we cannot have that $A\vec{v} = \vec{0}$. Indeed, if $A\vec{v} = \vec{0}$, then it follows that $A^\top A\vec{v} = \vec{0}$ which would contradict the fact that $A^\top A\vec{v} = \lambda \vec{v} \neq \vec{0}$. Therefore we have that $A\vec{v} \neq \vec{0}$.

Now suppose that,

$$AA^\top (A\vec{v}) = A(A^\top A\vec{v}) = A(\lambda \vec{v}) = \lambda A\vec{v}.$$

Therefore, $A\vec{v}$ is an eigenvector of $AA^\top$ corresponding to λ. Moreover, on the λ-eigenspaces, the map $\vec{v} \mapsto A\vec{v}$ has inverse $\vec{u} \mapsto \lambda^{-1} A^\top \vec{u}$.

Conversely, suppose that $AA^\top \vec{u} = \lambda \vec{u}$ for some $\lambda > 0$ and $\vec{u} \neq \vec{0}$. Then it follows that $A^\top \vec{u} \neq \vec{0}$ and,

$$A^\top A(A^\top \vec{u}) = A^\top (AA^\top \vec{u}) = A^\top (\lambda \vec{u}) = \lambda A^\top \vec{u}.$$

Therefore, $A^\top \vec{u}$ is an eigenvector of $A^\top A$ corresponding to λ. The corresponding positive-eigenvalue eigenspaces are thus isomorphic and have the same dimensions. Since both matrices are symmetric and hence diagonalizable, the two matrices have the same positive eigenvalues with the same algebraic multiplicities. The numbers of zero eigenvalues may differ because $A^\top A$ is $n \times n$ while $AA^\top$ is $m \times m$. □

Theorem 1.119 (Singular Value Decomposition Theorem.)**.** *Let $A \in \mathbb{M}_{m\times n}(\mathbb{R})$ have rank r.*

Then there exist orthogonal matrices $U \in \mathbb{M}_{m\times m}(\mathbb{R})$ and $V \in \mathbb{M}_{n\times n}(\mathbb{R})$ and a rectangular diagonal matrix $\Sigma \in \mathbb{M}_{m\times n}(\mathbb{R})$ such that $A = U\Sigma V^\top$.

The diagonal entries of Σ satisfy $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ and all remaining diagonal entries are zero. The columns of V are right singular vectors of A while the first r columns of U are the corresponding left singular vectors.

Proof. Since $A^\top A$ is real and symmetric, the Principal Axis Theorem guarantees an orthonormal basis $\{\vec{v}_1, \ldots, \vec{v}_n\}$ of $\mathbb{R}^n$ consisting of eigenvectors of $A^\top A$. Arrange the eigenvalues in non-increasing order such that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. Since $\operatorname{rank}(A^\top A) = \operatorname{rank}(A) = r$, there are exactly r positive eigenvalues. Therefore, $\lambda_1 \geq \cdots \geq \lambda_r > 0$ and $\lambda_{r+1} = \cdots = \lambda_n = 0$. Then, we define $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, \ldots, r$.

For each $i = 1, \ldots, r$ define,

$$\vec{u}_i = \frac{A\vec{v}_i}{\sigma_i}.$$

We showed previously that $\vec{u}_1, \ldots, \vec{u}_r$ form an orthonormal set in $\mathbb{R}^m$. We next show that these vectors span the range of A. Let $\vec{x} \in \mathbb{R}^n$. Since we know that $\vec{v}_1, \ldots, \vec{v}_n$ form an orthonormal basis of $\mathbb{R}^n$, there exist scalars $c_1, \ldots, c_n$ such that,

$$\vec{x} = c_1\vec{v}_1 + \cdots + c_n\vec{v}_n.$$

Applying A we get,

$$A\vec{x} = c_1A\vec{v}_1 + \cdots + c_nA\vec{v}_n.$$

For $i \leq r$, $A\vec{v}_i = \sigma_i\vec{u}_i$ and for $i > r$ we have that $A^\top A\vec{v}_i = \vec{0}$. Since $\text{null}(A^\top A) = \text{null}(A)$ then $A\vec{v}_i = \vec{0}$ for $i = r+1, \ldots, n$. Consequently, $A\vec{x} = c_1\sigma_1\vec{u}_1 + \cdots + c_r\sigma_r\vec{u}_r$ so $A\vec{x} \in \text{span}\{\vec{u}_1, \ldots, \vec{u}_r\}$. Since this holds for every $\vec{x} \in \mathbb{R}^n$ we conclude that $\text{range}(A) \subseteq \text{span}\{\vec{u}_1, \ldots, \vec{u}_r\}$.

Conversely, if for every $i = 1, \ldots, r$ we define,

$$\vec{u}_i = A\left(\frac{\vec{v}_i}{\sigma_i}\right) \iff \vec{u}_i \in \text{range}(A),$$

then we know that $\{\vec{u}_1, \ldots, \vec{u}_r\} \subseteq \text{range}(A)$ meaning that,

$$\text{range}(A) = \text{span}\{\vec{u}_1, \ldots, \vec{u}_r\}.$$

Since $\vec{u}_1, \ldots, \vec{u}_r$ form an orthonormal set, we may extend them to an orthonormal basis $\{\vec{u}_1, \ldots, \vec{u}_r, \vec{u}_{r+1}, \ldots, \vec{u}_m\}$ of $\mathbb{R}^m$. Define $U = [\vec{u}_1 \quad \cdots \quad \vec{u}_m] \in \mathbb{M}_{m\times m}(\mathbb{R})$ and $V = [\vec{v}_1 \quad \cdots \quad \vec{v}_n] \in \mathbb{M}_{n\times n}(\mathbb{R})$. Because their columns form orthonormal bases,

$$U^\top U = UU^\top = I_m, \quad V^\top V = VV^\top = I_n.$$

Therefore, U, V are orthogonal matrices. Consider the product AV,

$$AV = [A\vec{v}_1 \quad \cdots \quad A\vec{v}_n].$$

Since for every $i = 1, \ldots, r$, $A\vec{v}_i = \sigma_i\vec{u}_i$ and $A\vec{v}_i = \vec{0}$ for $i = r+1, \ldots, n$ then,

$$AV = [\sigma_1\vec{u}_1 \quad \cdots \quad \sigma_r\vec{u}_r \quad \vec{0} \quad \cdots \quad \vec{0}].$$

But this is precisely $U\Sigma$ so therefore $AV = U\Sigma$.

Multiplying on the right by $V^\top$ and using the fact that $VV^\top = I_n$ gives $AVV^\top = U\Sigma V^\top$.

Hence we have that $A = U\Sigma V^\top$ proving the singular value decomposition. □

Notice that the decomposition $A = U\Sigma V^\top$ describes the transformation A as three consecutive operations. For any $\vec{x} \in \mathbb{R}^n$, $A\vec{x} = U\Sigma V^\top\vec{x}$. First, $V^\top\vec{x}$ expresses $\vec{x}$ in the orthonormal basis of right singular vectors. Second, $\Sigma V^\top\vec{x}$ scales the i-th singular vector coordinate by σ_i. Directions corresponding to zero singular values are eliminated. Third, $U\Sigma V^\top\vec{x}$ expresses the resulting vector in the orthonormal basis of left singular vectors. Since U, V are orthogonal, they preserve lengths and angles so all changes in the lengths of the vectors are caused by Σ.

Definition 1.120. The factorization,

$$A = U\Sigma V^\top,$$

where $U \in \mathbb{M}_{m\times m}(\mathbb{R}), \Sigma \in \mathbb{M}_{m\times n}(\mathbb{R}), V \in \mathbb{M}_{n\times n}(\mathbb{R})$ is called the full singular value decomposition. If A has rank r, then only the first r singular values are positive.

The remaining columns of U and V do not contribute to the product.

Definition 1.121 (Reduced SVD)**.** The factorization,

$$A = U_r\Sigma_r V_r^\top,$$

is called the reduced singular value decomposition, also known as the compact or thin SVD. The dimensions are

$$(m \times r)(r \times r)(r \times n) = m \times n.$$

Since all diagonal entries of Σ_r are positive, Σ_r is invertible.

Theorem 1.122. *Let $A = U\Sigma V^\top$ have rank r. Every vector $\vec{x} \in \mathbb{R}^n$ can be written uniquely as,*

$$\vec{x} = \sum_{i=1}^{n} c_i\vec{v}_i, \quad c_i = \langle \vec{x}, \vec{v}_i\rangle.$$

Then we have $A\vec{x} = \sum_{i=1}^{r} \sigma_i c_i \vec{u}_i$ and $A\vec{x} = \sum_{i=1}^{r} \sigma_i \langle \vec{x}, \vec{v}_i\rangle \vec{u}_i$.

Proof. Since $\vec{v}_1, \ldots, \vec{v}_n$ form an orthonormal basis $\vec{x} = \sum_{i=1}^{n} c_i\vec{v}_i$, then applying A,

$$\begin{aligned} A\vec{x} &= A\left(\sum_{i=1}^{n} c_i\vec{v}_i\right) = \sum_{i=1}^{n} c_i A\vec{v}_i \\ &= \sum_{i=1}^{r} c_i\sigma_i\vec{u}_i = \sum_{i=1}^{r} \sigma_i\langle \vec{x}, \vec{v}_i\rangle\vec{u}_i, \end{aligned}$$

because $A\vec{v}_i = \vec{0}$ for $i > r$. This formulation describes the action of A completely:

$$\vec{x} \mapsto \langle \vec{x}, \vec{v}_i\rangle \mapsto \sigma_i\langle \vec{x}, \vec{v}_i\rangle \mapsto \sigma_i\langle \vec{x}, \vec{v}_i\rangle\vec{u}_i.$$

That is, the matrix extracts the component of $\vec{x}$ along each right singular vector, scales the component, then redirects it into the corresponding left singular direction and adds the results. □

Corollary 1.123. *If $\vec{x} = \sum_{i=1}^{n} c_i\vec{v}_i$ then,*

$$\|A\vec{x}\|^2 = \sum_{i=1}^{r} \sigma_i^2 c_i^2.$$

Proof. Recall that we had that $A\vec{x} = \sum_{i=1}^{r} \sigma_i c_i \vec{u}_i$. Because we know that $\vec{u}_1, \ldots, \vec{u}_r$ are orthonormal, we apply the generalized Pythagorean Theorem which gives,

$$\|A\vec{x}\|^2 = \sum_{i=1}^{r} \|\sigma_i c_i \vec{u}_i\|^2 = \sum_{i=1}^{r} \sigma_i^2 c_i^2.$$

□

Theorem 1.124. *If A has rank r then $A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^\top$.*

Proof. Consider the reduced SVD form for $A = U_r \Sigma_r V_r^\top$ such that we have $U_r \Sigma_r = [\sigma_1 \vec{u}_1 \quad \cdots \quad \sigma_r \vec{u}_r]$. Suppose that we define,

$$V_r^\top = \begin{bmatrix} \vec{v}_1^\top \\ \vdots \\ \vec{v}_r^\top \end{bmatrix} \quad \text{s.t.} \quad U_r \Sigma_r V_r^\top = \begin{bmatrix} \sigma_1 \vec{u}_1 & \cdots & \sigma_r \vec{u}_r \end{bmatrix} \begin{bmatrix} \vec{v}_1^\top \\ \vdots \\ \vec{v}_r^\top \end{bmatrix}.$$

Then, multiplying out we get,

$$A = \sigma_1 \vec{u}_1 \vec{v}_1^\top + \cdots + \sigma_r \vec{u}_r \vec{v}_r^\top.$$

Notice that each outer product $\vec{u}_i \vec{v}_i^\top$ is an $m \times n$ matrix of rank one. Indeed for any $\vec{x} \in \mathbb{R}^n$ we have that $\vec{u}_i \vec{v}_i^\top \vec{x} = (\vec{v}_i^\top \vec{x}) \vec{u}_i$ of which its entire range lies in $\text{span}\{\vec{u}_i\}$ which is one dimensional. □

The next theorem characterizes how the SVD provides orthonormal bases for the column space, row space, nullspace and left null space simultaneously.

Theorem 1.125. *Let $A \in \mathbb{M}_{m \times n}(\mathbb{R})$ have rank r with full SVD that is $A = U \Sigma V^\top$. Then,*

(i) $\text{range}(A) = \text{span}\{\vec{u}_1, \ldots, \vec{u}_r\}$,
(ii) $\text{null}(A^\top) = \text{span}\{\vec{u}_{r+1}, \ldots, \vec{u}_m\}$,
(iii) $\text{range}(A^\top) = \text{span}\{\vec{v}_1, \ldots, \vec{v}_r\}$,
(iv) $\text{null}(A) = \text{span}\{\vec{v}_{r+1}, \ldots, \vec{v}_n\}$.

Proof. We have already established that $\text{range}(A) = \text{span}\{\vec{u}_1, \ldots, \vec{u}_r\}$. Because $\vec{u}_1, \ldots, \vec{u}_m$ form an orthonormal basis of $\mathbb{R}^m$, the orthogonal complement of the range is $\text{range}(A)^\perp = \text{span}\{\vec{u}_{r+1}, \ldots, \vec{u}_m\}$. Since $\text{range}(A)^\perp = \text{null}(A^\top)$ then we obtain that $\text{null}(A^\top) = \text{span}\{\vec{u}_{r+1}, \ldots, \vec{u}_m\}$.

For every $i > r$ we know that $A\vec{v}_i = \vec{0}$ so it follows that $\vec{v}_i \in \text{null}(A)$. Therefore $\text{span}\{\vec{v}_{r+1}, \ldots, \vec{v}_n\} \subseteq \text{null}(A)$. The left-hand side has dimension $n - r$ so by Rank-Nullity theorem,

$$\dim \text{null}(A) = n - \text{rank}(A) = n - r.$$

Since one $(n - r)$ dimensional subspace is contained in another $n - r$ dimensional subspace they are equal. Finally we know that $\text{range}(A^\top) = \text{null}(A)^\perp$ and since the right singular vectors form an orthonormal basis,

$$\text{null}(A)^\perp = \text{span}\{\vec{v}_1, \ldots, \vec{v}_r\}.$$

Therefore $\text{range}(A^\top) = \text{span}\{\vec{v}_1, \ldots, \vec{v}_r\}$. Thus the SVD produces the orthogonal decompositions,

$$\mathbb{R}^n = \text{range}(A^\top) \oplus \text{null}(A), \qquad \mathbb{R}^m = \text{range}(A) \oplus \text{null}(A^\top).$$

□

Theorem 1.126. *The rank of a matrix equals the number of its positive singular values.*

That is, if $\sigma_1 \geq \cdots \geq \sigma_r > 0$ and all remaining singular values are zero, then $\text{rank}(A) = r$.

Proof. The range of A is $\text{range}(A) = \text{span}\{\vec{u}_1, \ldots, \vec{u}_r\}$. Because the vectors $\vec{u}_1, \ldots, \vec{u}_r$ are orthonormal, they are linearly independent. Therefore, they form a basis for $\text{range}(A)$. Hence, $\dim \text{range}(A) = r$.

By definition of rank, $\text{rank}(A) = \dim \text{range}(A)$ so $\text{rank}(A) = r$. □

The singular values are not merely algebraic quantities obtained from $A^\top A$ but also describe the maximal and minimal stretching produced by A.

Theorem 1.127. *Let σ_1 be the largest singular value of A. Then* $\sigma_1 = \max_{\|\vec{x}\|=1} \|A\vec{x}\|$.

The maximum is attained at $\vec{x} = \vec{v}_1$.

Proof. Suppose that $\vec{x} \in \mathbb{R}^n$ satisfies the unit vector condition i.e., $\|\vec{x}\| = 1$.

We write $\vec{x} = \sum_{i=1}^{n} c_i \vec{v}_i$. Then, since the right singular vectors are orthonormal, $\sum_i c_i^2 = 1$.

Moreover, since we know that $\sigma_i \leq \sigma_1$ for all $i = 1, \ldots, r$,

$$\begin{aligned} \|A\vec{x}\|^2 &= \sum_{i=1}^{r} \sigma_i^2 c_i^2 \\ &\leq \sigma_1^2 \sum_{i=1}^{r} c_i^2 \\ &\leq \sigma_1^2. \end{aligned}$$

Now, we choose $\vec{x} = \vec{v}_1$. Then, $\|\vec{v}_1\| = 1$ and $\|A\vec{v}_1\| = \|\sigma_1 \vec{u}_1\| = \sigma_1$.

Therefore the upper bound is attained at $\sigma_1 = \max_{\|\vec{x}\|=1} \|A\vec{x}\|$. □

Theorem 1.128. *For $k \leq r$,*

$$\sigma_k = \max_{\|\vec{x}\|=1;\ \vec{x} \perp \vec{v}_1, \ldots, \vec{v}_{k-1}} \|A\vec{x}\|.$$

Proof. Suppose that $\|\vec{x}\| = 1$ (i.e., is the unit vector) and $\vec{x} \perp \vec{v}_1, \ldots, \vec{v}_{k-1}$. Then the first $k-1$ singular-vector coordinates vanish so $\vec{x} = \sum_{i=k}^{n} c_i \vec{v}_i$ with $\sum_{i=k}^{n} c_i^2 = 1$. Then it follows,

$$\begin{aligned} \|A\vec{x}\|^2 &= \sum_{i=k}^{r} \sigma_i^2 c_i^2 \\ &\leq \sigma_k^2 \sum_{i=k}^{r} c_i^2, \qquad \sigma_i \leq \sigma_k, \\ &\leq \sigma_k^2. \end{aligned}$$

Thus, $\|A\vec{x}\| \leq \sigma_k$. Equality is attained by choosing $\vec{x} = \vec{v}_k$ and therefore,

$$\sigma_k = \max_{\|\vec{x}\|=1;\ \vec{x} \perp \vec{v}_1, \ldots, \vec{v}_{k-1}} \|A\vec{x}\|.$$

Notice that the first right singular vector gives the direction of greatest stretching and the second gives the direction of greatest stretching among unit vectors orthogonal to the first.

This continues successively. □

Theorem 1.129. *Suppose that $A \in \mathbb{M}_{m\times n}(\mathbb{R})$ has full column rank n. Then,*

$$\sigma_n = \min_{\|\vec{x}\|=1} \|A\vec{x}\|.$$

Proof. Let $\vec{x} = \sum_{i=1}^{n} c_i \vec{v}_i$ and $\sum_{i=1}^{n} c_i^2 = 1$.

Since A has full column rank and every singular value is positive, then,

$$\begin{aligned} \|A\vec{x}\|^2 &= \sum_{i=1}^{n} \sigma_i^2 c_i^2 \\ &\geq \sigma_n^2 \sum_{i=1}^{n} c_i^2 \\ &= \sigma_n^2. \end{aligned}$$

Equality is attained at $\vec{x} = \vec{v}_n$. Thus,

$$\sigma_n = \min_{\|\vec{x}\|=1} \|A\vec{x}\|.$$

If A is rank deficient, then the minimum is zero because there exists a unit vector in $\text{null}(A)$. □

Lastly, the singular values are uniquely determined by A but the singular vectors need not be unique.

Theorem 1.130. *The singular values of A including their multiplicities are uniquely determined.*

Proof. The squared singular values are the eigenvalues of the uniquely determined matrix $A^\top A$. The eigenvalues of a matrix, including their algebraic multiplicities are uniquely determined by its characteristic polynomial. Since each singular value is defined as the non-negative square root of an eigenvalue of $A^\top A$, the singular values are uniquely determined. □

However, it is important to note that the singular vectors have several ambiguities. If $A\vec{v}_i = \sigma_i \vec{u}_i$ then $A(-\vec{v}_i) = \sigma_i(-\vec{u}_i)$. Therefore, the pair $(-\vec{u}_i, -\vec{v}_i)$ is equally valid. If a singular value has multiplicity greater than one, then any orthonormal basis of the corresponding eigenspace of $A^\top A$ may be used as the right singular vectors. The associated left singular vectors rotate correspondingly.

Thus, the singular values are intrinsic, but the individual singular vectors may not be.

The SVD allows us to construct an inverse-like matrix when A is rectangular or singular. Suppose that we have the reduced SVD form $A = U_r \Sigma_r V_r^\top$ where $\Sigma_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$.

Since $\sigma_i > 0$ for $i = 1, \ldots, r$, define

$$\Sigma_r^{-1} = \operatorname{diag}\left(\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_r}\right).$$

Definition 1.131 (Moore-Penrose Pseudoinverse)**.** The pseudoinverse of A is denoted $A^+ = V_r \Sigma_r^{-1} U_r^\top$ or equivalently,

$$A^+ = \sum_{i=1}^{r} \frac{1}{\sigma_i} \vec{v}_i \vec{u}_i^\top.$$

The pseudoinverse reverses the action of A on the directions that survive the transformation.

Since $A\vec{v}_i = \sigma_i \vec{u}_i$ we have that $A^+ \vec{u}_i = \frac{1}{\sigma_i} \vec{v}_i$.

Directions in $\operatorname{null}(A^\top)$ are mapped to zero because they do not lie in the range of A.

Theorem 1.132. *Let A have reduced SVD $A = U_r \Sigma_r V_r^\top$. Then $AA^+ = U_r U_r^\top$ and $A^+ A = V_r V_r^\top$.*

Consequently, AA^+ is the orthogonal projection onto $\operatorname{range}(A)$ *while $A^+ A$ is the orthogonal projection onto* $\operatorname{range}(A^\top)$*, the row space of A.*

Proof. Using the reduced SVD,

$$AA^+ = (U_r \Sigma_r V_r^\top)(V_r \Sigma_r^{-1} U_r^\top).$$

Since the columns of V_r are orthonormal, $V_r^\top V_r = I_r$ so,

$$AA^+ = U_r \Sigma_r I_r \Sigma_r^{-1} U_r^\top = U_r U_r^\top.$$

Since the columns of U_r form an orthonormal basis for $\operatorname{range}(A)$, the matrix $U_r U_r^\top$ is the orthogonal projection onto $\operatorname{range}(A)$. Similarly,

$$A^+ A = (V_r \Sigma_r^{-1} U_r^\top)(U_r \Sigma_r V_r^\top) = V_r \Sigma_r^{-1} I_r \Sigma_r V_r^\top = V_r V_r^\top.$$

The columns of V_r form an orthonormal basis for $\operatorname{range}(A^\top)$.

So, $V_r V_r^\top$ is the orthogonal projection onto the row space. $\square$

Chapter 2

Multidimensional Statistics and Moments

Having developed the deterministic geometry of vectors and matrices, we now introduce uncertainty by allowing the coordinates of a vector to be random variables. A random vector $\vec{X} = (X_1, \ldots, X_p)^\top$ is therefore not a single fixed point in $\mathbb{R}^p$ but a random point whose distribution describes both the marginal behaviour of its components and their dependence through the covariance matrix.

Expectations form mean vectors, covariances form symmetric positive semi-definite matrices, linear transformations determine the distributions of portfolios and regression predictors, and orthogonal diagonalization separates correlated variation into uncorrelated directions.

Definition 2.1. Suppose that we have a vector $Y = [y_1 \quad \cdots \quad y_n]^\top$ containing random variables y_i for $i = 1, \ldots, n$. Such a vector is referred to as a random vector.

Theorem 2.2. *Let Y be a random vector in $\mathbb{R}^n$.*

Then, we specific it's characteristics as follows:

1. $\mathbb{E}[y_i] = \mu_i$,
2. $\mathrm{Var}(y_i) = \sigma_i^2$, *and*
3. $\mathrm{Var}[Y] = \mathbb{E}\left[(Y - \mathbb{E}[Y])(Y - \mathbb{E}[Y])^\top\right]$.

Remark 2.3. Let $Y \in \mathbb{R}^n$ and let $y_1, \ldots, y_n$ be independent and identically distributed random variables. Then, $\mathrm{Cov}(y_i, y_j) = 0$ for $i \neq j$ and $\mathrm{Var}(y_i) = \sigma^2$ for $i = j$.

More generally,

$$\mathrm{Var}(Y) = \begin{bmatrix} \mathrm{Var}(Y_1) & \mathrm{Cov}(Y_1, Y_2) & \mathrm{Cov}(Y_1, Y_3) & \cdots & \mathrm{Cov}(Y_1, Y_n) \\ \mathrm{Cov}(Y_2, Y_1) & \mathrm{Var}(Y_2) & \mathrm{Cov}(Y_2, Y_3) & \cdots & \mathrm{Cov}(Y_2, Y_n) \\ \mathrm{Cov}(Y_3, Y_1) & \mathrm{Cov}(Y_3, Y_2) & \mathrm{Var}(Y_3) & \cdots & \mathrm{Cov}(Y_3, Y_n) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(Y_n, Y_1) & \mathrm{Cov}(Y_n, Y_2) & \mathrm{Cov}(Y_n, Y_3) & \cdots & \mathrm{Var}(Y_n) \end{bmatrix} = \begin{bmatrix} \sigma^2 & 0 & 0 & \cdots & 0 \\ 0 & \sigma^2 & 0 & \cdots & 0 \\ 0 & 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_n,$$

where I_n is the $n \times n$ identity matrix in $\mathbb{M}_{n\times n}(\mathbb{R})$.

2.1 Covariance Matrices and Spectral Decomposition

Suppose the components are independent but not necessarily identically distributed. Thus, $\mathrm{Var}(Y_i) = \sigma_i^2$. Clearly, $\mathrm{Var}(Y) = \Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. The matrix is already diagonal so its corresponding eigenvalues are simply $\lambda_i = \sigma_i^2$ and the corresponding eigenvectors are simple the standard basis vectors in $\mathbb{R}^n$. This is the trivial case.

Now, consider the case that the covariance matrix has non-zero off-diagonal entries. Then, the entries of $\Sigma = \mathrm{Var}(Y)$ are $\sigma_{ij} = \mathrm{Cov}(Y_i, Y_j)$. By properties of symmetric matrices, $\mathrm{Cov}(Y_i, Y_j) = \mathrm{Cov}(Y_j, Y_i)$ by properties of covariances, namely, symmetry. It follows that $\Sigma^\top = \Sigma$. Therefore, by Spectral Decomposition Theorem, every real covariance matrix is orthogonally diagonalizable. Specifically, there exist an orthonormal basis of eigenvectors $q_1, \ldots, q_n$ with corresponding eigenvectors $\lambda_1, \ldots, \lambda_n$ such that $\Sigma q_i = \lambda_i q_i$. If we let $\mathrm{Col}(Q) = \mathrm{span}\{q_1, \ldots, q_n\}$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, then $\Sigma = Q\Lambda Q^\top$ where the columns of the matrix $Q \in \mathbb{M}_{m\times n}(\mathbb{R})$ span the orthonormal basis of eigenvectors.

Consider the 2×2 case namely,

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

To find the eigenvalues, we solve $\det(\Sigma - \lambda I_2) = 0$,

$$\begin{aligned} 0 &= \det \begin{bmatrix} \sigma_1^2 - \lambda & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 - \lambda \end{bmatrix} \\ &= (\sigma_1^2 - \lambda)(\sigma_2^2 - \lambda) - \sigma_{12}^2 \\ &= \lambda^2 - (\sigma_1^2 + \sigma_2^2)\lambda + \sigma_1^2 + \sigma_2^2 - \sigma_{12}^2. \end{aligned}$$

By application of the Quadratic Formula, we find eigenvalues,

$$\lambda_\pm = \frac{\sigma_1^2 + \sigma_2^2 \pm \sqrt{(\sigma_1^2 - \sigma_2^2)^2 + 4\sigma_{12}^2}}{2}.$$

For a given eigenvalue λ we solve $(\Sigma - \lambda I_2)v = 0$ so we maybe write $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ such that we can obtain,

$$\begin{bmatrix} \sigma_1^2 - \lambda & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

One possible eigenvector is therefore $v = \begin{bmatrix} \sigma_{12} & \lambda - \sigma_1^2 \end{bmatrix}$ because $(\sigma_1^2 - \lambda)\sigma_{12} + \sigma_{12}(\lambda - \sigma_1^2) = 0$.

After normalizing,

$$q = \frac{1}{\sqrt{\sigma_{12}^2 + (\lambda - \sigma_1^2)^2}} \begin{bmatrix} \sigma_{12} \\ \lambda - \sigma_1^2 \end{bmatrix}.$$

Applying this to λ_+ and λ_- gives two orthonormal eigenvectors q_+, q_- hence,

$$Q = \begin{bmatrix} q_+ & q_- \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{bmatrix}.$$

Theorem 2.4. *Let $A \in \mathbb{M}_{m\times n}(\mathbb{R})$, $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$.*

Then, the following properties hold,

(i) $\mathbb{E}[AY + b] = A \cdot \mathbb{E}[Y] + b$,
(ii) $\mathrm{Var}(Y + c) = \mathrm{Var}(Y)$
(iii) $\mathrm{Var}(AY) = A \cdot \mathrm{Var}(Y) \cdot A^\top$
(iv) $\mathrm{Var}(AY + b) = A \cdot \mathrm{Var}(Y) \cdot A^\top$

Theorem 2.5. *Let $Y \in \mathbb{R}^n$ be a random vector and $f : \mathbb{R}^n \to \mathbb{R}$.*

Then its gradient vector is,

$$\frac{df}{dY} = \left(\frac{df}{dY_1}, \ldots, \frac{df}{dY_n} \right).$$

Suppose we have some scalar vector $c \in \mathbb{R}^n$ such that $f = c^\top Y = \sum_{i=1}^{n} c_i Y_i$. Then, gradient vector is simply, $\frac{df}{dY} = c \in \mathbb{R}^n$. If A is symmetric, that is, $A = A^\top$ and f is the following operator,

$$f(Y) = Y^\top AY = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} Y_i Y_j,$$

then its corresponding gradient vector is,

$$\frac{df}{dY} = 2AY.$$

Proof. For the linear form $f(Y) = c^\top Y = \sum_{i=1}^{n} c_i Y_i$, we differentiate this formulation with respect to each coordinate Y_i so that,

$$\frac{\partial f}{\partial Y_k} = \frac{\partial}{\partial Y_k} \left(\sum_{i=1}^{n} c_i Y_i \right) = c_k.$$

In vector form we get,

$$\nabla_Y f = \begin{bmatrix} \frac{\partial f}{\partial Y_1} \\ \vdots \\ \frac{\partial f}{\partial Y_n} \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = c.$$

For the quadratic form,

$$f(Y) = Y^\top AY = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} Y_i Y_j.$$

We fix $k \in \{1, \ldots, n\}$ so we get,

$$\frac{\partial f}{\partial Y_k} = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \frac{\partial (Y_i, Y_j)}{\partial Y_k} = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \left(\frac{\partial Y_i}{\partial Y_k} Y_j + Y_i \frac{\partial Y_j}{\partial Y_k} \right).$$

Using the Kronecker delta we obtain the formulation,

$$\begin{aligned} \frac{\partial f}{\partial Y_k} &= \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} (\delta_{ik} Y_j + Y_i \delta_{jk}) \\ &= \sum_{j=1}^{n} a_{kj} Y_j + \sum_{i=1}^{n} a_{ik} Y_i \\ &= (AY)_k + (A^\top Y)_k. \end{aligned}$$

Hence for a general matrix A we have that $\nabla_Y (Y^\top AY) = (A + A^\top)Y$.

If A is symmetric then $A^\top = A$ so $\nabla_Y(Y^\top AY) = 2AY$ as required. □

Definition 2.6. Let $\vec{v} \in \mathbb{R}^n$ be an eigenvector corresponding to eigenvalue λ for matrix $A \in \mathbb{M}_{n\times n}(\mathbb{R})$. If $A^2 = A$, then A is idempotent.

Theorem 2.7. *Let $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ satisfy $A^2 = A$. Then every eigenvalue is either 0 or 1.*

Proof. Suppose that $A^2 = A$ and that $\lambda \in \mathbb{C}$ is an eigenvalue of A.

Then $A\vec{v} = \lambda\vec{v}$ for some corresponding eigenvector $\vec{v}$. Then,

$$A^2\vec{v} = A(A\vec{v}) = A(\lambda\vec{v}) = \lambda A\vec{v} = \lambda^2\vec{v}.$$

However, $A^2\vec{v} = A\vec{v} = \lambda\vec{v}$ too. Equating the two equations,

$$\lambda^2\vec{v} = \lambda\vec{v} \iff (\lambda^2 - \lambda)\vec{v} = 0 \iff \lambda(\lambda - 1)\vec{v} = 0.$$

Therefore $\lambda \in \{0, 1\}$ as required. □

Theorem 2.8. *Suppose $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ satisfies the idempotence condition $A^2 = A$.*

Then the eigenspaces,

$$E_0(A) = \ker(A), \qquad E_1(A) = \textit{Image}(A),$$

for eigenvalues 0 and 1, respectively.

Proof. By definition, the eigenspace corresponding to $\lambda = 0$,

$$\begin{aligned} E_0(A) &= \{\vec{x} \in \mathbb{R}^n; A\vec{x} = \vec{0}\vec{x}\} \\ &= \{\vec{x} \in \mathbb{R}^n; A\vec{x} = \vec{0}\} \\ &= \ker(A), \qquad \text{by definition.} \end{aligned}$$

For $E_1(A)$ suppose that $\vec{x} \in E_1(A)$. Then $A\vec{x} = \vec{x}$. $\vec{x}$ under A is $\vec{x}$ itself.

Therefore $\vec{x} \in \text{Image}(A)$. □

Theorem 2.9. *Let $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ be a matrix that satisfies $A^2 = A$. Then,*

$$R^n = \ker(A) + \textit{Image}(A).$$

Every vector $\vec{x} \in \mathbb{R}^n$ can be written uniquely as $\vec{x} = \vec{u} + \vec{w}$ where $\vec{u} \in \textit{Image}(A)$ and $\vec{w} \in \ker(A)$.

Proof. We prove existence then uniqueness.

Let $\vec{x} \in \mathbb{R}^n$ and express $\vec{x}$ as $\vec{x} = A\vec{x} + (I - A)\vec{x}$. Clearly, $A\vec{x} \in \text{Image}(A)$.

We now show that $(I - A)\vec{x} \in \ker(A)$. Indeed we have,

$$A(I - A)\vec{x} = (A - A^2)\vec{x} = 0,$$

since $A = A^2$. Therefore we conclude that $(I - A)\vec{x} \in \ker(A)$.

Thus, every vector has a decomposition $\vec{x} = A\vec{x} + (I - A)\vec{x}$ with,

$$A\vec{x} \in \text{Image}(A), \qquad (I - A)\vec{x} \in \ker(A).$$

To show uniqueness, we show that the intersection of $\ker(A)$ and $\text{Image}(A)$ is trivial.

Suppose that $\vec{z} \in \text{Image}(A) \cap \ker(A)$. Since $z \in \text{Image}(A)$ then there exists $\vec{y} \in \mathbb{R}^n$ such that $\vec{z} = A\vec{y}$. Because $A^2 = A$,

$$A\vec{z} = A(A\vec{y}) = A^2\vec{y} = A\vec{y} = \vec{z}.$$

But $\vec{z} \in \ker(A)$ so $A\vec{z} = 0$ therefore $\vec{z} = 0$. Hence,

$$\text{Image}(A) \cap \ker(A) = \{\vec{0}\}.$$

□

Theorem 2.10. *Every idempotent matrix is diagonalizable.*

More specifically, if $\text{rank}(A) = r$ *for* $A \in \mathbb{M}_{n\times n}(\mathbb{R})$*, then* A *is similar to* $\begin{bmatrix} I_r & 0 \\ 0 & 0_{n-r} \end{bmatrix}$.

Proof. Using the fact that $\mathbb{R}^n = \text{Image}(A) + \ker(A)$ then we can choose a basis $\{\vec{u}_1, \ldots, \vec{u}_r\}$ for $\text{Image}(A)$ and choose a basis $\vec{w}_1, \ldots, \vec{w}_{n-r}$ for $\ker(A)$. Because the sum is direct, the combined set is $B = \{\vec{u}_1, \ldots, \vec{u}_r, \vec{w}_1, \ldots, \vec{w}_{n-r}\}$ is a basis for $\mathbb{R}^n$.

For every $i = 1, \ldots, r$ we have that $u_i \in \text{Image}(A) = E_1(A)$ so $A\vec{u}_i = \vec{u}_i$. For every $j = 1, \ldots, n - r$ $\vec{w}_j \in \ker(A) = E_0(A)$ so $A\vec{w}_j = 0$. Therefore the matrix with respect to the basis B is,

$$[A]_B = \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-r} \end{bmatrix}.$$

Consequently, there exists an invertible matrix S such that,

$$A = S \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-r} \end{bmatrix} S^{-1}.$$

Since we know that $A^2 = A$ we have that $A^2 - A = 0$ or equivalently $A(A - I) = 0$. Thus the polynomial $p(t) = t(t - 1)$ annihilates A. Therefore the minimal polynomial $m_A(t)$ divides $t(t - 1)$. The polynomial $t(t - 1)$ has two distinct roots 0 and t hence m_A splits into distinct linear factors. Therefore, we conclude A is diagonalizable precisely when its minimal polynomial splits into distinct linear factors. Therefore A is diagonalizable. □

By the proof above, the following statements are equivalent,

(i) $A^2 = A$ (Idempotence)
(ii) A is diagonalizable and $\sigma(A) \subseteq \{0, 1\}$.

For the reverse direction, if we know that $A = SDS^{-1}$ and every diagonal entry of D is either 0 or 1, then $D^2 = D$. Therefore, we can conclude that,

$$A^2 = SD^2S^{-1} = SDS^{-1} = A.$$

Suppose we have that $\operatorname{rank}(A) = r$. Since A is similar to $\operatorname{diag}(I_r, 0_{n-r})$ then its characteristic polynomial is $C_A(\lambda) = \lambda^{n-r}(\lambda-1)^r$ up to the alternative sign convention $\det(A-\lambda I)$. The eigenvalue 1 has algebraic and geometric multiplicity r while the eigenvalue 0 has algebraic and multiplicity $n-r$. Therefore, the trace $\operatorname{tr}(A) = r = \operatorname{rank}(A)$. Indeed, the trace is the sum of the eigenvalues counted with algebraic multiplicity,

$$\operatorname{tr}(A) = \underbrace{1 + \cdots + 1}_{r \text{ times}} + \underbrace{0 + \cdots + 0}_{n-r \text{ times}} = r.$$

Also, we can say,

$$\det(A) = \prod_{i=1}^{n} \lambda_i = \begin{cases} 1, & r = n, \\ 0, & r < n. \end{cases}$$

In the case that $r = n$, A is invertible. Multiplying $A^2 = A$ by A^{-1} shows that $A = I$. Hence, $\det(A) = 1 \iff A = I$. Otherwise, $\det(A) = 0$, obviously.

Interestingly, suppose we have matrices $P_1 = A$ and $P_0 = I - A$ which are simply the projection operators associated with the eigenvalues 1 and 0, respectively. They satisfy $P_1^2 = P_!$ and $P_0^2 = P_0$ and $P_1P_0 = P_0P_1 = 0$ and $P_1 + P_0 = I$. By Theorem 1.141,

$$\operatorname{Image}(P_1) = E_1(A), \quad \operatorname{Image}(P_0) = E_0(A).$$

Therefore, we can decompose the identity matrix into,

$$I = P_1 + P_0,$$

while A has the spectral decomposition,

$$A = 1 \cdot P_1 + 0 \cdot P_0.$$

Since we have defined $P_1 = A$ and $P_0 = I - A$ then $A = 1 \cdot A + 0 \cdot (I - A)$. Although this expression appears trivial, it is the finite-dimensional spectral projection formula for an operator whose spectrum is $\{0, 1\}$. For any polynomial p, $p(A) = p(1)A + p(0)(I - A)$.

Proof. Since A acts as 1 on $\operatorname{Image}(A)$ and 0 on $\ker(A)$, then $p(A)$ acts as $p(1)$ on $\operatorname{Image}(A)$ and $p(0)$ on $\ker(A)$. Algebraically, because $A^k = A$ for every $k \geq 1$, if

$$p(t) = a_0 + a_1t + \cdots + a_mt^m,$$

then the polynomial on A becomes,

$$p(A) = a_0I + (a_1 + \cdots + a_m)A.$$

Therefore it follows that $p(A) = p(0)I + (p(1) - p(0))A$ and hence $p(A) = p(1)A + p(0)(I - A)$. □

Theorem 2.11. *Suppose that $A^2 = A$ and $A^\top = A$. Then* $\ker(A) = \mathit{Image}(A)^\perp$.

Consequently, $\mathbb{R}^n = \mathit{Image}(A)^\perp + \ker(A)$.

Proof. Let $\vec{u} \in \text{Image}(A)$ and $\vec{w} \in \ker(A)$. Since $\vec{u} \in \text{image}(A)$, there exists $\vec{x} \in \mathbb{R}^n$ such that $\vec{u} = A\vec{x}$. Then $\langle \vec{u}, \vec{w} \rangle = \langle A\vec{x}, \vec{w} \rangle$. Using the adjoint identity we have that $\langle A\vec{x}, \vec{w} \rangle = \langle \vec{x}, A^\top \vec{w} \rangle$.

Since $A^\top = A$, then $\langle \vec{x}, A^\top \vec{w} \rangle = \langle \vec{x}, A\vec{w} \rangle$.

Therefore, every vector in $\ker(A)$ is orthogonal to every vector in $\text{Image}(A)$ which gives $\ker(A) \subseteq \text{Image}(A)^\perp$. By Rank-Nullity theorem, $\dim \ker(A) = n - \text{rank}(A)$. Also,

$$\dim \text{Image}(A)^\perp = n - \dim \text{Image}(A) = n - \text{rank}(A).$$

The two subspaces have the same dimension and one is contained in the other.

Therefore, $\ker(A) = \text{Image}(A)^\perp$. □

Theorem 2.12. *If A is symmetric i.e., $A^\top = A$ and idempotent i.e., $A^2 = A$, then $A\vec{x}$ is the unique vector in Image(A) closest to $\vec{x}$.*

Proof. Let $\vec{u} \in U$. Then,

$$\vec{x} - \vec{u} = (\vec{x} - A\vec{x}) + (A\vec{x} - \vec{u}).$$

We have that $\vec{x} - A\vec{x} \in U^\perp$ and $A\vec{x} - \vec{u} \in U$. Therefore, we conclude that $\vec{x} - A\vec{x} \perp A\vec{x} - \vec{u}$.

By the Pythagorean Theorem,

$$\|\vec{x} - \vec{u}\|^2 = \|\vec{x} - A\vec{x}\|^2 + \|A\vec{x} - \vec{u}\|^2 \geq \|\vec{x} - A\vec{x}\|^2.$$

Therefore $\|\vec{x} - \vec{u}\| \geq \|\vec{x} - A\vec{x}\|$.

Equality holds precisely when $\|A\vec{x} - \vec{u}\|^2 = 0$ which is equivalent to when $\vec{u} = A\vec{x}$. □

Theorem 2.13 (Spectral Theorem for Symmetric Idempotent Matrices)**.** *Let $A \in \mathbb{M}_{n\times n}(\mathbb{R})$ satisfy $A^2 = A$ and $A^\top = A$. Let $r = \text{rank}(A)$. Then, there exists an orthogonal matrix Q such that,*

$$A = Q \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-r} \end{bmatrix} Q^\top.$$

Proof. Since A is real and symmetric, the spectral theorem guarantees that A has an orthonormal basis of eigenvectors. Therefore, there exists an orthogonal matrix Q and real diagonal matrix $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$ such that,

$$A = QDQ^\top.$$

Because $A^2 = A$, every eigenvalue A is either 0 or 1. Hence every diagonal entry of D is either 0 or 1. After re-ordering the eigenvectors, we may arrange the r eigenvectors corresponding to 1 first and the $n - r$ eigenvectors corresponding to 0 afterward. Therefore,

$$D = \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-1} \end{bmatrix}.$$

Consequently we obtain,

$$A = Q \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-1} \end{bmatrix} Q^\top.$$

Without invoking the full spectral theorem, we can also choose an orthonormal basis $\vec{u}_1, \ldots, \vec{u}_r$ of Image(A) and an orthonormal basis $\vec{v}_1, \ldots, \vec{v}_{n-r}$ of $\ker(A)$. Because we know that $\ker(A) = \text{Image}(A)^\perp$ the combined set $\{\vec{u}_1, \ldots, \vec{u}_r, \vec{v}_1, \ldots, \vec{v}_{n-r}\}$ is an orthonormal basis of $\mathbb{R}^n$. Define,

$$Q = \begin{bmatrix} \vec{u}_1 & \cdots & \vec{u}_r & \vec{v}_1 & \cdots & \vec{v}_{n-r} \end{bmatrix}.$$

Clearly, Q is orthogonal since $Q^\top Q = QQ^\top = I$. Because $A\vec{u}_i = \vec{u}_i$ and $A\vec{v}_j = 0$ we have that,

$$AQ = Q \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \implies A = Q \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^\top.$$

Therefore we have obtained the spectral decomposition both by invoking the Spectral Theorem for real symmetric matrices and by showing there exists an orthonormal basis from which we can obtain an orthogonal matrix. □

Theorem 2.14. *Let $\vec{u}_1, \ldots, \vec{u}_r$ be an orthonormal basis of Image(A). Then,*

$$A = \sum_{i=1}^{r} \vec{u}_i \vec{u}_i^\top.$$

Proof. Suppose that $U = \begin{bmatrix} \vec{u}_1 & \cdots & \vec{u}_r \end{bmatrix}$. Since the columns of U are orthonormal then $U^\top U = I_r$. From the orthogonal diagonalization,

$$A = Q \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^\top.$$

We see that only the first r columns contribute so $A = UU^\top$. Expanding the matrix product,

$$UU^\top = \begin{bmatrix} \vec{u}_1 & \cdots & \vec{u}_r \end{bmatrix} \begin{bmatrix} \vec{u}_1^\top \\ \vdots \\ \vec{u}_r^\top \end{bmatrix} = \sum_{i=1}^{r} \vec{u}_i \vec{u}_i^\top,$$

showing the rank-1 product as required. □

Notice that for any $\vec{x} \in \mathbb{R}^n$,

$$A\vec{x} = \sum_{i=1}^{r} \vec{u}_i \vec{u}_i^\top \vec{x} = \sum_{i=1}^{r} \langle \vec{x}, \vec{u}_i \rangle \vec{u}_i.$$

This is precisely the orthogonal projection formula.

Theorem 2.15. *Every symmetric idempotent matrix is positive semi-definite.*

Proof. Since $A = A^\top$ and $A^2 = A$ then $A = A^\top A$. Therefore for every $\vec{x} \in \mathbb{R}^n$,

$$\vec{x}^\top A\vec{x} = \vec{x}^\top A^\top A\vec{x} = (A\vec{x})^\top (A\vec{x}) = \|A\vec{x}\|^2.$$

Therefore it follows that $\vec{x}^\top A\vec{x} \geq 0$ and hence $A \geq 0$. Similarly, since $I - A$ is symmetric and idempotent, we know that $I - A \geq 0$. Therefore $0 \leq A \leq I$. Notice that also $\vec{x}^\top A\vec{x} = \|A\vec{x}\|^2$. □

2.2 Bivariate and Multivariate Normal Distribution

Suppose that we have two random variables $X_1, X_2 \sim \mathcal{N}(\mu, \Sigma)$ such that they share a mean vector and covariance matrix written as,

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

That is $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$. X is therefore a *Bivariate Normal Distribution.*

Theorem 2.16. *Let $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$.*

Their (joint) vector written as $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is said to be distributed as a Bivariate Normal,

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}; \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right),$$

where ρ is the covariance term between X_1, X_2 assuming they are not independent.

Their joint probability density function is written as,

$$f(X_1, X_2) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(X - \mu)^\top \Sigma^{-1}(X - \mu)\right\},$$

where Σ is a positive definite variance-covariance matrix between X_1, X_2. In the case that $X_1 \perp X_2$, the off-diagonal terms of the variance-covariance matrix are zero, so $\Sigma = \operatorname{diag}(\sigma_1^2, \sigma_2^2)$.

Example 2.17. We start from the simplest bivariate normal distribution: the standard bivariate normal distribution. Consider two *independent* standard normal random variables $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ and define the random vector of the two independent standard normal random variables $Z = \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}$.

$$\begin{aligned} \operatorname{Cov}(Z) &= \mathbb{E}\left[(Z - \mathbb{E}[Z])\,(Z - \mathbb{E}[Z])^\top\right] \\ &= \mathbb{E}[ZZ^\top] = \mathbb{E}\left[\begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}\begin{bmatrix} Z_1 & Z_2 \end{bmatrix}\right] \\ &= \mathbb{E}\left[\begin{bmatrix} Z_1^2 & Z_1Z_2 \\ Z_1Z_2 & Z_2^2 \end{bmatrix}\right] = \begin{bmatrix} \mathbb{E}[Z_1^2] & \mathbb{E}[Z_1Z_2] \\ \mathbb{E}[Z_1Z_2] & \mathbb{E}[Z_2^2] \end{bmatrix} = I_2. \end{aligned}$$

Clearly, since they are independent, their variance-covariance matrix is simply I_2,

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Therefore, simplifying their joint density,

$$\begin{aligned} f(Z_1, Z_2) &= \frac{1}{2\pi} \exp\left\{ -\frac{1}{2}(Z-\mu)^\top I_2 (Z-\mu) \right\} \\ &= \frac{1}{2\pi} \exp\left\{ -\frac{1}{2} Z^\top Z \right\} \\ &= \frac{1}{2\pi} \exp\left\{ -\frac{1}{2} \begin{bmatrix} Z_1 & Z_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \right\} \\ &= \frac{1}{2\pi} \exp\left\{ -\frac{1}{2} \left(Z_2^2 + Z_2^2 \right) \right\}. \end{aligned}$$

Note that in more general bivariate normal distributions, simplfying the joint probability density function often does *not* yield the convenient identity matrix simplification. The convenience of centering a matrix so that the two random variables become standard normal random variables is demonstrated in the example above.

In the case that $X_1 \not\perp X_2$ i.e., when they are not guaranteed to be independent, the off-diagonal terms are not zero since they have non-zero covariance i.e., $\mathrm{Cov}(X_1, X_2) = \rho\sigma_1\sigma_2 \neq 0$ where $\rho = \mathrm{Corr}(X_1, X_2) \in [-1, 1]$.

Theorem 2.18. *Let $X \sim \mathcal{N}(\mu, \Sigma)$ where μ is the mean vector of $X = \begin{bmatrix} X_1 & X_2 \end{bmatrix}^\top$ and Σ is the corresponding positive semi-definite variance-covariance matrix. Then, the joint pdf is,*

$$f_{X_1,X_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} \right] \right\}.$$

This is the probability density of a joint bivariate random normal distribution where X_1, X_2 are not necessarily independent and Σ is a positive semi-definite matrix.

We show that the covariance matrix is necessarily a positive semi-definite matrix.

Theorem 2.19. *Let $X \in \mathbb{R}^n$ be a random vector with mean vector $\mu = \mathbb{E}[X] \in \mathbb{R}^n$ and covariance matrix,*

$$\Sigma = \mathrm{Cov}(X) = \mathbb{E}[(X-\mu)(X-\mu)^\top] \in \mathbb{M}_{n\times n}(\mathbb{R}).$$

For any deterministic scalar coefficient vector $\vec{a} \in \mathbb{R}^n$, the scalar random variable $\vec{a}^\top X$ satisfies $\mathbb{E}[\vec{a}^\top X] = \vec{a}^\top \mu$ and $\mathrm{Var}(\vec{a}^\top X) = \vec{a}^\top \Sigma \vec{a}$. Consequently $\vec{a}^\top \Sigma \vec{a} \geq 0$ for every $\vec{a}^\top \in \mathbb{R}^n$ and therefore every covariance matrix is positive semi-definite.

Proof. Note that the first identity regarding the expectation of the linear combination follows directly by linearity of expectation of a random vector. Therefore, we can conclude that $\mathbb{E}[\vec{a}^\top X] = \vec{a}^\top \mu$ where $\mu = \mathbb{E}[X]$. By definition, the variance is computed as,

$$\begin{aligned}
\operatorname{Var}(\vec{a}^\top X) &= \mathbb{E}\left[\left(\vec{a}^\top X - \mathbb{E}[\vec{a}^\top X]\right)^2\right] \\
&= \mathbb{E}\left[\left(\vec{a}^\top X - \vec{a}^\top \mu\right)^2\right] \\
&= \mathbb{E}\left[\left(\vec{a}^\top (X - \mu)\right)^2\right].
\end{aligned}$$

$\vec{a}^\top (X - \mu)$ is a scalar in $\mathbb{R}$ so it is equal to its transpose,

$$\vec{a}^\top (X - \mu) = \left(\vec{a}^\top (X - \mu)\right)^\top = (X - \mu)^\top \vec{a}.$$

Substituting the equivalence above we get,

$$\begin{aligned}
\operatorname{Var}(\vec{a}^\top X) &= \mathbb{E}\left[(\vec{a}^\top (X - \mu))\left(\vec{a}^\top (X - \mu)\right)\right] \\
&= \mathbb{E}\left[(\vec{a}^\top (X - \mu))\left(\vec{a}^\top (X - \mu)\right)^\top\right] \\
&= \mathbb{E}\left[\vec{a}^\top (X - \mu)(X - \mu)^\top \vec{a}\right].
\end{aligned}$$

$\vec{a}$ is deterministic therefore it moves outside the expectation,

$$\begin{aligned}
\operatorname{Var}(\vec{a}^\top X) &= \vec{a}^\top \mathbb{E}\left[(X - \mu)(X - \mu)^\top\right] \vec{a} \\
&= \vec{a}^\top \operatorname{Cov}(X) \vec{a} \\
&= \vec{a}^\top \Sigma \vec{a}.
\end{aligned}$$

The rank-1 expression of the matrix formulation above is simply,

$$\begin{aligned}
\vec{a}^\top \Sigma \vec{a} &= \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} \begin{bmatrix} \operatorname{Cov}(X_1, X_2) & \cdots & \operatorname{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \operatorname{Cov}(X_n, X_1) & \cdots & \operatorname{Cov}(X_n, X_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} a_i \operatorname{Cov}(X_i, X_j) a_j \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \operatorname{Cov}(X_i, X_j).
\end{aligned}$$

By right and left-linearity of covariances,

$$\begin{aligned}\operatorname{Var}\left(\sum_{i=1}^{n} a_i X_i\right) &= \operatorname{Cov}\left(\sum_{i=1}^{n} a_i X_i, \sum_{j=1}^{n} a_j X_j\right) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \operatorname{Cov}(X_i, X_j) \\ &= \vec{a}^{\top}\Sigma\vec{a} \geq 0,\end{aligned}$$

since variances are always non-negative. Therefore it follows that $\vec{a}^{\top}\Sigma\vec{a} = \operatorname{Var}(\vec{a}^{\top}X) \geq 0$ for every $\vec{a} \in \mathbb{R}^n$. By definition this requires that Σ is positive semi-definite. Moreover, $\vec{a}^{\top}\Sigma\vec{a} = 0$ if and only if $\operatorname{Var}(\vec{a}^{\top}X) = 0$ which ocurrs if and only if $\vec{a}^{\top}X$ is constant almost surely. Since $\mathbb{E}[\vec{a}^{\top}X] = \vec{a}^{\top}\mu$, this constant must be $\vec{a}^{\top}\mu$. Hence the following two statements are equivalent,

$$\vec{a}^{\top}\Sigma\vec{a} = 0 \iff \vec{a}^{\top}(X - \mu) = 0 \quad a.s.$$

Thus, Σ is positive definite precisely when no non-zero linearly combination of centered components of X is almost surely zero. We conclude that Σ is positive semi-definite. □

Theorem 2.20 (Spectral Theorem for Covariance Matrices)**.** *Let $X \in \mathbb{R}^n$ be a random vector with finite second moments and mean vector $\mu = \mathbb{E}[X]$ with corresponding covariance matrix,*

$$\Sigma = \operatorname{Cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^{\top}].$$

Then Σ is a real symmetric positive semi-definite matrix. Consequently, there exists an orthonormal basis of eigenvectors $\vec{q}_1, \ldots, \vec{q}_n \in \mathbb{R}^n$ and corresponding non-negative eigenvalues $\lambda_1, \ldots, \lambda_n \geq 0$ such that $\Sigma\vec{q}_i = \lambda_i\vec{q}_i$ for $i = 1, \ldots, n$. If,

$$Q = \begin{bmatrix}\vec{q}_1 & \cdots & q_n\end{bmatrix}, \quad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

then Q is an orthogonal matrix and,

$$\Sigma = Q\Lambda Q^{\top} = \sum_{i=1}^{n} \lambda_i \vec{q}_i \vec{q}_i^{\top}.$$

Moreover the rotated and centered random vector $Y = Q^{\top}(X - \mu)$ has mean vector zero and diagonal covariance matrix $\operatorname{Cov}(Y) = \Lambda$ thus if $Y_i = \vec{q}_i^{\top}(X - \mu)$ then $\operatorname{Var}(Y_i) = \lambda_i$ and $\operatorname{Cov}(Y_i, Y_j) = 0$ whenever $i \neq j$. Therefore the eigenvectors of Σ determine mutually orthogonal directions along which the centered random vector varies while the corresponding eigenvalues determine the amount of variance in each direction. Furthermore $\lambda_i = 0$ if and only if $\vec{q}_i^{\top}(X - \mu) = 0$ almost surely. Hence if $\operatorname{rank}(\Sigma) = r$ then X lies almost surely in the subspace $\mu + \operatorname{span}\{\vec{q}_i : \lambda > 0\}$.

If Σ is positive definite so that every $\lambda_i > 0$, then $Z = \lambda^{1/2}Q^{\top}(X - \mu)$ satisfies $\operatorname{Cov}(Z) = I_n$.

Proof. It follows that Σ is symmetric. By definition,

$$\Sigma = \mathbb{E}\left[(X-\mu)(X-\mu)^\top\right].$$

Then by definition it follows that,

$$\begin{aligned}
\Sigma^\top &= \left(\mathbb{E}[(X-\mu)(X-\mu)^\top]^\top\right) \\
&= \mathbb{E}\left[\left((X-\mu)(X-\mu)^\top\right)^\top\right] \\
&= \mathbb{E}\left[\left((X-\mu)^\top\right)^\top (X-\mu)^\top\right] \\
&= \mathbb{E}\left[(X-\mu)(X-\mu)^\top\right] \\
&= \Sigma,
\end{aligned}$$

as required. So Σ is real and symmetric. We have already shown that for every $\vec{\in}\mathbb{R}^n$ that $\vec{a}^\top\Sigma\vec{a} = \mathrm{Var}(\vec{a}^\top X) \geq 0$. Therefore it follow that Σ is positive semi-definite. We now show that every eigenvalue of the covariance matrix Σ is non-negative. Suppose that $\vec{q_i} \neq \vec{0}$ is an eigenvector of Σ with corresponding eigenvalue λ_i. Then, $\Sigma\vec{q_i} = \lambda_i\vec{q_i}$. Premultiplying both sides by $\vec{q_i}^\top$ gives,

$$\vec{q_i}^\top\Sigma\vec{q_i} = \vec{q_i}^\top(\lambda_i\vec{q_i}) = \lambda_i\vec{q_i}^\top\vec{q_i} = \lambda_i\|\vec{q_i}\|^2.$$

Since Σ is positive semi-definite it follows that $\vec{q_i}^\top\Sigma\vec{q_i} \geq 0$. Therefore $\lambda_i\|\vec{q_i}\|^2 \geq 0$. Since $\vec{q_i} \neq \vec{0}$ then it follows that the norm $\|\vec{q_i}\|^2 > 0$ as well. Dividing through $\|\vec{q_i}\|^2$ so we obtain $\lambda_i \geq 0$.

Equivalently, using the variance identity,

$$\lambda_i\|\vec{q_i}\|^2 = \vec{q_i}^\top\Sigma\vec{q_i} = \mathrm{Var}(\vec{q_i}^\top X) \geq 0.$$

Thus, an eigenvalue of the covariance matrix is precisely the variance of X when projected onto the corresponding unit eigenvector. Since Σ is a real symmetric matrix, the spectral theorem tells us that it possesses an orthonormal basis of eigenvectors namely $\vec{q}_1, \ldots, \vec{q}_n$ therefore we know that,

$$\vec{q_i}^\top\vec{q_j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Because the columns of Q are orthonormal, $Q^\top Q = I_n$. Since Q is a suqare, this also gives $QQ^\top = I_n$ and hence $Q^{-1} = Q^\top$. Thus Q is an orthogonal matrix.

The eigenvalue equations can be written therefore as,

$$\Sigma\begin{bmatrix}\vec{q}_1 & \cdots & \vec{q}_n\end{bmatrix} = \begin{bmatrix}\Sigma\vec{q}_1 & \cdots & \Sigma\vec{q}_n\end{bmatrix} = \Sigma Q = \begin{bmatrix}\lambda_1\vec{q}_1 & \cdots & \lambda_n\vec{q}_n\end{bmatrix}.$$

Notice that using $\Sigma\vec{q_i} = \lambda_i\vec{q_i}$ this becomes,

$$Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n) = Q\Lambda.$$

Therefore $\Sigma Q = Q\Lambda$ and right-multiplying gives $\Sigma Q Q^\top = Q\Lambda Q^\top$. Since $QQ^\top = I_n$ then $\Sigma = Q\Lambda Q^\top$. Hence each matrix $\vec{q}_i\vec{q}_i^\top$ is the orthogonal projection matrix onto the one-dimensional subspace, since we can write Σ as the rank-1 projection,

$$\Sigma = \sum_{i=1}^{n} \lambda_i \vec{q}_i \vec{q}_i^\top.$$

Thus the covariance matrix is a weighted sum of orthogonal projection matrices with weights equal to the variance λ_i. Consider the centered rotated vector $Y = Q^\top(X - \mu)$ with expectation,

$$\begin{aligned}
\mathbb{E}[Y] &= \mathbb{E}\left[Q^\top(X-\mu)\right] \\
&= Q^\top \mathbb{E}[X-\mu] \\
&= Q^\top\left(\mathbb{E}[X]-\mu\right) \\
&= Q^\top(\mu-\mu) = \vec{0}.
\end{aligned}$$

Therefore given that $\mathbb{E}[Y] = \vec{0}$, its covariance matrix is,

$$\begin{aligned}
\mathrm{Cov}(Y) &= \mathbb{E}\left[(Y-\mathbb{E}[Y])(Y-\mathbb{E}[Y])^\top\right] \\
&= \mathbb{E}[YY^\top] \\
&= \mathbb{E}\left[Q^\top(X-\mu)\left(Q^\top(X-\mu)\right)^\top\right] \\
&= \mathbb{E}\left[Q^\top(X-\mu)(X-\mu)^\top Q\right] \\
&= Q^\top \mathbb{E}\left[(X-\mu)(X-\mu)^\top\right] Q \\
&= Q^\top \Sigma Q.
\end{aligned}$$

Substituting our diagonalization of $\Sigma = Q\Lambda Q^\top$ we get,

$$\mathrm{Cov}(Y) = Q^\top(Q\Lambda Q^\top)^\top Q = (Q^\top Q)\Lambda(Q^\top Q) = I_n \Lambda I_n = \Lambda.$$

Therefore $\mathrm{Cov}(Y) = \Lambda$. Since we know that,

$$Y = Q^\top(X-\mu) = \begin{bmatrix} \vec{q}_1^\top \\ \vdots \\ \vec{q}_n^\top \end{bmatrix} (X-\mu).$$

Therefore the components of Y are $Y_i = \vec{q}_i^\top(X-\mu)$. The diagonal entries of $\mathrm{Cov}(Y) = \Lambda$ give $\mathrm{Var}(Y_i) = \mathrm{Var}(\vec{q}_i^\top(X-\mu)) = \lambda_i$ whereas the diagonal entries give $\mathrm{Cov}(Y_i, Y_j) = 0$ for $i \neq j$.

We verify this as,

$$\begin{aligned}
\mathrm{Cov}(Y_i, Y_j) &= \mathrm{Cov}\left(\vec{q}_i^\top(X-\mu), \vec{q}_j^\top(X-\mu)\right) \\
&= \vec{q}_i^\top \Sigma \vec{q}_j = \vec{q}_i^\top(\lambda_j \vec{q}_j) = \lambda_j \vec{q}_i^\top \vec{q}_j = 0,
\end{aligned}$$

since we know that $\vec{q}_i^{\top}\vec{q}_j = 0$ whenever $i \neq j$. Thus, orthogonal diagonalization does not merely simplify the matrix algebra but constructs new random variables $Y_1, \ldots, Y_m$ that are pairwise uncorrelated. Consider the eigenvalue $\lambda_i = 0$. Since,

$$\lambda_i = \mathrm{Var}(\vec{q}_i^{\top}(X - \mu)),$$

we have that $\lambda_i = 0$ if and only if $\mathrm{Var}(\vec{q}_i^{\top}(X - \mu)) = 0$. A random variable has variance zero if and only if it is constant almost surely. But $\mathbb{E}[\vec{q}_i^{\top}(X - \mu)] = \vec{q}_i^{\top}\mathbb{E}[X - \mu] = 0$. Therefore, the constant must be zero. So, $\lambda_i = 0$ if and only if $\vec{q}_i^{\top}(X - \mu) = 0$ almost surely.

Suppose that $\mathrm{rank}(\Sigma) = r$. Then exactly r eigenvalues are positive. Suppose that $\lambda_1, \ldots, \lambda_r > 0$ and $\lambda_{r+1} = \cdots = \lambda_n = 0$. Since $\vec{q}_i^{\top}(X - \mu) = 0$ almost surely for $i = r+1, \ldots, n$, the centered random vector has no component in those directions. Because we have that,

$$I_n = QQ^{\top} = \sum_{i=1}^{n} \vec{q}_i\vec{q}_i^{\top}.$$

We can write that,

$$\begin{aligned} X - \mu &= I_n(X - \mu) \\ &= \sum_{i=1}^{n} \vec{q}_i\vec{q}_i^{\top}(X - \mu) \\ &= \sum_{i=1}^{r} \vec{q}_i\vec{q}_i^{\top}(X - \mu) + \sum_{i=r+1}^{n} \vec{q}_i\vec{q}_i^{\top}(X - \mu) \\ &= \sum_{i=1}^{r} \left(\vec{q}_i^{\top}(X - \mu)\right)\vec{q}_i. \end{aligned}$$

Therefore $X - \mu \in \mathrm{span}\{\vec{q}_1, \ldots, \vec{q}_r\}$ almost surely. Equivalently, we see that $X \in \mu + \mathrm{span}\{\vec{q}_1, \ldots, \vec{q}_r\}$ almost surely. Thus, a singular covariance matrix means that the random vector does not genuinely vary in all n dimensions. Finally, suppose that Σ is positive definite, then every eigenvalue is strictly positive $\lambda_i > 0$ for $i = 1, \ldots, n$. Therefore,

$$\Lambda^{-1/2} = \mathrm{diag}\left(\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_n}}\right).$$

The above is well define so we define $Z = \Lambda^{-1/2}Q^{\top}(X - \mu)$. Its expectation is $\mathbb{E}[Z] = \Lambda^{-1/2}Q^{\top}\mathbb{E}[X - \mu] = \vec{0}$. Its covariance matrix is therefore computed as,

$$\begin{aligned} \mathrm{Cov}(Z) &= \Lambda^{-1/2}Q^{\top}\Sigma Q\Lambda^{-1/2} \\ &= \Lambda^{-1/2}\left(Q^{\top}\Sigma Q\right)\Lambda^{-1/2} \\ &= \Lambda^{-1/2}\Lambda\Lambda^{-1/2} \\ &= \mathrm{diag}\left(\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_n}}\right)\mathrm{diag}(\lambda_1, \ldots, \lambda_n)\,\mathrm{diag}\left(\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_n}}\right) \\ &= \mathrm{diag}(1, \ldots, 1) = I_n. \end{aligned}$$

Hence, $\mathrm{Cov}(Z) = I_n$, as required. □

Using the properties of the covariance matrix, we derive the closed form of the general density,

$$f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(X-\mu)^\top \Sigma^{-1}(X-\mu)\right\}.$$

Computing the determinant of the covariance matrix,

$$|\Sigma| = \begin{vmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{vmatrix} = \sigma_1^2\sigma_2^2 - \rho^2\sigma_1^2\sigma_2^2 = \sigma_1^2\sigma_2^2(1-\rho^2), \qquad |\Sigma|^{1/2} = \sigma_1\sigma_2\sqrt{1-\rho^2}.$$

Using the formula of the inverse of a 2×2 matrix,

$$\Sigma^{-1} = \frac{1}{ac-bd}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)}\begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}.$$

Expanding the quadratic form in the exponent we get,

$$\begin{aligned}
(X-\mu)^\top\Sigma^{-1}(X-\mu) &= \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)}\begin{bmatrix} X_1-\mu_1 \\ X_2-\mu_2 \end{bmatrix}^\top \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}\begin{bmatrix} X_1-\mu_1 \\ X_2-\mu_2 \end{bmatrix} \\
&= \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)}\begin{bmatrix} X_1-\mu_1 & X_2-\mu_2 \end{bmatrix}\begin{bmatrix} \sigma_2^2(X_1-\mu_1) - \rho\sigma_1\sigma_2(X_2-\mu_2) \\ -\rho\sigma_1\sigma_2(X_1-\mu_1) + \sigma_1^2(X_2-\mu_2) \end{bmatrix} \\
&= \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)}\left[\sigma_2^2(X_1-\mu_1)^2 - 2\rho\sigma_1\sigma_2(X_1-\mu_1)(X_2-\mu_2) + \sigma_1^2(X_2-\mu_2)^2\right] \\
&= \frac{\sigma_2^2(X_1-\mu_1)^2 - 2\rho\sigma_1\sigma_2(X_1-\mu_1)(X_2-\mu_2) + \sigma_1^2(X_2-\mu_2)^2}{\sigma_1^2\sigma_2^2(1-\rho^2)} \\
&= \frac{1}{(1-\rho^2)}\left[\frac{(X_1-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(X_1-\mu_1)(X_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(X_2-\mu_2)^2}{\sigma_2^2}\right]
\end{aligned}$$

Substituting the expanded quadratic term and the root-determinant back into the original equation,

$$f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(X_1-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(X_1-\mu_1)(X_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(X_2-\mu_2)^2}{\sigma_2^2}\right]\right\}.$$

We can recover the case where $\text{Corr}(X_1, X_2) = \rho = 0$, which yields,

$$\begin{aligned}
f_{X_1,X_2}(x_1,x_2) &= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left\{-\frac{1}{2}\left[\frac{(X_1-\mu_1)^2}{\sigma_1^2} + \frac{(X_2-\mu_2)^2}{\sigma_2^2}\right]\right\} \\
&= \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left\{-\frac{(X_1-\mu_1)^2}{2\sigma_1^2}\right\} \cdot \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left\{-\frac{(X_2-\mu_2)^2}{2\sigma_2^2}\right\} \\
&= f_{X_1}(x_1) \cdot f_{X_2}(x_2).
\end{aligned}$$

Consider the case where instead of having just two normal random variables $(X_1, X_2)^\top$ in a vector, we instead extend to the k-dimensional case where the vector X is now $(X_1, \ldots, X_k)^\top \sim \mathcal{N}(\mu, \Sigma)$.

Starting with the covariance matrix, we assume the case where the individual X_i's for $i = 1, \ldots, k$ are mutually uncorrelated i.e., for $i \neq j$, $X_i \perp X_j$.

Then, the determinant and inverse of the covariance matrix respectively become,

$$|\Sigma|^{1/2} = \det\left(\begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_k^2 \end{bmatrix}\right)^{1/2} = (\sigma_1^2 \cdot \sigma_2^2 \cdots \sigma_k^2)^{1/2} = \prod_{i=1}^{k} \sigma_i^2, \qquad \Sigma^{-1} = \begin{bmatrix} \frac{1}{\sigma_1^2} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_k^2} \end{bmatrix}$$

Substituting our matrix definitions we get,

$$\begin{aligned} f(x_1, \ldots, x_k) &= \frac{1}{(2\pi)^{k/2} \cdot \det(\Sigma)^{1/2}} \exp\left\{ -\frac{1}{2} \begin{bmatrix} X_1 - \mu_1 \\ \vdots \\ X_k - \mu_k \end{bmatrix}^\top \begin{bmatrix} \frac{1}{\sigma_1^2} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_k^2} \end{bmatrix} \begin{bmatrix} X_1 - \mu_1 \\ \vdots \\ X_k - \mu_k \end{bmatrix} \right\} \\ &= \frac{1}{(2\pi)^{k/2} \left(\prod_{i=1}^{k} \sigma_i^2 \right)^{1/2}} \exp\left\{ -\frac{1}{2} \begin{bmatrix} X_1 - \mu_1 & \cdots & X_k - \mu_k \end{bmatrix} \begin{bmatrix} \frac{X_1 - \mu_1}{\sigma_1^2} \\ \vdots \\ \frac{X_k - \mu_k}{\sigma_k^2} \end{bmatrix} \right\} \\ &= \frac{1}{\left(\sqrt{2\pi}\right)^k \prod_{i=1}^{k} \sigma_i} \exp\left\{ -\frac{1}{2} \left[(X_1 - \mu_1) \cdot \left(\frac{X_1 - \mu_1}{\sigma_1^2} \right) + \cdots + (X_k - \mu_k) \cdot \left(\frac{X_k - \mu_k}{\sigma_k^2} \right) \right] \right\}. \end{aligned}$$

Simplifying the product in the denominator and using exponent laws,

$$\begin{aligned} = \frac{1}{\prod_{i=1}^{k} \sqrt{2\pi}\sigma_i} \exp\left\{ -\frac{1}{2} \sum_{i=1}^{k} \frac{(X_i - \mu_i)^2}{\sigma_i^2} \right\} &= \prod_{i=1}^{k} \left(\frac{1}{\sqrt{2\pi}\sigma_i} \right) \left(\prod_{i=1}^{k} \exp\left\{ -\frac{(X_i - \mu_i)^2}{2\sigma_i^2} \right\} \right) \\ &= \prod_{i=1}^{k} \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left\{ -\frac{(X_i - \mu_i)^2}{2\sigma_i^2} \right\} \\ &= \prod_{i=1}^{k} f_{X_i}(x_i). \end{aligned}$$

However, it is unlikely that X_i's are perfectly mutually uncorrelated. In the case that $\rho \neq 0$, we cannot simply the covariance matrix into a convenient diagonal matrix with $\sigma_1^2, \ldots, \sigma_k^2$ as its diagonals for the k-dimensional case. We will denote the (i, j)-th entry of the covariance matrix as Σ_{ij} and, similarly, denote the (i, j)-th entry of the inverse covariance matrix as $(\Sigma^{-1})_{ij}$.

Again, starting with the definition we have that,

$$
\begin{aligned}
f_X(x) &= \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(X-\mu)^\top \Sigma^{-1}(X-\mu)\right\} \\
&= \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\left\{ -\frac{1}{2} \begin{bmatrix} X_1-\mu_1 \\ \vdots \\ X_k-\mu_k \end{bmatrix}^\top \begin{bmatrix} \left(\Sigma^{-1}\right)_{11} & \left(\Sigma^{-1}\right)_{12} & \cdots & \left(\Sigma^{-1}\right)_{1k} \\ \left(\Sigma^{-1}\right)_{21} & \left(\Sigma^{-1}\right)_{22} & \cdots & \left(\Sigma^{-1}\right)_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \left(\Sigma^{-1}\right)_{k1} & \left(\Sigma^{-1}\right)_{k2} & \cdots & \left(\Sigma^{-1}\right)_{kk} \end{bmatrix} \begin{bmatrix} X_1-\mu_1 \\ \vdots \\ X_k-\mu_k \end{bmatrix} \right\} \\
&= \frac{1}{(2\pi)^{k/2}\,|\Sigma|^{1/2}} \exp\left\{ -\frac{1}{2} \begin{bmatrix} X_1-\mu_1 & \cdots & X_k-\mu_k \end{bmatrix} \begin{bmatrix} \sum_{j=1}^{k} \left(\Sigma^{-1}\right)_{1j}(x_j-\mu_j) \\ \vdots \\ \sum_{j=1}^{k} \left(\Sigma^{-1}\right)_{kj}(x_j-\mu_j) \end{bmatrix} \right\} \\
&= \frac{1}{(2\pi)^{k/2}\,|\Sigma|^{1/2}} \exp\left\{ -\frac{1}{2} \left[(X_1-\mu_1)\sum_{j=1}^{k} \left(\Sigma^{-1}\right)_{1j}(X_j-\mu_j) + \cdots + (X_k-\mu_k)\sum_{j=1}^{k} \left(\Sigma^{-1}\right)_{kj}(X_j-\mu_j) \right] \right\} \\
&= \frac{1}{(2\pi)^{k/2}\,|\Sigma|^{1/2}} \exp\left\{ -\frac{1}{2} \sum_{i=1}^{k}\sum_{j=1}^{k} \left(\Sigma^{-1}\right)_{ij}(X_i-\mu_i)(X_j-\mu_j) \right\}.
\end{aligned}
$$

The final sum inside the exponential can be split into diagonal and off-diagonal sums,

$$
\begin{aligned}
f_{\mathbf{X}}(\mathbf{x}) &= \frac{1}{(2\pi)^{k/2}|\mathbf{\Sigma}|^{1/2}} \exp\left\{ -\frac{1}{2} \left[\sum_{i=1}^{k} (\mathbf{\Sigma}^{-1})_{ii}(x_i-\mu_i)^2 + \sum_{i=1}^{k}\sum_{\substack{j=1 \\ j\neq i}}^{k} (\mathbf{\Sigma}^{-1})_{ij}(x_i-\mu_i)(x_j-\mu_j) \right] \right\} \\
&= \frac{1}{(2\pi)^{k/2}|\mathbf{\Sigma}|^{1/2}} \exp\left\{ -\frac{1}{2} \left[\sum_{i=1}^{k} (\mathbf{\Sigma}^{-1})_{ii}(x_i-\mu_i)^2 + 2\sum_{i=1}^{k-1}\sum_{j=i+1}^{k} (\mathbf{\Sigma}^{-1})_{ij}(x_i-\mu_i)(x_j-\mu_j) \right] \right\}.
\end{aligned}
$$

Now that we have the probability density function for the multivariate normal, we can obtain its moment generating function, that is, we need to evaluate $\mathbb{E}\left[\exp\left(t^\top X\right)\right]$ for some scalar vector $t \in \mathbb{R}^n$. Suppose we have a k-dimensional multivariate normal random vector.

By definition and simplifying, the mgf of X is,

$$\begin{aligned}
M_X(t) &= \mathbb{E}\left[\exp\left(t^\top X\right)\right] \\
&= \int_{\mathbb{R}^k} \exp\left(t^\top X\right) f_X(x)dx \\
&= \int_{\mathbb{R}^k} \exp\left(t^\top X\right) \frac{1}{(2\pi)^{k/2}\left|\Sigma\right|^{1/2}} \exp\left\{-\frac{1}{2}\left(X-\mu\right)^\top \Sigma^{-1}\left(X-\mu\right)\right\} dx \\
&= \frac{1}{(2\pi)^{k/2}\left|\Sigma\right|^{1/2}} \int_{\mathbb{R}^k} \exp\left\{t^\top X - \frac{1}{2}\left(X-\mu\right)^\top \Sigma^{-1}\left(X-\mu\right)\right\} dx.
\end{aligned}$$

We let $Y = X - \mu$ be the centered X matrix so that completing the square is convenient,

$$\begin{aligned}
&= \frac{1}{(2\pi)^{k/2}\left|\Sigma\right|^{1/2}} \int_{\mathbb{R}^k} \exp\left\{t^\top (Y+\mu) - \frac{1}{2}Y^\top\Sigma^{-1}Y\right\} \\
&= \exp\left(t^\top\mu\right) \int_{\mathbb{R}^k} \frac{1}{(2\pi)^{k/2}\left|\Sigma\right|^{1/2}} \exp\left\{t^\top Y - \frac{1}{2}Y^\top\Sigma^{-1}Y\right\} dy \\
&= \exp\left(t^\top\mu\right) \int_{\mathbb{R}^k} \frac{1}{(2\pi)^{k/2}\left|\Sigma\right|^{1/2}} \exp\left\{-\frac{1}{2}\left(Y^\top\Sigma^{-1}Y - 2t^\top Y\right)\right\} dy \\
&= \exp\left(t^\top\mu\right) \int_{\mathbb{R}^k} \frac{1}{(2\pi)^{k/2}\left|\Sigma\right|^{1/2}} \exp\left\{-\frac{1}{2}\left(Y^\top\Sigma^{-1}Y - 2t^\top\Sigma\Sigma^{-1}Y + t^\top\Sigma\Sigma^{-1}\Sigma t - t^\top\Sigma t\right)\right\} dy \\
&= \exp\left(t^\top\mu + \frac{1}{2}t^\top\Sigma t\right) \int_{\mathbb{R}^k} \frac{1}{(2\pi)^{k/2}\left|\Sigma\right|^{1/2}} \exp\left\{-\frac{1}{2}\left(Y-\Sigma t\right)^\top\Sigma^{-1}\left(Y-\Sigma t\right)\right\} dy.
\end{aligned}$$

Observe that the quantity under the integral in the last line is precisely the probability density function of a new multivariate normal distribution specifically of a random vector Y with mean vector Σt and covariance matrix Σ or $Y \sim \mathcal{N}_k\left(\Sigma t, \Sigma\right)$. Because the integral of any valid probability density function over its entire support $\mathbb{R}^k$ must equal 1, the entire integral term evaluates to 1 and drops out of the equation. Therefore the MVN density simplifies to the following,

$$= \exp\left(t^\top\mu + \frac{1}{2}t^\top\Sigma t\right) \cdot 1 = \exp\left(t^\top\mu + \frac{1}{2}t^\top\Sigma t\right).$$

Theorem 2.21 (Transformations under matrices)**.** *Let $A \in \mathbb{M}_{m\times k}(\mathbb{R})$ and $C \in \mathbb{R}^m$ and let $X \sim \mathcal{N}_k(\mu, \Sigma)$. Then, the transformed MVN random variable X is,*

$$AX + c \sim \mathcal{N}_m(A\mu + c, A\Sigma A^\top).$$

Proof. We use the MGF method to prove Theorem 1.153.

We knowe that the MGF of a MVN random variable $X \sim \mathcal{N}_k(\mu, \Sigma)$ is,

$$M_X(t) = \exp\left(t^\top\mu + \frac{1}{2}t^\top\Sigma t\right).$$

We wish to find $Y = AX + c$ for an $m \times 1$ vector t. We compute this as,

$$\begin{aligned} M_Y(t) &= \mathbb{E}\left[\exp\left(t^\top Y\right)\right] \\ &= \mathbb{E}\left[\exp\left(t^\top (AX + c)\right)\right] \\ &= \mathbb{E}\left[\exp\left(t^\top AX + t^\top c\right)\right] \\ &= \exp\left(t^\top c\right) \cdot \mathbb{E}\left[\exp\left(\left(A^\top c\right)^\top X\right)\right]. \end{aligned}$$

Notice that $\mathbb{E}\left[\exp\left(\left(A^\top t\right)^\top X\right)\right]$ is simply the MGF of X evaluated at the vector $\left(A^\top t\right)$.

Substituting this into the known MGF of X we get,

$$\begin{aligned} M_Y(t) &= \exp\left(t^\top c\right) \cdot M_X\left(A^\top t\right) \\ &= \exp\left(t^\top c\right) \cdot \exp\left(\left(A^\top t\right)^\top \mu + \frac{1}{2}\left(A^\top t\right)^\top \Sigma \left(A^\top t\right)\right) \\ &= \exp\left(t^\top c + t^\top A\mu + \frac{1}{2} t^\top A\Sigma A^\top t\right) \\ &= \exp\left(t^\top (A\mu + c) + \frac{1}{2} t^\top \left(A\Sigma A^\top\right) t\right). \end{aligned}$$

This is precisely the MGF of a multivariate normal distribution with a new mean vector and covariance matrix. Therefore, $AX + c \sim \mathcal{N}_m(A\mu + c, A\Sigma A^\top)$. □

Theorem 2.22. *Let $X \in \mathbb{R}^k$ be a k-dimensional multivariate normal random vector with corresponding mean vector μ and covariance matrix Σ. Assume Σ is symmetric positive definite, that is, $\Sigma = \Sigma^\top$ and $\vec{a}^\top \Sigma \vec{a} > 0$ for all $a \in \mathbb{R}^k$ excluding $\vec{0}_k$. Then,*

$$(X - \mu)^\top \Sigma^{-1} (X - \mu) \sim \chi^2_k.$$

Proof. Indeed since our corresponding covariance matrix is symmetric and positive definite, by Spectral Theorem, there exists an orthogonal matrix $Q \in \mathbb{R}^{k \times k}$ and a diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_k)$ for $\lambda_i > 0$ for all $i = 1, \ldots, k$ such that $\Sigma = Q\Lambda Q^\top$.

Note that $Q^\top Q = QQ^\top = I_k$ by properties of orthogonal matrices.

Then the root and inverse-roots of our covariance matrices are,

$$\Sigma^{1/2} = Q\Lambda^{1/2} Q^\top, \quad \Sigma^{-1/2} = Q\Lambda^{-1/2} Q^\top,$$

where the diagonal matrices,

$$\Lambda^{1/2} = \operatorname{diag}\left(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_k}\right), \quad \Lambda^{-1/2} = \operatorname{diag}\left(\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_k}}\right),$$

are well defined. Consequently,

$$\begin{aligned}\Sigma^{-1/2}\Sigma\Sigma^{-1/2} &= Q\Lambda^{-1/2}Q^\top Q\Lambda Q^\top Q\Lambda^{-1/2}Q^\top \\ &= Q\Lambda^{-1/2}\Lambda\Lambda^{-1/2}Q^\top \\ &= QI_kQ^\top \\ &= I_k.\end{aligned}$$

Similarly, we get that,

$$\begin{aligned}\left(\Sigma^{-1/2}\right)^\top \Sigma^{-1/2} &= \Sigma^{-1/2}\Sigma^{-1/2} \\ &= Q\Lambda^{-1/2}Q^\top Q\Lambda^{-1/2}Q^\top \\ &= Q\Lambda^{-1}Q^\top \\ &= \Sigma^{-1}.\end{aligned}$$

Define the standardized normal random vector $Z = \Sigma^{-1/2}(X-\mu)$. By Theorem 1.153, any affine transformation of a multivariate normal random vector is again a multivariate normal,

$$\begin{aligned}Z &\sim \text{MVN}\left(\Sigma^{-1/2}(\mu-\mu), \Sigma^{-1/2}\Sigma\left(\Sigma^{-1/2}\right)^\top\right) \\ &= \text{MVN}\left(0, \Sigma^{-1/2}\Sigma\Sigma^{-1/2}\right) \\ &= \text{MVN}(0, I_k),\end{aligned}$$

where I_k is the $k \times k$ identity matrix.

Since Z is jointly multivariate normal, it follows that for,

$$Z = \begin{bmatrix} Z_1 \\ \vdots \\ Z_k \end{bmatrix} \iff Z_1, \ldots, Z_k \sim \mathcal{N}(0,1).$$

Expanding our original identity into a rank-1 vector formulation we obtain,

$$\begin{aligned}(X-\mu)^\top\Sigma^{-1}(X-\mu) &= (X-\mu)^\top\left(\Sigma^{-1}\right)^\top\Sigma^{-1/2}(X-\mu) \\ &= \left[\Sigma^{-1/2}(X-\mu)\right]^\top\left[\Sigma^{-1/2}(X-\mu)\right] \\ &= Z^\top Z = \sum_{i=1}^{k} Z_i^2 \sim \chi_k^2.\end{aligned}$$

The sum of k independent standard normal random variables is a chi-squared random variable with k degrees of freedom since each individual squared standard normal random variable possesses 1 degree of freedom. Therefore, $(X-\mu)^\top\Sigma^{-1}(X-\mu) \sim \chi_k^2$. □

2.3 Sampling Distributions and Multidimensional Asymptotic Theory

Up until now, we have been doing our matrix computations with respect to a population vector or population matrix. However, in reality, we do not have access to the entire population. Therefore, we utilize statistics such at the sample mean and sample covariance matrix which are functions of a random sample and are therefore random variables or random matrices. Under multivariate normality, these sampling distributions can often be derived exactly because linear transformations remain multivariate normal and orthogonal Gaussian components are independent.

Definition 2.23 (Random Sample)**.** Let P be a probability distribution in $\mathbb{R}^p$. A collection of random vectors $X_1, \ldots, X_n \in \mathbb{R}^p$ is called a random sample of size n from P if the random vectors are mutually independent and have same distribution P. We denote this as,

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P.$$

To normalize some of our notation convention, if the population distribution is multivariate normal, we will from now on write this as $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}_p(\mu, \Sigma)$ where $\mu \in \mathbb{R}^p$ is the population mean vector And $\Sigma \in \mathbb{R}^{p \times p}$ is the population covariance matrix. We assume that every covariance matrix is symmetric and positive semi-definite, that is, $\Sigma = \Sigma^\top$ and $a^\top \Sigma a \geq 0$ for every $a \in \mathbb{R}^p$. We will remove the arrow to denote vectors from now on since it is obvious we are working in dimensions higher than scalar values.

When an expression involving Σ^{-1} is used, we additionally assume that Σ is positive definite, that is, $a^\top \Sigma a > 0$ for every $a \neq 0$. This guarantees that our covariance matrices are non-singular and that Σ^{-1} and $\Sigma^{-1/2}$ exist.

Definition 2.24. Let $X_1, \ldots, X_n$ be a random sample. A statistic is a measurable function $T = T(X_1, \ldots, X_n)$ that does not explicit depend on unknown population parameters.

Because T is a function of random observations, T is itself random. The probability distribution of T induced by the joint distribution of the sample is therefore called the sampling distribution of T.

For example, you are most likely familiar with the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$ which is a statistic, while $\bar{X} - \mu$ is generally not regarded as a statistic when μ is unknown since it explicitly contains an unknown population parameter.

Definition 2.25. Let $X_1, \ldots, X_p \in \mathbb{R}^p$. The sample mean vector is defined by,

$$\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i.$$

Remark 2.26. If we write $X_i = \begin{bmatrix} X_{i1} & \cdots & X_{ip} \end{bmatrix}$ then we have that $\bar{X} = \begin{bmatrix} \bar{X}_1 & \cdots & \bar{X}_p \end{bmatrix}$ and therefore the sample mean vector is,

$$\bar{X}_j = \frac{1}{n}\sum_{i=1}^n X_{ij}.$$

Therefore, each coordinate of the sample mean vector $\bar{X}$ is the ordinary scalar sample mean of the corresponding coordinates of the observations.

Definition 2.27 (Sample Covariance Matrix)**.** The sample covariance matrix is defined by,

$$S = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})^\top \iff (n-1)S = \sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})^\top.$$

Note the two representations are equivalent.

The (j,l)-th entry of sample covariance matrix S is,

$$S_{jl} = \frac{1}{n-1}\sum_{i=1}^{n}(X_{ij} - \bar{X}_j)(X_{il} - \bar{X}_l).$$

Through this representation, the sample variance of the j-th component is simply,

$$S_{jj} = \frac{1}{n-1}\sum_{i=1}^{n}(X_{ij} - \bar{X}_j)^2.$$

Observe that since every outer product $(X_i - \bar{X})(X_i - \bar{X})^\top$ is symmetric and positive semi-definite, S is also symmetric and positive semi-definite. Indeed for every $a \in \mathbb{R}^p$,

$$\begin{aligned} a^\top S a &= \frac{1}{n-1}\sum_{i=1}^{n} a^\top (X_i - \bar{X})(X_i - \bar{X})^\top a \\ &= \frac{1}{n-1}\sum_{i=1}^{n}\left[a^\top (X_i - \bar{X})\right]^2 \geq 0. \end{aligned}$$

Theorem 2.28. *Suppose that $X_1, \ldots, X_n \in \mathbb{R}^p$ are independent random vector satisfying $\mathbb{E}[X_i] = \mu$ and $\mathrm{Cov}(X_i) = \Sigma$. Then $\mathbb{E}[\bar{X}] = \mu$ and $\mathrm{Cov}(\bar{X}) = \frac{1}{n}\Sigma$.*

Proof. By linearity of expectation, $\mathbb{E}[\bar{X}] = \mu$ follows. $\bar{X}$ is therefore an unbiased estimate.

An important identity for the covariance of the sample mean is $\bar{X} - \mu = \dfrac{1}{n}\sum_{i=1}^{n}(X_i - \mu)$.

Therefore, the sample mean covariance follows as,

$$\begin{aligned} \mathrm{Cov}(\bar{X}) &= \mathbb{E}\left[(\bar{X} - \mu)(\bar{X} - \mu)^\top\right] \\ &= \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)\right)\left(\frac{1}{n}\sum_{j=1}^{n}(X_j - \mu)\right)^\top\right] \\ &= \frac{1}{n^2}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{n}(X_i - \mu)(X_j - \mu)^\top\right] \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\mathbb{E}\left[(X_i - \mu)(X_j - \mu)^\top\right]. \end{aligned}$$

When $i = j$, $\mathbb{E}[(X_i - \mu)(X_i - \mu)^\top] = \Sigma$. Otherwise, independence gives,

$$\mathbb{E}[(X_i - \mu)(X_j - \mu)^\top] = \mathbb{E}[X_i - \mu]\mathbb{E}[(X_j - \mu)^\top] = 0 \cdot 0^\top = 0 \in \mathbb{R}^p.$$

Consequently, the results above give,

$$\mathrm{Cov}(\bar{X}) = \frac{1}{n}\sum_{i=1}^{n}\Sigma = \frac{n}{n^2}\Sigma = \frac{1}{n}\Sigma.$$

That is, averaging n independent observations preserves the population mean while reducing the covariance matrix by the factor $\frac{1}{n}$. □

Theorem 2.29 (Fundamental Decomposition of Total Sum of Squares)**.** *For any vectors $X_1, \ldots, X_n \in \mathbb{R}^p$ and mean vector $\mu \in \mathbb{R}^p$,*

$$\sum_{i=1}^{n}(X_i - \mu)(X_i - \mu)^\top = \sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})^\top + n(\bar{X} - \mu)(\bar{X} - \mu)^\top.$$

Proof. Exercise to the reader. □

Theorem 2.30. *Suppose that $X_1, \ldots, X_n$ are independent and identically distributed with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Cov}(X_i) = \Sigma$. Then we have that $\mathbb{E}[S] = \Sigma$.*

Proof. Using the decomposition in Theorem 1.161 we have that,

$$(n-1)S = \sum_{i=1}^{n}(X_i - \mu)(X_i - \mu)^\top - n(\bar{X} - \mu)(\bar{X} - \mu)^\top.$$

Taking expectations we get,

$$\begin{aligned}(n-1)\mathbb{E}[S] &= \sum_{i=1}^{n}\mathbb{E}\left[(X_i - \mu)(X_i - \mu)^\top\right] - n\mathbb{E}[(\bar{X} - \mu)(\bar{X} - \mu)^\top] \\ &= \sum_{i=1}^{n}\Sigma - n\,\mathrm{Cov}(\bar{X}) \\ &= n\Sigma - n\left(\frac{1}{n}\Sigma\right) \\ &= n\Sigma - \Sigma = (n-1)\Sigma.\end{aligned}$$

Dividing by $n-1$ yields $\mathbb{E}[S] = \Sigma$. The denominator $n-1$ rather than n compensates for the loss of one degree of freedom caused by estimating μ with $\bar{X}$. □

We showed earlier that the quadratic form in the exponential of a multivariate normal follows a chi-squared distribution with k degrees of freedom. Here, we normalize some notation when using chi-squared distributions with respect to random vectors and matrices.

Definition 2.31. Let $Z_1, \ldots, Z_\nu \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. Then, the random variable,

$$Q = \sum_{j=1}^{\nu} Z_j^2 \sim \chi^2_\nu,$$

where ν is the dimension of the Gaussian subspace whose squared Euclidean norm is being computed.

The chi-squared distribution satisfies $\mathbb{E}[Q] = \nu$ and $\mathrm{Var}(Q) = 2\nu$.

Definition 2.32 (Projection onto the constant-vector subspace.)**.** Suppose we have the ones-vector $\mathbf{1}_n \in \mathbb{R}^n$. Define $P = \frac{1}{n}\mathbf{1}_n^\top \mathbf{1}_n$ and $H = I_n - P$. The matrix P is the orthogonal projection onto $\text{span}\{\mathbf{1}_n\}$ while H is the orthogonal projection onto $\text{span}\{\mathbf{1}_n\}^\perp$.

There are some convenient properties that the projection and centering matrices satisfy.

Theorem 2.33. *The matrices P and H satisfy,*

$$P^\top = P, \quad P^2 = P, \quad H^\top = H, \quad H^2 = H.$$

and $PH = HP = 0$. Moreover, $\text{rank}(P) = 1$ *and* $\text{rank}(H) = n - 1$.

Proof. First, we prove symmetric by directly computing the transpose,

$$P^\top = \left(\frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\right)^\top = \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top = P.$$

Next, we show idempotence of the projection matrix,

$$\begin{aligned} P^2 &= \left(\frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\right)\left(\frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\right) \\ &= \frac{1}{n^2}\mathbf{1}_n\left(\mathbf{1}_n^\top\mathbf{1}_n\right)\mathbf{1}_n^\top \\ &= \frac{1}{n^2}\mathbf{1}_n(n)\mathbf{1}_n^\top \\ &= \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top = P. \end{aligned}$$

Thus, P is symmetric and idempotent by definition.

Now, since we have that $H = I_n - P$, it is valid to write its transpose as,

$$H^\top = I_n - P^\top = I_n - P = H.$$

Similarly, the idempotence condition follows by direct computation,

$$H^2 = (I_n - P)^2 = I_n - 2P + p^2 = I_n - 2P + P = I_n - P = H.$$

Furthermore, applying the projection to the centering matrix we obtain,

$$PH = P(I_n - P) = P - P^2 = P - P = 0.$$

Similarly, $HP = 0$. Because P projects onto the one-dimensional space spanned by $\mathbf{1}_n$, it follows that $\text{rank}(P) = 1$. Moreover, since H projects onto its orthogonal complement whose dimension is $n - 1$, then $\text{rank}(H) = n - 1$. □

You may be familiar with the asymptotic normality results regarding scalar normal samples and their variances and means. However, it is common for 3rd and 4th year students to skip the proof as it requires application of coordinate orthogonality.

Theorem 2.34. *Let* $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$ *for* $\sigma^2 > 0$.

Utilize classical definitions of sample mean and sample covariance. Then,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim \mathcal{N}(0,1), \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

provided that $\bar{X} \perp S^2$.

Proof. Suppose we have some standardize observation vector in $\mathbb{R}^n$,

$$Z = \frac{1}{\sigma} \begin{bmatrix} X_1 - \mu \\ \vdots \\ X_n - \mu \end{bmatrix}.$$

Given that X_i are independent $\mathcal{N}(\mu, \sigma^2)$ random variables, define the standard normal random observation vector as $Z \sim \mathcal{N}_n(0, I_n)$. Choose an orthogonal matrix $Q \in \mathbb{M}_{n\times n}(\mathbb{R})$ such that the first row of such matrix is,

$$q_1^\top = \frac{1}{\sqrt{n}} \mathbf{1}_n^\top.$$

The remaining rows $q_2^\top, \ldots, q_n^\top$ are chosen via a method that yields a valid orthonormal basis for $\text{span}\{\mathbf{1}_n\}^\perp$. Define $U = QZ$. Given that Q is orthogonal, it obviously satisfies,

$$Q^\top Q = QQ^\top = I_n.$$

Therefore $\mathbb{E}[U] = Q\mathbb{E}[Z] = 0$ and,

$$\text{Cov}(U) = Q\,\text{Cov}(Z)Q^\top = QI_nQ^\top = QQ^\top = I_n.$$

It follows that $U \sim \mathcal{N}_n(0, I_n)$ i.e., is an n-dimensional multivariate normal distribution with mean 0 and covariance matrix equal to the $n \times n$ identity matrix. Therefore, $U_1, \ldots, U_n$ are independent standard normal random variables. The first coordinate is,

$$\begin{aligned} U_1 = q_1^\top Z &= \frac{1}{\sqrt{n}} \mathbf{1}_n^\top \cdot \frac{1}{\sigma} \begin{bmatrix} X_1 - \mu \\ \vdots \\ X_n - \mu \end{bmatrix} \\ &= \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^{n} (X_i - \mu) \qquad \text{rank-1 representation,} \\ &= \frac{1}{\sigma\sqrt{n}} (n\bar{X} - n\mu) \\ &= \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}. \end{aligned}$$

Hence it follows that $U_1 = \dfrac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim \mathcal{N}(0,1)$.

Now, consider the residual sum of squares.

Using definitions of our hat matrices well-defined in Theorem 1.165, we can write this as,

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \bar{X})^2 = Z^\top H Z, \qquad \text{where } H = I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top.$$

Because the rows of Q form an orthonormal basis with first direction parallel to $\mathbf{1}_n$,

$$QHQ^\top = \begin{bmatrix} 0 & 0 \\ 0 & I_{n-1} \end{bmatrix}.$$

Since $U = QZ$ we have that $Z = Q^\top U$ so therefore,

$$\begin{aligned} Z^\top H Z &= U^\top Q H Q^\top U \\ &= U^\top \begin{bmatrix} 0 & 0 \\ 0 & I_{n-1} \end{bmatrix} U = \sum_{j=2}^{n} U_j^2. \end{aligned}$$

Therefore,

$$\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{j=2} U_j^2 \sim \chi^2_{n-1}.$$

Finally, $\bar{X}$ is a function only of U_1 while S^2 is a function only of $U_2, \ldots, U_n$. Since $U_1 \perp (U_2, \ldots, U_n)$ it follows that $\bar{X} \perp S^2$. The $n-1$ degrees of freedom arise because the residual vector lies in the $(n-1)$ dimensional subspace orthogonal to $\mathbf{1}_n$. □

Definition 2.35 (Student-t Distribution)**.** Let $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_\nu$ with $Z \perp V$.

The random variable,

$$T = \frac{Z}{\sqrt{V/\nu}} \sim T \sim t_\nu,$$

follows a Student-t distribution with ν degrees of freedom.

Theorem 2.36 (Studentized Sample Mean)**.** *Let* $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$ *for* $\sigma^2 > 0$. *Then,*

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}.$$

Proof. By Theorem 1.166, $Z = \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim \mathcal{N}(0,1)$ and $V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$.

Normality of the original sample implies that $\bar{X} \perp S^2$ and hence $Z \perp V$.

Observe that we can write,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} = \frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{S/\sigma} = \frac{Z}{\sqrt{S^2/\sigma^2}} = \frac{Z}{\sqrt{\frac{1}{n-1} \cdot \frac{(n-1)S^2}{\sigma^2}}} = \frac{Z}{\sqrt{V/(n-1)}}.$$

By definition of the Student-t distribution,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}.$$

Therefore, the Student-t distribution appears when the unknown population standard deviation σ in a standardized Gaussian statistic is replaced by the random sample standard deviation S. □

Definition 2.37. Let $U \sim \chi^2_r$ and $V \sim \chi^2_s$ where $U \perp V$. Then,

$$F = \frac{U/r}{V/s},$$

has an F distribution with r numerator degrees of freedom and s degrees of freedom $F \sim F_{r,s}$.

Theorem 2.38. *Suppose we have $X_1, \ldots, X_m \overset{i.i.d.}{\sim} \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_n \overset{i.i.d.}{\sim} \mathcal{N}(\mu_Y, \sigma_Y^2)$ where the two samples are independent. Let S_X^2 and S_Y^2 denote their sample variances. Then,*

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{m-1,n-1},$$

i.e., the F distribution with $n-1$ by $m-1$ degrees of freedom.

Proof. By Theorem 1.168, the normal sampling distributions of the sample variances,

$$U = \frac{(m-1)S_X^2}{\sigma_X^2} \sim \chi^2_{m-1}, \qquad V = \frac{(n-1)S_Y^2}{\sigma_Y^2} \sim \chi^2_{n-1}.$$

Because the two samples are independent $U \perp V$ are independent,

$$\frac{U/(m-1)}{V/(n-1)} = \frac{\frac{1}{m-1} \cdot \frac{(m-1)S_X^2}{\sigma_X^2}}{\frac{1}{n-1} \cdot \frac{(n-1)S_Y^2}{\sigma_Y^2}} = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{m-1,n-1},$$

which is precisely the form of an $F_{m-1,n-1}$ distribution. □

Theorem 2.39. *Let $Z_1, \ldots, Z_\nu \overset{i.i.d.}{\sim} \mathcal{N}_p(0, \Sigma)$ where Σ is symmetric and positive definite. Define, $W = \sum_{i=1}^{\nu} Z_i Z_i^\top$. Then W is said to have a Wishart distribution with scale matrix Σ and ν degrees of freedom $W \sim W_p(\Sigma, \nu)$. The Wishard distribution is the multivariate generalization of the chi-squared distribution. In the case $p = 1$, $W = \sum_{i=1}^{\nu} Z_i^2$ and therefore $\frac{W}{\Sigma} \sim \chi^2_\nu$.*

Theorem 2.40 (Quadratic Projections of Wishart)**.** *Suppose that $W \sim \mathcal{W}_p(\Sigma, \nu)$.*

Then for every non-zero $a \in \mathbb{R}^p$,

$$\frac{a^\top W a}{a^\top \Sigma a} \sim \chi^2_\nu.$$

Proof. By the definition of the Wishart distribution,

$$W = \sum_{i=1}^{\nu} Z_i Z_i^\top, \quad Z_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_p(0, \Sigma).$$

Therefore the quadratic form simplifies to,

$$a^\top W a = a^\top \left(\sum_{i=1}^{\nu} Z_i Z_i^\top \right) a = \sum_{i=1}^{\nu} a^\top Z_i Z_i^\top a = \sum_{i=1}^{\nu} \left(a^\top Z_i \right)^2.$$

Since $Z_i \sim \mathcal{N}_p(0, \Sigma)$, the the scalar linear combination satisfies $a^\top Z_i \sim \mathcal{N}(0, a^\top \Sigma a)$ hence,

$$\frac{a^\top Z_i}{\sqrt{a^\top \Sigma a}} \sim \mathcal{N}(0, 1).$$

The random variables are independent across i so therefore,

$$\frac{a^\top W a}{a^\top \Sigma a} = \sum_{i=1}^{\nu} \left(\frac{a^\top Z_i}{\sqrt{a^\top \Sigma a}} \right)^2 \sim \chi^2_\nu.$$

Thus every 1-dimensional projection of a Wishart matrix produces a chi-squared random variable. □

Theorem 2.41. *Let* $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \mathcal{N}_p(\mu, \Sigma)$ *where* Σ *is a positive definite matrix.*

Then $(n-1)S \sim \mathcal{W}_p(\Sigma, n-1)$*. Moreover,* $\bar{X} \perp S$*.*

Proof. Let $X \sim \mathbb{M}_{n \times p}(\mathbb{R})$ be a data matrix.

Let $Q \in \mathbb{M}_{n \times n}(\mathbb{R})$ be an orthogonal matrix whose first row is,

$$q_1^\top = \frac{1}{\sqrt{n}} \mathbf{1}_n^\top.$$

Define the transformed data matrix $Y = QX$ and write the rows of Y as $Y = \begin{bmatrix} Y_1^\top \\ \vdots \\ Y_n^\top \end{bmatrix}$.

The first transformed row is,

$$\begin{aligned} Y_1^\top = q_1^\top X &= \frac{1}{\sqrt{n}} \mathbf{1}_n^\top X \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i^\top = \sqrt{n} \bar{X}^\top. \end{aligned}$$

Therefore $Y_1 = \sqrt{n} X$. Because Q is orthogonal and the rows of X are independent multivariate normal random vectors with common covariance Σ, the transformed rows $Y_1, \ldots, Y_n$ are jointly multivariate normal. Their means satisfy,

$$\mathbb{E}[Y_j] = \sum_{i=1}^{n} q_{ji} \mathbb{E}[X_i] = \left(\sum_{i=1}^{n} q_{ji} \right) \mu.$$

For $j = 1$, $\sum_i q_{1i} = \sqrt{n}$ so $\mathbb{E}[Y_1] = \sqrt{n}\mu$. For $j \geq 2$ the row q_j is orthogonal to $\mathbf{1}_n$ and therefore we have that $\sum_i q_{ji} = 0$. Hence $\mathbb{E}[Y_j] = 0$ for $j = 2, \ldots, n$. For the cross-covariance between Y_j and Y_l,

$$\operatorname{Cov}(Y_j, Y_l) = \operatorname{Cov}\left(\sum_{i=1}^n q_{ji} X_i, \sum_{r=1}^n q_{lr} X_r\right) = \sum_{i=1}^n \sum_{r=1}^n q_{ji} q_{lr} \operatorname{Cov}(X_i, X_r).$$

By independence we know that $\operatorname{Cov}(X_i, X_r) = 0$ when $i \neq r$ so,

$$\operatorname{Cov}(Y_j, Y_l) = \sum_{i=1}^n q_{ji} q_{li} \Sigma = \left(\sum_{i=1}^n q_{ji} q_{li}\right) \Sigma = \left(q_j^\top q_l\right) \Sigma.$$

Since the rows of Q are orthonormal $q_j^\top q_l = 1$ for $j = l$, otherwise when $j \neq l$, $q_j^\top q_l = 0$. Therefore, $\operatorname{Cov}(Y_j) = \Sigma$ and $\operatorname{Cov}(Y_j, Y_l) = 0$ for $j \neq l$. Since $Y_1, \ldots, Y_n$ are jointly multivariate normal, zero cross-covariance implies independence. Consequently we have that $Y_1, \ldots, Y_n$ are independent with $Y_1 \sim \mathcal{N}_p(\sqrt{n}\mu, \Sigma)$ and $Y_j \sim \mathcal{N}_p(0, \Sigma)$ for $j = 2, \ldots, n$. We define,

$$P = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top, \quad H = I_n - P.$$

By Theorem 1.162, the sample sum of squares matrix is given as,

$$(n-1)S = \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^\top = X^\top H X.$$

Because Q diagonalizes the projection matrix H, we know that,

$$QHQ^\top = \begin{bmatrix} 0 & 0 \\ 0 & I_{n-1} \end{bmatrix}.$$

Since $X = Q^\top Y$, compute,

$$(n-1)S = Y^\top Q H Q^\top Y = Y^\top \begin{bmatrix} 0 & 0 \\ 0 & I_{n-1} \end{bmatrix} Y = \sum_{j=2}^n Y_j Y_j^\top.$$

The random vectors $Y_2, \ldots, Y_n$ are independent and each has distribution $\mathcal{N}_p(0, \Sigma)$. Therefore by definition $(n-1)S \sim \mathcal{W}_p(\Sigma, n-1)$. Moreover, $\bar{X} = \frac{1}{\sqrt{n}} Y_i$ depends only on Y_1 while S depends on $Y_2, \ldots, Y_n$. Since $Y_1 \perp Y_2, \ldots, Y_n$, then $\bar{X} \perp S$, as required. □

Corollary 2.42 (Sampling Distribution of Directional Sample Variance)**.** *For every non-zero $a \in \mathbb{R}^n$,*

$$\frac{(n-1)a^\top S a}{a^\top \Sigma a} \sim \chi^2_{n-1}.$$

Proof. By Theorem 1.161 we know that $(n-1)S \sim \mathcal{W}_p(\Sigma, n-1)$ which shows that the quadratic-projection property of the Wishart distribution gives,

$$\frac{a^\top[(n-1)S]a}{a^\top \Sigma a} \sim \chi^2_{n-1}.$$

Hence it follows that,

$$\frac{(n-1)a^\top \Sigma a}{a^\top \Sigma a} \sim \chi^2_{n-1}.$$

□

Definition 2.43 (Hotelling's T^2 statistic.)**.** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}_p(\mu, \Sigma)$. For a hypothesized population mean vector μ_0, Hotelling's one-sample T^2 statistic is written as,

$$T^2 = n(\bar{X} - \mu_0)^\top S^{-1}(\bar{X} - \mu_0).$$

To ensure that S^{-1} exists almost surely, it is sufficient to assume that $n > p$ and Σ is positive definite.

Theorem 2.44 (Normal inverse-Wishart Quadratic Form)**.** *Suppose that $Z \sim \mathcal{N}_p(0, I_p)$ and $W \sim \mathcal{W}_p(I_p, \nu)$ where W has ν degrees of freedom where Z and W are independent and $\nu \geq p$.*

Then,

$$\frac{\nu - p + 1}{p} Z^\top W^{-1} Z \sim F_{p,\nu-p+1}.$$

Proof. Because $W \sim \mathcal{W}_p(I_p, \nu)$, then there exist a $\nu \times p$ random matrix A with independent standard normal entries such that $W = A^\top A$. Suppose that $a \in \mathbb{R}^\nu$ is the first column and $B \in \mathbb{R}^{\nu \times (p+1)}$ contains the remaining columns. Then we can write,

$$A = \begin{bmatrix} a & B \end{bmatrix}.$$

Then the following form,

$$W = \begin{bmatrix} a^\top a & a^\top B \\ B^\top a & B^\top B \end{bmatrix}.$$

Notice that the Schur complement of $B^\top B$ in W is,

$$C = a^\top a - a^\top B(B^\top B)^{-1} B^\top a = a^\top [I_\nu - B(B^\top B)^{-1} B^\top] a.$$

Define $P = B(B^\top B)^{-1} B^\top$ as the projection matrix. Almost surely, B has rank $p-1$. Therefore, P_B is the orthogonal projection onto the column space of B and $I_\nu - P_B$ is a symmetric idempotent matrix with rank $\nu - (p-1) = \nu - p + 1$ by Rank-Nullity theorem. The vector a is independent of B and conditional on B so therefore $A \sim \mathcal{N}_\nu(0, I_\nu)$. Because we know that $I_\nu - P_B$ is an orthogonal projection of rank $\nu - p + 1$, then it follows that,

$$C = a^\top (I_\nu - P_B) a \sim \chi^2_{\nu-p+1}.$$

The block inverse formula gives $\left(W^{-1}\right)_{11} = \frac{1}{C}$. Next, we write $R^2 = Z^\top Z$. Since we know that $Z \sim \mathcal{N}_p(0, I_p)$, we know that $R^2 \sim \chi^2_p$. Conditional on a non-zero realization where $Z = z$, we choose

an orthogonal matrix O_z such that $O_z \frac{z}{\|z\|} = e_1$. Then $O_z z = \|z\| e_1$. By orthogonal invariance of the standard Wishart distribution it follows that $O_z W O_z^\top \stackrel{d}{=} W$.

Therefore, conditional on $Z = z$,

$$\begin{aligned} z^\top W^{-1} z &= (O_z z)^\top (O_z W O_z^\top)^{-1} (O_z z) \\ &= \|z\|^2 e_1^\top (O_z W O_z^\top)^{-1} e_1 \\ &\stackrel{d}{=} \|z\|^2 e_1^\top W^{-1} e_1 \\ &= \|z\|^2 \left(W^{-1}\right)_{11} = \frac{\|z\|^2}{C}. \end{aligned}$$

Consequently we have that $Z^\top W^{-1} Z \stackrel{d}{=} \frac{R^2}{C}$ where $R^2 \sim \chi^2_p$ and $C \sim \chi^2_{\nu-p+1}$ and $R^2 \perp C^2$ because $Z \perp W$. It follows that,

$$\frac{\nu - p + 1}{p} Z^\top W^{-1} Z \stackrel{d}{=} \frac{R^2/p}{C/(\nu - p + 1)} \sim F_{p,\nu-p+1},$$

as required. □

The preceding resuls are exact because the underlying observations are normally distributed. When the population is not normal, the exact finite-sample disetributions of the sample mean $\bar{X}$ and the sample standard deviation S, and their associated quadratic forms usually cease to be normal, Wishart, t or F distributed. Nevertheless, under finite-moment assumptions, the multivariate central limit theorem provides asymptotic sampling distributions.

Definition 2.45 (Convergence in Distribution)**.** A sequence of random vectors $Y_n \in \mathbb{R}^n$ is said to converge in distribution to a random vector $Y \in \mathbb{R}^p$ written as $Y_n \xrightarrow{d} Y$ if $\mathbb{E}[f(Y_n)] \to \mathbb{E}[f(Y)]$ for every bounded continuous function $f : \mathbb{R}^p \to \mathbb{R}$. Equivalently, convergence in distribution may be characterized through convergence of multivariate characteristic functions.

Theorem 2.46 (Multivariate Central Limit Theorem)**.** *Let $X_1, X_2, \ldots$ be independent and identically distributed random vectors in $\mathbb{R}^p$ satisfying $\mathbb{E}[X_i] = \mu$ and $\mathrm{Cov}(X_i) = \Sigma$ where every entry of the covariance matrix Σ is finite. Then,*

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} \mathcal{N}_p(0, \Sigma).$$

Proof. Fix an arbitrary vector $a \in \mathbb{R}^p$. Therefore $a^\top X_1, a^\top X_2, \ldots$ are independent and identically distributed scalar random variables satisfying $\mathbb{E}[a^\top X_i] = a^\top \mu$ and,

$$\begin{aligned} \mathrm{Var}(a^\top X_i) &= \mathbb{E}\left[\left(a^\top X_i - a^\top \mu\right)^2\right] \\ &= \mathbb{E}\left[a^\top (X_i - \mu)(X_i - \mu)^\top a\right] \\ &= a^\top \mathbb{E}\left[(X_i - \mu)(X_i - \mu)^\top\right] a \\ &= a^\top \Sigma a. \end{aligned}$$

By the scalar central limit theorem

$$\sqrt{n}\left(a^\top \bar{X} - a^\top \mu\right) \xrightarrow{d} \mathcal{N}(0, a^\top \Sigma a).$$

Observe that $\sqrt{n}(a^\top \bar{X} - a^\top \mu) = a^\top \sqrt{n}(\bar{X} - \mu)$. Therefore for every $a \in \mathbb{R}^p$,

$$a^\top \sqrt{n}(\bar{X} - \mu) \xrightarrow{d} \mathcal{N}(0, a^\top \Sigma a).$$

If $Z \sim \mathcal{N}_p(0, \Sigma)$ then $a^\top Z \sim \mathcal{N}(0, a^\top \Sigma a)$ by Theorem 1.153 for every $a \in \mathbb{R}^p$. Hence,

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} Z \sim \mathcal{N}_p(0, \Sigma).$$

□

Corollary 2.47 (Asymptotic Chi-squared of the Standardized Sample Mean)**.** *Suppose the assumptions of the multivariate central limit theorem hold and that Σ is positive definite. Then,*

$$n(\bar{X} - \mu)\Sigma^{-1}(\bar{X} - \mu) \xrightarrow{d} \chi^2_p.$$

Proof. By Theorem 1.178,

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} \mathcal{N}_p(0, \Sigma).$$

Multiplying by the inverse root of the covariance matrix,

$$\sqrt{n}\Sigma^{-1/2}\left(\bar{X} - \mu\right) \xrightarrow{d} \mathcal{N}_p(0, I_p).$$

Let $Z_n = \sqrt{n}\Sigma^{-1/2}(\bar{X} - \mu)$ then $Z_n \xrightarrow{d} Z$ for $Z \sim \mathcal{N}_p(0, I_p)$.

Consider the mapping $g(z) = z^\top z$ is continous. By Continuous Mapping Theorem,

$$Z_n^\top Z_n \xrightarrow{d} Z^\top Z.$$

Now given,

$$Z_n^\top Z_n = n(\bar{X} - \mu)^\top \Sigma^{-1}(\bar{X} - \mu).$$

While we know that $Z^\top Z \sim \chi^2_p$ therefore,

$$n(\bar{X} - \mu)^\top \Sigma^{-1}(\bar{X} - \mu) \xrightarrow{d} \chi^2_p,$$

as required. □

Theorem 2.48 (Consistency of the Sample Covariance matrix)**.** *Suppose that $X_1, X_2, \ldots$ are independent and identically distributed with $\mathbb{E}[\|X_i\|^2] < \infty$. Then $S \xrightarrow{p} \Sigma$.*

Proof. We define the sample covariance matrix as,

$$S = \frac{n}{n-1}\left[\frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top = \overline{XX}^\top\right].$$

By the Weak Law of Large Numbers,

$$\frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top \xrightarrow{p} \mathbb{E}[XX^\top].$$

Also, it follows that $\bar{X} \xrightarrow{p} \mu$ and therefore by continuity of the outer-product mapping, $\overline{XX}^\top \xrightarrow{p} \mu\mu^\top$ where μ is the population mean vector. Moreover, notice that $\frac{n}{n-1} \to 1$. Hence,

$$S \xrightarrow{p} \mathbb{E}[XX^\top] - \mu\mu^\top = \mathbb{E}[(X-\mu)(X-\mu)^\top] = \Sigma.$$

□

Corollary 2.49 (Asymptotic Chi-squared Statistic)**.** *Suppose that $\mathbb{E}[\|X_i\|^2] < \infty$ and that Σ is a positive definite matrix. Then,*

$$n(\bar{X}-\mu)^\top S^{-1}(\bar{X}-\mu) \xrightarrow{d} \chi^2_p.$$

Since $S \xrightarrow{p} \Sigma$ and Σ is positive definite, continuity of the matrix inversion on a seto non-singular matrices gives that S^{-1} converges in probability to Σ^{-1}. Also, by Theorem 1.178, we know that,

$$\sqrt{n}(\bar{X}-\mu) \xrightarrow{d} \mathcal{N}_p(0,\Sigma).$$

Therefore, by Slutsky's Theorem,

$$n(\bar{X}-\mu)^\top S^{-1}(\bar{X}-\mu) = \left[\sqrt{n}(\bar{X}-\mu)\right]^\top S^{-1}\left[\sqrt{n}(\bar{X}-\mu)\right] \xrightarrow{d} Z^\top \Sigma^{-1} Z,$$

where $Z \sim \mathcal{N}_p(0,\Sigma)$ since $\Sigma^{-1/2}\Sigma \sim \mathcal{N}_p(0, I_p)$ we obtain,

$$Z^\top \Sigma^{-1} Z = \|\Sigma^{-1/2} Z^2\|^2 \sim \chi^2_p.$$

Therefore, $\sqrt{n}(\bar{X}-\mu)^\top S^{-1}(\bar{X}-\mu) \xrightarrow{d} \chi^2_p$.

Under exact multivariate normaly, the finite-sample transformation of this statistic is Hotelling's T^2 distribution. Without normality, the same quadratic statistic is asymptotically chi-squared.

Part II

Classical Linear Regression

Suppose that you are a quantitative researcher on a systematic futures trading desk that trades E-mini S&P 500 futures on an intraday basis. Over the past twenty years, the desk has collected a large panel of raw and derived market features, including limit order book variables, trade-flow statistics, volatility measures, and signals constructed from related asset classes. These observations may be represented by a feature matrix whose columns correspond to individual predictors and whose rows correspond to successive points in time.

In practice, storing and processing the entire twenty-year history within a single matrix may be computationally inefficient, and the data would ordinarily be partitioned, compressed, or processed incrementally. For the moment, however, we set these computational considerations aside and assume that the complete feature matrix can be manipulated directly.

The primary statistical question is whether the available intraday limit order book and cross-asset features contain information about a specified response variable. Such a response might be the return of the E-mini S&P 500 futures contract over the next minute, the future change in its mid-price, or the number of ticks by which the contract moves over a fixed prediction horizon. More generally, we wish to determine whether some linear combination of the observed predictors has a systematic relationship with the future target.

For now, we also set aside the martingale assumptions commonly imposed on discounted asset prices under risk-neutral valuation. The objective here is not to construct a risk-neutral pricing model, but rather to study predictive relationships under the empirical probability measure.

Nevertheless, financial time-series data introduce several important complications, including temporal dependence, non-stationarity, overlapping observations, and forward-looking bias. These issues will be treated carefully in later sections through appropriate time-series cross-validation and out-of-sample evaluation procedures.

We begin with the classical linear regression model and the method of ordinary least squares.

Chapter 3

Ordinary Least Squares

3.1 Least Squares and Orthogonalization

We start by setting up the notion of least squares. Consider an $n \times p$ feature matrix, where n is the number of observations i.e., the number of timestamps (necessarily disjoint and unique), and p is the number of features. We have a vector of responses (targets) $Y \in \mathbb{R}^n$ and a vector of coefficients $\beta \in \mathbb{R}^p$ and some error $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ where σ^2 is a constant.

Therefore, our linear model is well-defined in this theorem.

Theorem 3.1. *Let $z_1, \ldots, z_n$ denote the original observations where each z_i may contain one or more raw explanatory variables. Let $\phi_1, \ldots, \phi_p$ be arbitrary real-valued transformations of those explanatory variables and define the design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ by $X_{ij} = \phi_j(x_j)$. Suppose that,*

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n), \quad \sigma^2 > 0,$$

where $Y \in \mathbb{R}^n$ and $\beta \in \mathbb{R}^p$. Provided that the feature transformations $\phi_1, \ldots, \phi_p$ are fixed and do not depend on the unknown coefficient vector β, the model is linear in its parameters, regardless of how non-linear the transformations ϕ_j may be in the original explanatory variables.

Proof. The design matrix has the form,

$$X = \begin{bmatrix} \phi_1(z_1) & \phi_2(z_1) & \cdots & \phi_p(z_1) \\ \phi_1(z_2) & \phi_2(z_2) & \cdots & \phi_p(z_2) \\ \cdots & \cdots & \ddots & \cdots \\ \phi_1(z_n) & \phi_2(z_n) & \cdots & \phi_p(z_n) \end{bmatrix}, \qquad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}.$$

Consequently, $X\beta$ becomes,

$$X\beta = \begin{bmatrix} \phi_1(z_1) & \phi_2(z_1) & \cdots & \phi_p(z_1) \\ \phi_1(z_2) & \phi_2(z_2) & \cdots & \phi_p(z_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(z_n) & \phi_2(z_n) & \cdots & \phi_p(z_n) \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^{p} \beta_j \phi_j(z_1) \\ \sum_{j=1}^{p} \beta_j \phi_j(z_2) \\ \vdots \\ \sum_{j=1}^{p} \beta_j \phi_j(z_n) \end{bmatrix}.$$

For each observation i,

$$Y_i = \sum_{j=1}^{p} \beta_j \phi_j(z_i) + \varepsilon_i.$$

The expression is possibly highly non-linear as a function of the raw observation z_i since the functions ϕ_j may involve powers, logarithms, exponentials, trigonometric functions, splines or other transformations. Nevertheless, each unknown parameter β_j appears only to the first power and multiplies a quantity $\phi_j(z_i)$ that is fixed once that data has been observed.

Consider the systematic component of the model by the mapping $m : \mathbb{R}^p \to \mathbb{R}^n$ where $m(\beta) = X\beta$. Let $\beta, \gamma \in \mathbb{R}^p$ be arbitrary coefficient vectors and let $a, b \in R$. Then, by linearity,

$$\begin{aligned} m(a\beta + b\gamma) &= X(a\beta + b\gamma) \\ &= X(a\beta) + X(b\gamma) \\ &= aX\beta + bX\gamma \\ &= am(\beta) + bm(\gamma). \end{aligned}$$

Therefore, $m(a\beta + b\gamma) = am(\beta) + bm(\gamma)$ for every $\beta, \gamma \in \mathbb{R}^p$ and every $a, b \in \mathbb{R}$. Hence, m is a linear transformation from the parameter space $\mathbb{R}^p$ into the observation space $\mathbb{R}^n$. The same conclusion can be seen componentwise. As a rank-1 expansion, observe that,

$$\begin{aligned} m_i(a\beta + b\gamma) &= \sum_{j=1}^{p} (a\beta_j + b\gamma_j)\phi_j(z_i) \\ &= a \sum_{j=1}^{p} \beta_j \phi_j(z_i) + b \sum_{j=1}^{p} \gamma_j \phi_j(z_i) \\ &= a\, m_i(\beta) + b\, m_i(\gamma). \end{aligned}$$

Because this identity holds for every component $i = 1, \ldots, n$, the entire vector-valued mapping is linear for any non-linear transformations ϕ_j for $j = 1, \ldots, p$ applied to our features. □

Moreover, it is important to note that all probability statements regarding our model are done with respect to Y and ε and *not* the feature matrix X. We refer to our model being fixed in X, that is, our observations are fixed, and any probabilistic statements are conditional on ε and our targets.

Instead of simply listing the set of assumptions, which, in academia, is often referred to as the Gauss-Markov assumptions, let's decompose the ordinary least squares model and see how each assumption follows from interpretations of the ordinary least squares formulation.

Consider our specified linear regression model given as,

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n),$$

where $X \in \mathbb{M}_{n \times p}(\mathbb{R})$, $Y \in \mathbb{R}^n$, $\beta \in \mathbb{R}^p$ and $\varepsilon \in \mathbb{R}^n$. Denote the least squares estimate as $\hat{\beta} \in \mathbb{R}^p$ and the least squares estimator $\hat{\beta} \in \mathbb{R}^p$ such that $\hat{\beta}$ is the solution to the least squares problem i.e., minimizing the residual sum of squares with respect to β in $Y = X\beta + \varepsilon$.

Theorem 3.2. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$, $Y \in \mathbb{R}^n$ and $\beta \in \mathbb{R}^p$. A vector $\hat{\beta} \in \mathbb{R}^p$ is referred to as the least squares solution that minimizes the Euclidean squared norm $\|Y - X\beta\|_2^2$ if and only if it satisfies the normal equations,*

$$X^\top X\hat{\beta} = X^\top Y.$$

If the design matrix has full column rank such that $\operatorname{rank}(X) = p$*, then the Gram matrix $X^\top X$ is positive definite and invertible, yielding the least squares estimate,*

$$\hat{\beta} = \left(X^\top X\right)^{-1} X^\top Y.$$

Proof. We aim to prove Theorem 3.2.

($\Longleftarrow$) Suppose that $\hat{\beta}$ satisfies the normal equations, that is, $X^\top X\hat{\beta} = X^\top Y$. Then, the equation re-arranges in $X^\top(Y - X\hat{\beta}) = 0$. By definition, $Y - X\hat{\beta}$ lies in the nullspace of $X^\top$, denoted $\text{null}(X^\top)$. By the Fundamental Theorem of Linear Algebra, the nullspace of $X^\top$ is the complement of the column space of X, so it follows that $(Y - X\hat{\beta}) \in \text{Col}(X)^\perp$.

Suppose we have another arbitrary vector $\beta \in \mathbb{R}^p$. Then, the residual $Y - X\beta$ decomposes as,

$$\begin{aligned} Y - X\beta &= (Y - X\hat{\beta}) + (X\hat{\beta} - X\beta) \\ &= (Y - X\hat{\beta}) + X(\hat{\beta} - \beta). \end{aligned}$$

$X(\hat{\beta} - \beta)$ is a linear combination of vectors in $\text{Col}(X)$ therefore $X(\hat{\beta} - \beta) \in \text{Col}(X)$. Since $(Y - X\hat{\beta}) \in \text{Col}(X)^\perp$ and $X(\hat{\beta} - \beta) \in \text{Col}(X)$, it follows that they are orthogonal i.e., $(Y - X\hat{\beta}) \perp X(\hat{\beta} - \beta)$.

Thus, by Pythagorean Theorem in $\mathbb{R}^n$,

$$\begin{aligned} \|Y - X\beta\|_2^2 &= \|(Y - X\hat{\beta}) + X(\hat{\beta} - \beta)\|_2^2 \\ &= \|Y - X\hat{\beta}\|_2^2 + \|X(\hat{\beta} - \beta)\|_2^2. \end{aligned}$$

Since the norm of any vector in an inner product space is non-negative i.e., $\|X(\hat{\beta} - \beta)\|_2^2 \geq 0$,

$$\|Y - X\beta\|_2^2 \geq \|Y - X\hat{\beta}\|_2^2.$$

Therefore, we conclude that $\hat{\beta}$ is the global minimizer.

($\Longrightarrow$) Conversely, assume that $\hat{\beta}$ minimizes the squared norm $\|Y - X\beta\|_2^2$. By the Orthogonal Projection theorem in a finite-dimensional vector space, the shortest distance from a vector Y to a subspace $\text{Col}(X)$ is achieved when the error vector is strictly orthogonal to the subspace. Thus, $Y - X\hat{\beta}$ must be orthogonal to every column vector X. Mathematically, this requires that the dot product of every column of X with the residual to be zero which we can write as,

$$\begin{aligned} X^\top(Y - X\hat{\beta}) &= 0 \\ X^\top Y - X^\top X\hat{\beta} = 0 &\iff X^\top X\hat{\beta} = X^\top Y. \end{aligned}$$

Solving for $\hat{\beta}$ gives the least squares estimate $\hat{\beta} = \left(X^\top X\right)^{-1} X^\top Y$. □

The classical derivation of the least squares estimate often does not invoke the Orthogonal Projection theorem or the Fundamental Theorem of Linear Algebra. More often than not, it is easier to simply

expand the residual sum of squares into matrix form then set it's gradient to zero. Again, suppose that $X \in \mathbb{M}_{n\times p}(\mathbb{R}), Y \in \mathbb{R}^n$ and $\beta \in \mathbb{R}^p$.

Then the residual sum of squares is written and expanded as,

$$\begin{aligned}\|Y - X\beta\|_2^2 &= (Y - X\beta)^\top (Y - X\beta) \\ &= Y^\top Y - Y^\top X\beta - \beta^\top X^\top Y + \beta^\top X^\top X\beta.\end{aligned}$$

Since a scalar is equal to its own transpose, we re-write the second term as,

$$Y^\top X\beta = \left(Y^\top X\beta\right)^\top = \beta^\top X^\top \left(Y^\top\right)^\top = \beta^\top X^\top Y.$$

Substituting we get,

$$= Y^\top Y - 2\beta^\top X^\top Y + \beta^\top X^\top X\beta.$$

Taking the derivative with respect to the coefficient vector and setting to 0,

$$\frac{\partial}{\partial\beta}\|Y - X\beta\|_2^2 = -2X^\top Y + 2X^\top X\beta = 0,$$

so we are left with,

$$-2X^\top Y = -2X^\top X\beta \iff \hat{\beta} = (X^\top X)^{-1}X^\top Y.$$

For an arbitrary matrix $\mathbb{M}_{n\times p}(\mathbb{R})$, $X^\top X$ is always symmetric regardless of whether or not X has full column rank. This simply follows by expanding the transpose of the gram matrix,

$$(X^\top X)^\top = (X^\top)^\top X^\top = X^\top X.$$

So by definition, $X^\top X$ is symmetric.

Observe that in the normal equation for the least squares estimate we take the inverse of this matrix. For $X^\top X$ to be invertible, we require that it's determinant is non-zero, it has full rank, or it's nullspace is trivial. Let's analyze it's nullspace.

Suppose that we have a vector $\vec{v} \in \text{null}(X)$. Then, by definition, $X\vec{v} = 0$ which implies that $X^\top X\vec{v} = 0$. Therefore $\vec{v} \in \text{null}(X^\top X)$. Similarly, suppose that $\vec{w} \in \text{null}(X^\top X)$ which implies that $X^\top X\vec{w} = 0$. Therefore, the quadratic form $\vec{w}^\top X^\top X\vec{w} = 0$ which implies that it's ℓ_2 norm $\|X\vec{w}\|_2^2 = 0$. Therefore, $X\vec{w} = 0$ so $\vec{w} \in \text{null}(X)$ as well.

Since we have shown inclusion, therefore $\text{null}(X^\top X) = \text{null}(X)$. By the Rank-Nullity Theorem, it follows that $\text{rank}(X^\top X) = \text{rank}(X)$. Since $X^\top X$ is a $p \times p$ matrix, it is invertible if and only if $\text{rank}(X^\top X) = p$. Therefore, $X^\top X$ is invertible if and only if $\text{rank}(X) = p$. Note that showing that $\text{rank}(X^\top X) = \text{rank}(X)$ is sufficient since a full rank matrix X possesses a trivial nullspace.

Since the design matrix is a linear transformation $X : \mathbb{R}^p \to \mathbb{R}^n$, by the Rank-Nullity Theorem explicitly, it follows that $\text{rank}(X) + \text{nullity}(X) = \dim \mathbb{R}^p$. Clearly, the only case where $\text{rank}(X) = \dim \mathbb{R}^p$ is when $\text{nullity}(X) = \dim \text{null}(X) = 0$ or when $\text{null}(X) = \{\vec{0}\}$.

These distinctions are important.

For a linear system $Ax = b$, a solution exists if and only if $b \in \text{Col}(A)$. Therefore, the normal equations $X^\top X\beta = X^\top Y$ have a solution if and only if $X^\top Y \in \text{Col}(X^\top X)$. Observe that $\text{Col}(X^\top X) \subseteq \text{Col}(X^\top)$. Because every vector in $\text{Col}(X^\top X)$ has the form $X^\top Xv = X^\top(Xv)$ for some $v \in \mathbb{R}^p$ and is therefore in $\text{Col}(X^\top)$. Moreover $\text{rank}(X^\top X) = \text{rank}(X) = \text{rank}(X^\top)$. Therefore, $\text{Col}(X^\top X)$ and $\text{Col}(X^\top)$ are subspaces of the same space $\mathbb{R}^p$, one is contained in the other, and they have the same dimension therefore $\text{Col}(X^\top X) = \text{Col}(X^\top)$. Since $X^\top Y$ is by definition an element of $\text{Col}(X^\top)$ it follows that $X^\top Y \in \text{Col}(X^\top) = \text{Col}(X^\top X)$.

Consequently, the normal equations are always consistent regardless of whether X has full column rank. Thus, a least-squares solution always exists.

When X is rank deficient, the solution need not be unique.

Theorem 3.3. *Let $\hat{\beta}$ be the least squares estimate that minimizes the least squares objective $\|Y - X\beta\|_2^2$ for some design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and response vector $Y \in \mathbb{R}^n$. Then,*

$$\mathbb{E}[\hat{\beta} \mid X] = \beta.$$

That is, $\hat{\beta}$ is an unbiased estimate of the true β.

Proof. We aim to prove Theorem 3.3.

There are two components to this. First, the unbiasedness property of the least squares estimate and the requirement that our errors ε are normally distributed with mean 0 and variance $\sigma^2 I_n$.

We prove unbiasedness first. First, substitute the normal definition for $\hat{\beta}$ and $Y = X\beta + \varepsilon$,

$$\begin{aligned}
\hat{\beta} &= (X^\top X)^{-1}X^\top Y \\
&= (X^\top X)^{-1}X^\top(X\beta + \varepsilon) \\
&= \cancel{(X^\top X)^{-1}X^\top X}\beta + (X^\top X)^{-1}X^\top\varepsilon \\
&= \beta + (X^\top X)^{-1}X^\top\varepsilon.
\end{aligned}$$

Taking expectations,

$$\begin{aligned}
\mathbb{E}[\hat{\beta} \mid X] &= \mathbb{E}[\beta] + (X^\top X)^{-1}X^\top\mathbb{E}[\varepsilon] \\
&= \beta + (X^\top X)^{-1}X^\top\mathbb{E}[\varepsilon].
\end{aligned}$$

In order for $\hat{\beta}$ to be an unbiased estimate of β, we require that $\mathbb{E}[\hat{\beta} \mid X] = \beta$ which occurs when $(X^\top X)^{-1}X^\top\mathbb{E}[\varepsilon \mid X] = 0$. Since X is full column rank, then $X^\top X$ is an invertible $p \times p$ matrix so multiplying both sides by $X^\top X$,

$$\begin{aligned}
(X^\top X)^{-1}X^\top\mathbb{E}[\varepsilon \mid X] = 0 &\iff (X^\top X)(X^\top X)^{-1}X^\top\mathbb{E}[\varepsilon \mid X] = (X^\top X)\cdot 0 \\
&\iff X^\top\mathbb{E}[\varepsilon \mid X] = 0.
\end{aligned}$$

Therefore, $\mathbb{E}[\varepsilon \mid X]$ must lie in the nullspace of $X^\top$. Otherwise, we could decompose $\mathbb{E}[\varepsilon \mid X]$ into an orthogonal component and a component spanning the columns of X. Suppose we could decompose $\mathbb{E}[\varepsilon \mid X]$ such that $g(X) = X \cdot c + v$ where $X^\top v = 0$ and $c \neq 0$. Plugging this into $X^\top\mathbb{E}[\varepsilon \mid X] = 0$,

$$X^\top(Xc + v) = 0 \iff X^\top Xc + X^\top v = 0 \iff X^\top Xc = 0.$$

Since $X^\top X$ is positive definite, $X^\top Xc = 0$ implies that $c = 0$.

Therefore $\mathbb{E}[\varepsilon \mid X]$ cannot contain any component that projects onto X and we conclude that $\mathbb{E}[\varepsilon \mid X] = 0$. Substituting this property into our original expectation of $\hat{\beta}$,

$$\mathbb{E}[\hat{\beta} \mid X] = \beta + (X^\top X)^{-1}X^\top \cdot \vec{0} = \beta.$$

Therefore $\hat{\beta}$ is an unbiased estimate of β. □

To prove $\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2(X^\top X)^{-1}$, we must define another theorem first regarding ε.

Theorem 3.4. *Let $Y = X\beta + \varepsilon$ such that $\mathbb{E}[\varepsilon \mid X] = 0$ and $\mathrm{Var}(\varepsilon \mid X) = \Sigma$. The ordinary least squares estimator $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ minimizes estimation variance among all linear unbiased estimators if and only if $\Sigma = \sigma^2 I_n$.*

Proof. Let $\tilde{\beta} = CY$ be any linear estimator of β where $C \in \mathbb{M}_{p\times n}(\mathbb{R})$ depending on X. For $\tilde{\beta}$ to be unbiased for any parameter vector β,

$$\begin{aligned}\mathbb{E}[\tilde{\beta} \mid X] &= \mathbb{E}[C(X\beta + \varepsilon) \mid X] \\ &= CX\beta + C\mathbb{E}[\varepsilon \mid X] \\ &= CX\beta,\end{aligned}$$

since we know that $\mathbb{E}[\varepsilon \mid X] = 0$. For $\mathbb{E}[\tilde{\beta} \mid X] = \beta$ to hold for all $\beta \in \mathbb{R}^p$, we must define $CX = I_p$. Suppose we let $C = (X^\top X)^{-1}X^\top + D$ where $D \in \mathbb{M}_{p\times n}(\mathbb{R})$ representing the deviation from the OLS matrix. Plugging this into the unbiased constraint that $CX = I_p$,

$$\begin{aligned}\left((X^\top X)^{-1}X^\top + D\right)X = I_p &\iff (X^\top X)^{-1}(X^\top X) + DX = I_p \\ &\iff I_p + DX = I_p \\ &\iff DX = 0_{p\times p}.\end{aligned}$$

Thus, unbiasedness requires that all the rows of D lie in the nullspace of $X^\top$.

The conditional variance of $\tilde{\beta} = CY$ is $\mathrm{Var}(\tilde{\beta} \mid X) = C\Sigma C^\top = \sigma^2 CC^\top$.

$$\begin{aligned}\mathrm{Var}(\tilde{\beta} \mid X) &= \sigma^2\left[(X^\top X)^{-1}X^\top + D\right]\left[(X^\top X)^{-1} + D\right]^\top \\ &= \sigma^2[(X^\top X)^{-1}X^\top + D][X(X^\top X)^{-1} + D^\top] \\ &= \sigma^2[(X^\top X)^{-1}X^\top X(X^\top X)^{-1} + (X^\top X)^{-1}X^\top D^\top + DX(X^\top X)^{-1} + DD^\top].\end{aligned}$$

Substituting $DX = 0$ and $X^\top D^\top = (DX)^\top = 0$,

$$\begin{aligned}\mathrm{Var}(\tilde{\beta} \mid X) &= \sigma^2[(X^\top X)^{-1} + 0 + 0 + DD^\top] \\ &= \sigma^2(X^\top X)^{-1} + \sigma^2 DD^\top \\ &= \mathrm{Var}(\hat{\beta} \mid X) + \sigma^2 DD^\top.\end{aligned}$$

For any non-zero vector $a \in \mathbb{R}^p$,

$$a^\top(\sigma^2 DD^\top)a = \sigma^2\|D^\top a\|_2^2 \geq 0.$$

Therefore $\sigma^2 DD^\top$ is a positive semi-definite matrix. The variance matrix $\mathrm{Var}(\tilde{\beta} \mid X)$ is strictly minimized if and only if $DD^\top = 0$ or $D \in \mathbb{M}_{p\times n}(\mathbb{R})$. This proves that $C = (X^\top X)^{-1}X^\top$ making the ordinary least squares the unique minimum variance linear unbiased estimator when $\Sigma = \sigma^2 I_n$.

Consider an arbitrary covariance matrix $\Sigma \neq \sigma^2 I_n$. Calculating the variance of $\tilde{\beta} = \left((X^\top X)^{-1}X^\top + D\right)Y$ using the general variance formula,

$$\begin{aligned}\mathrm{Var}(\tilde{\beta} \mid X) &= [(X^\top X)^{-1}X^\top + D]\Sigma[X(X^\top X)^{-1} + D^\top] \\ &= (X^\top X)^{-1}X^\top \Sigma X(X^\top X)^{-1} + (X^\top X)^{-1}X^\top \Sigma D^\top + D\Sigma X(X^\top X)^{-1} + D\Sigma D^\top.\end{aligned}$$

For ordinary least squares, where $D = 0$, to remain the variance-minimizing choice, the linear cross-term containing D must vanish for all metrics D satisfying $DX = 0$. That is $D\Sigma X = 0_{p\times p}$ for all D such that $DX = 0$. The condition that $DX = 0$ implies that $D(\Sigma X) = 0$ meaning that every column of ΣX must lie in the column space of X. Moreover, X must form an invariant subspace of Σ, that is, $\Sigma X = XQ$ for some $p \times p$ matrix Q.

For this condition to hold universally across any arbitrary full-rank design matrix X, Σ must map every subspace of $\mathbb{R}^n$ to itself. By standard linear algebra, the only matrices that leave every subspace invariant are scalar multiples of the identity matrix. Therefore, $\Sigma = \sigma^2 I_n$. □

Theorem 3.5. *Let $\hat{\beta}$ be the ordinary least squares estimate that minimizes the least squares objective $\|Y - X\beta\|_2^2$ for some design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and response vector $Y \in \mathbb{R}^n$. Then,*

$$\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X^\top X)^{-1}.$$

Proof. Using the results from the proofs of Theorem 3.3 and 3.4,

$$\begin{aligned}\mathrm{Var}(\hat{\beta} \mid X) &= \mathrm{Var}\left((X^\top X)^{-1}X^\top Y, \middle| , X\right) \\ &= (X^\top X)^{-1}X^\top \mathrm{Var}(Y \mid X)\left((X^\top X)^{-1}X^\top\right)^\top \\ &= (X^\top X)^{-1}X^\top \mathrm{Var}(X\beta + \varepsilon \mid X)X(X^\top X)^{-1} \\ &= (X^\top X)^{-1}X^\top \mathrm{Var}(\varepsilon \mid X)X(X^\top X)^{-1} \\ &= (X^\top X)^{-1}X^\top \left(\sigma^2 I_n\right)X(X^\top X)^{-1}, \qquad \text{by Theorem 3.4} \\ &= \sigma^2 (X^\top X)^{-1}.\end{aligned}$$

□

Theorem 3.6 (Gauss-Markov Assumptions)**.** *Let $\hat{\beta}$ be the least squares estimate that minimizes $\|Y - X\beta\|_2^2$ for some design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and response vector $Y \in \mathbb{R}^n$. Then, the model,*

$$Y = X\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0, \quad \mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n,$$

is linear in its parameters where X is a design matrix with full column rank.

Proof. Theorems 3.2 through 3.5 prove these assumptions are necessary and sufficient. □

Theorem 3.7 (Gauss-Markov Theorem)**.** *Consider the linear regression model $Y = X\beta + \varepsilon$, where $Y \in \mathbb{R}^n$ is an observable response vector, $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ is a fixed design matrix with full column rank ($rank(X) = p$), $\beta \in \mathbb{R}^p$ is an unknown parameter vector, and $\varepsilon \in \mathbb{R}^n$ is an unobservable random error vector. Suppose the conditions in Theorem 3.7 hold:*

1. ***Strict Exogeneity:*** $\mathbb{E}[\varepsilon \mid X] = 0$
2. ***Spherical Errors:*** $Var(\varepsilon \mid X) = \sigma^2 I_n \quad$ *for some scalar* $\sigma^2 > 0$

Then, the Ordinary Least Squares (OLS) estimator

$$\hat{\beta}_{OLS} = (X^\top X)^{-1} X^\top Y$$

is the ***Best Linear Unbiased Estimator (BLUE)*** *of β. That is, for any other linear estimator $\tilde{\beta} = CY$ (where $C \in \mathbb{M}_{p\times n}(\mathbb{R})$) that is conditionally unbiased ($\mathbb{E}[\tilde{\beta} \mid X] = \beta$ for all $\beta \in \mathbb{R}^p$), the difference between their conditional covariance matrices is positive semi-definite:*

$$Var(\tilde{\beta} \mid X) - Var(\hat{\beta}_{OLS} \mid X) \succeq 0.$$

Proof. Use Theorem 3.4 and 3.5 and their corresponding proofs. □

Now that we have established the geometric structure of what the least squares estimates are, it remains for us to characterize how to conduct inference using ordinary least squares i.e., obtain fitted values and characterize the distribution.

Consider the fitted response vector $\hat{Y} = X\hat{\beta}$. Expanding using the formula for $\hat{\beta}$,

$$\hat{Y} = X\hat{\beta} = X(X^\top X)^{-1} X^\top Y.$$

We obtain a fascinating matrix that precedes our response vector Y, specifically,

$$H = X(X^\top X)^{-1} X^\top,$$

which we'll define as the hat matrix since it puts the hat on the fitted values that we obtain. It is more precisely referred to as the orthogonal projection matrix that projects any vector in $\mathbb{R}^n$ onto the column space of a design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$.

The hat matrix possesses some important properties.

Theorem 3.8. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and $Y \in \mathbb{R}^n$. Then,*

$$H = X(X^\top X)^{-1} X^\top,$$

defined as the hat matrix, is idempotent and symmetric.

Proof. Both proofs follows algebraically.

For idempotence, we show that $H^2 = H$.

$$\begin{aligned} H^2 &= \left(X(X^\top X)^{-1}X^\top\right)^2 \\ &= X(X^\top X)^{-1}X^\top \left(X(X^\top X)^{-1}X^\top\right)^\top \\ &= X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top \\ &= X(X^\top X)^{-1}X^\top \\ &= H. \end{aligned}$$

For symmetry, we show that $H^\top = H$,

$$\begin{aligned} H^\top &= \left(X(X^\top X)^{-1}X^\top\right)^\top \\ &= X^\top \left((X^\top X)^{-1}\right)^\top \left(X^\top\right)^\top \\ &= X(X^\top X)^{-1}X^\top \\ &= H. \end{aligned}$$

Therefore, since $H^2 = H$ and $H^\top = H$, then H is idempotent and symmetric. □

This is quite significant as it means that applying the same linear transformation (i.e., the orthogonal projection matrix) twice does nothing. That is, once we orthogonally project a vector in $\mathbb{R}^n$ onto a column space spanned by the columns of our design matrix X, applying the same projection does nothing the second time. As soon as $Y - X\hat{\beta}$, which it is the first time, it cannot be orthogonal again once it is already orthogonal.

By the spectral theorem, clearly, H is orthogonally diagonalizable. That is, because H is symmetric and real, we can decompose it as $Q\Lambda Q^\top$ where Q is an orthogonal matrix such that its columns form an orthonormal basis, and Λ is a diagonal matrix containing the real eigenvalues of our hat matrix H. By idempotence and the spectral decomposition, we previously established that the eigenvalues of the hat matrix must be either 1 or 0 since it must be true that $Q\Lambda^2 Q^\top = Q\Lambda Q^\top$ which is only true for eigenvalues that satisfy the condition $\lambda^2 = \lambda$.

Clearly, the only eigenvalues that satisfy such a condition are $\{0, 1\}$

Now that we have a rigorous geometric setup for least squares, we can now perform inference on our least squares estimates and characterize the sampling distributions of our estimates.

Assume the typical least squares model $Y = X\beta + \varepsilon$. We obtain our fitted values by letting,

$$\hat{Y} = X\hat{\beta} = X(X^\top X)^{-1}X^\top Y = HY.$$

Then, the error between our predictions $\hat{Y}$ and the actual responses is,

$$r = \hat{Y} - Y = HY - Y = (I - H)Y.$$

This is our residual vector.

Theorem 3.9. *Define,*

$$H = X(X^\top X)^{-1}X^\top,$$

as the hat matrix. Then $(I - H)$ is idempotent and symmetric.

Proof. Since the identity matrix I is symmetric and the hat matrix H is also symmetric,

$$(I - H)^\top = I^\top - H^\top = I - H.$$

Therefore, $I - H$ is also symmetric.

To prove idempotence we simply expand,

$$\begin{aligned}(I - H)^2 &= (I - H)(I - H)\\ &= I \cdot I - I \cdot H - H \cdot I + H^2\\ &= I^2 - 2I \cdot H + H^2\\ &= I - 2H - H, \qquad \text{by idempotence of } H\\ &= I - H.\end{aligned}$$

Therefore $(I - H)$ is idempotent and symmetric. □

Recall that our residuals are orthogonal to the subspace spanned by our columns of the design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$. Since X is assumed to be full rank, there are precisely $\text{rank}(X) = p$ columns in this subspace. Because we are splitting an n dimensional space into the column space of X and its orthogonal complement, the n dimensional residual vector $r = (I - H)Y$ can only move within an $(n - p)$-dimensional hyperplane.

Theorem 3.10. *Suppose $r = (I - H)Y \in \mathbb{R}^n$ is our n-dimensional residual vector resulting from taking the difference of our fitted least squares estimates and the actual response vector $Y \in \mathbb{R}^n$. Then, for our design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$*

$$X^\top r = 0, \qquad \hat{Y}^\top r = 0.$$

Proof. This theorem shows that the residual vector is orthogonal to our design matrix and our fitted response vector. Substituting $r = (I - H)Y$ into $X^\top r$,

$$\begin{aligned}X^\top r &= X^\top (I - H)Y\\ &= X^\top (I - X(X^\top X)^{-1}X^\top)Y\\ &= \left(X^\top - X^\top X(X^\top X)^{-1}X^\top\right)Y\\ &= \left(X^\top - X^\top\right)Y = 0.\end{aligned}$$

Similarly, substituting $\hat{Y} = X\hat{\beta}$ and $r = (I - H)Y$ into the second equality,

$$\begin{aligned}
\hat{Y}^\top r &= (X\hat{\beta})^\top (I - H)Y \\
&= X^\top \hat{\beta}^\top (I - X(X^\top X)^{-1}X^\top)Y \\
&= \left(X^\top \hat{\beta}^\top - \hat{\beta}^\top X^\top X(X^\top X)^{-1}X^\top\right) Y \\
&= \left(X^\top \hat{\beta}^\top - X^\top \hat{\beta}^\top\right) Y \\
&= 0.
\end{aligned}$$

Therefore, r is indeed orthogonal to X and $\hat{Y}$. □

For the matrix case, the zero mean and constant variance properties of the residual vector hold as well. By properties of $I - H$, the hat matrix,

$$\begin{aligned}
\mathbb{E}[r] &= \mathbb{E}[(I - H)Y] \\
&= (I - H)\mathbb{E}[Y] \\
&= (I - H)X\beta \\
&= X\beta - X(X^\top X)^{-1}X^\top X\beta \\
&= X\beta - X\beta = 0.
\end{aligned}$$

For the variance, the same substitution follows,

$$\begin{aligned}
\operatorname{Var}(r) &= \operatorname{Var}((I - H)Y) \\
&= (I - H)\operatorname{Var}(Y)(I - H)^\top \\
&= (I - H)\sigma^2 I(I - H)^\top \\
&= \sigma^2(I - H)(I - H), \qquad \text{by Theorem 3.9} \\
&= \sigma^2(I - H).
\end{aligned}$$

Remark 3.11. When we regress Y on X, we are regressing a vector Y in $\mathbb{R}^n$ on a subspace spanned by a full column rank matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$. That is, we are decomposing the "total" information contained in the vector Y into two mutually orthogonal subspaces. Constructing the hat matrix $H = X(X^\top X)^{-1}X^\top$ gives its complementary residual marker matrix $M_X = (I_n - H_X)$.

Applying these matrices to Y yields a decomposition

$$HM_X = (I_n - H_X)Y \iff Y = HM_X + H_X Y.$$

$H_X Y$ is precisely $\hat{Y}$ representing the synthesized information which is precisely the projection of Y onto $\operatorname{Col}(X)$ which is completely redundant to the subspace spanned by the columns of our design matrix X. Letting $e = M_X Y$, instead, represents the remaining information. Because M_X orthogonally projects onto $\operatorname{null}(X^\top)$, it is orthogonal to every vector in the column space of X and represents the shortest euclidean path from Y to the hyperplane $\operatorname{Col}(X)$.

3.2 Finite-Sample distribution theory

Up until now, we have made rudimentary necessary and sufficient assumptions regarding the variance and expectation of our least squares model. However, the linear regression model $Y = X\beta + \varepsilon$ does not by itself imply any distribution of either Y or ε but rather only states that the observed response vector Y can be decomposed into a systematic component $X\beta$ and an error component ε.

Recall the usual assumptions that are stated conditionally on the design matrix. We first showed that $\mathbb{E}[\varepsilon \mid X] = 0$ we necessary and sufficient, which implied that $\mathbb{E}[Y \mid X] = X\beta$. Thus, the zero conditional mean assumption says that $X\beta$ is the conditional mean of Y which rules our systematic information remaining in the error term once X is known.

Separately, we also showed that $\text{Var}(\varepsilon \mid X) = \sigma^2 I_n$ which means that $\text{Var}(Y \mid X) = \sigma^2 I_n$ because $X\beta$ is fixed once we condition on our design matrix X. It is important to note that this covariance assumption does not require normality and the constant variance assumption only states that every error has the same conditional variance and that distinct errors are conditionally uncorrelated.

Thus, zero conditional covariance across observations allows the errors to satisfy the conditions while being skewed, heavy-tailed, discrete, or otherwise non-normal. Thus, to obtain finite-sample distribution theorem, we impose a normality assumption.

The classical normal linear model assumes that,

$$\varepsilon \mid X \sim \mathcal{N}_n(0, \sigma^2 I_n).$$

Because $X\beta$ is a fixed vector conditional on X, adding it to a multivariate normal vector shifts the mean without changing the covariance matrix. Consequently, conditional on X, our responses,

$$Y \mid X = X\beta + \varepsilon \mid X \sim \mathcal{N}_n(X\beta, \sigma^2 I_n).$$

Thus, we see that $Y \mid X$ is a multivariate normal with mean $X\beta$ and covariance matrix $\sigma^2 I_n$ when we impose the conditional normality assumption on our errors. The converse holds as well. Since $\varepsilon = Y - X\beta$, then if $Y \mid X \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$, then subtracting the fixed conditional mean $X\beta$ gives $\varepsilon \mid X \sim \mathcal{N}_n(0, \sigma^2 I_n)$. Therefore, conditional on X, $\varepsilon \mid X \sim \mathcal{N}_n(0, \sigma^2 I_n)$ if and only if $Y \mid X \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$. Conditional on X, the vector $X\beta$ is fixed so the normal-shift argument works directly. Marginally, however, $Y = X\beta + \varepsilon$ contains the random term $X\beta$.

Even when $\varepsilon \mid X$ is normal, the unconditional distribution of Y need not be normal unless additional assumptions are imposed on X. It is very possible that the unconditional distribution may be a mixture of normal distributions whose means vary with X.

In this section, I aim to derive the sampling distributions of our least squares estimates and inference under normality assumptions. First, we start with an elementary theorem.

We often do not know the population variance and are thus left with estimating it.

Theorem 3.12. *Let $\sigma^2 \in \mathbb{R}$ such that $\sigma^2 > 0$. Then the estimator,*

$$\hat{\sigma}^2 = \frac{1}{n-(p+1)} \sum_{i=1}^{n} r^\top r,$$

is an unbiased estimator of σ^2.

Proof. We show that $\mathbb{E}[\hat{\sigma}^2 \mid X] = \sigma^2$.

Recall that hat matrix $H = X(X^\top X)^{-1}X^\top$ be the projection matrix onto $\text{Col}(X)$ and let $M = I_n - H$ be the residual-maker matrix. Since $r = Y - X\hat{\beta} = MY$ and $Y = X\beta + \varepsilon$, we can expand MX as,

$$MX = \left(I_n - X(X^\top X)^{-1}X^\top\right) X = X - X(X^\top X)^{-1}X^\top X = X - X = 0.$$

Therefore, the residual can be re-written as,

$$r = M(X\beta + \varepsilon) = MX\beta + M\varepsilon = M\varepsilon.$$

Then, the expectation of the estimator becomes,

$$\begin{aligned}
\mathbb{E}[\hat{\sigma}^2] &= \mathbb{E}\left[\frac{1}{n-(p+1)} r^\top r \mid X\right] \\
&= \frac{1}{n-(p+1)} \mathbb{E}[r^\top r \mid X] \\
&= \frac{1}{n-(p+1)} \mathbb{E}[(M\varepsilon)^\top (M\varepsilon) \mid X] \\
&= \frac{1}{n-(p+1)} \mathbb{E}[\varepsilon^\top M^\top M\varepsilon \mid X] \\
&= \frac{1}{n-(p+1)} \mathbb{E}[\varepsilon^\top M\varepsilon \mid X], \qquad \text{by symmetry of } M.
\end{aligned}$$

Since a scalar is equal to its trace,

$$\varepsilon^\top M\varepsilon = \text{tr}(\varepsilon M\varepsilon) = \text{tr}(M\varepsilon\varepsilon^\top).$$

Thus the expectation of the estimator follows as,

$$\begin{aligned}
\mathbb{E}[\hat{\sigma}^2 \mid X] &= \frac{1}{n-(p+1)} \mathbb{E}[\text{tr}(M\varepsilon\varepsilon^\top)\top X] \\
&= \frac{1}{n-(p+1)} \text{tr}\left(M\mathbb{E}[\varepsilon\varepsilon^\top \mid X]\right)
\end{aligned}$$

Because $\mathbb{E}[\varepsilon \mid X] = 0$ and $\text{Var}(\varepsilon \mid X) = \sigma^2 I_n$,

$$\mathbb{E}[\varepsilon\varepsilon^\top \mid X] = \text{Var}(\varepsilon \mid X) + \mathbb{E}[\varepsilon \mid X]\mathbb{E}[\varepsilon \mid X]^\top = \text{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

It follows that,

$$\begin{aligned}\mathbb{E}[\hat{\sigma}^2 \mid X] &= \frac{1}{n-(p+1)} \operatorname{tr}(M\sigma^2 I_n) \\ &= \frac{\sigma^2}{n-(p+1)} \operatorname{tr}(M) = \frac{\sigma^2}{n-(p+1)} \operatorname{tr}(I_n - H) = \frac{\sigma^2}{n-(p+1)} \left[\operatorname{tr}(I_n) - \operatorname{tr}(H)\right].\end{aligned}$$

Since H is the orthogonal projection matrix onto $\mathrm{Col}(X)$ and X has $p+1$ linearly independent columns, then its trace,

$$\operatorname{tr}(H) = \operatorname{rank}(H) = \operatorname{rank}(X) = p+1.$$

Then,

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{\sigma^2}{n-(p+1)} \cdot (n-(p+1)) = \sigma^2.$$

□

Remark 3.13. The fitted hat matrix is defined as,

$$H = X(X^\top X)^{-1} X^\top,$$

and projects a vector $Y \in \mathbb{R}^n$ onto to the column space of $X \in \mathbb{M}_{n\times p}(\mathbb{R})$. If X has $p+1$ linearly independent columns including the intercept then,

$$\dim \mathrm{Col}(X) = \operatorname{rank}(X) = p+1.$$

We know that H has eigenvalues $\{0,1\}$ meaning that it has exactly $p+1$ eigenvalues equal to 1 corresponding to the $p+1$ dimensions in $\mathrm{Col}(X)$. Therefore the trace of H is simply,

$$\operatorname{tr}(H) = \sum_{i=1}^{p+1} \lambda_i(H) = p+1.$$

The residual-maker matrix is $M = I_n - H$, which projects Y onto the orthogonal complement of $\mathrm{Col}(X)$. Consequently,

$$\operatorname{tr}(M) = \operatorname{tr}(I_n - H) = \operatorname{tr}(I_n) - \operatorname{tr}(H) = n - (p+1.)$$

Thus $\operatorname{tr}(M) = n-(p+1)$ which is precisely the number of degrees of freedom. Y has n independent directions so estimating $p+1$ regression coefficients uses $p+1$ directions of variation. The residual vector must satisfy $X^\top r = 0$ so it is restricted to the orthogonal complement of the $p+1$-dimensional column space of X. That residual subspace has dimension $n-(p+1)$.

Theorem 3.14. *Let $X \in \mathbb{M}_{n\times(p+1)}(\mathbb{R})$ be our fixed real design matrix satisfying the condition that $\operatorname{rank}(X) = p+1 < n$. Under normality conditions, assume that $Y \mid X \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$ where $\beta \in \mathbb{R}^{p+1}$ and $\sigma^2 > 0$. Define $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ as our least squares estimate and,*

$$\hat{\sigma}^2 = \frac{1}{n-(p+1)} r^\top r = \frac{1}{n-(p+1)}(Y - X\hat{\beta})(Y - X\hat{\beta}).$$

Then,

(i) $\hat{\beta} \mid X \sim \mathcal{N}_{p+1}(\beta, \sigma^2(X^\top X)^{-1})$,
(ii) $\hat{\beta}$ *and* $\hat{\sigma}^2$ *are independent conditional on* X,
(iii) $\dfrac{(n-(p+1))\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-(p+1)}$.

Proof. Assume that $Y \mid X \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$ is a multivariate normal conditional random variable.

Proof of (i). Of course, taking expectations yields,

$$\mathbb{E}[Y \mid X] = \mathbb{E}[X\beta + \varepsilon \mid X] = X\beta + \mathbb{E}[\varepsilon \mid X] = X\beta,$$

since $\mathbb{E}[\varepsilon \mid X] = 0$. Moreover, taking variances gives us,

$$\operatorname{Var}(Y \mid X) = \operatorname{Var}(X\beta + \varepsilon \mid X) = \operatorname{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

By properties of multivariate normal,

$$\begin{aligned}
\mathbb{E}[\hat{\beta} \mid X] &= \mathbb{E}[(X^\top X)^{-1}X^\top Y \mid X] \\
&= (X^\top X)^{-1}X^\top \mathbb{E}[Y \mid X] \\
&= (X^\top X)^{-1}X^\top \mathbb{E}[X\beta + \varepsilon \mid X] \\
&= \mathbb{E}[\beta \mid X] = \beta.
\end{aligned}$$

Moreover,

$$\begin{aligned}
\operatorname{Var}(\hat{\beta} \mid X) &= \operatorname{Var}((X^\top X)^{-1}X^\top Y \mid X) \\
&= \operatorname{Var}((X^\top X)^{-1}X^\top (X\beta + \varepsilon) \mid X) \\
&= \operatorname{Var}((X^\top X)^{-1}X^\top X\beta + (X^\top X)^{-1}X^\top \varepsilon \mid X) \\
&= \operatorname{Var}(\beta + (X^\top X)^{-1}X^\top \varepsilon \mid X) \\
&= \operatorname{Var}((X^\top X)^{-1}X^\top \varepsilon \mid X) \\
&= (X^\top X)^{-1}X^\top \operatorname{Var}(\varepsilon \mid X)\left[(X^\top X)^{-1}X^\top\right]^\top \\
&= (X^\top X)^{1}X^\top \operatorname{Var}(\varepsilon \mid X)X(X^\top X)^{-1} \\
&= \operatorname{Var}(\varepsilon \mid X)(X^\top X)^{-1} \\
&= \sigma^2 I_n (X^\top X)^{-1} \\
&= \sigma^2 (X^\top X)^{-1}.
\end{aligned}$$

Therefore, it follows, under normality conditions on X, $\hat{\beta} \sim \mathcal{N}_n(\beta, \sigma^2(X^\top X)^{-1})$.

Proof of (ii). We show that $\text{Cov}(\hat{\beta}, r) = 0$. Assume that $\hat{\beta} \mid X \sim \mathcal{N}_n(\beta, \sigma^2(X^\top X)^{-1})$ and recall that we previously let our residuals $r = (I - H)Y \sim \mathcal{N}(0, \sigma^2(I - H))$ where H is our hat matrix.

The covariance between our least squares estimates and the residuals is,

$$\begin{aligned}
\text{Cov}(\hat{\beta}, r) &= \text{Cov}\left((X^\top X)^{-1}X^\top Y, (I - H)Y\right) \\
&= (X^\top X)^{-1}X^\top \text{Var}(Y)(I - H)^\top, \qquad \text{by commutativity of covariance,} \\
&= \sigma^2(X^\top X)^{-1}X^\top(I - X(X^\top X)^{-1}X^\top) \\
&= \sigma^2(X^\top X)^{-1}X^\top - \sigma^2 \cdot X^\top X(X^\top X)^{-1}(X^\top X)^{-1}X^\top \\
&= \sigma^2(X^\top X)^{-1}X^\top - \sigma^2 \cdot (X^\top X)^{-1}X^\top \\
&= 0 \in \mathbb{M}_{(p+1)\times n}.
\end{aligned}$$

Since $\text{Cov}(\hat{\beta}, r) = 0$ implies they are independent, and since the sample variance estimator $\hat{\sigma}^2$ is a function of the residuals, therefore it follows that our least squares estimates are independent of the sample variance estimator $\hat{\sigma}^2$. Therefore, we write $\hat{\beta} \perp \hat{\sigma}^2$.

Proof of (iii). Using the definition of our sample variance-covariance estimator,

$$\hat{\sigma}^2 = \frac{1}{n - (p + 1)} r^\top r,$$

recall that $r = (I - H)Y \sim \mathcal{N}(0, \sigma^2(I - H))$. Re-arranging for $r^\top r$,

$$r^\top r = \frac{\hat{\sigma}^2}{n - (p + 1)}.$$

By sampling distribution theory, we know that,

$$\frac{(n - (p + 1))\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-(p+1)}.$$

We can re-arrange $r^\top r$ into,

$$\frac{r^\top r}{\sigma^2} = \left(\frac{r}{\sigma}\right)^\top \left(\frac{r}{\sigma}\right), \qquad \frac{r}{\sigma} = \frac{(I - H)Y}{\sigma} \sim \mathcal{N}(0, I - H).$$

Recall that for a symmetric and idempotent matrix $I - H$, we can find an orthogonal matrix P and diagonal matrix Λ with the eigenvalues of $I - H$ such that $I - H = P\Lambda P^\top$ or alternatively where $P^\top(I - H)P = \Lambda$. Define Z such that,

$$\begin{aligned}
Z &\sim \mathcal{N}(0, P^\top(I - H)P) \\
&\sim \mathcal{N}(P^\top P\Lambda P^\top P) \\
&\sim \mathcal{N}(0, \Lambda).
\end{aligned}$$

Then, $Z_1, \ldots, Z_{n-(p+1)}$ are independently and identically distribution standard multivariate normal random variables such that $Z_{n-p}, \ldots, Z_n$ are all zero matrices. We can write,

$$Z = P^\top \left(\frac{r}{\sigma}\right) \iff PZ = PP^\top \left(\frac{r}{\sigma}\right) = \frac{r}{\sigma}.$$

Therefore,

$$\left(\frac{r}{\sigma}\right)^\top \left(\frac{r}{\sigma}\right) = (PZ)^\top (PZ) = Z^\top P^\top PZ = Z^\top Z \sim \chi^2_{n-(p+1)},$$

since $Z \sim \mathcal{N}(0, (I - H))$ and χ^2 are the product of two standard normal random variables. □

Up until now, we have essentially only considered relationship between all p predictors in our design matrix and the response vector $Y \in \mathbb{R}^n$. However, there are instances in which we may be interested in a particular predictor. Suppose you are a prediction market quant, and you have a design matrix with p predictors derived from President Donald Trump's twitter feed. Suppose you wish to regress the response vector Y on these p predictors but are only interested in a single predictor (e.g., Trump's tendency to exaggerate) on the movements of certain aerospace stock movements. Suppose that particular predictor is at position j such that $1 \leq j \leq p$ in your design matrix. We are interested in performing inference on that single β_j.

Recall under normality assumptions, we previously had that $\hat{\beta} \mid X \sim \mathcal{N}_{p+1}(\beta, \sigma^2(X^\top X)^{-1})$. To visualize this completely, the parameters of the $(p+1)$-dimensional multivariate normal we are referring to visually is,

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \Bigg| \; X \sim \mathcal{N}_{p+1} \left(\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \begin{bmatrix} \text{Var}(\hat{\beta}_0 \mid X) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid X) & \cdots & \text{Cov}(\hat{\beta}_0, \hat{\beta}_p \mid X) \\ \text{Cov}(\hat{\beta}_1, \hat{\beta}_0 \mid X) & \text{Var}(\hat{\beta}_1 \mid X) & \cdots & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p \mid X) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\hat{\beta}_p, \hat{\beta}_0 \mid X) & \text{Cov}(\hat{\beta}_p, \hat{\beta}_1 \mid X) & \cdots & \text{Var}(\hat{\beta}_p \mid X) \end{bmatrix} \right).$$

Notice to obtain the j-th least squares estimate corresponding to the j-th predictor, we simply need the j-th index of β and j-th diagonal element of $\sigma^2(X^\top X)^{-1}$. Therefore, under normal singularity, we have that $\hat{\beta}_j \mid X \sim \mathcal{N}(\beta_j, \sigma^2 \Sigma_{jj})$ where Σ_{jj} is the j-th diagonal element $(X^\top X)^{-1}$.

We aim to test the hypothesis that $H : \beta_j = \beta_{j,0}$ so we need a discrepancy measure. We know that $\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2 \Sigma_{jj})$, then we standardize to obtain,

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2 \Sigma_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{\Sigma_{jj}}} \sim \mathcal{N}(0, 1).$$

However, since in practice we usually never know the true population mean, we aim to replace it with $\hat{\sigma}^2$ which adds degrees of freedom due to estimating $\hat{\sigma}^2$. Recall from sampling theory,

$$\frac{(n - (p+1))\hat{\sigma}^2}{\sigma^2} = \frac{(n-p-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p-1}.$$

By now, you should see that we aim to derive a Student-t distributed test statistic T which is simply a standard normal random variables divided by the root of chi-squared random variable scaled down by its degrees of freedom. Thus,

$$\sqrt{\frac{\frac{(n-p-1)\hat{\sigma}^2}{\sigma^2}}{n-p-1}} = \sqrt{\frac{\hat{\sigma}^2}{\sigma^2}} = \frac{\hat{\sigma}}{\sigma}.$$

Therefore,

$$\frac{\dfrac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\Sigma_{jj}}} \sim \mathcal{N}(0,1)}{\dfrac{\hat{\sigma}}{\sigma} \sim \dfrac{1}{n-p-1}\chi^2_{n-p-1}} = \frac{\hat{\beta}_j - \hat{\beta}_j}{\sqrt{\hat{\sigma}^2\Sigma_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\Sigma_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{\mathrm{SE}(\beta_j)} \sim t_{n-p-1}.$$

In the case that we wish to test the null hypothesis that there is no relationship i.e., $H : \beta_j = 0$, then we arrive at the formulation of the standardized coefficient or the *Z-score*,

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\Sigma_{jj}}} = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{\Sigma_{jj}}} = z_j.$$

If $\hat{\sigma} := \sigma$, we would arrive back at a standard normal random variable.

Theorem 3.15. *Suppose that $\hat{\beta} \in \mathbb{R}^p$, $Y \in \mathbb{R}^n$ and $X \in \mathbb{M}_{n\times(p+1)}(\mathbb{R})$ such that $\hat{\beta}$ minimizes the least squares problem $\|Y - X\beta\|_2^2$ for some vector β in $\mathbb{R}^p$. Then, a test statistic for β_j for some $1 \leq j \leq p$,*

$$T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\Sigma_{jj}}} \sim t_{n-p-1}.$$

At the $(1-\alpha)\%$ confidence level, a $100(1-\alpha)\%$ confidence interval for β_j is,

$$\mathbb{P}\left(-t_{1-\frac{\alpha}{2},n-p-1} \leq \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\Sigma_{jj}}} \leq t_{1-\frac{\alpha}{2},n-p-1}\right) = 1-\alpha.$$

That is, if we continuously drew new samples, computed $\hat{\beta}_j$, and constructed a confidence interval using this same procedure each, approximately 95% of those intervals would contain the fixed true coefficient β_j. 95% is the long run success rate of the sampling process.

Remark 3.16. We can re-arrange the formulation of the $100(1-\alpha)\%$ confidence interval for β_j,

$$\mathbb{P}\left(\hat{\beta}_j - t_{1-\frac{\alpha}{2},n-p-1}\cdot\hat{\sigma}\sqrt{\Sigma_{jj}} \leq \beta_j \leq \hat{\beta}_j + t_{1-\frac{\alpha}{2},n-p-1}\cdot\hat{\sigma}\sqrt{\Sigma_{jj}}\right) = 1-\alpha.$$

This gives us a more explicit interval by construction,

$$\left[\hat{\beta}_j - t_{1-\frac{\alpha}{2},n-p-1}\cdot\hat{\sigma}\sqrt{\Sigma_{jj}}, \hat{\beta}_j + t_{1-\frac{\alpha}{2},n-p-1}\cdot\hat{\sigma}\sqrt{\Sigma_{jj}}\right].$$

Geometrically, inference on a single β_j is fundamentally a projection problem. Using our prior notions of the hat matrix, residual-maker matrix and the multivariate normality assumption of the least squares estimate $\hat{\beta}$, in order to perform inference on a single β geometrically, we isolate the exact one-dimensional direction corresponding to the predictor j. To keep dimensions consistent, let the design matrix be $X \in \mathbb{M}_{n\times(p+1)}$ (including the intercept coefficient) and the let $\beta, \hat{\beta} \in \mathbb{R}^{p+1}$. Thus, we can write the design matrix completely as the single component x_j and the remaining component with the j-th component removed i.e., X_{-j},

$$X = \begin{bmatrix} x_j & X_{-j} \end{bmatrix}$$

where $x_j \in \mathbb{R}^n$ is indeed the predictor whose coefficient interests us and X_{-j} contains every other column, including the intercept when $j \neq 0$. Thus, we can write the vector correspondingly as,

$$\beta = \begin{bmatrix} \beta_j & \beta_{-j} \end{bmatrix}^\top.$$

Expanding the model into a rank-1 equation we get,

$$Y = X\beta + \varepsilon = \begin{bmatrix} x_j & X_{-j} \end{bmatrix} \begin{bmatrix} \beta_j \\ \beta_{-j} \end{bmatrix} + \varepsilon = x_j\beta_j + X_{-j}\beta_{-j} + \varepsilon.$$

We are not simply concerned with whether or not x_j is related to Y but whether the component of x_j that cannot be explained by the other predictors is related to the part of Y that cannot be explained by those predictors. Consider the projection matrix onto the column space of X_{-j} as,

$$H_{-j} = X_{-j}(X_{-j}^\top X_{-j})^{-1}X_{-j}^\top,$$

and is complementary residual-maker matrix as,

$$M_{-j} = I_n - H_{-j}.$$

Denote the residualizes x_j against all the other predictors as $\tilde{x}_j = M_{-j}x_j$. Because M_{-j} projects onto $\text{Col}(X_{-j})^\perp$, it follows that,

$$X_{-j}^\top \tilde{x}_j = X_{-j}^\top M_{-j} x_j = 0.$$

Substituting this into our previous formulation of x_j,

$$x_j = H_{-j}x_j + M_{-j}x_j = \hat{x}_j + \tilde{x}_j,$$

where $\hat{x}_j$ is the part of x_j explained by the other predictors while $\tilde{x}_j$ is the new direction contributed by x_j. The normal equations for the partitioned model are given as,

$$\begin{bmatrix} x_j^\top x_j & x_j^\top X_{-j} \\ X_{-j}^\top x_j & X_{-j}^\top X_{-j} \end{bmatrix} \begin{bmatrix} \hat{\beta}_j \\ \hat{\beta}_{-j} \end{bmatrix} = \begin{bmatrix} x_j^\top Y \\ X_{-j}^\top Y \end{bmatrix}.$$

Notice that the second linear equation yielded by the system above gives,

$$X_{-j}^\top x_j \hat{\beta}_j + X_{-j}^\top X_{-j}\hat{\beta}_{-j} = X_{-j}^\top Y \iff \hat{\beta}_{-j} = (X_{-j}^\top X_{-j})^{-1}X_{-j}^\top(Y - x_j\hat{\beta}_j).$$

Substituting this into our decomposition of x_j,

$$\begin{aligned} x_j^\top Y &= x_j^\top x_j\hat{\beta}_j + x_j^\top X_{-j}\hat{\beta}_{-j} \\ &= x_j^\top x_j\hat{\beta}_j + x_j^\top X_{-j}(X_{-j}^\top X_{-j})^{-1}X_{-j}^\top(Y - x_j\hat{\beta}_j). \end{aligned}$$

Solving for $\hat{\beta}_j$ we get,

$$\begin{aligned} x_j^\top \left[I_n - X_{-j}(X_{-j}^\top X_{-j})^{-1}X_{-j}^\top\right] x_j\hat{\beta}_j &= x_j^\top \left[I_n - X_{-j}(X_{-j}^\top X_{-j})^{-1}X_{-j}^\top\right] Y \\ x_j^\top M_{-j}x_j\hat{\beta}_j &= x_j^\top M_{-j}Y \\ \implies \hat{\beta}_j &= \frac{x_j^\top M_{-j}Y}{x_j^\top M_{-j}x_j}. \end{aligned}$$

By the idempotence and symmetry properties of our residual-maker matrix,

$$x_j^\top M_{-j}x_j = (M_{-j}x_j)^\top (M_{-j}x_j) = \tilde{x}_j^\top \tilde{x}_j = \|\tilde{x}_j\|_2^2.$$

Likewise, $x_j^\top M_{-j}Y = \tilde{x}_j^\top \tilde{Y}$ where $\tilde{Y} = M_{-j}Y$. Hence,

$$\hat{\beta}_j = \frac{x_j^\top M_{-j}Y}{x_j^\top M_{-j}x_j} = \frac{\tilde{x}_j^\top \tilde{Y}}{\tilde{x}_j^\top \tilde{x}_j}.$$

Notice this is precisely the coefficient obtained by regressing the residualized response $\tilde{Y}$ on the residualized predictor $\tilde{x}_j$. Applying the removed residual-maker matrix to the original model,

$$\begin{aligned} M_{-j}Y &= M_{-j}(x_j\beta_j + X_{-j}\beta_{-j} + \varepsilon) \\ &= M_{-j}x_j\beta_j + M_{-j}X_{-j}\beta_{-j} + M_{-j}\varepsilon, \qquad \text{by Orthogonality of } M_{-j}, \\ &= \tilde{x}_j\beta_j + M_{-j}\varepsilon. \end{aligned}$$

Therefore, solving for $\hat{\beta}_j$,

$$\hat{\beta}_j = \frac{\tilde{x}_j^\top (\tilde{x}_j\beta_j + M_{-j}\varepsilon)}{\tilde{x}_j^\top \tilde{x}_j} = \beta_j + \frac{\tilde{x}_j^\top M_{-j}\varepsilon}{\tilde{x}_j^\top \tilde{x}_j}.$$

Since $\tilde{x}_j = M_{-j}x_j$ lies in range(M_{-j}) then $x_j^\top M_{-j} = \tilde{x}_j^\top$. Consequently, their difference,

$$\hat{\beta}_j - \beta_j = \frac{\tilde{x}_j^\top \varepsilon}{\tilde{x}_j^\top \tilde{x}_j}, \qquad \varepsilon \mid X \sim \mathcal{N}_n(0, \sigma^2 I_n),$$

such that the scalar $\tilde{x}_j^\top \varepsilon$ satisfies the normality $\tilde{x}_j^\top \varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 \tilde{x}_j^\top \tilde{x}_j)$.

It follows that,

$$\hat{\beta}_j - \beta_j \mid X \sim \mathcal{N}\left(0, \frac{\sigma^2}{\tilde{x}_j^\top \tilde{x}_j}\right) \iff \text{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{x_j^\top M_{-j}x_j},$$

which must equal the j-th diagonal entry of the usual covariance matrix. This diagonal identity can be written following the Schur complement of the partitioned Gram matrix,

$$\left[(X^\top X)^{-1}\right]_{jj} = \frac{1}{x_j^\top M_{-j}x_j} = \frac{1}{\|\tilde{x}_j\|_2^2},$$

which quantifies what determines the uncertainty in $\hat{\beta}_j$ given as $\text{SE}(\hat{\beta}_j) = \frac{\hat{\sigma}}{\|M_{-j}x_j\|_2}$.

If x_j has a large component orthogonal to all other predictors, then the ℓ_2 norm $\|M_{-j}x_j\|_2$ is large and the standard error is small. The data contains a strong independent direction along which β_j can be identified. If x_j is nearly contained in $\text{Col}(X_{-j})$ then $M_{-j}x_j \approx 0$ so its squared ℓ_2 normal is approximately 0. Therefore, the standard errors of $\hat{\beta}_j$ becomes very large implying multicollinearity i.e., if there is almost no direction unique to predictor j, the model cannot distinguish its effect from the effects of the other predictors.

The resulting test statistic is,

$$T = \frac{\hat{\beta}_j - \beta_{j,0}}{\hat{\sigma}\sqrt{\left[(X^\top X)^{-1}\right]_{jj}}} = \frac{\hat{\beta}_j - \beta_{j,0}}{\frac{\hat{\sigma}}{\|M_{-j}x_j\|_2}} = \frac{(\hat{\beta}_j - \beta_{j,0})\|M_{-j}x_j\|_2}{\hat{\sigma}},$$

which estimates the displacement along the unique direction supplied by x_j relative to the residual noise level. In parameter-space, suppose we have the j-th standard basis vector $e_j \in \mathbb{R}^{p+1}$, then $\hat{\beta}_j = e_j^\top \hat{\beta}_j$ and $\beta_j = e_j^\top \beta$ and $\text{Var}(e_j^\top \hat{\beta} \mid X) = \sigma^2 e_j^\top (X^\top X)^{-1} e_j$. So, the null hypothesis $H_0 : \beta_j = \beta_{j,0}$ is the hyperplane $\{\beta \in \mathbb{R}^{p+1} : e_j^\top \beta = \beta_{j,0}\}$. The statistic,

$$T = \frac{e_j^\top \hat{\beta} - \beta_{j,0}}{\hat{\sigma}\sqrt{e_j^\top (X^\top X)^{-1} e_j}},$$

is therefore the signed standardized distance from the estimator vector $\hat{\beta}$ to that null hyperplane. This interpretation gives us the *Frisch-Waugh-Lovell Theorem.*

Theorem 3.17 (Frisch-Waugh-Lovell Theorem.)**.** *Let $Y = x_j\beta_j + X_{-j}\beta_{-j} + \varepsilon$ where,*

$$X = \begin{bmatrix} x_j & X_{-j} \end{bmatrix},$$

has full column rank. Define $M_{-j} = I_n - X_{-j}(X_{-j}^\top X_{-j})^{-1}X_{-j}^\top$. Then the least-squares coefficient $\hat{\beta}_j$ from the full regression of Y on x_j and X_{-j} equals the least-squares coefficient from the residualized regression,

$$M_{-j}Y = (M_{-j}x_j)\beta_j + \varepsilon.$$

More explicitly,

$$\hat{\beta}_j = \frac{(M_{-j}x_j)^\top (M_{-j}Y)}{(M_{-j}x_j)^\top (M_{-j}x_j)}.$$

Now that we have characterized the geometric and statistical properties of performing inference on our least squares estimates and a single estimate β_j, what if we choose to predict an entirely new response at a specified predictor vector. Assume the normal setup where $X \in \mathbb{M}_{n\times(p+1)}$, $Y = X\beta + \varepsilon$ and $\varepsilon \mid X \sim \mathcal{N}_n(0, \sigma^2 I_n)$ where X contains an intercept and p predictors. Assume that $\text{rank}(X) = p + 1 < n$. Consider a new observation with predictor vector.

$$a_* = \begin{bmatrix} 1 \\ x_{*1} \\ \vdots \\ x_{*p} \end{bmatrix} \in \mathbb{R}^{p+1}.$$

The new response follows the same regression model where $Y_* = a_*^\top \beta + \varepsilon_*$ where $\varepsilon_* \sim \mathcal{N}(0, \sigma^2)$ is independent of the errors used to fit the original linear regression. Thus, the quantity $a_*^\top \beta$ is the unknown conditional mean response at the new predictor vector while Y_* is the actual future response. The future response need be distinct since it also contains a new random error ε_*. Since β is unknonw, we estimate it by oridinary least squares to obtain $\hat{\beta}$.

Our point prediction for the future response is therefore $\hat{Y}_* = a_*^\top \hat{\beta}$.

We now wish to quantify, in euclidean space, how far could Y_* be from $\hat{Y}_*$, our fitted least squares prediction. Therefore, we study the prediction error quantified as $Y_* - \hat{Y}_*$.

Using $Y_* = a_*^\top \beta + \varepsilon_*$ and $\hat{Y}_* = a_*^\top \hat{\beta}$ we obtain,

$$Y_* - \hat{Y}_* = a_*^\top \beta + \varepsilon_* - a_*^\top \hat{\beta} = \varepsilon_* - a_*^\top (\hat{\beta} - \beta).$$

From the formulation, we observe two sources of prediction error: one from the new observation noise and the other from coefficient-estimation error. Even if we somehow knew β perfectly, the actual future observation would still differ from its mean because of ε_*.

Recall previously we expanded $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$ which means that the difference $\hat{\beta} - \beta = (X^\top X)^{-1} X^\top \varepsilon$. Multiplying by our predicto vector $a_*^\top$ we see,

$$a_*^\top (\hat{\beta} - \beta) = a_*^\top (X^\top X)^{-1} X^\top \varepsilon.$$

Since this is simply a linear transformation of a multivariate normal vector, it is also normally distributed with variance,

$$\begin{aligned}
\operatorname{Var}\left(a_*^\top (\hat{\beta} - \beta) \mid X, a_*\right) &= \operatorname{Var}\left(a_*^\top (X^\top X)^{-1} X^\top \varepsilon \mid X, a_*\right) \\
&= a_*^\top (X^\top X)^{-1} X^\top \operatorname{Var}(\varepsilon \mid X) X (X^\top X)^{-1} a_* \\
&= a_*^\top (X^\top X)^{-1} X^\top \left(\sigma^2 I_n\right) X (X^\top X)^{-1} a_* \\
&= \sigma^2 a_*^\top (X^\top X)^{-1} X^\top X (X^\top X)^{-1} a_* \\
&= \sigma^2 a_*^\top (X^\top X)^{-1} a_*.
\end{aligned}$$

Consequently, $a_*^\top \hat{\beta} \mid X, a_* \sim \mathcal{N}(a_*^\top \beta, \sigma^2 a_*^\top (X^\top X)^{-1} a_*)$. Given that we have found that $Y_* - \hat{Y}_* = \varepsilon_* - a_*^\top (\hat{\beta} - \beta)$, conditional on X and a_*, we know that $\operatorname{Var}(\varepsilon_*) = \sigma^2$ and $\operatorname{Var}\left(a_*^\top (\hat{\beta} - \beta)\right) = \sigma^2 h_*$. Furthermore, ε_* is independent of the training sample and therefore independent of $\hat{\beta} - \beta$.

Thus the covariance term is zero,

$$\operatorname{Cov}\left(\varepsilon_*, a_*^\top (\hat{\beta} - \beta)\right) = 0.$$

As a result, we characterize the variance of our prediction error as,

$$\begin{aligned}\operatorname{Var}\left(Y_* - \hat{Y}_* \mid X, a_*\right) &= \operatorname{Var}\left(\varepsilon_* - a_*^\top(\hat{\beta} - \beta) \mid X, a_*\right) \\ &= \operatorname{Var}(\varepsilon_*) + \operatorname{Var}\left(a_*^\top(\hat{\beta} - \beta)\right) - 2\operatorname{Cov}\left(\varepsilon_*, a_*^\top(\hat{\beta} - \beta)\right) \\ &= \sigma^2 + \sigma^2 h_* \\ &= \sigma^2(1 + h_*) \\ &= \sigma^2\left[1 + a_*^\top(X^\top X)^{-1}a_*\right].\end{aligned}$$

Since both terms in the prediction error are normally distributed and independent,

$$Y_* - \hat{Y}_* \mid X, a_* \sim \mathcal{N}(0, \sigma^2[1 + a_*^\top(X^\top X)^{-1}a_*]).$$

Therefore, a test statistic for our predictions if σ were known would be,

$$\frac{Y_* - \hat{Y}_*}{\sigma\sqrt{1 + a_*^\top(X^\top X)^{-1}a_*}} \sim \mathcal{N}(0, 1).$$

Notice that the term σ^2 comes from the new, irreducible observation noise ε_*. Thus, the term $\sigma^2 h_* = \sigma^2 a_*^\top(X^\top X)^{-1}a_*$ comes from the uncertainty in the estimate coefficients $\hat{\beta}$ that is $\sigma^2(1 + h_*)$ is the new observation plus the estimated regression surface represented by $\sigma^2 h_*$. The 1 appears because we are predicting an actual future observation rather than merely estimating the mean response. However, again, we often do not know σ^2 so we can estimate it from training residuals.

Recall the sample variance-covariance estimator,

$$\hat{\sigma}^2 = \frac{1}{n - (p+1)}(Y - X\hat{\beta})^\top(Y - X\hat{\beta}) = \frac{1}{n - p - 1}r^\top r.$$

Under normality distribution theory we know that $\frac{(n-p-1)\hat{\sigma}^2}{\sigma^2}\chi^2_{n-p-1}$. Independence follows required for a Student-t statistic given that the residual variance $\hat{\sigma}^2 \perp \hat{\beta}$ conditional on our design matrix X by Theorem 3.14, and the future error ε_* is independent of the entire training sample. Therefore,

$$\frac{Y_* - \hat{Y}_*}{\sigma\sqrt{1 + h_*}} \perp \frac{(n - p - 1)\hat{\sigma}^2}{\sigma^2}.$$

By construction, we form the test statistic as,

$$T = \frac{\dfrac{Y_* - \hat{Y}_*}{\sigma\sqrt{1 + h_*}}}{\sqrt{\dfrac{(n - p - 1)\hat{\sigma}^2/\sigma^2}{n - p - 1}}} = \frac{\dfrac{Y_* - \hat{Y}_*}{\sigma\sqrt{1 + h_*}}}{\hat{\sigma}/\sigma} = \frac{Y_* - \hat{Y}_*}{\hat{\sigma}\sqrt{1 + h_*}} = \frac{Y_* - \hat{Y}_*}{\hat{\sigma}\sqrt{1 + a_*^\top(X^\top X)^{-1}a_*}} \sim t_{n-p-1}.$$

To construct the prediction interval, let $t_{1-\alpha/2,n-p-1}$ be the $1-\alpha/2$ quantile of the Student-t distribution with $n-p-1$ degrees of freedom. Then,

$$\mathbb{P}\left(-t_{1-\alpha/2,n-p-1} \leq \frac{Y_* - \hat{Y}_*}{\hat{\sigma}\sqrt{1+h_*}} \leq t_{1-\alpha/2,n-p-1}\right) = 1-\alpha$$

$$\mathbb{P}\left(-t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+h_*} \leq Y_* - \hat{Y}_* \leq t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+h_*}\right) = 1-\alpha$$

$$\mathbb{P}\left(\hat{Y}_* - t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+h_*} \leq Y_* \leq \hat{Y}_* + t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+h_*}\right) = 1-\alpha$$

$$\mathbb{P}\left(\hat{Y}_* - t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+a_*^\top(X^\top X)^{-1}a_*} \leq Y_* \leq \hat{Y}_* + t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+a_*^\top(X^\top X)^{-1}a_*}\right) = 1-\alpha.$$

Therefore, the $100(1-\alpha)\%$ prediction interval is simply,

$$\hat{Y}_* \pm t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+a_*^\top(X^\top X)^{-1}a_*}.$$

Theorem 3.18 (Prediction Interval)**.** *Assume the classical normal linear regression model*

$$Y = X\beta + \varepsilon, \qquad \varepsilon \mid X \sim \mathcal{N}_n(0, \sigma^2 I_n),$$

and let $\hat{Y}_* = a_*^\top \hat{\beta}$ denote the predicted response at a new predictor vector a_*. Then, a $100(1-\alpha)\%$ prediction interval for the future response Y_* is

$$\hat{Y}_* \pm t_{1-\alpha/2,n-p-1}\hat{\sigma}\sqrt{1+a_*^\top(X^\top X)^{-1}a_*}.$$

3.3 Analysis of Variance and Multicollinearity

Up until now, we have characterized the ordinary least squares under normality as a sufficient and sound model. However, in practice, that is often not the case. More specifically, we have yet to characterize how much of the variation in our response vector Y is actually captured by our regression model, and how much is left unexplained. Simply looking at our residuals will not provide us with a sufficient view of how well our model explains variation.

Consider the sample variance-covariance estimator as defined before,

$$\hat{\sigma}^2 = \frac{1}{n-p-1}(Y - X\hat{\beta})^\top (Y - X\hat{\beta}).$$

Previously we denoted our residuals $r := Y - X\hat{\beta}$ as lying in the orthogonal column space $\text{Col}(X)^\perp$. Suppose that we have estimated the mean response such that the error between the mean response and our actual responses is,

$$Y - \overline{Y} - I_n.$$

It follows that we can decompose this error into two components,

$$\begin{aligned} Y - \overline{Y} I_n &= Y - \hat{Y} + \hat{Y} - \hat{Y} I_n \\ &= (Y - \hat{Y}) + (\hat{Y} + \overline{Y} I_n). \end{aligned}$$

The first term $Y - \hat{Y}$ represents the error between the actual response vector and our fitted least squares response i.e., the unexplained variation in our response. The second term, $\hat{Y} - \overline{Y} I_n$ is the variation that is explained by our regression. Taking ℓ_2 norms of this equality,

$$\begin{aligned} \|Y - \overline{Y} I_n\|_2^2 &= \|(Y - \hat{Y}) + (\hat{Y} - \overline{Y} I_n)\|_2^2 \\ &= \|Y - \hat{Y}\|_2^2 + \|\hat{Y} - \overline{Y} I_n\|_2^2 + 2(\hat{Y} - \overline{Y} I_n)^\top (Y - \hat{Y}), \end{aligned}$$

where the third term on the 2nd line is the covariation of the two components. Recall that $Y - \hat{Y}$ lies in the orthogonal subspace $\text{Col}(X)^\perp$ so $X^\top (Y - \hat{Y}) + 0$. Since $\hat{Y} - \overline{Y} I_n \in \text{Col}(X)$ and $\hat{Y} \in \text{Col}(X)$ then we conclude that $(\hat{Y} - \overline{I}_n)^\top (Y - \hat{Y}) = 0$ meaning that $\hat{Y} - \overline{Y} I_n$ is orthogonal to $Y - \hat{Y}$. Therefore the cross variation term is zero and our total variation is decomposed as,

$$\|Y - \overline{Y} I_n\|_2^2 = \|Y - \hat{Y}\|_2^2 + \|\hat{Y} - \overline{Y} I_n\|_2^2.$$

The term on the left is the total variation of response Y about its mean. The first term on the right is the variation explained by the fitted regression and the last term on the right is the variation left unexplained. The rank-1 formulations of the euclidean norms is,

$$\text{SST} = \sum_{i=1}^n (Y_i - \overline{Y})^2, \qquad \text{SSR} = \sum_{i=1}^n (\hat{Y}_i - \overline{Y})^2, \qquad \text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

Definition 3.19. Let $Y \in \mathbb{R}^n$ denote our response vector and $\hat{Y} = X\hat{\beta}$ be our fitted response under the classical ordinary least squares setup. Then,

$$\|Y - \overline{Y}I_n\|_2^2 = \underbrace{\|Y - \hat{Y}\|_2^2}_{=SSE} + \underbrace{\|\hat{Y} - \overline{Y}I_n\|_2^2}_{=SSR},$$

where the first component is the sum of squared errors (SSE) and the second component is sum of squared regression (SSR). SSE represents the amount of unexplained variation after regression, and SSR represents the amount of variation in our response explained by our regression of Y on X.

Summing the two gives us the total sum of squares,

$$SST = SSE + SSR.$$

Suppose we decompose our design matrix into intercept and non-intercept predictors such that Z is the submatrix that contains the p non-intercept predictors and $\beta \in \mathbb{R}^p$ contains their coefficients. Then,

$$X = \begin{bmatrix} \mathbf{1}_n & Z \end{bmatrix}, \qquad \theta = \begin{bmatrix} \beta_0 \\ \beta \end{bmatrix}.$$

The regression hypothesis is that $H_0 : \beta = 0$. Observe that the intercept β_0 is not being tested so that under the null hypothesis, the model becomes $Y = \beta_0 \mathbf{1}_n + \varepsilon$. The conditional mean of Y is constant and none of the predictors therefore explain the variation of the response Y, beyond the intercept.

Recall our hat matrix H again for the full model. Since X contains $p+1$ linearly independent columns, the rank of H is also simply just $p+1$. The fitted response $\hat{Y} = HY$ and the residual vector is $r = Y - \hat{Y} = (I_n - H)Y$. Taking the intercept-only model projects onto the one-dimensional space spanned y the constant vector $\mathbf{1}_n$. Therefore, its projection matrix is $H_0 = \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$.

Re-arranging gives,

$$H_0 Y = \overline{Y}\mathbf{1}_n.$$

The full-regression space has dimension $p+1$ while the intercept-only space has dimension of 1. Thus, the part of the full regression space added by the p predictors has dimension $(p+1) - 1 = p$. Notice that the projection matrix onto this added predictor space is $H - H_0$ which has rank,

$$\text{rank}(H - H_0) = p.$$

Since H has rank $p+1$, then the rank of our residual-maker matrix is simply just $n - \text{rank}(H) = n - p - 1$. Now, let's give some convenient alternative formulations of our sum of squares,

$$SST = Y^\top(I_n - H_0)Y, \qquad SSR = Y^\top(H - H_0)Y, \qquad SSE = Y^\top(I_n - H)Y.$$

Notice that the previous total sum of squares decomposition follows because,

$$I_n - H_0 = (H - H_0) + (I_n - H).$$

The corresponding degrees of freedom satisfy,

$$n - 1 = p + (n - p - 1).$$

The total variation has $n-1$ degrees of freedom because estimating the sample mean removes one degree of freedom. The predictors explain variation in a p-dimensional subspace leaving $n-p-1$ residual dimensions.

Theorem 3.20. *Let SST be the total sum of squares resulting from the classical ordinary least squares setup. Then under the null hypothesis $H_0 : \beta = 0$, for $\beta \in \mathbb{R}^{p+1}$,*

$$\frac{SSE}{\sigma^2} \sim \chi^2_{n-p-1}, \qquad \frac{SSR}{\sigma^2} \sim \chi^2_p.$$

Proof. Consider the following setup, $X = \begin{bmatrix} \mathbf{1}_n & Z \end{bmatrix}$, $\theta = \begin{bmatrix} \beta_0 \\ \beta \end{bmatrix}$, where X is decomposed into an n-dimensional constant vector and Z is a submatrix containing the p non-intercept terms. θ contains a $\beta \in \mathbb{R}^p$ corresponding to the p non-intercept predictors in our design matrix, and a constant intercept coefficient β_0. Thus our linear model is, $Y = X\theta + \varepsilon$. By definition,

$$SSE = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = Y^\top (I_n - H) Y.$$

Substituting $Y = X\theta + \varepsilon$,

$$SSE = (X\theta + \varepsilon)^\top (I_n - H)(X\theta + \varepsilon).$$

Because H projects onto $\mathrm{Col}(X)$ and $HX = X$, then $(I_n - H)X = 0$. Therefore, we are left with,

$$SSE = \varepsilon^\top (I_n - H)\varepsilon.$$

Since $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, define $W = \frac{\varepsilon}{\sigma}$ to be the standardized systematic errors of our regression. Then, by standardization, W is a multivariate standard normal, denoted $W \sim \mathcal{N}_n(0, I_n)$ and thus,

$$\frac{SSE}{\sigma^2} = \left(\frac{\varepsilon}{\sigma}\right)^\top (I_n - H) \left(\frac{\varepsilon}{\sigma}\right) = W^\top (I_n - H) W.$$

Clearly, $I_n - H$ is a symmetric projection matrix of rank $n-p-1$ so therefore, rotating its coordinates, it has the form,

$$I_n - H = Q \begin{bmatrix} I_{n-p-1} & 0 \\ 0 & 0 \end{bmatrix} Q^\top$$

for some orthogonal matrix Q. If we let $U = U^\top Q$, then $U \sim \mathcal{N}_n(0, I_n)$ so,

$$\frac{SSE}{\sigma^2} = U^\top \begin{bmatrix} I_{n-p-1} & 0 \\ 0 & 0 \end{bmatrix} U = U_1^2 + \cdots + U_{n-p-1}^2.$$

This is clearly the sum of $n-p-1$ squared standard normal random variables so it follows that, $\frac{SSE}{\sigma^2}$ follows a χ^2_{n-p-1}. Note that this result does not require the null hypothesis and holds for every β. Thus, SSE only contains projected noise after the residual projection $I_n - H$ removes the entire mean vector $X\theta$. Similarly, we can write the regression sum of squares as,

$$SSR = Y^\top (H - H_0) Y.$$

The hat matrix $H - H_0$ projects onto the p-dimensional space added by the predictors beyond the intercept. Therefore $\mathrm{rank}(H - H_0) = p$. Under the null hypothesis, $H_0 : \beta = 0$, our linear model

is simply $Y = \beta_0 \mathbf{1}_n + \varepsilon$. Because $H - H_0$ projects orthogonally away from the intercept direction, $(H - H_0)\mathbf{1}_n = 0$. Thus,

$$SSR = (\beta_0 \mathbf{1}_n + \varepsilon)^\top (H - H_0)(\beta_0 \mathbf{1}_n + \varepsilon) = \varepsilon^\top (H - H_0)\varepsilon.$$

Therefore, by a similar decomposition,

$$\frac{SSR}{\sigma^2} = W^\top (H - H_0)W, \qquad W = \frac{\varepsilon}{\sigma} \sim \mathcal{N}_n(0, I_n).$$

Since $H - H_0$ is a symmetric projection matrix of rank p, $\frac{SSR}{\sigma^2} \sim \chi^2_p$ under H_0. □

Under the null hypothesis H_0, the predictors contain no genuine signal meaning the apparent variation explained by the predictors is therefore just Gaussian noise projected onto a p-dimensional subspace. The squared length of a standard Gaussian vector in p dimensions by definition follows a χ^2_p distribution.

Definition 3.21. Under $H_0 : \beta = 0$ and by Theorem 3.20,

$$F = \frac{SSR/p}{SSE/(n-p-1)} \sim F_{p,n-p-1}.$$

The total coefficient of determination is therefore,

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

Recall previously we had seen that Y is projected onto two orthogonal subspaces. Namely, these subspaces are the column space of X denoted $\text{Col}(X)$ and the subspace orthogonal to the column space of X denoted $\text{Col}(X)^\perp$. Recall that our fitted response is therefore in $\text{Col}(X)$ and the residuals are in $\text{Col}(X)^\perp$ which have $p + 1$ and $n - p - 1$ dimensions respectively.

The term "finding alpha" was primarily coined from the fact that simply testing for significance/relationships are not sufficient. Suppose that instead of just testing $H_0 : \beta = 0$ i.e., there are not relationships between our predictors and the response vector across all p predictors, we were interested in the relationship between a particular subset or a particular linear combination of our predictors and the response Y. We can do so by setting a specific subset of our predictors to 0 using a $(p - q + 1) \times (p + 1)$ matrix,

$$A = \begin{bmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix},$$

for some integer $1 \leq q \leq p$ that sets $\beta_q = \beta_{q+1} = \cdots \beta_p = 0$, effectively "silencing" a $p - q$ subset of our predictors. By then testing the full model against this new restricted model, we effectively test whether or not these specific predictors bring new information beyond what the other/existing predictors explain. We formalize the following framework.

Theorem 3.22. *Let $X \in \mathbb{M}_{n\times(p+1)}$ be our design matrix with p predictors and 1 intercept term. Let Y be the corresponding response matrix and $\beta \in \mathbb{R}^p$. Then $\hat{Y} = X\hat{\beta}$ is the fitted response vector where $\hat{\beta}$ is the least squares estimate. Let $A \in \mathbb{M}_{l\times(p+1)}$ be our constraint matrix.*

Let the resulting fitted response from the constrained regression,

$$\hat{Y}_A = X\left(\hat{\beta} - (X^\top X)^{-1}A^\top\left[A(X^\top X)^{-1}A^\top\right]^{-1}A\hat{\beta}\right).$$

Then $\hat{Y}$ and $\hat{Y}_A$ satisfy,

$$\frac{\|\hat{Y} - \hat{Y}_A\|^2}{\sigma^2} \sim \chi^2_l, \qquad \|\hat{Y} - \hat{Y}_A\|^2 = SSE_A - SSE,$$

where SSE_A is the sum of squared errors (unexplained variation after constrained regression) of the constrained regression. Consequently, we have that,

$$\|\hat{Y} - \hat{Y}_A\|^2 \perp \hat{\sigma}^2 = \frac{1}{n-p-1}(Y-\hat{Y})^\top(Y-\hat{Y}).$$

The F statistic is then defined as,

$$F = \frac{\|\hat{Y} - \hat{Y}_A\|^2/l}{\frac{1}{n-p-1}(Y-\hat{Y})^\top(Y-\hat{Y})} \sim F_{l,n-p-1}.$$

Proof. We aim to show that the residual error between our fitted full least squares model and our constrained least squares model is the difference between their respective SSE's. Assume we are performing inference under the null $H_0 : A\beta = 0$.

We begin with the analytical solution for the restricted parameter estimator,

$$\hat{\beta}_A = \hat{\beta} - (X^\top X)^{-1}A^\top[A(X^\top X)^{-1}A^\top]^{-1}A\hat{\beta},$$

which imposes a penalty term to the unrestricted ordinary least squares estimator. To consutrct the restricted fitted values, we multiply the design matrix by the restricted estimator containing the constraint matrix A,

$$\begin{aligned}
\hat{Y}_A &= X\hat{\beta}_A \\
&= X\left(\hat{\beta} - (X^\top X)^{-1}A^\top\left[A(X^\top X)^{-1}A^\top\right]^{-1}A\hat{\beta}\right) \\
&= X\hat{\beta} - X(X^\top X)^{-1}A^\top\left[A(X^\top X)^{-1}A^\top\right]^{-1}A\hat{\beta} \\
&= \hat{Y} - X(X^\top X)^{-1}A^\top\left[A(X^\top X)^{-1}A^\top\right]^{-1}A\hat{\beta}.
\end{aligned}$$

Define the restricted sum of squared errors as the squared Euclidean inner product of its residuals. To bridge the restricted and unrestricted models, we can inject the unrestricted fitted values by simultaneously adding and subtracting them inside the quadratic form,

$$\begin{aligned} SSE_A &= (Y - \hat{Y}_A)^\top (Y - \hat{Y}_A) \\ &= \Big((Y - \hat{Y}) + (\hat{Y} - \hat{Y}_A)\Big)^\top \Big((Y - \hat{Y}) + (\hat{Y} - \hat{Y}_A)\Big) \\ &= (Y - \hat{Y})^\top (Y - \hat{Y}) + 2(Y - \hat{Y})^\top (\hat{Y} - \hat{Y}_A) + (\hat{Y} - \hat{Y}_A)^\top (\hat{Y} - \hat{Y}_A). \end{aligned}$$

Notice the first term is precisely the definition of the unrestricted sum of squared errors and the third term is the definition of the squared Euclidean norm of the difference in predictions which is precisely $\|\hat{Y} - \hat{Y}_A\|^2$. The unrestricted residuals are guaranteed to be orthogonal to the entire column space of the design matrix meaning that the cross term is therefore also 0. We are left with,

$$\begin{aligned} SSE_A &= (Y - \hat{Y})^\top (Y - \hat{Y}) + 2(Y - \hat{Y})^\top (\hat{Y} - \hat{Y}_A) + (\hat{Y} - \hat{Y}_A)^\top (\hat{Y} - \hat{Y}_A) \\ &= SSE + 0 + \|\hat{Y} - \hat{Y}_A\|^2. \end{aligned}$$

Therefore, it follows that,

$$SSE_A - SSE = \|\hat{Y} - \hat{Y}_A\|^2.$$

Recall that previously we saw that,

$$\hat{Y}_A = \hat{Y} - X(X^\top X)^{-1} A^\top \left[A(X^\top X)^{-1} A^\top\right]^{-1} A\hat{\beta},$$

which means the difference between the unrestricted and restricted fitted response vectors is precisely,

$$X(X^\top X)^{-1} A^\top \left[A(X^\top X)^{-1} A^\top\right]^{-1} A\hat{\beta}.$$

Taking squared Euclidean norms,

$$\|\hat{Y} - \hat{Y}_A\|^2 = (A\hat{\beta})^\top \left[A(X^\top X)^{-1} A^\top\right]^{-1} (A\hat{\beta}).$$

By Theorem 3.14, recall that the unrestricted parameter estimator follows a multivariate normal distribution centered on the population parameter $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (X^\top X)^{-1})$. Applying a fixed linear transformation constraint matrix A to this random vector gives,

$$\begin{aligned} A\hat{\beta} &\sim \mathcal{N}(A\beta, \sigma^2 A(X^\top X)^{-1} A^\top) \\ &\sim \mathcal{N}(0, \sigma^2 A(X^\top X)^{-1} A), \qquad \text{under } H_0 : A\beta = 0. \end{aligned}$$

Dividing by σ^2,

$$\begin{aligned} \frac{\|\hat{Y} - \hat{Y}_A\|^2}{\sigma^2} &= (A\hat{\beta})^\top \left[\sigma^2 A(X^\top X)^{-1} A^\top\right]^{-1} (A\hat{\beta}) \\ &= (A\hat{\beta}) \Sigma_A^{-1} (A\hat{\beta}) \sim \chi^2_l. \end{aligned}$$

Because the constraint matrix A has full row rank, the quadratic form by definition follows a chi-squared distribution with degrees of freedom equal to the covariance matrix.

By our proof above,

$$U = \frac{\|\hat{Y} - \hat{Y}_A\|^2}{\sigma^2} \sim \chi^2_l,$$

and scaling the residual sum of squares forms an independent chi-squared random variable,

$$V = \frac{1}{\sigma^2}(Y - \hat{Y})^\top (Y - \hat{Y}) \sim \chi^2_{n-p-1}.$$

Taking the ratio of U/l and $V/(n-p-1)$,

$$F = \frac{U/l}{V/(n-p-1)} = \frac{\frac{\|\hat{Y} - \hat{Y}_A\|^2}{\sigma^2 \cdot l}}{\frac{1}{\sigma^2(n-p-1)}(Y - \hat{Y})^\top (Y - \hat{Y})} = \frac{\|\hat{Y} - \hat{Y}_A\|^2/l}{\frac{(Y-\hat{Y})^\top (Y-\hat{Y})}{n-p-1}} \sim F_{l,n-p-1},$$

giving us the F statistic. □

In practice, we use the F statistic derived above when we test the null hypothesis $H_0 : \beta = 0$ against the alternative hypothesis $H_A : \beta \neq 0$ where $\beta \neq 0$ means that at least one of the added l predictors is non-zero. Under H_0, the reduced model is considered sufficient i.e., the added predictors do not change the conditional mean of Y. We reject the upper tail of the F distribution.

At significance level α, we define the critical value, $F_{1-\alpha,n-p-1}$, by the probability,

$$\mathbb{P}(F_{l,n-p-1} \leq F_{1-\alpha,n-p-1}) = 1 - \alpha.$$

We reject H_0 at significance level α if $F_{\text{obs}} > F_{1-\alpha,n-p-1}$ where F_{obs} is computed from the formulation given in Theorem 3.22. Intuitively, the F statistic is computing the proportion of the additional variation explained (if any) per added predictor(s) to the unexplained variation per residual degree of freedom. Under H_0, the additional predictors have zero population coefficients, so any reduction in the sum of squared errors (unexplained variation) comes only from fitting random noise.

The $F_{l,n-p-1}$ distribution describes how large this ratio can be from noise alone. A large F_{obs} means that the numerator is large relative to the estimated noise variance $\hat{\sigma}^2$. This is evidence that the extra reduction in residual error is too large to plausibly be explained by random noise under H_0. A small F statistic is not evidence against H_0 which is why there is no lower-tail rejection region.

Let's return to the normal equation formulation for ordinary least squares for a moment. Recall that, for a design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ with full column rank i.e., $\text{rank}(X) = p$, we can obtain our least squares estimate vector as,

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y,$$

where Y is our response vector in $\mathbb{R}^n$. In scalar form, the normal equations become,

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y = \frac{\sum_{i=1}^{n} x_i \cdot y_i}{\sum_{i=1}^{n} x_i \cdot x_i}.$$

Notice that the identity on the far right hand side of the equality above is simply the ratio of inner products. Thus, the least squares estimates simplify to,

$$\hat{\beta} = \frac{\sum_{i=1}^{n} x_i \cdot y_i}{\sum_{i=1}^{n} x_i \cdot x_i} = \frac{\langle x, y \rangle}{\langle x, x \rangle}.$$

Note that we require that our input predictors in our design matrix X are orthogonal. Consider the case that the j-th predictor for some $1 \leq j \leq p$ where p is the number of predictors in our design matrix, is not orthogonal. We can expand the expression for the j-th column as,

$$\begin{aligned} x_j^\top y &= x_j^\top X \hat{\beta}. \\ &= x_j^\top \left(\hat{\beta}_1 x_1 + \cdots + \hat{\beta}_j x_j + \cdots + \hat{\beta}_p + x_p \right) \\ &= \hat{\beta}_1 \langle x_j, x_1 \rangle + \cdots + \hat{\beta}_j \langle x_j, x_j \rangle + \cdots + \hat{\beta}_p \langle x_j, x_p \rangle. \end{aligned}$$

In the case that X has columns that are mutually orthogonal, then $\langle x_j, x_j \rangle = 0$ for whenever $j \neq k$. Thus, all the cross terms disappear simplifying,

$$\begin{aligned} \langle x_j, y \rangle &= \cancel{\hat{\beta}_1 \langle x_j, x_1 \rangle} + \cdots + \hat{\beta}_j \langle x_j, x_j \rangle + \cdots + \cancel{\hat{\beta}_p \langle x_j, x_p \rangle} \\ \iff \hat{\beta}_j \langle x_j, x_j \rangle = \langle x_j, y \rangle &\implies \hat{\beta}_j = \frac{\langle x_j, y \rangle}{\langle x_j, x_j \rangle}, \end{aligned}$$

allowing us to compute each coefficient independently by projecting y onto its corresponding predictors. If the columns are not orthogonal, then the cross-terms do not vanish and become,

$$\hat{\beta}_j \langle x_j, x_j \rangle + \sum_{k \neq j} \hat{\beta}_k \langle x_j, x_k \rangle = \langle x_j, y \rangle.$$

Re-arranging, our j-th least squares coefficient would be,

$$\hat{\beta}_j = \frac{\langle x_j, y \rangle - \sum_{k \neq j} \hat{\beta}_k \langle x_j, x_k \rangle}{\langle x_j, x_j \rangle}.$$

Thus, $\langle x_j, y \rangle$ not only contains the relationship between x_j and y but also variation associated with the other predictors that overlap with x_j. Clearly, it is useful when our predictors are orthogonal. However, such a property is not necessarily attainable in real-world data. Thus, we derive a orthogonalization method that allows us to simultaneously regress our predictors on our targets.

Consider the design matrix,

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_p \end{bmatrix} \in \mathbb{M}_{n \times p}(\mathbb{R}),$$

where the columns $x_1, \ldots, x_p$ are linearly independent. We wish to solve $\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^p} \|y - X\beta\|_2^2$. The idea of successive orthogonalization is to use Gram-Schmidt to replace the correlated predictors $x_1, \ldots, x_p$ with orthogonal predictors $z_1, \ldots, z_p$ spanning the entire column space.

Starting with $z_1 = x_1$.

For $j \geq 2$ we remove from x_j its projections onto all earlier orthogonal directions,

$$\begin{aligned} z_j &= x_j - \sum_{k=1}^{j-1} \text{proj}_{z_k}(z_j) \\ &= x_j - \sum_{k=1}^{j-1} \frac{\langle z_k, x_j \rangle}{\langle z_k, z_k \rangle} z_k. \end{aligned}$$

By Gram-Schmidt successive orthogonalization we obtain,

$$\begin{aligned} z_1 &= x_1, \\ z_2 &= x_2 - \frac{\langle z_1, x_2 \rangle}{\langle z_1, z_1 \rangle} z_1, \\ z_3 &= x_3 - \frac{\langle z_1, x_2 \rangle}{\langle z_1, z_1 \rangle} z_1 - \frac{\langle z_2, x_3 \rangle}{\langle z_2, z_2 \rangle} z_2, \\ &\vdots \\ z_p &= x_p - \sum_{k=1}^{p-1} \frac{\langle z_k, x_p \rangle}{\langle z_k, z_k \rangle} z_k. \end{aligned}$$

By construction, $\langle z_j, z_k \rangle = 0$ for when $j \neq k$. Observe that for fixed $m < j$,

$$\begin{aligned} \langle z_m, z_j \rangle &= \left\langle z_m, x_j - \sum_{k=1}^{j-1} \frac{\langle z_k, x_j \rangle}{\langle z_k, z_k \rangle} z_k \right\rangle \\ &= \langle z_m, x_j \rangle - \sum_{k=1}^{j-1} \frac{\langle z_k, x_j \rangle}{\langle z_k, z_k \rangle} \langle z_m, z_k \rangle \\ &= \langle z_m, x_j \rangle - \frac{\langle z_m, x_j \rangle}{\langle z_m, z_m \rangle} \langle z_m, z_m \rangle \\ &= 0. \end{aligned}$$

Note that the third equality follows because the z_k's constructed before z_j are mutually orthogonal so every term in the sum vanishes except for when $k = m$. Moreover, each z_j is a linear combination of $x_1, \ldots, x_j$ so $\text{span}(z_1, \ldots, z_j) \subseteq \text{span}(x_1, \ldots, x_j)$.

Conversely, re-arranging the Gram-Schmidt equation gives,

$$x_j = z_j + \sum_{k=1}^{j-1} \frac{\langle z_k, x_j \rangle}{\langle z_k, z_k \rangle} z_k,$$

so clearly x_j lies in $\text{span}(z_1, \ldots, z_j)$. Thus, $\text{span}(x_1, \ldots, x_j) = \text{span}(z_1, \ldots, z_j)$ and thus $\text{Col}(X) = \text{Col}(Z)$ where $Z \in \mathbb{M}_{n \times p}(\mathbb{R})$ contains the orthogonalized $x_1, \ldots, x_p$.

Therefore, regressing y on X and regressing y on Z produce the same fitted vector because both regression project y onto the same subspace. We can write the fitted vector in the orthogonal basis as,

$$\hat{y} = Z\hat{\gamma} = \sum_{j=1}^{p} \hat{\gamma}_j z_j.$$

The least squares problem becomes $\hat{\gamma} = \arg\min_{\gamma \in \mathbb{R}^p} \|y - Z\gamma\|_2^2$ so the normal equations can be re-written as $Z^\top Z\hat{\gamma} = Z^\top y$. Notice, since the columns of Z are orthogonal, $Z^\top Z$ is a diagonal matrix,

$$Z^\top Z = \operatorname{diag}(\langle z_1, z_1\rangle, \langle z_2, z_2\rangle, \ldots, \langle z_p, z_p\rangle).$$

Hence, the equations are separable,

$$\hat{\gamma}_1\langle z_1, z_1\rangle = \langle z_1, y\rangle, \ldots, \hat{\gamma}_p\langle z_p, z_p\rangle = \langle z_p, y\rangle.$$

Re-arranging we get that,

$$\hat{\gamma}_j = \frac{\langle z_j, y\rangle}{\langle z_j, z_j\rangle}, \qquad \hat{y} = \sum_{j=1}^{p} \frac{\langle z_j, y\rangle}{\langle z_j, z_j\rangle} z_j.$$

Thus, we have constructed regression by successively orthogonalization, that is, by projecting y independently onto each newly constructed orthogonal direction. Recall our residual-maker after regressing on the first $j-1$ orthogonal directions. We can write this as,

$$M_{j-1} = I - \sum_{k=1}^{j-1} \frac{z_k z_k^\top}{z_k^\top z_k}.$$

Then, the j-th orthogonalized predictor is given as,

$$z_j = M_{j-1} z_j = z_j - \sum_{k=1}^{j-1} \frac{z_k^\top x_j}{z_k^\top z_k} z_k.$$

Similarly, after fitting the first $j-1$ directions, the residual response,

$$r_{j-1} = M_{j-1} y = y - \sum_{k=1}^{j-1} \frac{z_k^\top y}{z_k^\top z_k} z_k.$$

Regressing this residual response on the residualized predictor z_j,

$$\begin{aligned}
\hat{\gamma}_j &= \frac{\langle z_j, r_{j-1}\rangle}{\langle z_j, z_j\rangle} \\
&= \frac{\left\langle z_j, y - \sum_{k=1}^{j-1} \frac{\langle z_k, y\rangle}{\langle z_k, z_k\rangle} z_k \right\rangle}{\langle z_j, z_j\rangle} = \frac{\langle z_j, y\rangle - \sum_{k=1}^{j-1} \frac{\langle z_k, y\rangle}{\langle z_k, z_k\rangle}\langle z_j, z_k\rangle}{\langle z_j, z_j\rangle} \\
&= \frac{\langle z_j, y\rangle}{\langle z_j, z_j\rangle}.
\end{aligned}$$

The final equality follows from the fact that $\langle z_j, z_k \rangle = 0$ for $k < j$. The coefficients $\hat{\gamma}$ are coefficients on the orthogonal predictors z_j not generally the original coefficients $\hat{\beta}$. Because each x_j is a linear combination of $z_1, \ldots, z_j$, there exists an upper triangular matrix R such that $X = ZR$.

More explicitly we can write each predictor as,

$$x_j = z_j + \sum_{k=1}^{j-1} r_{kj} z_k, \qquad r_{kj} = \frac{\langle z_k, x_j \rangle}{\langle z_k, z_k \rangle}.$$

Therefore, the decomposition follows as $X\beta = ZR\beta$. From this decomposition, the same fitted vector can be written as $\hat{y} = X\hat{\beta} = Z\hat{\gamma}$ where then we have that, $Z\hat{\gamma} = ZR\hat{\beta}$ so $\hat{\gamma} = R\hat{\beta}$ because the columns of Z are linearly independent. Hence, it follows that $R\hat{\beta} = \hat{\gamma}$. Since R is upper triangular, it follows that the original coefficients are recovered by backward substitution. That is,

$$\begin{aligned}
\hat{\beta} &= \frac{\hat{\gamma}_p}{r_{pp}} \\
\hat{\beta}_{p-1} &= \frac{\hat{\gamma}_{p-1} - r_{p-1,p} \cdot \hat{\beta}_p}{r_{p-1,p-1}}, \\
&\vdots \\
\hat{\beta}_j &= \frac{\hat{\gamma}_j - \sum_{k=j+1}^{p} r_{jk} \hat{\beta}_k}{r_{jj}}.
\end{aligned}$$

For the unnormalized Gram-Schmidt construction above, we let $r_{jj} = 1$ so the coefficients are simply,

$$\hat{\beta}_j = \hat{\gamma}_j - \sum_{k=j+1}^{p} r_{jk} \hat{\beta}_k.$$

If we now normalize each orthogonal direction z_j such that we let $q_1, \ldots, q_p$ be the orthonormal directions from $z_1, \ldots, z_p$ then it follows that,

$$q_j^\top q_k = \begin{cases} 1, & j = k, \\ 0, & j \neq k \end{cases}.$$

Suppose that $Q \in \mathbb{M}_{n \times (p+1)}(\mathbb{R})$ is the matrix that contains these orthonormal vectors as columns. Then, Gram-Schmidt gives the QR factorization such that $X = QR$ where $R = Q^\top X$ is an upper triangular matrix. Then, the least-squares problem can be re-formulated as,

$$\|y - X\beta\|_2^2 = \|y - QR\beta\|_2^2.$$

The normal equations are re-written as,

$$\begin{aligned}
X^\top X\hat{\beta} = X^\top y &\iff R^\top Q^\top QR\hat{\beta} = R^\top Q^\top y \\
R^\top R\hat{\beta} = R^\top Q^\top y &\iff R\hat{\beta} = Q^\top y.
\end{aligned}$$

Therefore it follows that $R\hat{\beta} = Q^\top y$. Then, the vector $Q^\top y$ contains the independent projections of y onto the orthonormal direction. Solving the upper-triangular system then converts those orthogonal-basis coefficients back into coefficients on the original predictors.

Therefore, regression by successive orthogonalization is equivalent. Thus, we see that Gram-Schmidt provides us with an alternative computation of least squares. When predictors overlap, we saw that the off-diagonal terms can be non-zero, making it so that the coefficient must be determined simultaneously. After orthogonalization, we see that $Q^\top Q = I$ meaning the transformed regression contains no off-diagonal interactions.

Each γ_j therefore measures the projection of y onto one distinct directions.

Recall from our formulation using Gram-Schmidt. q_j ultimately represents only the genuinely new direction contributed by x_j after removing what earlier predictors already contained. This makes multicollinearity visible, that is, when x_j contributes little new information, $\|\tilde{q}_j\| = 0$. When it contributes no new information, $\|\tilde{q}_j\|$ is exactly 0.

Here, we expressed our predictors, the x_j's, as linear combinations of the z_k's for $k \leq j$. Since all the z_j's are orthogonal, they form an orthogonal basis for $\text{Col}(X)$ meaning that the least squares projection onto this subspace spanning our z_j's is simply $\hat{y}$.

Algorithm 1 Successive Orthogonalization of the Predictors

Input: $X = [x_1, \ldots, x_p] \in \mathbb{R}^{n \times p}$ with linearly independent columns
Output: Orthogonal predictors $z_1, \ldots, z_p$
$z_1 \leftarrow x_1$
for $j = 2, \ldots, p$ **do**
    $z_j \leftarrow x_j - \sum_{k=1}^{j-1} \frac{\langle z_k, x_j \rangle}{\langle z_k, z_k \rangle} z_k$
end for
return $z_1, \ldots, z_p$

From our orthogonalization representation, the last coefficient $\hat{\beta}_p$ is simply $\langle z_p, y \rangle / \langle z_p, z_p \rangle$. Using the formulation of our classical linear model, that is $y = X\beta + \varepsilon$ given that $\text{Var}(\varepsilon \mid X) = \sigma^2 I$. Because we know that z_p is orthogonal to $x_1, \ldots, x_{p-1}$ and x_p can be written as $x_p = z_p$ plus a linear combination of the earlier directions, we re-arrange to get that $\langle z_p, X\beta \rangle = \beta_p \langle z_p, z_p \rangle$.

Therefore, our least squares estimate can be expanded into,

$$\hat{\beta}_p = \frac{\langle z_p, y \rangle}{\langle z_p, z_p \rangle} = \frac{\langle z_p, X\beta + \varepsilon \rangle}{\langle z_p, z_p \rangle} = \beta_p + \frac{\langle z_p, \varepsilon \rangle}{\langle z_p, z_p \rangle}.$$

So, the randomness of $\hat{\beta}_p$ comes from the second term,

$$\text{Var}(\hat{\beta}_p \mid X) = \text{Var}\left(\frac{\langle z_p, \varepsilon \rangle}{\langle z_p, z_p \rangle} \middle| X \right) = \frac{1}{\langle z_p, z_p \rangle^2} \text{Var}(z_p^\top \varepsilon \mid X).$$

Substituting and simplfying we get,

$$\text{Var}(z_p^\top \varepsilon \mid X) = z_p^\top \text{Var}(\varepsilon \mid X) z_p = z_p^\top (\sigma^2 I) z_p = \sigma^2 z_p^\top z_p = \sigma^2 \langle z_p, z_p \rangle.$$

Finally, substituting,

$$\text{Var}(\hat{\beta}_p \mid X) = \frac{\sigma^2}{\langle z_p, z_p \rangle} = \frac{\sigma^2}{\|z_p\|^2}.$$

z_p is the part of x_p that remains after removing everything explained by $x_1, \ldots, x_{p-1}$. Thus, $\langle z_p, z_p \rangle = \|z_p\|^2$ measures how much unique variation predictor x_p has. If x_p is almost completely explained by the other predictors, z_p is tiny meaning that $\|z_p\|^2$ decreases which inflates $\text{Var}(\hat{\beta}_p)$.

This leads into our discussion regarding multicollinearity.

Consider a matrix that has perfectly multicollinear columns. More specifically, when the columns of our design matrix are linearly dependent meaning there exists scalars $c_i \neq 0$ for which $c_1 \cdot x_1 + \cdots c_p \cdot x_p = 0$. Suppose that we are interested in the coefficient on x_p. Through successive orthogonalization, we can remove from x_p everything that can already be explained by the other predictors. If $X_{-p} = \begin{bmatrix} x_1 & \cdots & x_{p-1} \end{bmatrix}$ then the residualized predictor is precisely $z_p = M_{-p} x_p$ where

$$M_{-p} = I - P_{-p} = I - X_{-p}(X_{-p}^\top X_{-p})^{-1} X_{-p}^\top.$$

z_p is precisely the component of x_p that is orthogonal to the space spanned by the other predictors meaning z_p represents the variation in x_p that is unique to x_p. From our formulation of $\text{Var}(\hat{\beta}_p \mid X)$, this immediately exposes the multicollinearity problem.

Suppose that x_p is nearly a linear combination of all the other predictors.

Then, there exists coefficients $a_1, \ldots, a_{p-1}$ such that,

$$x_p \approx a_1 x_1 + \cdots + a_{p-1} x_{p-1}.$$

When we project x_p onto the span of the other predictors, almost all of x_p is therefore removed meaning that $z_p = x_p - P_{-p} x_p \approx 0$. Consequently, $\|z_p\|^2 \approx 0$ meaning that $\text{Var}(\hat{\beta}_p \mid X) = \frac{\sigma^2}{\|z_p\|^2} \gg 1$. More precisely, OLS is trying to determine the effect of x_p on y while holding the other predictors fixed but can only identify such an effect using variation in x_p that cannot already be explained by those other predictors. When x_p is highly collinear with them, very little such variation remains meaning even a small amount of noise can produce a large change in the estimated coefficient.

Consider the limiting case of multicollinearity.

If $x_p \in \text{span}(x_1, \ldots, x_{p-1})$, then $z_p = 0$. The variance expression would then have a zero denominator $\frac{\sigma^2}{\|z_p\|^2} = \frac{\sigma^2}{0}$ which is not finite. This corresponds to perfect multicollinearity i.e., when X loses full column rank, β_p is no longer uniquely identifiable.

Definition 3.23 (Variance Inflation Factor(s))**.** Suppose that X is a full column rank matrix containing an intercept and let x_j be a non-intercept predictor. Let R_j^2 denote the coefficient of determination from the auxiliary least-squares regression of x_j on all remaining columns of our design matrix, denoted X_{-j}. The *Variance Inflation Factor* associated with x_j is,

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

measuring the factor by which the conditional variance of $\hat{\beta}_j$ is increased because x_j is linearly associated with the remaining predictors, relative to a predictor having the same total centered variation but orthogonal to the remaining predictors.

Proof. This is a computational proof aiming to derive the variance inflation factor for a classical ordinary least squares setup. Assume a classical linear model of the form,

$$y = X\beta + \varepsilon,$$

with assumptions $\mathbb{E}[\varepsilon \mid X] = 0$ and $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n$ where X has full column rank and contains an intercept. Fix a non-intercept predictor x_j and write the full design matrix as,

$$X = \begin{bmatrix} x_j & X_{-j} \end{bmatrix},$$

where X_{-j} contains the intercept and every predictor other than x_j. To isolate the component of x_j that cannot be explained by the other predictors, we define projection matrix less the j predictors and its associated residual-maker matrix less the j predictor as,

$$P_{-j} = X_{-j}(X_{-j}^\top X_{-j})^{-1} X_{-j}^\top, \qquad M_{-j} = I - P_{-j},$$

and let $z_j = M_{-j} x_j$. Thus, z_j is the residual obtained by regressing x_j on all the other predictors. From the inner-product representation of the partial regression coefficient we define $\hat{\beta}_j = \dfrac{\langle z_j, y \rangle}{\langle z_j, x_j \rangle}$.

Since $x_j = P_{-j} x_j + z_j$ and z_j is orthogonal to $\mathrm{Col}(X_{-j})$ we have that,

$$\langle z_j, x_j \rangle = \langle z_j, P_{-j} x_j + z_j \rangle = \langle z_j, P_{-j} x_j \rangle + \langle z_j, z_j \rangle = \langle z_j, z_j \rangle.$$

Consequently, our j-th least squares estimate,

$$\begin{aligned} \hat{\beta}_j &= \frac{z_j^\top (X\beta + \varepsilon)}{z_j^\top z_j} \\ &= \frac{z_j^\top X_{-j}\beta_{-j} + \beta_j z_j^\top x_j + z_j^\top \varepsilon}{z_j^\top z_j} \\ &= \frac{0 + \beta_j z_j^\top z_j + z_j^\top \varepsilon}{z_j^\top z_j} \\ &= \beta_j + \frac{z_j^\top \varepsilon}{z_j^\top z_j}. \end{aligned}$$

Conditional on our full design matrix, taking variances we get,

$$\begin{aligned} \mathrm{Var}(\hat{\beta}_j \mid X) &= \frac{1}{(z_j^\top z_j)^2} \mathrm{Var}(z_j^\top \varepsilon \mid X) = \frac{1}{(z_j^\top z_j)^2} z_j^\top \mathrm{Var}(\varepsilon \mid X) z_j \\ &= \frac{\sigma^2 z_j^\top z_j}{(z_j^\top z_j)^2} = \frac{\sigma^2}{z_j^\top z_j}. \end{aligned}$$

Because X_{-j} contains an intercept, the auxiliary regression $x_j = X_{-j}\delta + z_j$ has coefficient of determination given as,

$$R^2 = 1 - \frac{\mathrm{SSE}_j}{\mathrm{SST}_j},$$

where $SSE_j = z_j^\top z_j$ and $SST_j = \sum_i (x_{ij} - \bar{x}_j)^2$. Hence, it follows that,

$$R_j^2 = 1 - \frac{z_j^\top z_j}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} \implies 1 - R_j^2 = \frac{z_j^\top z_j}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

Solving for $z_j^\top z_j$ we get,

$$z_j^\top z_j = (1 - R_j^2) \sum_{i=1}^n (x_{ij} - \bar{z}_j)^2.$$

Substituting this back into the variance formulation,

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{z_j^\top z_j} = \frac{\sigma^2}{(1 - R_j^2) \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} = \frac{1}{1 - R_j^2} \cdot \frac{\sigma^2}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

Consider the variance if x_j were orthogonal to all of the other non-intercept predictors. Then, none of its centered variation would be explained by them meaning that $R_j^2 = 0$ making,

$$\mathrm{Var}_\perp(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

Taking proportions, we get that,

$$\frac{\mathrm{Var}(\hat{\beta}_j \mid X)}{\mathrm{Var}_\perp(\hat{\beta}_j \mid X)} = \frac{\frac{1}{1 - R_j^2} \cdot \frac{\sigma^2}{\sum_i (x_{ij} - x_j)^2}}{\frac{\sigma^2}{\sum_i (x_{ij} - \bar{x}_j)^2}} = \frac{1}{1 - R_j^2}.$$

□

Theorem 3.24. *Under classical linear model assumptions, $y = X\beta + \varepsilon$, $\mathbb{E}[\varepsilon \mid X] = 0$ and $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n$, suppose that X has full column rank and contains an intercept. For any non-intercept predictor x_j,*

$$\mathrm{Var}(\hat{\beta}_j \mid X) = VIF_j \cdot \frac{\sigma^2}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}, \qquad VIF_j = \frac{1}{1 - R_j^2}.$$

Equivalently, if $z_j = M_{-j} z_j$ is the component of x_j orthogonal to the remaining predictors then,

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\|z_j\|^2} = \frac{1}{1 - R_j^2} \cdot \frac{\sigma^2}{\|x_j - \bar{x}_j I_n\|^2}.$$

3.4 Decompositions of Ordinary Least Squares

3.4.1 QR Decomposition

Perhaps you have already observed that from the ordinary least squares formulation, we must compute the inverse of $(X^\top X)^{-1}$. When X is a full rank, we can guarantee that $X^\top X$ has a unique inverse. However, when X is not full rank, we are unable to provide a unique inverse which makes the problem ill-conditioned. More specifically, when X is full rank, the quadratic form as we saw before,

$$\vec{x}^\top (X^\top X)\vec{v} = (X\vec{v})^\top (X\vec{v}) = \|X\vec{v}\|.$$

When X is full rank, $X\vec{v} = 0$ has only one solution i.e., the trivial solution where $\vec{v} = \vec{0}$. When the columns of X are not linearly independent, there exists a non-zero vector $\vec{v}$ in the nullspace implying that 0 is an eigenvalue of $X^\top X$ meaning that the determinant is 0 therefore making it singular and non-invertible. QR decomposition solves this issue by decomposing the normal equations into a composition of an orthogonal matrix and an upper triangular matrix.

Suppose we have a design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ with full column rank. We decompose the design matrix into a full QR decomposition such that,

$$X = QR, \qquad Q \in \mathbb{M}_{n\times n}(\mathbb{R}), \quad R \in \mathbb{M}_{n\times p}(\mathbb{R}),$$

where Q is an orthogonal matrix satisfying $Q^\top Q = QQ^\top = I_n$ and R is an upper triangular matrix. To further clarify the decomposition, we isolate the components that contribute to the column space of our design matrix X and partition Q and R into block matrices. Consider,

$$Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}, \quad R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

where $Q_1 \in \mathbb{M}_{n\times p}(\mathbb{R})$ has columns that form an orthonormal basis for the column space of X and $Q_2 \in \mathbb{M}_{n\times(n-p)}(\mathbb{R})$ forms an orthonormal basis for the orthogonal complement of $\text{Col}(X)$, specifically $\text{null}(X^\top)$. The matrix $R_1 \in \mathbb{M}_{p\times p}(\mathbb{R})$ is a square, upper triangular matrix. Because X has full column rank i.e., $\text{rank}(X) = p$, all diagonal elements of R_1 are non-zero. Therefore R_1 ise strictly invertible.

Theorem 3.25. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ is a full rank design matrix. Then, its QR decomposition is,*

$$X = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_1 \\ 0 \end{bmatrix} = QR.$$

where Q_1 is a submatrix with columns that form an orthonormal basis for the column space of X, Q_2 has columns that form an orthonormal basis for the nullspace of $X^\top$ and R_1 is a p-dimensional square, upper triangular matrix.

Proof. By construction, as required. □

Note the bottom block of the upper triangular matrix R is an $(n-p)\times p$ zero matrix.

Recall that the goal of ordinary least squares is to find the parameter vector $\hat{\beta}$ that minimizes the sum of squared residuals which is represented by the squared Euclidean norm ℓ_2 norm. We begin directly with this objective function and replace the design matrix X with its QR factorization,

$$\text{RSS}(\beta) = \|Y - X\beta\|_2^2 = \|Y - QR\beta\|_2^2.$$

By properties of the Euclidean norm and that $\|\vec{v}\|_2^2 = \|Q^\top \vec{v}\|_2^2$,

$$\begin{aligned}
\|Y - X\beta\|_2^2 &= \|Y - QR\beta\|_2^2 \\
&= \|Q^\top (Y - QR\beta)\|_2^2 \\
&= \|Q^\top Y - Q^\top QR\beta\|_2^2 \\
&= \|Q^\top Y - I_n R\beta\|_2^2 \\
&= \|Q^\top Y - R\beta\|_2^2.
\end{aligned}$$

Substituting the partitioned forms of $Q^\top$ and R,

$$\begin{aligned}
\|Q^\top Y - R\beta\|_2^2 &= \left\| \begin{bmatrix} Q_1^\top \\ Q_2^\top \end{bmatrix} Y - \begin{bmatrix} R_1 \\ 0 \end{bmatrix} \beta \right\|_2^2 \\
&= \left\| \begin{bmatrix} Q_1^\top Y - R_1\beta \\ Q_2^\top Y - 0 \end{bmatrix} \right\|_2^2 \\
&= \left\| \begin{bmatrix} Q_1^\top Y - R_1\beta \\ Q_2^\top Y \end{bmatrix} \right\|_2^2 \\
&= \left\| Q_1^\top Y - R_1\beta \right\|_2^2 + \left\| Q_2^\top Y \right\|_2^2.
\end{aligned}$$

Taking argument minimums,

$$\text{RSS}_{\text{QR}}(\beta) = \arg\min_\beta \left\| Q_1^\top Y - R_1\beta \right\|_2^2 + \arg\min_\beta \left\| Q_2^\top Y \right\|_2^2$$

Notice that the term $\|Q_2^\top Y\|_2^2$ represents the projection of the response vector Y onto the orthogonal complement of X. Since it contains no β term, it is an irreducible minimum sum of squared errors. In contrast, the first term $\|Q_1^\top Y - R_1\beta\|_2^2$ contains our parameter β. Since R_1 is a $p \times p$ invertible matrix by our full rank assumption for least squares, the linear system $R_1\beta = Q_1^\top Y$ has a unique, exact solution. We can achieve the theoretical minimum of 0 by setting the term exactly to zero.

$$\begin{aligned}
\|Q_1^\top Y - R_1\hat{\beta}\|_2^2 = 0 &\iff Q_1^\top Y - R_1\hat{\beta} = 0 \\
&\iff R_1\hat{\beta} = Q_1^\top Y \\
&\iff \hat{\beta} = R_1^{-1} Q_1^\top Y.
\end{aligned}$$

Given that R_1 is an upper triangular matrix, it is unnecessary for us to compute the inverse. Instead, suppose that $c = Q_1^\top Y$ where $c = \begin{bmatrix} c_1 & c_2 & \cdots & c_p \end{bmatrix}^\top$. Because $R_1 \in \mathbb{M}_{p\times p}(\mathbb{R})$ is a strictly upper triangular form from the reduced QR decomposition, the system $R_1\hat{\beta} = c$ expands to,

$$\begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ 0 & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_{pp} \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_p \end{bmatrix}.$$

By evaluating the matrix multiplication from the bottom row where $i = p$ upwards to the top row where $i = 1$, we can actually isolate each $\hat{\beta}_i$ sequentially. Such a system resolves to,

$$\begin{aligned} r_{pp}\hat{\beta}_p = c_p \iff \hat{\beta}_p &= \frac{c_p}{r_{pp}} \\ r_{p-1,p-1}\hat{\beta}_{p-1} + r_{p-1,p}\hat{\beta}_p = c_{p-1} \iff \hat{\beta}_{p-1} &= \frac{c_{p-1} - r_{p-1,p}\hat{\beta}_p}{r_{p-1,p-1}} \\ &\vdots \\ r_{ii}\hat{\beta}_i + \sum_{j=i+1}^{p} r_{ij}\hat{\beta}_j = c_i \iff \hat{\beta}_i &= \frac{1}{r_{ii}}\left(c_i - \sum_{j=i+1}^{p} r_{ij}\hat{\beta}_j\right), \end{aligned}$$

for $i = p-1, p-2, \ldots, 1$. This is precisely back-substitution.

We successfully bypass the need for computing R_1^{-1} explicity. Recall that matrix inverse is an $O(p^3)$ operation and is prone to floating-point errors. By back-substitution, this problem reduces to an $O(p^2)$ problem. Solving the system iteratively via back-substitution is much more numerically stable than computing the inverse and multiplying by the vector c from above.

Note that we still, however, have not resolved the problem where our design matrix is not necessarily full rank. QR decompositions still requires that the design matrix X and therefore R_1 to have full column rank. Inspecting the denominator of our derived back-substitution step,

$$\hat{\beta}_i = \frac{1}{r_{ii}}\left(c_i - \sum_{j=i+1}^{p} r_{ij}\hat{\beta}_j\right)$$

if the design matrix is rank-deficient with collinear features $p > n$, the matrix R_1 will not be full rank. Consequently, at least one of the diagonal elements r_{ii} will be exactly zero or extremely close to zero in floating-point arithmetic. If r_{ii} is exactly 0, the formula thereby attempts a division by zero and the algorithm will fail because there are infinitely many solutions to the least squares problem.

Suppose that X is now a rank-deficient matrix where $\operatorname{rank}(X) = k < p$. We introduce the permutation matrix $P \in \mathbb{M}_{p\times p}(\mathbb{R})$. We introduce the notion of column pivoting in order to re-order the columns of X at each step of the decomposition such that the column with the largest norm is selected as the pivot guaranteeing the diagonal elements of the resulting upper triangular matrix are sorted in descending magnitude i.e., $|r_{11}| \geq |r_{22}| \geq \cdots \geq |r_{kk}| > 0$, while the remaining diagonal entries are identically zero.

Theorem 3.26. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ be a design matrix with $\operatorname{rank}(X) = k < p$. Then, there exists a permutation matrix $P \in \mathbb{M}_{p\times p}(\mathbb{R})$, an orthogonal matrix $Q \in \mathbb{M}_{n\times n}(\mathbb{R})$ and an upper triangular matrix $R \in \mathbb{M}_{n\times p}(\mathbb{R})$ such that,*

$$XP = QR = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} \\ 0 & 0 \end{bmatrix},$$

where $Q_1 \in \mathbb{M}_{n\times k}(\mathbb{R})$ forms an orthonormal basis for the column space of $X, R_{11} \in \mathbb{M}_{k\times k}(\mathbb{R})$ is a strictly non-singular, upper traignular matrix and $R_{12} \in \mathbb{M}_{k\times(p-k)}(\mathbb{R})$.

Proof. By Gram-Schmidt construction with a greedy column selection algorithm, the first k independent columns are pivoted to the leading block. Since $\operatorname{rank}(X) = k$, the remaining $p - k$ columns are linearly dependent on the first k columns forcing the bottom right block of R to be exactly the zero matrix. □

We return to the ordinary least squares objective function. Since P is an orthogonal permutation matrix, $PP^\top = I_p$, we can inject this identity into our design matrix formulation,

$$\operatorname{RSS}(\beta) = \|Y - X\beta\|_2^2 = \|Y - XPP^\top\beta\|_2^2 = \|Y - QR(P^\top\beta)\|_2^2.$$

Let $\theta = P^\top\beta$. Because $P^\top$ simply reorders the entries of β, θ represents our parameter vector sorted to align with the pivoted columns of X. We partition θ conformably with R such that $\theta = \begin{bmatrix} \theta_1 & \theta_2 \end{bmatrix}^\top$ where $\theta_1 \in \mathbb{R}^k$ corresponds to the full-rank block R_{11} and $\theta_2 \in \mathbb{R}^{p-k}$ corresponds to the linearly dependent columns. Applying the orthogonal transformation $Q^\top$ to the residual vector, we obtain,

$$\begin{aligned}
\|Y - X\beta\|_2^2 &= \|Q^\top Y - R\theta\|_2^2 \\
&= \left\| \begin{bmatrix} Q_1^\top \\ Q_2^\top \end{bmatrix} Y - \begin{bmatrix} R_{11} & R_{22} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \right\|_2^2 \\
&= \left\| \begin{bmatrix} Q_1^\top Y - (R_{11}\theta_1 + R_{12}\theta_2) \\ Q_2^\top Y - 0 \end{bmatrix} \right\|_2^2 \\
&= \|Q_1^\top Y - R_{11}\theta_1 - R_{12}\theta_2\|_2^2 + \|Q_2^\top Y\|_2^2.
\end{aligned}$$

If $c_1 = Q_1^\top Y$, then the theoretical minimum is achieved when the first term is strictly zero i.e., when $R_{11}\theta_1 + R_{12}\theta_2 = c_1$. Because the system is underdetermined, we have k equations but p unknowns where $p > k$, so there are infinitely many solutions for θ. To obtain a unique basic solution, we can contrain the free variables associated with the linearly dependent columns to zero and set $\hat{\theta}_2 = 0$.

Observe that under this unique solution we have a collapsed system into a full-rank system,

$$R_{11}\hat{\theta}_1 + R_{12}(0) = c_1 \iff R_{11}\hat{\theta}_1 = c_1 \iff \hat{\theta}_1 = R_{11}^{-1}c_1.$$

More importantly, because R_{11} is a $k \times k$ upper triangular matrix with exclusively non-zero diagonal entries by construction, we can once again apply the $O(k^2)$ back-substitution algorithm to solve for $\hat{\theta}_1$ without explicitly computing R_{11}^{-1}.

Once $\hat{\theta}_1$ is computed iteratively, we reconstruct the full permuted parameter vector $\hat{\theta} = \begin{bmatrix}\hat{\theta}_1 & 0\end{bmatrix}^\top$. Finally, we multiply by the permutation matrix to reverse the sorting and recover our original parameter vector $\hat{\beta}$ by re-arranging we get,

$$P^\top \hat{\beta} = 0 \iff \hat{\beta} = P\hat{\theta} \iff \hat{\beta} = P \begin{bmatrix}\hat{\theta}_1 \\ 0\end{bmatrix}.$$

By isolating the linearly independent subspace into R_{11}, column pivoting ensures numerical stability and extracts a sparse basic solution even when the initial design matrix X is perfectly collinear.

3.4.2 Singular Value Decomposition of OLS

While QR decomposition with column pivoting extracts a full-rank submatrix to compute a solution i.e., by selecting an arbitrary basic solution by setting $p - r$ elements to zero, rendering such a result is highly sensitive to the permutation matrix P we choose. Furthermore, in the presence of severe multicollinearity where singular values are near zero rather than identically zero, QR pivoting is numerically unstable. To better handle near-collinearity and guarantee a unique, optimal solution when $\text{rank}(X) < p$, we require a decomposition that explicitly isolates and quantifies the rank of the matrix. Specifically, when infinitely many solutions exist, we seek the unique solution that minimizes the Euclidean norm of the parameter vector $\|\hat{\beta}\|_2$. We will do so by applying a Singular Value Decomposition as characterized in *Theorem 1.119*.

We start by obtaining the vectors need to construct matrix V and U. We form the square, symmetric matrix $M = X^\top X \in \mathbb{M}_{p\times p}(\mathbb{R})$ and find the eigenvalues $\lambda_1, \ldots, \lambda_p$ and their corresponding orthonormal eigenvectors $\vec{v}_1, \ldots, \vec{v}_p$ such that $(X^\top X)\vec{v}_j = \lambda_j \vec{v}_j$ for $j = 1, \ldots, p$. Since the eigenvalues of a square, symmetric real matrix $X^\top X$ are guaranteed to be real and non-negative, we obtain our singular values by taking the square roots of these eigenvalues, ordering them from largest to smallest,

$$\sigma_j = \sqrt{\lambda_j}, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0.$$

The eigenvectors, $\vec{v}_1, \ldots, \vec{v}_p$ are orthonormal, so stacking them gives us our $p \times p$ matrix V. Since $\vec{v}_j$ are orthogonal eigenvectors of a symmetric matrix, V automatically satisfies $V^\top V = VV^\top = I_p$. Note that if $\{\vec{v}_1, \ldots, \vec{v}_k\}$ for $k < p$ are not orthogonal/orthonormal, we can extend to an orthonormal basis using the Gram-Schmidt orthogonalization procedure to obtain exactly p vectors that form an orthonormal basis. Thus, we have formed our matrix V.

Once we have our right-singular vectors, $\vec{v}_j$, and singular values σ_j, we can find the left-singular values $\vec{u}_j$ via $X\vec{v}_j = \sigma_j \vec{u}_j$ where for every j where $\sigma_j > 0$ we compute,

$$\vec{u}_j = \frac{1}{\sigma_j} X\vec{v}_j.$$

Stacking these vectors, we obtain the $n \times p$ matrix U. Similarly, $U^\top U = UU^\top = I_p$ since these vectors are guaranteed to be mutually orthonormal because,

$$\begin{aligned}\vec{u}_i^\top \vec{u}_j &= \left(\frac{X\vec{v}_i}{\sigma_i}\right)^\top \left(\frac{X\vec{v}_j}{\sigma_j}\right) \\ &= \frac{\vec{v}_i^\top X^\top X \vec{v}_j}{\sigma_i \sigma_j} = \frac{\vec{v}_i^\top (\lambda_j \vec{v}_j)}{\sigma_i \sigma_j} = \frac{\sigma_j^2 (\vec{v}_i^\top \vec{v}_j)}{\sigma_i \sigma_j} \\ &= \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j \end{cases}.\end{aligned}$$

Thus, our design matrix, not necessarily symmetric nor orthogonal, is successive orthogonalized via the decomposition,

$$\begin{aligned}X &= U\Sigma V^\top \\ &= \underbrace{\begin{bmatrix} | & | & & | \\ \vec{u}_1 & \vec{u}_2 & \cdots & \vec{u}_p \\ | & | & & | \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_p \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} - & \vec{v}_1^\top & - \\ - & \vec{v}_2^\top & - \\ & \vdots & \\ - & \vec{v}_p^\top & - \end{bmatrix}}_{V^\top},\end{aligned}$$

where $U^\top U = I_p$ and $V^\top V = VV^\top = I_p$ where we solve the systems,

$$\begin{aligned}X\vec{v}_j &= U\Sigma V^\top \vec{v}_j = U\Sigma \vec{e}_j = U(\sigma_j \vec{e}_j) = \sigma_j u_j, \\ X^\top \vec{u}_j &= V\Sigma U^\top \vec{u}_j = V\Sigma \vec{e}_j = \sigma_j \vec{v}_j,\end{aligned}$$

to obtain the right and left singular vectors for $j = 1, \ldots, p$. Recall the least squares objective,

$$\hat{\beta} = \arg\min_\beta \|y - X\beta\|_2^2.$$

We can decompose our target response into a projection onto the column space and its orthogonal complement but using our $U \in \mathbb{M}_{p\times p}(\mathbb{R})$ matrix which has columns that form an orthonormal basis for the column space of our design matrix, that is,

$$y = UU^\top y + (I - UU^\top)y.$$

Note that expanding this simply gives us our y response vector back,

$$y = UU^\top y + (I - UU^\top)y = \cancel{UU^\top y} + y - \cancel{UU^\top y} = y.$$

Expanding the first component that projects y onto $\text{Col}(X)$,

$$\begin{aligned}y &= UU^\top y + (I - UU^\top)y \\ &= \sum_{j=1}^{p} (\vec{u}_j^\top y)\vec{u}_j + (I - UU^\top)y.\end{aligned}$$

Now we shift our attention to $X\beta$ in the least squares objective. After substituting $X := U\Sigma V^\top$,

$$\begin{aligned} X\beta &= U\Sigma V^\top \beta, \qquad \text{letting } \theta := V^\top \beta, \\ &= U\Sigma V^\top V\theta = U\Sigma\theta \\ &= \sum_{j=1}^{p} \sigma_j \theta_j \vec{u}_j. \end{aligned}$$

We use the substituting $\theta = V^\top \beta$ so that $\beta = V\theta = \sum_{j=1}^{p} \theta_j \vec{v}_j$ which also conveniently allows us to cancel $V^\top V = I_p$ in the 2nd line. Therefore, subtracting $y - X\beta$,

$$\begin{aligned} y - X\beta &= \sum_{j=1}^{p} (\vec{u}_j^\top y)\vec{u}_j - \sum_{j=1}^{p} \sigma_j \theta_j \vec{u}_j + (I - UU^\top)y \\ &= \sum_{j=}^{p} (\vec{u}_j^\top y - \sigma_j \theta_j)\vec{u}_j + (I - UU^\top)y. \end{aligned}$$

Taking squared ℓ_2 norms of the difference,

$$\begin{aligned} \|y - X\beta\|_2^2 &= \left\| \sum_{j=1}^{p} (\vec{u}_j^\top y - \sigma_j \theta_j)\vec{u}_j + (I - UU^\top)y \right\|_2^2 \\ &= \sum_{j=1}^{p} (\vec{u}_j^\top y - \sigma_j \theta_j)^2 + \left\| (I - UU^\top)y \right\|_2^2. \end{aligned}$$

Notice that the second term of the decomposition above is independent of θ, so our minimization problem can be re-written into,

$$\hat{\theta}_j = \arg\min_{\theta_j} (\vec{u}_j^\top y - \sigma_j \theta_j)^2,$$

which does not contain the second term in its objective. Taking partial derivatives with respect to θ_j,

$$\begin{aligned} \frac{\partial}{\partial \theta_j} \left(\vec{u}_j^\top y - \sigma_j \theta_j \right)^2 = -2\sigma_j \left(\vec{u}_j^\top y - \sigma_j \theta_j \right) = 0 \\ \iff \vec{u}_j^\top y - \sigma_j \hat{\theta}_j = 0 \\ \iff \sigma_j \hat{\theta}_j = \vec{u}_j^\top y \\ \iff \hat{\theta}_j = \frac{\vec{u}_j^\top y}{\sigma_j}. \end{aligned}$$

Therefore, the $\hat{\theta}$ that minimizes the least squares problem,

$$\hat{\theta} = \begin{bmatrix} \frac{\vec{u}_1^\top y}{\sigma_1} \\ \vdots \\ \frac{\vec{u}_p^\top y}{\sigma_p} \end{bmatrix} = \begin{bmatrix} 1/\sigma_1 & 0 & \cdots & 0 \\ 0 & 1/\sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sigma_p \end{bmatrix} \begin{bmatrix} \vec{u}_1^\top \\ \vec{u}_2^\top y \\ \vdots \\ \vec{u}_p^\top y \end{bmatrix} = \Sigma^{-1} U^\top y.$$

Therefore, $\hat{\beta} = V\hat{\theta} = V\Sigma^{-1}U^\top y$ or equivalently, $\hat{\beta} = \sum_{j=1}^{p} \frac{\vec{u}_j^\top y}{\sigma_j}\vec{v}_j$.

Substituting our decomposition $X := U\Sigma V^\top$ into our normal system,

$$\begin{aligned}
\hat{\beta} &= V\hat{\theta} = V\Sigma^{-1}U^\top y \\
&= V\Sigma^{-2}\Sigma U^\top y = V\Sigma^{-2}I_p\Sigma U^\top y = V\Sigma^{-2}V^\top V\Sigma U^\top y \\
&= V\Sigma^{-2}V^\top(U\Sigma V^\top)^\top y = V\Sigma^{-2}V^\top X^\top y = (V\Sigma^2 V^\top)^{-1}X^\top y \\
&= (V\Sigma^\top U^\top U\Sigma V^\top)^{-1}X^\top y \\
&= \left[(U\Sigma V^\top)^\top(U\Sigma V^\top)\right]^{-1}X^\top y \\
&= (X^\top X)^{-1}X^\top y.
\end{aligned}$$

Therefore, our least squares estimate is re-written as,

$$\hat{\beta} = (X^\top X)^{-1}X^\top y = V\Sigma^{-1}U^\top y.$$

Now, we can derive our fitted response where $\hat{y} = X\hat{\beta}$,

$$\begin{aligned}
\hat{y} = X\hat{\beta} &= U\Sigma V^\top V\Sigma^{-1}U^\top y = UU^\top y \\
&= \sum_{j=1}^{p}(\vec{u}_j^\top y)\vec{u}_j.
\end{aligned}$$

Therefore $\hat{y} = UU^\top y$. Since the columns of U form an orthonormal basis for the column space of X we have that $\mathrm{Col}(U) = \mathrm{Col}(X)$ therefore meaning that $UU^\top$ is the orthogonal projection matrix onto $\mathrm{Col}(X)$. Therefore $\hat{y} = UU^\top y$ says that the fitted response is the orthogonal projection of y onto the column space of the design matrix. That is, for each orthonormal direction $\vec{u}_j$, the scalar $\vec{u}_j^\top y$ is the component of y along $\vec{u}_j$ and multiplying by $\vec{u}_j$ gives the corresponding projected vector. Summing these components gives the entire projection. Our residual vector then becomes,

$$\vec{\varepsilon} = y - \vec{y} = (I - UU^\top)y.$$

By definition, this vector is orthogonal to U. Therefore $\vec{\varepsilon} \perp \mathrm{Col}(X)$. We can verify this,

$$\begin{aligned}
U^\top\hat{\varepsilon} &= U^\top(I - UU^\top)y \\
&= (U^\top U^\top UU^\top)y \\
&= (U^\top - U^\top)y \\
&= 0,
\end{aligned}$$

as required. Assume least squares setup conditional on our design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$,

$$y = X\beta + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0, \quad \mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

Substituting our definition for our responses under linearity,

$$\begin{aligned}\hat{\beta} &= (X^\top X)^{-1} X^\top (X\beta + \varepsilon) \\ &= V\Sigma^{-1}U^\top (X\beta + \varepsilon) \\ &= V\Sigma^{-1}U^\top X\beta + V\Sigma^{-1}U^\top \varepsilon \\ &= V\Sigma^{-1}U^\top U\Sigma V^\top \beta + V\Sigma^{-1}U^\top \varepsilon \\ &= VV^\top \beta + V\Sigma^{-1}U^\top \varepsilon \\ &= \beta + V\Sigma^{-1}U^\top \varepsilon.\end{aligned}$$

Taking variances, we get the variance of our least squares estimate under SVD,

$$\begin{aligned}\mathrm{Var}(\hat{\beta}) &= \mathrm{Var}(V\Sigma^{-1}U^\top \varepsilon) \\ &= V\Sigma^{-1}U^\top (\sigma^2 I_n) U\Sigma^{-1}V^\top \\ &= \sigma^2 V\Sigma^{-1}(U^\top U)\Sigma^{-1}V^\top \\ &= \sigma^2 V\Sigma^{-2}V^\top \\ &= \sigma^2 \sum_{j=1}^{p} \frac{1}{\sigma_j^2} \vec{v}_j \vec{v}_j^\top.\end{aligned}$$

Specifically, along the j eigenvector (direction), the variance is,

$$\mathrm{Var}(\vec{v}_j^\top \hat{\beta}) = \vec{v}_j^\top \mathrm{Var}(\hat{\beta})\vec{v}_j = \sigma^2 \vec{v}_j^\top V\Sigma^{-2}V^\top \vec{v}_j = \frac{\sigma^2}{\sigma_j^2}.$$

Recall that we could write our least squares estimate as,

$$\hat{\beta} = \sum_{j=1}^{p} \frac{\vec{u}_j^\top y}{\sigma_j} y.$$

Thus, $\vec{u}_j^\top y$ is the amount of the response lying in the j-th orthogonal design matrix direction. OLS then divides that quantity by σ_j. When σ_j is large, this division is benign. But, if σ_j is very small, OLS must divide by a very small number and consequently even a tiny noise complement $\vec{u}_j^\top \varepsilon$ becomes magnified enormously in the coefficient direction $\vec{v}_j$,

$$\vec{v}_j(\hat{\beta} - \beta) = \frac{\vec{u}_j^\top \varepsilon}{\sigma_j}, \qquad \mathrm{Var}(\vec{v}_j^\top \hat{\beta}) = \frac{\sigma^2}{\sigma_j^2}.$$

Near-linear dependent among the columns of X manifests itself as one or more very small singular values. The corresponding right singular vector $\vec{v}_j$ describes the particular linear combination of predictors that is nearly unidentified. OLS attempts to estimate the coefficient component along precisely that direction by $1/\sigma_j$ so the noise becomes amplified.

More importantly, the same singular value that inflates sampling variance also causes the OLS solutions to be extremely sensitive to small perturbations in our response vector y. It is appropriate now to look at the condition number κ that compares the most amplified direction to the least amplified direction and gives a measure of ill-conditioning of our OLS solution.

3.4.3 Ill-conditioning and Singularity

Assume the same setup previously where we have a full column rank design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ with SVD $X = U\Sigma V^\top$ for $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p > 0$. Then, recall we could decompose our least squares estimate as,

$$\hat{\beta} = V\Sigma^{-1}U^\top y = X^+ y = \sum_{j=1}^{p} \frac{\vec{u}_j^\top y}{\sigma_j}\vec{v}_j.$$

Taking the first Euclidean norm of the OLS estimator, given that V is an orthogonal matrix,

$$\|\hat{\beta}\|_2^2 = \|V\Sigma^{-1}U^\top y\|_2^2 = \|\Sigma^{-1}U^\top y\|_2^2 = \left\| \begin{bmatrix} 1/\sigma_1 & & 0 \\ & \ddots & \\ 0 & & 1/\sigma_p \end{bmatrix} \begin{bmatrix} \vec{u}_1^\top y \\ \vdots \\ \vec{u}_p^\top y \end{bmatrix} \right\|_2^2 = \sum_{j=1}^{p} \frac{(\vec{u}_j^\top y)^2}{\sigma_j^2}.$$

By our ordering convention of singular values, $\sigma_p \leq \sigma_j$ for every j meaning that $\dfrac{1}{\sigma_j^2} \leq \dfrac{1}{\sigma_p^2}$,

$$\|\hat{\beta}\|_2^2 = \sum_{j=1}^{p} \frac{(\vec{u}_j^\top y)^2}{\sigma_j^2} \leq \frac{1}{\sigma_p^2}\sum_{j=1}^{p}(\vec{u}_j^\top y)^2 = \frac{1}{\sigma_p^2}\|U^\top y\|_2^2 \leq \frac{1}{\sigma_p^2}\|y\|_2^2.$$

Therefore, the squared norm of our least squares estimate is clearly bounded by the norm of the response vector scaled by the smallest singular value. Suppose that now we perturb our response from y to $y + \delta y$. Then the change in our least squares estimate can be formulated as,

$$\hat{\beta} + \delta\hat{\beta} = X^+(y + \delta y) = X^+ y + X^+(\delta y) = \hat{\beta} + X^+\delta y,$$

where $X^+ = V\Sigma^{-1}U^\top$. Our perturbation,

$$\delta\hat{\beta} = X^+\delta y = V\Sigma^{-1}U^\top\delta y = \sum_{j=1}^{p} \frac{\vec{u}_j^\top \delta y}{\sigma_j}\vec{v}_j.$$

Taking ℓ_2 squared norms,

$$\begin{aligned} \|\delta\hat{\beta}\|_2^2 &= \left\| \sum_{j=1}^{p} \frac{\vec{u}_j^\top \delta y}{\sigma_j}\vec{v}_j \right\|_2^2 = \sum_{j=1}^{p} \frac{(\vec{u}_j^\top \delta y)^2}{\sigma_j^2} \\ &\leq \frac{1}{\sigma_p^2}\sum_{j=1}^{p}(\vec{u}_j^\top \delta y)^2 = \frac{1}{\sigma_p^2}\|U^\top \delta y\|_2^2 \leq \frac{1}{\sigma_p^2}\|\delta y\|_2^2. \end{aligned}$$

Now, we have a bound for a perturbation in our response. Notice that this bound is also sharp. For some scalar $c \in \mathbb{R}$, if we let $\delta y = c\vec{u}_p$ where $\vec{u}_p$ is the direction in the row space of y where perturbations are amplified the most in coefficient space,

$$\begin{aligned} \delta\hat{\beta} &= V\Sigma^{-1}U^\top(c\vec{u}_p) \\ &= cV\Sigma^{-1}e_p = \frac{c}{\sigma_p}\vec{v}_p. \end{aligned}$$

Taking norms, the proportion of amplification is quantified as,

$$\frac{\|\delta\hat{\beta}\|_2}{\|\delta y\|_2} = \frac{|c|/\sigma_p}{|c|} = \frac{1}{\sigma_p}.$$

By contrast, if $\delta y = c\vec{u}_1$, then $\delta\hat{\beta} = \frac{c}{\sigma_1}\vec{v}_1$. Hence, the amplification factors across singular directions range from $\frac{1}{\sigma_1}$ $\frac{1}{\sigma_p}$. The ration between the greatest and smallest amplification is therefore,

$$\frac{1/\sigma_p}{1/\sigma_1} = \frac{\sigma_1}{\sigma_p}.$$

Thus, we arrive at the formulation for the spectral condition number,

$$\kappa_2(X) = \|X\|_2\|X^+\|_2 = \frac{\sigma_1}{\sigma_p}.$$

Indeed because we know that $\|X\|_2 = \|U\Sigma V^\top\|_2 = \|\Sigma\|_2 = \sigma_1$ and $\|X^+\|_2 = \frac{1}{\sigma_p}$,

$$\kappa_(X) = \sigma_1 \cdot \frac{1}{\sigma_p} = \frac{\sigma_{\max}(X)}{\sigma_{\min}(X)}.$$

To interpret this as a relative-error bound, suppose that the signal lies exactly in $\mathrm{Col}(X)$ so $y = X\beta$,

$$\|y\|_2 = \|X\beta\|_2 \le \|X\|_2\|\beta\|_2 = \sigma_1\|\beta\|_2.$$

Therefore, $\|\beta\|_2 \ge \dfrac{\|y\|_2}{\sigma_1}$.

Combining this with the bound $\|\delta\beta\|_2 \le \dfrac{1}{\sigma_p}\|\delta y\|_2$ gives,

$$\frac{\|\delta\beta\|_2}{\|\beta\|_2} = \frac{\frac{1}{\sigma_p}\|\delta y\|_2}{\frac{1}{\sigma_1}\|y\|_2} = \frac{\sigma_1}{\sigma_p}\frac{\|\delta y\|_2}{\|y\|_2} = \kappa_2(X)\cdot\frac{\|\delta y\|_2}{\|y\|_2}.$$

Therefore, we get the bound,

$$\frac{\|\delta\beta\|_2}{\|\beta\|_2} \le \kappa_2(X)\cdot\frac{\|\delta y\|_2}{\|y\|_2}.$$

Returning to our linear model $y = X\beta + \varepsilon$ and using the fact that $\mathrm{Var}(\vec{u}_j^\top\varepsilon) = \sigma^2$,

$$\hat{\beta} - \beta = V\Sigma^{-1}U^\top\varepsilon = \sum_{j=1}^{p}\frac{\vec{u}_j^\top\varepsilon}{\sigma_j}\vec{v}_j \iff \|\hat{\beta}-\beta\|_2^2 = \sum_{j=1}^{p}\frac{(\vec{u}_j^\top\varepsilon)^2}{\sigma_j^2}$$

$$\iff \mathbb{E}\|\hat{\beta}-\beta\|_2^2 = \sum_{j=1}^{p}\frac{\mathbb{E}[(\vec{u}_j^\top\varepsilon)^2]}{\sigma_j^2} = \sigma^2\sum_{j=1}^{p}\frac{1}{\sigma_j^2}.$$

Since $\sigma_j \ge \sigma_p$, this expectation is bounded by,

$$\mathbb{E}\|\hat{\beta}-\beta\|_2^2 \le \sigma^2\sum_{j=1}^{p}\frac{1}{\sigma_p^2} = \frac{p\sigma^2}{\sigma_p^2}.$$

At the level of the covariance matrix,

$$\mathrm{Var}(\hat{\beta}) = \sigma^2 V\Sigma^{-2}V^\top \implies \|\mathrm{Var}(\hat{\beta})\|_2 = \sigma^2\|\Sigma^{-2}\|_2 = \frac{\sigma^2}{\sigma_p^2}.$$

With our fitted response,

$$\begin{aligned}\hat{y} = UU^\top y &\iff \delta\hat{y} = UU^\top \delta y \\ &\iff \|\delta\hat{y}\|_2 = \|UU^\top\delta y\|_2 \le \|UU^\top\|_2\|\delta y\|_2 = \|\delta y\|_2.\end{aligned}$$

The projection of y on to $UU^\top y$ has operator norm 1 so it does not amplify Euclidean perturbation at all. However, the inverse mapping $\hat{y} \mapsto \hat{\beta} = V\Sigma^{-1}U^\top\hat{y}$ contains $\Sigma^{-1} = \mathrm{diag}(1/\sigma_1, \ldots, 1/\sigma_p)$. So, when σ_p is tiny, there exists a direction $\vec{v}_p$ in coefficient space in which a large change in coefficients produces only a tiny change in the fitted values i.e., $X\vec{v}_p = \sigma_p\vec{u}_p$. If $\sigma_p \approx 0$, then $X\vec{v}_p \approx 0$. Therefore for some scalar c,

$$X(\beta + c\vec{v}_p) = X\beta + cX\vec{v}_p = X\beta + c\sigma_p\vec{u}_p \approx X\beta,$$

even when c itself is large. The data therefore cannot reliably distinguish coefficient vectors that differ substantially along nearly-null directions of X. We see this with the conditioning number $\kappa_2(X) = \frac{\sigma_1}{\sigma_p}$ where when $\sigma_p \to 0$ makes $\kappa_2(X) \to \infty$ and simultaneously we get that,

$$\|X^+\|_2 = \frac{1}{\sigma_p} \to \infty, \qquad \|\mathrm{Var}(\hat{\beta})\|_2 = \frac{\sigma^2}{\sigma_p^2} \to \infty.$$

Moreover, if you solve OLS through normal equations, then,

$$X^\top X = V\Sigma^2 V^\top,$$

so its largest and smallest eigenvalues are simply $\lambda_{\max}(X^\top X) = \sigma_1^2$ and $\lambda_{\min}(X^\top X) = \sigma_p^2$. Consequently we get that,

$$\kappa_2(X^\top X) = \frac{\lambda_{\max}(X^\top X)}{\lambda_{\min}(X^\top X)} = \frac{\sigma_1^2}{\sigma_p^2} = \left(\frac{\sigma_1}{\sigma_p}\right)^2 = \kappa_2(X)^2.$$

So, forming $X^\top X$ squares the condition number meaning that if X is already ill-conditioned, the normal eqautions make that numerical conditioning substantially worse. Naturally, we ask, can we modify the estimation problem so that directions associated with very small singular values are no longer allowed to receive arbitrarily large amplification?

Part III

Shrinkage Methods

Chapter 4

Ridge & LASSO Regression

We saw that previously, if we do SVD on the feature matrix X such that $X = U\Sigma V^\top$ where the columns of U form an orthonormal basis for X and the columns of V form an orthonormal basis for X, then along the j-th right singular vector of X, or an eigenvector of $X^\top X$, we can write that,

$$X^\top X = V\Sigma^2 V^\top.$$

From this formulation, we showed that the variance of the least squares estimate along the j-th right singular vector of X is,

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1} = \sigma^2 V \Sigma^{-2} V^\top.$$

Therefore, projecting the coefficient estimate onto direction $\vec{v}_j$,

$$\text{Var}(\vec{v}_j^\top \hat{\beta}) = \sigma^2 \vec{v}_j^\top V \Sigma^{-2} V^\top \vec{v}_j = \frac{\sigma^2}{\sigma_j^2}.$$

Therefore, the variance of the OLS coefficient estimate along the j-th right singular vector direction $\vec{v}_j$ is equal to $\frac{\sigma^2}{\sigma_j^2}$. In the case that a column(s) in our feature matrix are approximately linear combinations of the other columns, then there is some coefficient direction such as $\vec{v}_j$ for which $X\vec{v}_j \approx 0$ meaning that changing the coefficients in this direction barely change the fitted values $X\beta$. Taking norms, we see that $\|X\vec{v}_j\| = \sigma_j$ so that when $X\vec{v}_j \approx 0$ means that $\sigma_j \approx 0$. So as $\sigma_j \to 0$, then $\frac{\sigma^2}{\sigma_j^2} \to \infty$ precisely quantifying the instability created by multicollinearity.

In the perfect collinearity case, when σ_j is exactly 0, the Gram matrix $X^\top X$ is singular, meaning that one or more singular values is 0 implying that the ordinary inverse of $X^\top X$ does not exist. Therefore, OLS no longer has a unique coefficient solution. More generally, even when singular values are merely close to zero, the resulting OLS estimates can exhibit substantial variance and numerical instability as shown in the previous section.

This motivates the introduction of shrinkage estimators which deliberately regularize the coefficient estimates to improve stability in the presence of collinearity and ill-conditioning. In this section, we rigorously introduce and derive several such methods, beginning with ridge regression and subsequently considering alternative forms of regularization.

4.1 Ridge & ℓ_2 Regularization

Ridge regression is the first shrinkage estimator method we will discuss and investigate. Published in 1970 by *Arthur E. Hoerl* and *Robert W. Kennard*, ridge regression is introduced as a *"Biased Estimation"* method for *"Nonorthogonal Problems"* with an applications seminar paper published in conjunction with their core work.

Ridge regression imposes an ℓ_2 penalty on the size of the coefficients where the penalty term, λ, is a tunable parameter through cross-validation. The ridge estimation problem can be written as,

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta \in \mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}, \tag{4.1.1}$$

with its closed form solution being,

$$\hat{\beta}_{\text{ridge}} = \left(X^\top X + \lambda I_n \right)^{-1} X^\top y.$$

Theorem 4.1. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ be a feature matrix and let $y \in \mathbb{R}^n$ be a response vector. Then, the closed-form ridge solution to the minimization problem in* ***Equation (1)*** *is,*

$$\hat{\beta}_{ridge} = (X^\top X + \lambda I)^{-1} X^\top y. \tag{4.1.2}$$

Proof. We aim to derive (4.1.2) from (4.1.1).

Expanding the minimization constraints in the curly braces,

$$\begin{aligned} \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2 &= (y - X\beta)^\top (y - X\beta) + \lambda\beta^\top\beta \\ &= y^\top y - y^\top X\beta - X^\top\beta^\top y + X^\top\beta^\top\beta X + \lambda\beta^\top\beta \\ &= y^\top y - 2X^\top\beta^\top y + \beta^\top\beta(X^\top X + \lambda I_n), \end{aligned}$$

by symmetry of scalar transpose. Taking the derivative wrt β,

$$\nabla_\beta = -2X^\top y + 2\beta(X^\top X + \lambda I_n). \tag{4.1.3}$$

Setting the gradient in (4.1.3) to 0 and solving for β we get,

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I_n)^{-1} X^\top y.$$

□

Remark 4.2. In practice, ridge is often applied to centered inputs such that each x_{ij} for $i \neq j$ is replaced with $x_{ij} - \bar{x}_j$. It is important to note that the ridge penatly is imposed only on the slope coefficients and not on the intercept β_0 when we decide to include it. Penalizing the intercept would destroy the desired location invariance property of the regression model. That is, translating a predictor by a constant should merely change the intercept while leaving the estimated slopes and fitted values unchanged. The intercept depends on the arbitrary choice of origin, shrinking it toward zero would cause the ridge solution to depend on howe the variables are centered rather than solely on their relationships with the response.

Notice that the choice of the penalty term enters the component with which we take the inverse of the gram matrix $X^\top X$ where we are adding a positive constant to the diagonal of $X^\top X$ before inversion. This ensures that non-singularity is almost surely impossible even when X is not full column rank.

Theorem 4.3. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and $\lambda \in \mathbb{R}$. Then,*

$$X^\top X + \lambda I_n,$$

is invertible and non-singular, almost surely for $\lambda > 0$.

Proof. We will prove theorem 4.3 two ways: singularity asymptotic argument and norm argument.

We start with the singular value argument. Suppose that $X = U\Sigma V^\top$ for a typical singular value decomposition of our design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$. Then, the expansion of the penalized gram matrix,

$$\begin{aligned} X^\top X + \lambda I_p &= V\Sigma^\top U^\top U\Sigma V^\top + \lambda I_p \\ &= V\Sigma^\top \Sigma V^\top + \lambda V V^\top \\ &= V(\Sigma^\top \Sigma + \lambda I_p)V^\top. \end{aligned}$$

Let the eigenvalues of $X^\top X$ be $\sigma_1^2, \sigma_2^2, \ldots, \sigma_p^2$ where each $\sigma_j^2 \geq 0$. Therefore, the eigenvalues of $X^\top X + \lambda I_p$ are $\sigma_j^2 + \lambda$ for $j = 1, \ldots, p$. Hence, every eigenvalue is strictly positive. Equivalently,

$$\det(X^\top X + \lambda I_p) = \prod_{j=1}^{p} (\sigma_j^2 + \lambda) > 0,$$

which implies that $X^\top X + \lambda I_p$ is non-singular and therefore invertible. In particular, consider the limiting case in which collinearity becomes increasingly severe. As $\sigma_j \to 0$ we have that $\sigma_j + \lambda \to \lambda > 0$. Thus, even when a singular value approaches zero, the corresponding eigenvalue of the penalized gram matrix remains bounded below by λ.

Therefore, the singularity of $X^\top X$ is removed by addition of the ridge penalty.

For the norm argument, suppose for a contradiction that $X^\top X + \lambda I_p$ is singular. Then there exists some non-zero vector $\vec{v} \in \mathbb{R}^p$ such that $(X^\top X + \lambda I_p)\vec{v} = 0$. Left-multiplying by $\vec{v}^\top$,

$$\begin{aligned} 0 &= \vec{v}^\top (X^\top X + \lambda I_p)\vec{v} \\ &= \vec{v}^\top X^\top X \vec{v} + \lambda \vec{v}^\top \vec{v} \\ &= (X\vec{v})^\top (X\vec{v}) + \lambda \vec{v}^\top \vec{v} \\ &= \|X\vec{v}\|_2^2 + \lambda \|\vec{v}\|_2^2. \end{aligned}$$

But, $\|X\vec{v}\|_2^2 \geq 0$ and since $\vec{v} \neq 0$ and $\lambda > 0$, then $\lambda\|\vec{v}\|_2^2 > 0$. Therefore we see that,

$$\|X\vec{v}\|_2^2 + \lambda \|\vec{v}\|_2^2 > 0,$$

which contradicts $\|X\vec{v}\|_2^2 + \lambda\|\vec{v}\|_2^2 = 0$. Therefore, $\ker(X^\top X + \lambda I_p) = \{0\}$ so $X^\top X + \lambda I_p$ is non-singular and therefore invertible. This is deterministically true, always.

Alternatively, suppose that $\lambda > 0$ and $\vec{v}$ is an eigenvector corresponding to the eigenvalue λ'. Here λ represents our tunable penalty term and λ' is the eigenvalue with corresponding eigenvector $\vec{v}$.

Then it follows that,

$$\begin{aligned}(X^\top X + \lambda I_p)\vec{v} = \lambda'\vec{v} &\iff \vec{v}^\top(X^\top X + \lambda I_p)\vec{v} = \vec{v}^\top(\lambda'\vec{v}) \\ &\iff \vec{v}^\top X^\top X\vec{v} + \lambda\vec{v}^\top\vec{v} = \lambda'\vec{v}^\top\vec{v} \\ &\iff (X\vec{v})^\top(X\vec{v}) + \lambda\vec{v}^\top\vec{v} = \lambda'\vec{v}^\top\vec{v} \\ &\iff \|X\vec{v}\|^2 + \lambda\|\vec{v}\|^2 = \lambda'\|\vec{v}\|^2 \\ &\iff \frac{\|X\vec{v}\|^2}{\|\vec{v}\|^2} + \lambda = \lambda', \qquad \|\vec{v}\|^2 > 0.\end{aligned}$$

Since,

$$\lambda' = \frac{\|X\vec{v}\|^2}{\|\vec{v}\|^2} + \lambda,$$

of which both are greater than or equal to 0, we have that $\lambda' \geq \lambda > 0$.

Thus every eigenvalue of $X^\top X + \lambda I_p$ is strictly positive so the matrix has no zero eigenvalue and therefore the penalized gram matrix is invertible. □

Theorem 4.4. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and $y \in \mathbb{R}^n$ for $\lambda > 0$. The ridge regression estimator,*

$$\hat{\beta}_{ridge} = \arg\min_{\beta\in\mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\},$$

is equivalent, for an appropriate $t \geq 0$ to the constraint optimization problem,

$$\hat{\beta}_\lambda^{ridge} = \arg\min_{\beta\in\mathbb{R}^p} \|y - X\beta\|_2^2, \qquad s.t. \quad \|\beta\|_2^2 \leq t.$$

Thus, ridge restricts the coefficient vector to the solution space $\mathcal{B}_2(t) = \{\beta \in \mathbb{R}^p : \|\beta\|_2^2 \leq t\}$ which is an ℓ_2-ball centered at the origin. Moreover, because $\lambda > 0$, we know that $X^\top X + \lambda I_p$ is positive definite, so the ridge objective is strictly convex and hence has a unique solution.

Proof. Let the ridge estimator be defined as it is in *Theorem 4.4*. Define $t = \|\hat{\beta}_\lambda\|_2^2$. We show that $\hat{\beta}_\lambda$ also minimizes the residual sum of squares over the ℓ_2-ball such that $\{\beta : \|\beta\|_2^2 \leq t\}$. Suppose, for contradiction that it does not. Then there exists some $\tilde{\beta}$ satisfying,

$$\|\tilde{\beta}\|_2^2 \leq t = \|\hat{\beta}_\lambda\|_2^2 \quad \text{s.t.} \quad \|y - X\tilde{\beta}\|_2^2 < \|y - X\hat{\beta}_\lambda\|_2^2.$$

Because we know that $\lambda > 0$ then $\lambda\|\tilde{\beta}\|_2^2 \leq \lambda\|\hat{\beta}_\lambda\|_2^2$,

$$\|y - X\tilde{\beta}\|_2^2 + \lambda\|\tilde{\beta}\|_2^2 < \|y - X\hat{\beta}_\lambda\|_2^2 + \lambda\|\hat{\beta}_\lambda\|_2^2.$$

However, this contradicts the definition of $\hat{\beta}_\lambda$ as the minimizer of the ridge objective. Hence no such $\tilde{\beta}$ can exists. The penalized ridge problem is equivalent to least squares constrained to an ℓ_2 ball. □

4.2 Singular Value Decomposition of Ridge

Our SVD of OLS previously showed that directions that correspond to near-zero singular values result in numerically unstable/non-unique solutions of the least squares estimate. To understand how

ridge estimation resolves this problem, we derive it's SVD and compare it to OLS. Recall that for a given design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ with corresponding response vector $y \in \mathbb{R}^n$, we can derive an SVD $X = U\Sigma V^\top$ where the columns of U form an orthonormal basis for the columns of X and V has columns that form an orthonormal basis for the rows of X i.e., the column space of $X^\top$.

Then, the gram matrix is derived as,

$$\begin{aligned} X^\top X &= (U\Sigma V^\top)^\top (U\Sigma V^\top) \\ &= U^\top \Sigma^\top (V^\top)^\top U\Sigma V^\top \\ &= U^\top U V \Sigma^\top \Sigma V^\top. \end{aligned}$$

Recall that Σ is a diagonal matrix whose entries are the non-negative singular values, then multiplying it by it's transpose is equivalent to squaring the diagonal entries,

$$\Sigma^\top \Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_p \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_p \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_p^2 \end{bmatrix} = \Sigma^2.$$

Therefore, since U is orthogonal, the gram matrix simplifies to,

$$X^\top X = V\Sigma^2 V^\top.$$

Adding λI_p is equivalent to adding λ to the diagonals of the squared Σ matrix so we have that,

$$X^\top X + \lambda I_p = V \operatorname{diag}(\sigma_1^2 + \lambda, \ldots, \sigma_p^2) V^\top.$$

Applying the inverse to the penalized gram matrix we obtain,

$$\begin{aligned} X^\top X + \lambda I_p &= V^{-1} \left[\operatorname{diag}(\sigma_1^2 + \lambda, \ldots, \sigma_p^2 + \lambda)\right]^{-1} (V^\top)^{-1} \\ &= V \operatorname{diag}\left(\frac{1}{\sigma_1^2 + \lambda}, \ldots, \frac{1}{\sigma_p^2 + \lambda}\right) V^\top \end{aligned}$$

The remaining $X^\top y$ component under SVD is computed as,

$$X^\top y = (U\Sigma V^\top)^\top y = U^\top \Sigma^\top V y = V\Sigma^\top U^\top y.$$

Putting the penalized gram matrix with the remaining y component,

$$\begin{aligned}
\hat{\beta}_\lambda &= (X^\top X + \lambda I_p)^{-1} X^\top y \\
&= V \operatorname{diag}\left(\frac{1}{\sigma_1^2 + \lambda}, \ldots, \frac{1}{\sigma_p^2 + \lambda}\right) \underbrace{\cancel{V^\top V}}_{=I_p} \Sigma^\top U^\top y \\
&= V \operatorname{diag}\left(\frac{1}{\sigma_1^2 + \lambda}, \ldots, \frac{1}{\sigma_p^2 + \lambda}\right) \Sigma^\top U^\top y \\
&= V \operatorname{diag}\left(\frac{\sigma_1}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p}{\sigma_p^2 + \lambda}\right) U^\top y.
\end{aligned}$$

Our fitted ridge solution is therefore computed as,

$$\begin{aligned}
X\hat{\beta}_\lambda &= X(X^\top X + \lambda I)^{-1} X^\top y \\
&= U\Sigma \cancel{V^\top V} \operatorname{diag}\left(\frac{\sigma_1}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p}{\sigma_p^2 + \lambda}\right) U^\top y \\
&= \cancel{UV^\top V} \operatorname{diag}(\sigma_1, \ldots, \sigma_p) \cdot \operatorname{diag}\left(\frac{\sigma_1}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p}{\sigma_p^2 + \lambda}\right) U^\top y \\
&= U \cdot \operatorname{diag}\left(\frac{\sigma_1^2}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p^2}{\sigma_p^2 + \lambda}\right) U^\top y \\
&= \sum_{j=1}^{p} \vec{u}_j \cdot \frac{\sigma_j^2}{\sigma_j^2 + \lambda} \cdot \vec{u}_j^\top y,
\end{aligned}$$

where the vectors $\vec{u}_j$ for $j = 1, \ldots, p$ are the orthonormal columns of matrix U for our SVD. Similar to OLS, we compute the coordinates of our response vector y with respect to this orthonormal basis then shrink the coefficients by factors of $\frac{\sigma_j^2}{\sigma_j^2 + \lambda}$. Under the minimization problem, greater shrinkage is applied to coordinates of basis vectors with smaller σ_j^2 i.e., $\sigma_j^2 \approx 0$ since σ_j^2 is always non-negative.

Fixing our design matrix X, we compute the variance of the ridge estimate as,

$$\begin{aligned}
\operatorname{Var}(\hat{\beta}_\lambda) &= \operatorname{Var}\left((X^\top X + \lambda I_p)^{-1} X^\top y\right) \\
&= (X^\top X + \lambda I_p)^{-1} X^\top \operatorname{Var}(y) X (X^\top X + \lambda I_p)^{-1} \\
&= \sigma^2 (X^\top X + \lambda I_p)^{-1} X^\top X (X^\top X + \lambda I_p)^{-1}.
\end{aligned}$$

Substituting our SVD,

$$\operatorname{Var}(\hat{\beta}_\lambda) = \sigma^2 V \cdot \operatorname{diag}\left(\frac{\sigma^2}{(\sigma_1^2 + \lambda)^2}, \ldots, \frac{\sigma_p^2}{(\sigma_p^2 + \lambda)^2}\right) V^\top.$$

Theorem 4.5. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ be a design matrix and $y \in \mathbb{R}^n$ be a corresponding response vector. Then, we obtain an SVD such that $X = U\Sigma V^\top$ where the columns of U, namely u_j's for $j = 1, \dots, p$ form an orthonormal basis for the columns of X and the columns of V form an orthonormal basis for the row space of X i.e., the column space of $X^\top$.*

For some ridge estimate that satisfies the ridge minimization problem (4.1.1)*,*

$$\hat{\beta}_\lambda = V \operatorname{diag}\left(\frac{\sigma_1}{\sigma_1^2+\lambda}, \dots, \frac{\sigma_p}{\sigma_p^2+\lambda}\right) U^\top y,$$

is the SVD of the Ridge estimate with respect to (4.1)*.*

Proof. Proof follows from the previous derivations. □

Theorem 4.6. *For a design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and response vector $y \in \mathbb{R}^n$, the ridge estimate that satisfies* (4.1.1) *is a biased estimate,*

$$\mathbb{E}[\hat{\beta}_\lambda] = (X^\top X + \lambda I)^{-1} X^\top X \beta.$$

Proof. Taking expectations of the ridge estimate conditioning on our design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$,

$$\begin{aligned}\mathbb{E}[\hat{\beta}_\lambda \mid X] &= \mathbb{E}[(X^\top X + \lambda I)^{-1} X^\top y \mid X] \\ &= (X^\top X + \lambda I)^{-1} X^\top \mathbb{E}[y \mid X] \\ &= (X^\top X + \lambda I)^{-1} X^\top X\beta \neq \beta.\end{aligned}$$

Therefore, we conclude that ridge is biased.

Alternatively, for some $\lambda > 0$ using $X^\top X = V\Sigma^2 V^\top$ under SVD,

$$\mathbb{E}[\hat{\beta}_\lambda \mid X] = V \cdot \operatorname{diag}\left(\frac{\sigma_1^2}{\sigma_1^2+\lambda}, \dots, \frac{\sigma_p^2}{\sigma_p^2+\lambda}\right) V^\top \beta.$$

□

4.3 Principal Component Analysis of Ridge

As you may have noticed, doing SVD on a centered design matrix is equivalent to *Principal Component Analysis*. The columns of V, namely $\vec{v}_j$'s form an orthonormal basis for the row space of X and represent the principal components/directions of X. That is, given the eigendecomposition of,

$$X^\top X = V\Sigma^2 V^\top,$$

taking $z_j = X\vec{v}_j$ gives us the sample variance along the j-th principal component.

Theorem 4.7. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_p\}$ be an orthonormal basis for $\mathrm{Col}(X)$. Centered X and define $z_{ij} = x_i^\top \vec{v}_j$ for $j = 1, \ldots, p$. Define $\bar{z}_j = \frac{1}{n}\mathbf{1}^\top X\vec{v}_j = 0$. Then, the sample variance along the j-th principal component $\vec{v}_j$ is,*

$$\mathrm{Var}(\boldsymbol{z}_j) = \mathrm{Var}(X\vec{v}_j) = \frac{\sigma_j^2}{n-1}.$$

Proof. Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and $\vec{v}_j \in \mathcal{B}$ where $\mathcal{B}$ is a p-dimensional orthonormal basis for $\mathrm{Col}(X)$. Then, $z_j = X\vec{v}_j$ is the vector of principal-component scores of all observations along direction $\vec{v}_j$. The variance along $\vec{v}_j$ is obtained from the squared norm of this vector.

Define the sample covariance matrix as,

$$S = \frac{1}{n-1}X^\top X, \qquad X^\top X = V\Sigma^2 V^\top.$$

Since each ridge singular vector $\vec{v}_j$ satisfies the system $X^\top X\vec{v}_j = \sigma_j^2\vec{v}_j$, we project every observation $x_i^\top$ onto $\vec{v}_j$. Define the i-th projected coordinate as $z_{ij} = x_i^\top \vec{v}_j$ such that stacking all n projected coordinates yields,

$$z_j = \begin{bmatrix} x_1^\top\vec{v}_j \\ x_2^\top\vec{v}_j \\ \vdots \\ x_n^\top\vec{v}_j \end{bmatrix} = \begin{bmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_n^\top \end{bmatrix} \vec{v}_j = X\vec{v}_j.$$

$z_j = X\vec{v}_j$ is precisely the j-th principal component score vector. Assuming that X is a centered design matrix, then z_j is also centered. Define,

$$\bar{z}_j = \frac{1}{n}\mathbf{1}^\top X\vec{v}_j = 0.$$

Therefore, the sample variance is computed as,

$$\begin{aligned}
\mathrm{Var}(z_j) &= \frac{1}{n-1}z_j^\top z_j \\
&= \frac{1}{n-1}(X\vec{v}_j)^\top(X\vec{v}_j) \\
&= \frac{1}{n-1}\vec{v}_j^\top X^\top X\vec{v}_j \\
&= \frac{1}{n-1}\vec{v}_j^\top(\sigma_j^2\vec{v}_j) = \frac{\sigma_j^2}{n-1}\vec{v}_j^\top\vec{v}_j = \frac{\sigma_j^2}{n-1},
\end{aligned}$$

since $\vec{v}_j$'s are orthogonal, $\vec{v}_j^\top\vec{v}_j = 1$. Thus $z_j = X\vec{v}_j$ is the vector of observations expressed along the j-th principal direction while $\mathrm{Var}(z_j) = \frac{\sigma_j^2}{n-1}$ is the sample variance explained along that direction. Equivalently, we can derive the variance directly from the covariance matrix for easier computation,

$$\mathrm{Var}(X\vec{v}_j) = \vec{v}_j^\top S\vec{v}_j = \frac{1}{n-1}\vec{v}_j^\top X^\top X\vec{v}_j = \frac{\sigma_j^2}{n-1}.$$

□

From *Theorem 4.7* above, we often list principal components in descending order with the largest principal component z_1 corresponding to the direction with the largest sample variance amongst all linear combinations of the columns of X. Generically, we can also derive the variance along a particular projected principal component $\vec{v}_j$.

Theorem 4.8. *Let $\hat{\beta}_\lambda$ be the ridge estimate that satisfies* (4.1.1) *and $\vec{v}_j$ be the j-th column vector of the orthonormal matrix that forms the SVD for our design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$. Then,*

$$\text{Var}(\vec{v}_j\hat{\beta}_\lambda) = \frac{\sigma^2 \cdot \sigma_j^2}{(\sigma_j^2+\lambda)^2},$$

is the variance of the ridge estimate along the j-th eigendirection.

Proof. We aim to derive a generic formulation for $\text{Var}(\vec{v}_j\hat{\beta}_\lambda)$, that is the variance along a particular direction of our ridge estimate. Recall the closed-form SVD of the ridge estimate,

$$\hat{\beta}_\lambda = V \cdot \text{diag}\left(\frac{\sigma_1}{\sigma_1^2+\lambda}, \dots, \frac{\sigma_p}{\sigma_p^2+\lambda}\right) U^\top y.$$

We obtain the full covariance matrix as,

$$\begin{aligned}
\text{Var}(\hat{\beta}_\lambda \mid X) &= \text{Var}\left((X^\top X + \lambda I)^{-1} X^\top y \mid X\right) \\
&= (X^\top X + \lambda I_p)^{-1} X^\top \cdot \text{Var}(y \mid X) \cdot X(X^\top X + \lambda I_p)^{-1} \\
&= \sigma^2 (X^\top X + \lambda I_p)^{-1} X^\top X (X^\top X + \lambda I_p)^{-1}.
\end{aligned}$$

Substituting the SVD of the inverse penalized gram matrix we obtained previously,

$$\begin{aligned}
\text{Var}(\hat{\beta}_\lambda \mid X) = \sigma^2 &\left[V \,\text{diag}\left(\frac{1}{\sigma_1^2+\lambda}, \dots, \frac{1}{\sigma_p^2+\lambda}\right) V^\top\right] \\
&\cdot \left[V \,\text{diag}(\sigma_1^2, \dots, \sigma_p^2) V^\top\right] \cdot \left[V \,\text{diag}\left(\frac{1}{\sigma_1^2+\lambda}, \dots, \frac{1}{\sigma_p^2+\lambda}\right) V^\top\right] \\
= \sigma^2 V \,\text{diag}&\left(\frac{1}{\sigma_1^2+\lambda}, \dots, \frac{1}{\sigma_p^2+\lambda}\right) V^\top V \cdot \text{diag}(\sigma_1^2, \dots, \sigma_p^2) V^\top V \,\text{diag}\left(\frac{1}{\sigma_1^2+\lambda}, \dots, \frac{1}{\sigma_p^2+\lambda}\right) V^\top \\
= \sigma^2 V \,\text{diag}&\left(\frac{\sigma_1^2}{(\sigma_1^2+\lambda)^2}, \dots, \frac{\sigma_p^2}{(\sigma_p^2+\lambda)^2}\right) V^\top.
\end{aligned}$$

Projecting this covariance matrix onto $\vec{v}_j$, by properties of scalar linear combinations,

$$\begin{aligned}\mathrm{Var}(\vec{v}_j\hat{\beta}_\lambda) &= \vec{v}_j^\top \mathrm{Var}(\hat{\beta}_\lambda \mid X)\vec{v}_j \\ &= \sigma^2 \vec{v}_j^\top V \,\mathrm{diag}\left(\frac{\sigma_1^2}{(\sigma_1^2+\lambda)^2}, \dots \frac{\sigma_p^2}{(\sigma_p^2+\lambda)^2}\right) V^\top \vec{v}_j \\ &= \sigma^2 \vec{e}_j^\top \,\mathrm{diag}\left(\frac{\sigma_1^2}{(\sigma_1^2+\lambda)^2}, \dots, \frac{\sigma_p^2}{(\sigma_p^2+\lambda)^2}\right) \vec{e}_j \\ &= \sigma^2 \cdot \frac{\sigma_j^2}{(\sigma_j^2+\lambda)^2},\end{aligned}$$

since $\vec{v}_j$'s for an orthonormal basis, we know that $V^\top \vec{v}_j = \vec{e}_j$ and similarly $\vec{v}_j^\top V = \vec{e}_j$ where $\vec{e}_j$ is the j-th standard basis vector. Therfore, the variance of the ridge coefficient estimate along the j-th principal component/right-singular vector direction $\vec{v}_j$ is,

$$\mathrm{Var}(\vec{v}_j\hat{\beta}_\lambda \mid X) = \frac{\sigma^2\sigma_j^2}{(\sigma_j^2+\lambda)^2} = \frac{\sigma^2}{\sigma_j^2}\left(\frac{\sigma_j^2}{\sigma_j^2+\lambda}\right)^2 = \mathrm{Var}(\vec{v}_j^\top \hat{\beta}_{\mathrm{OLS}} \mid X) \cdot \left(\frac{\sigma_j^2}{\sigma_j^2+\lambda}\right)^2.$$

□

Remark 4.9. By the proof above we see that ridge reduces the OLS variance along each principal direction by the square of its corresponding ridge shrinkage factor, where directions corresponding to smaller σ_j^2 (near-zero) values have greater shrinkage. This is why ridge resolves the multicollinearity problem. The directions that are most unstable under OLS (directions associated with very small singular values) are precisely the directions whose sampling variance ridge suppresses most strongly.

Moreover, along the j-th right singular vector direction, we know that $\vec{v}_j^\top \hat{\beta}_\lambda = \frac{\sigma_j^2}{\sigma_j^2+\lambda}\vec{v}_j^\top \hat{\beta}_{\mathrm{OLS}}$ assuming that $\sigma_j > 0$. For any finite $\lambda > 0$ and any non-zero singular value $\sigma_j > 0$,

$$0 < \frac{\sigma_j^2}{\sigma_j^2+\lambda} < 1.$$

Consequently, ridge makes the coefficient coordinate smaller but the multiplier itself is never exactly zero since $\vec{v}_j^\top \hat{\beta}_{\mathrm{OLS}} \neq 0$ meaning that $\vec{v}_j^\top \hat{\beta}_\lambda \neq 0$. Only in the limit $\lambda \to \infty$ do we obtain $\frac{\sigma_j^2}{\sigma_j^2+\lambda} \to 0$. Of course, if $\sigma_j = 0$ then the shrinkage is precisely zero so ridge sets the component in that completely unidentified nullspace direction to zero. But for identifiable directions with $\sigma_j > 0$, finite ridge regularization does not annihilate the direction.

Consider the ridge hat matrix,

$$H_\lambda = X(X^\top X + \lambda I_p)^{-1} X^\top.$$

For an arbitrary linear smoother $\hat{y} = Hy$ where H is a corresponding projection matrix, the effective degrees of freedom are defined by $\mathrm{df}(\lambda) = \mathrm{tr}(H_\lambda)$. The degrees of freedom of ridge are,

$$\begin{aligned}\mathrm{df}(\lambda) &= \mathrm{tr}\left[U \,\mathrm{diag}\left(\frac{\sigma_1^2}{\sigma_1^2+\lambda}, \dots, \frac{\sigma_p^2}{\sigma_p^2+\lambda}\right) U^\top\right] \\ &= \mathrm{tr}\,\mathrm{diag}\left(\frac{\sigma_1^2}{\sigma_1^2+\lambda}, \dots, \frac{\sigma_p^2}{\sigma_p^2+\lambda}\right) \\ &= \sum_{j=1}^{p} \frac{\sigma_j^2}{\sigma_j^2+\lambda}.\end{aligned}$$

Thus each singular vector direction contributes $\frac{\sigma_j^2}{\sigma_j^2+\lambda}$ of one effective degrees of freedom. OLS, however, for each identifiable direction contributes one full degree of freedom. So for ridge, each direction contributes only a fractional degree of freedom, $0 \leq \frac{\sigma_j^2}{\sigma_j^2+\lambda} < 1$.

Theorem 4.10. *Given the ridge estimate $\hat{\beta}_\lambda$, the degrees of freedom are,*

$$\mathrm{df}(\lambda) = \frac{\sigma_j^2}{\sigma_j^2+\lambda}.$$

The term under the sum also appears under sampling properties of ridge estimates. Suppose we strengthen the linear-model assumptions to linearity $y = X\beta + \varepsilon$ with $\varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 I_n)$ as we did in OLS. Then conditional on X, $y \mid X \sim \mathcal{N}(X\beta, \sigma^2 I_n)$. Since ridge is a linear transformation of our response y, the ridge estimate is a linear transformation of a multivariate normal vector which is again multivariate normal. Deriving each component individually, the mean is simply,

$$\begin{aligned}\mathbb{E}[\hat{\beta}_\lambda \mid X] &= \mathbb{E}[(X^\top X + \lambda I_p)^{-1} X^\top y \mid X] \\ &= (X^\top X + \lambda I_p)^{-1} X^\top \mathbb{E}[y \mid X] \\ &= (X^\top X + \lambda I_p)^{-1} X^\top X \beta.\end{aligned}$$

Using $X^\top X = V \Sigma^2 V^\top$ we obtain,

$$\mathbb{E}[\hat{\beta}_\lambda \mid X] = V \,\mathrm{diag}\left(\frac{1}{\sigma_1^2+\lambda}, \dots, \frac{1}{\sigma_p^2+\lambda}\right) \Sigma^2 V^\top \beta = V^\top \,\mathrm{diag}\left(\frac{\sigma_1^2}{\sigma_1^2+\lambda}, \dots, \frac{\sigma_p^2}{\sigma_p^2+\lambda}\right) V^\top \beta.$$

Similarly, the conditional covariance is computed as,

$$\begin{aligned}\mathrm{Var}(\hat{\beta}_\lambda \mid X) &= \mathrm{Var}\left((X^\top X + \lambda I_p)^{-1} X^\top y \mid X\right) \\ &= (X^\top X + \lambda I_p)^{-1} X^\top \cdot \mathrm{Var}(y \mid X) \cdot X (X^\top X + \lambda I_p)^{-1} \\ &= \sigma^2 (X^\top X + \lambda I_p)^{-1} X^\top X (X^\top X + \lambda I_p)^{-1} \\ &= \sigma^2 V \,\mathrm{diag}\left(\frac{\sigma_1^2}{(\sigma_1^2+\lambda)^2}, \dots, \frac{\sigma_p^2}{(\sigma_p^2+\lambda)^2}\right) V^\top.\end{aligned}$$

Therefore, the full conditional sampling distribution is,

$$\hat{\beta}_\lambda \mid X \sim \mathcal{N}_p \left(V \operatorname{diag}\left(\frac{\sigma_j^2}{\sigma_j^2 + \lambda} \right) V^\top \beta,\ \sigma^2 V \operatorname{diag}\left(\frac{\sigma_j^2}{(\sigma_j^2 + \lambda)^2} \right) V^\top \right).$$

Theorem 4.11. *Let $\hat{\beta}_\lambda$ be the ridge estimate that minimizes* (4.1.1)*. Let $\vec{v}_j$ be the j-th column vector of the orthonormal basis of the row space of our design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ in our ridge minimization problem. Then, $\vec{v}_j^\top \hat{\beta}_\lambda$ has distribution,*

$$\vec{v}_j^\top \hat{\beta}_\lambda \mid X \sim \mathcal{N}\left(\frac{\sigma_j^2}{\sigma_j^2 + \lambda} \vec{v}_j^\top \beta, \frac{\sigma^2 \sigma_j^2}{(\sigma_j^2 + \lambda)^2} \right),$$

under normality assumptions for ridge regression.

Proof. Take the scalar coordinate $\vec{v}_j^\top \hat{\beta}_\lambda$. Because a linear combination of a multivariate normal vector is also normal, then the conditional distribution $\vec{v}_j^\top \hat{\beta}_\lambda \mid X$ is a univariate normal.

Thus, taking expectations,

$$\begin{aligned}
\mathbb{E}[\vec{v}_j^\top \hat{\beta}_\lambda \mid X] &= \vec{v}_j^\top \mathbb{E}[\hat{\beta}_\lambda \mid X] \\
&= \vec{v}_j^\top V \operatorname{diag}(\sigma_1, \ldots, \sigma_p) V^\top \beta \\
&= \vec{e}_j^\top \operatorname{diag}(d_1, \ldots, d_p) V^\top \beta \\
&= \sigma_j \cdot \vec{v}_j^\top \beta \\
&= \frac{\sigma_j^2}{\sigma_j^2 + \lambda} \vec{v}_j^\top \beta.
\end{aligned}$$

Taking variances, similarly, we get,

$$\operatorname{Var}(\vec{v}_j^\top \hat{\beta}_\lambda \mid X) = \vec{v}_j^\top \operatorname{Var}(\hat{\beta}_\lambda \mid X) \vec{v}_j = \sigma^2 \cdot \frac{\sigma_j^2}{(\sigma_j^2 + \lambda)^2}.$$

Therefore, we can conclude that,

$$\vec{v}_j^\top \mid \hat{\beta}_\lambda \sim \mathcal{N}\left(\frac{\sigma_j^2}{\sigma_j^2 + \lambda} \vec{v}_j^\top \beta, \frac{\sigma^2 \sigma_j^2}{(\sigma_j^2 + \lambda)^2} \right).$$

□

4.4 LASSO and Orthogonal Closed-Forms

Recall that previously with ridge regression, we applying greater shrinkage by a factor of $\sigma_j^2/(\sigma_j^2+\lambda)$ for $\sigma_j > 0$ for σ_j^2 that are close (i.e., are close to zero). Therefore, for all identifiable directions where $\sigma_j > 0$, ridge performs shrinkage while retaining all the predictors in the design matrix. However, suppose that instead of performing shrinkage, we would like to obtain a selection of our features in our design matrix to reduce our feature set to become less redundant. In order to do so, we will need a "shrinkage" method that shrinks coefficients exactly down to zero under collinearity conditions.

Suppose again that we have a design matrix $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ and suppose that X and y have been centered so that we do not need to explicitly include an intercept. We define the LASSO estimator to be the solution to the minimization problem,

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta\in\mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda\|\beta\|_1 \right\}.$$

where in particular we have that $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$ is the regularization term that penalizes the absolute value of the coefficients. Expanding the squared error term we obtain,

$$\begin{aligned}\hat{\beta}_{\text{LASSO}} &= \arg\min_{\beta} \left\{ y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta + \lambda\sum_{j=1}^p |\beta_j| \right\} \\ &= \arg\min_{\beta} \left\{ -2\beta^\top X^\top y + \beta^\top X^\top X\beta + \lambda\sum_{j=1}^p |\beta_j| \right\},\end{aligned}$$

which follows because $y^\top y$ does not depend on β so the optimization problem in the last line is equivalent. Hence, when our predictors $x_k^\top x_k \neq 0$, the coefficient β_j interacts with all of the other coefficients such that the optimization problem does not separate into p independent problems.

However, we are able to derive a closed-form solution when our predictors are orthogonal. Taking $z_1 = x_1$ we can apply the Gram-Schmidt orthogonalization procedure to obtain,

$$z_j = x_j - \sum_{k=1}^{j-1} \frac{z_k^\top x_j}{z_k^\top z_k} \cdot z_k,$$

such that the resulting vectors satisfy $z_k^\top z_k = 0$ for $j \neq k$. If we define the matrix Z such that its columns are our z_j's for $j = 1, \ldots, p$, then, $Z^\top Z = \text{diag}(\|z_1\|_2^2, \ldots, \|z_p\|_2^2)$. Normalizing each direction by taking $q_j = \frac{z_j}{\|z_j\|_2}$ then makes the matrix Q orthogonal, that is $Q^\top Q = I_p$.

Theorem 4.12. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ be a design matrix, not necessarily with full column rank, and let $y \in \mathbb{R}^n$ be a corresponding response vector. Then, the LASSO estimate is defined as the solution that satisfies the minimization problem,*

$$\hat{\beta}_{LASSO} = \arg\min_{\beta\in\mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda\|\beta\|_1 \right\}.$$

Remark 4.13. Note the optimization problem in **Theorem 5.1** does not have a closed-form solution.

Suppose that we have applied Gram-Schmidt to our predictors x_k for $k = 1, \ldots, p$ such that $x_k^\top x_k = 0$ for $j \neq k$. That is, for any off-diagonal terms in our design matrix i.e., coordinate for (x_j, x_k) for $j \neq k$, the entries are zero. Therefore, X is a diagonal matrix and the Gram matrix is,

$$\begin{aligned}
X^\top X &= \begin{bmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_p^\top \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \cdots & x_p \end{bmatrix} \\
&= \begin{bmatrix} x_1^\top x_1 & x_1^\top x_2 & \cdots & x_1^\top x_p \\ x_2^\top x_1 & x_2^\top x_2 & \cdots & x_2^\top x_p \\ \vdots & \vdots & \ddots & \vdots \\ x_p^\top x_1 & x_p^\top x_2 & \cdots & x_p^\top x_p \end{bmatrix} \\
&= \begin{bmatrix} \|x_1\|_2^2 & 0 & \cdots & 0 \\ 0 & \|x_2\|_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \|x_p\|_2^2 \end{bmatrix} \\
&= \operatorname{diag}\left(\|x_1\|_2^2, \ldots, \|x_p\|_2^2 \right),
\end{aligned}$$

Excluding the penalty term, the two terms involving y and β have closed forms,

$$\begin{aligned}
\beta^\top X^\top X \beta &= \sum_{j=1}^{p} \beta_j \cdot \|x_j\|_2^2 \cdot \beta_j = \sum_{j=1}^{p} \beta_j^2 \cdot \|x_j\|_2^2 \\
\beta^\top X^\top y &= \sum_{j=1}^{p} \beta_j \|x_j\|_2^2 \cdot y = \sum_{j=1}^{p} \|x_j\|_2^2 \cdot \beta_j \cdot y.
\end{aligned}$$

Substituting the terms into our expanded minimization problem,

$$\begin{aligned}
\beta^\top X^\top X - 2X^\top \beta^\top y + \lambda \|\beta\| &= \sum_{j=1}^{p} \beta_j^2 \cdot \|x_j\|_2^2 - 2 \sum_{j=1}^{p} \beta_j \cdot \|x_j\|_2^2 \cdot y + \lambda |\beta| \\
&= \left(\sum_{j=1}^{p} \beta_j^2 \cdot \|x_j\|_2^2 - 2\beta_j \cdot \|x_j\|_2^2 y \right) + \lambda |\beta| \\
&= \sum_{j=1}^{p} \beta_j^2 \cdot \|x_j\|_2^2 - 2\beta_j \cdot \|x_j\|_2^2 + \lambda |\beta_j|.
\end{aligned}$$

Therefore, the LASSO optimization problem under orthogonality has closed form,

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \beta_j^2 \cdot \|x_j\|_2^2 - 2\beta_j \|x_j\|_2^2 y + \lambda \|\beta_j\| \right\}.$$

Theorem 4.14. *Let $X \in \mathbb{M}_{n\times p}(\mathbb{R})$ be a design matrix with orthogonal columns $x_k^\top x_j = 0$ for $j \neq k$. Then, the LASSO closed-form optimization is,*

$$\hat{\beta}_{LASSO} = \arg\min_{\beta} \left\{ \beta_j^2 \cdot \|x_j\|_2^2 - 2\beta_j \|x_j\|_2^2 y + \lambda \|\beta_j\| \right\}.$$

Notice that while the absolute-value function is not differentiable at zero, we can consider three cases under which we can derive three separate formulations for our LASSO estimate. That is, we can evaluate β_j for when it is $\beta_j > 0$, $\beta_j < 0$ and $\beta_j = 0$.

Consider when $\beta_j > 0$. Then the absolute-value function is simply $|\beta_j| = \beta_j$ so the one-dimensional LASSO objective we derived above becomes,

$$\|x_j\|_2^2 \beta_j^2 - 2\beta_j x_j^\top y + \lambda \beta_j.$$

Since this expression is differentiable for $\beta_j > 0$, we differentiate wrt β_j and set it equal to zero,

$$\begin{aligned}
\frac{\partial}{\partial \beta_j} \left(\|x_j\|_2^2 \cdot \beta_j^2 - 2\beta_j x_j^\top y + \lambda \beta_j \right) = 0 &\iff 2\|x_j\|_2^2 \beta_j - 2x_j^\top y + \lambda = 0 \\
&\iff 2\|x_j\|_2^2 \beta_j = 2x_j^\top y - \lambda \\
&\iff \|x_j\|_2^2 \beta_j = x_j^\top - \frac{\lambda}{2} \\
&\iff \beta_j = \frac{x_j^\top y - \lambda/2}{\|x_j\|_2^2},
\end{aligned}$$

which is only valid if and only if the term in the last line is greater than 0. Since the squared ℓ_2 norm $\|x_j\|_2^2 > 0$ for any non-zero predictor, the denominator is strictly positive. Hence,

$$x_j^\top y - \frac{\lambda}{2} > 0 \iff x_j^\top y > \frac{\lambda}{2}.$$

Now suppose that $\beta_j < 0$. Then, $|\beta_j| = -\beta_j$ and therefore the one-dimensional LASSO objective becomes,

$$\|x_j\|_2^2 \beta_j^2 - 2\beta_j x_j^\top y - \lambda \beta_j.$$

Taking the derivative wrt β_j of the above and setting to zero,

$$\begin{aligned}
\frac{\partial}{\partial \beta_j} \left(\|x_j\|_2^2 \beta_j^2 - 2\beta_j x_j^\top y - \lambda \beta_j \right) = 0 &\iff 2\|x_j\|_2^2 \beta_j - 2x_j^\top y - \lambda = 0 \\
&\iff 2\|x_j\|_2^2 \beta_j = 2x_j^\top y + \lambda \\
&\iff \|x_j\|_2^2 \beta_j = x_j^\top y + \frac{\lambda}{2} \\
&\iff \beta_j = \frac{x_j^\top y + \frac{\lambda}{2}}{\|x_j\|_2^2}.
\end{aligned}$$

Again, this candidate solution is valid only when it satisfies the assumption under which is was derived, that is $\beta_j < 0$. Since $\|x_j\|_2^2 > 0$ this is equivalent to $x_j^\top y + \frac{\lambda}{2} < 0$ which also means that $x_j^\top y < -\frac{\lambda}{2}$. The remaining case is when $\beta_j = 0$, where the absolute-value function is not differentiable. Therefore we cannot simply take an ordinary derivative and substitute $\beta_j = 0$.

We therefore shall make note of the following theorem.

Theorem 4.15. *The subdifferential of the absolute-value function evaluated at β_j is given by:*

$$\partial|\beta_j| = \begin{cases} \{1\}, & \beta_j > 0, \\ [-1,1], & \beta_j = 0, \\ \{-1\}, & \beta_j < 0. \end{cases}$$

Proof. Evaluating at the three intervals gives our result. □

At $\beta_j = 0$, therefore, the absolute-value function $|\beta_j|$ is not differentiable. We therefore cannot simply take an ordinary derivative and substitute $\beta_j = 0$. Instead, by applying Theorem 5.4, we can reduce our objective to,

$$\|x_j\|_2^2\beta_j^2 - 2\beta_j x_j^\top y + \lambda|\beta_j| \iff 0 \in 2\|x_j\|_2^2\beta_j - 2x_j^\top y + \lambda\partial|\beta_j|.$$

When $\beta_j = 0$ we know that $\partial|\beta_j| = [-1,1]$ so it follows that,

$$0 \in 2\|x_j\|_2^2(0) - 2x_j^\top y + \lambda[-1,1] = -2x_j^\top y + [-\lambda, \lambda].$$

For zero to belong to this interval, there must exist some $s \in [-1,1]$ such that $-2x_j^\top y + \lambda s = 0$. Equivalently, this is simply where $s = \frac{2x_j^\top y}{\lambda}$. Since $s \in [-1,1]$ we require that,

$$-\frac{\lambda}{2} \leq \frac{2x_j^\top y}{\lambda} \leq 1 \iff -\frac{\lambda}{2} \leq x_j^\top y \leq \frac{\lambda}{2}.$$

Therefore, our closed-form LASSO estimator for mutually orthogonal predictors is simply,

$$\hat{\beta}_{\lambda,j} = \begin{cases} \dfrac{x_j^\top y - \lambda/2}{\|x_j\|_2^2}, & x_j^\top y > \dfrac{\lambda}{2}, \\ 0, & -\dfrac{\lambda}{2} \leq x_j^\top y \leq \dfrac{\lambda}{2}, \\ \dfrac{x_j^\top y + \lambda/2}{\|x_j\|_2^2}, & x_j^\top y < -\dfrac{\lambda}{2} \end{cases}$$

where the predictors x_j that comprise of the columns of our design matrix X are orthogonal or orthogonalized via the Gram-Schmidt orthogonalization procedure. Thus, we make note of the following theorem under orthogonalization.

Theorem 4.16. *Let $X = [\,\vec{x}_1, \ldots, \vec{x}_p\,] \in \mathbb{M}_{n\times p}(\mathbb{R})$ be a design matrix whose columns are nonzero and mutually orthogonal, so that $\vec{x}_j^\top \vec{x}_k = 0$ for all $j \neq k$.*

Equivalently, $X^\top X = \operatorname{diag}\Big(\|\vec{x}_1\|_2^2, \ldots, \|\vec{x}_p\|_2^2\Big)$. Let $y \in \mathbb{R}^n$ and let $\lambda \geq 0$.

Consider the LASSO optimization problem $\hat{\beta}_\lambda = \arg\min_{\beta\in\mathbb{R}^p}\left\{\|y - X\beta\|_2^2 + \lambda\|\beta\|_1\right\}$. *Then the objective function is separable across the coordinates* $\beta_1, \ldots, \beta_p$, *and the unique minimizer* $\hat{\beta}_\lambda = (\hat{\beta}_{\lambda,1}, \ldots, \hat{\beta}_{\lambda,p})^\top$ *satisfies, for each* $j \in \{1, \ldots, p\}$,

$$\hat{\beta}_{\lambda,j} = \begin{cases} \dfrac{\vec{x}_j^\top y - \lambda/2}{\|\vec{x}_j\|_2^2}, & \vec{x}_j^\top y > \dfrac{\lambda}{2}, \\ 0, & -\dfrac{\lambda}{2} \le \vec{x}_j^\top y \le \dfrac{\lambda}{2}, \\ \dfrac{\vec{x}_j^\top y + \lambda/2}{\|\vec{x}_j\|_2^2}, & \vec{x}_j^\top y < -\dfrac{\lambda}{2}. \end{cases}$$

4.5 Ridge vs. LASSO Regression

Consider the Ridge and LASSO optimization problems for a moment,

$$\hat{\beta}_{\text{ridge}} = \arg\min_\beta \left\{\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2\right\},$$
$$\hat{\beta}_{\text{LASSO}} = \arg\min_\beta \left\{\|y - X\beta\|_2^2 + \lambda\|\beta\|_1\right\}.$$

As we saw before, LASSO only has a closed-form solution when the predictors exhibit orthogonality because it imposes a penalty equivalent to the absolute-value of the coefficient. In the contrary, Ridge regression always has a solution, regardless of whether or not our design matrix X has full column rank since, in contrast to LASSO, imposes an ℓ_2 penalty on the size of the coefficients, making it always differentiable. These characteristicis of LASSO and Ridge have impacts on their physical/geometric properties starting with how they shrink coefficients.

Consider the constraint forms of Ridge and LASSO,

$$\hat{\beta}_{\text{ridge}} = \arg\min_\beta \left\{\|y - X\beta\|_2^2 : \sum_{j=1}^p \beta_j^2 \le t\right\},$$
$$\hat{\beta}_{\text{LASSO}} = \arg\min_\beta \left\{\|y - X\beta\|_2^2 : \sum_{j=1}^p |\beta_j| \le t\right\},$$

where t controls the size of the admissible coefficient region. Thus, a smaller value of t corresponds to stronger regularization. In two dimensions, with coefficients β_1 and β_2, the Ridge constraint is equivalent to, $\beta_1^2 + \beta_2^2 \le t$ which forms a circular/ellipsoid region centered at $(0, 0)$. More generally, we refer to the solution space of Ridge as an ℓ_2-ball. In contrast, the LASSO constraint is instead, $|\beta_1| + |\beta_2| \le t$, which forms a circular region centered at the origin. More generally, we refer to it as an ℓ_1-ball. Notice that the ridge boundary is smooth everywhere whereas the LASSO boundary has corners lying directly on the coordinate axes.

We can generalize the concept of ℓ_q ball sparsity in the following theorem.

Theorem 4.17 (ℓ_q-Ball)**.** *Let $q > 0$ and let $t \geq 0$.*

For $\beta = (\beta_1, \ldots, \beta_p)^\top \in \mathbb{R}^p$, the ℓ_q-ball of radius t centered at the origin is defined by

$$B_q(t) = \left\{\beta \in \mathbb{R}^p : \|\beta\|_q \leq t\right\} = \left\{\beta \in \mathbb{R}^p : \left(\sum_{j=1}^{p} |\beta_j|^q\right)^{1/q} \leq t\right\}.$$

Equivalently,

$$B_q(t) = \left\{\beta \in \mathbb{R}^p : \sum_{j=1}^{p} |\beta_j|^q \leq t^q\right\}.$$

For $q \geq 1$, $\|\cdot\|_q$ is a norm and $B_q(t)$ is a convex set.

Now, consider the ordinary least-squares objective defined as $\|y - X\beta\|_2^2$. Expanding around the unconstrained OLS solution we can write the least squares objective as,

$$\|y - X\beta\|_2^2 = \|y - X\beta\|_2^2 + (\beta - \hat{\beta}_{\text{OLS}})^\top X^\top X(\beta - \hat{\beta}_{\text{OLS}}).$$

Fixing the value of the residual sum of squares gives contours satisfying the equation,

$$(\beta - \hat{\beta}_{\text{OLS}})^\top X^\top X(\beta - \hat{\beta}_{\text{OLS}}) = c.$$

These are precisely ellipses centered at the OLS solution. Without regularization, we can simply move toward the center of these ellipses until we reach $\hat{\beta}_{\text{OLS}}$. With a constraint, however, we are only allowed to choose coefficient vectors inside the permitted region. Therefore, we expand the least-squares ellipse inward until it first touches the constraint region.

For Ridge, the regularized solution occurs when one of the elliptical least-squares contours is tangent to the circular boundary defined by $\beta_1^2 + \beta_2^2 = t$. There is nothing geometrically unique when $\beta_1 = 0$ or $\beta_2 = 0$ since they are simply ordinary smooth points on the circle. There is therefore essentially no reason for the tangency point to occur exactly at one of the coordinate aces.

In analytical solutions, Ridge typically moves the solution toward the origin while keeping both coordinates non-zero that is $(\hat{\beta}_1^{\text{OLS}}, \hat{\beta}_2^{\text{OLS}}) \to (\hat{\beta}_1^{\text{ridge}}, \hat{\beta}_2^{\text{ridge}})$ with $|\hat{\beta}_j^{\text{ridge}}| < |\hat{\beta}_j^{\text{OLS}}|$ thus producing shrinkage but not sparsity. Algebraically, we previously saw for some diagonal $X^\top X$ gram matrix,

$$\hat{\beta}_{\lambda,j}^{\text{ridge}} = \frac{x_j^\top y}{\|x_j\|_2^2 + \lambda} = \frac{\|x_j\|_2^2}{\|x_j\|_2^2 + \lambda} \cdot \hat{\beta}_{\text{OLS}}.$$

Therefore, if $\hat{\beta}_j^{\text{OLS}} \neq 0$ then ordinarily the ridge solution $\hat{\beta}_{\lambda,j}^{\text{ridge}} \neq 0$. The coefficient gets smaller and smaller as λ increases but only approaches zero asymptotically. Thus, Ridge shrinks toward zero rather than thresholding at zero as $\lambda \to \infty$.

Comparing this to LASSO